EDUTERIA · 72nd BPSC PRELIMS · PHYSICS

PHYSICS

Consolidated Class Notes — A4 Edition

All 19 lectures of Physics by Sakshi Ma'am — re-arranged topic-wise, duplicates merged, every chart & diagram kept, and every BPSC physics PYQ (56–59th → 71st) mapped to its unit.

16 UNITS + PYQ TOPIC-PRIORITY MAP + FULL PYQ APPENDIX

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Re-arranged, de-duplicated & PYQ-prioritised edition

Source: 19 lecture PDFs in “physic notes eduteria/” (84 pages) · Physics by Sakshi Ma'am · Eduteria
PYQ data: 56–59th (2015) paper + BPSC master question bank 60–62nd … 71st (2025)
Blue boxes = added by the editor (not in the lectures) · coral strips = source corrections Print-ready A4 · 2026

About this edition

These are the **complete class-wise physics notes of Sakshi Ma'am (Eduteria, 72nd BPSC batch)** — all 19 lecture PDFs — re-arranged into 16 topic-wise units for A4 printing. **Every fact, table, formula, example, chart and diagram of the original notes is kept**; only the repetition is gone. Lecture cover pages, watermarks, running headers/footers and the contact strip were removed.

Where a topic appeared in more than one lecture, the most complete version is kept and the repeats are merged into it; a green **"De-dup note"** strip records exactly what was merged or moved, so nothing is lost silently. Formulas that came out garbled in the source typesetting were re-typed from the rendered pages. Obvious slips in the lectures are fixed *visibly* in coral **"Source correction"** strips — the original wording is always quoted.

What was added by the editor is always marked. Anything that is not in the lecture PDFs — an extra data table, a derivation the notes skipped, a definition needed to make a PYQ answerable — sits in a **blue "ADDED BY EDITOR" box**. Amber **"PYQ lens"** strips show how BPSC has actually asked the concept, quoting the exam and question number.

#	Source lecture (PDF in "physic notes eduteria/")	Pages	Now in
1	01 Introduction to Physics (course plan + light intro + reflection)	3	Front matter, Units 2 & 4
2	02 Light Part-01 (colour mixing, plane & spherical mirrors)	3	Units 2 & 4
3	03 Light Part-02 (mirror formula, refraction, TIR)	3	Units 2 & 3
4	04 Light Part-03 (lenses, power, human eye, defects)	3	Unit 3
5	05 Light Part-04 (dispersion, scattering, wave optics, instruments)	4	Unit 4
6	06 Wave (EM waves, EM spectrum table)	2 (+1 blank)	Unit 5
7	07 Sound Wave	4	Units 5 & 6
8	08 Heat-01 (heat, transfer modes, specific/latent heat)	3	Unit 7
9	09 Temperature (state change, scales, thermometers)	3	Unit 7
10	10 Heat Part-3 (expansion, radiation laws, thermodynamics)	4	Unit 8
11	11 Kinematics	3	Unit 9
12	12 Motion Part-02 (graphs, Newton's laws, Kepler, friction)	5	Units 9, 10 & 11
13	13 Circular Motion & Gravitation	3	Unit 11
14	14 Gravitation & Mechanical Properties of Fluids	4	Units 11 & 12
15	15 Electricity Part-01 (charge, Coulomb, field, potential, current)	2	Unit 13
16	16 Electricity Part-02 (Ohm's law, resistance, combinations)	2	Unit 13
17	17 Electricity Part-03 (power, effects, instruments, devices, lamps, safety)	4	Unit 14
18	18 Magnetism	3	Unit 15
19	19 Unit & Dimension, Photoelectric, WPE, Nuclear, Measurement	6	Units 1, 10 & 16

Lecture-1 course plan (as printed in the source — kept for reference)

Block	Topics announced in Lecture 1
Light (1)	Nature and properties of light, reflection of light and mirrors, types of mirrors and their uses.
Light (2)	Refraction of light, total internal reflection, lenses and their types, uses of lenses and correction of eye defects.
Light (3)	Scattering of light, dispersion of light, polarization, interference and their examples.
Waves	Types of waves, mechanical waves and their properties, non-mechanical waves (electromagnetic waves), their properties and uses.
Sound waves	Speed of sound waves, echo, Mach number, shock waves, Doppler effect.
Heat (1)	Heat and temperature, thermometer and its scale, methods of heat conduction, thermal expansion and its examples.

Block	Topics announced in Lecture 1
Heat (2)	Anomalous behaviour of water, specific heat, latent heat, evaporation and humidity (relative humidity), precipitation.
Motion	Basic elements of motion – distance, displacement, speed, velocity, and acceleration, force and types of force, Newton's laws of motion.
Types of motion	Linear motion, uniform circular motion, rotational motion, projectile motion, periodic motion, oscillatory motion, simple harmonic motion, etc.
Universal gravitation	Gravitational force and gravity, Newton's law of gravitation, gravitational acceleration, changes in the value of g, orbital velocity, escape velocity, Kepler's planetary laws.
Mechanical properties of fluids	Cohesive and cohesive forces, fluid pressure and Pascal's law, surface tension and capillarity, streamline flow and Bernoulli's theorem, buoyancy and Archimedes' principle.
Electricity (1)	Charge, electric field, electric field intensity, electric potential, potential difference, electric capacitance and capacitors, electric current, Ohm's law and resistance.
Electricity (2)	Electric power, commercial units (kilowatt-hour units), electromagnetic induction, electric motors and generators.
Magnetism	Magnetic field, magnetic flux, magnetic permeability, magnetic susceptibility, types of magnetic materials, geomagnetism.

↻ **De-dup note:** Several items of this printed plan (Mach number, shock waves, Doppler effect, anomalous expansion of water, humidity, capacitors, electromagnetic induction, flux, permeability, geomagnetism) never received their own lecture; the missing ones that BPSC has asked about are supplied in clearly marked blue editor boxes inside the relevant unit.

Major duplicate topics resolved

- ✔ Scattering of light & Rayleigh's law (Lecture 1 p. 3 and Lecture 5 p. 2) → kept once in Unit 4; the Lecture-1 formula strip merged there.
- ✔ Reflection of light — Lecture 1 prints the laws on p. 3 and again on p. 4 (regular/diffused) and Lecture 2 repeats the plane-mirror image properties → one sequence in Unit 2.
- ✔ Periscope & kaleidoscope — Lecture 2 (uses of plane mirror) and Lecture 5 (optical instruments) → kept in Unit 4, cross-referenced from Unit 2.
- ✔ Mirror formula & magnification — Lecture 3 opens with them; moved next to spherical mirrors (Unit 2) so the mirror story reads in one place.
- ✔ Lens formula & magnification are printed in Lecture 4 for both convex and concave lenses → one formula block in Unit 3.
- ✔ Heat-transfer modes — Lecture 8 (detailed) and Lecture 9 (comparison table) → both kept together in Unit 7 (table + details).
- ✔ Temperature definition, SI unit and the temperature-scales table — printed in Lecture 8, Lecture 9 p. 2 and again in Lecture 9 p. 4 → one master table (5 scales) in Unit 7.
- ✔ Velocity of sound: Newton's formula and Laplace correction are each printed twice on Lecture 7 p. 3 → one derivation ladder in Unit 6.
- ✔ Kinematics: the distance/displacement and speed/velocity definitions are written twice on Lecture 11 p. 2 → once, plus the comparison table.
- ✔ Newton's first law is stated three times in Lecture 12 (statement, 'Galileo's law of inertia', inertia of rest example printed twice) → one statement + three inertia cases in Unit 10.
- ✔ Kepler's laws (Lecture 12 p. 5) sit between momentum and friction in the source; moved to gravitation (Unit 11).
- ✔ Escape velocity is introduced twice on Lecture 14 p. 2 (and a stray fragment repeats under Pascal's law on p. 3) → one section in Unit 11.
- ✔ Oersted's discovery is narrated twice on Lecture 18 pp. 2–3 → one numbered list in Unit 15.
- ✔ Fleming's rules: the paragraph and the two hand-rule lists on Lecture 18 p. 4 → one table in Unit 15.
- ✔ Lecture 19 prints the dimensional-formula introduction and the 'how to write a dimensional formula' question twice (pp. 3–4) → once in Unit 1; both dimension tables kept (they list different quantities).
- ✔ Lecture 17 lists 'Advantages of CFLs' under LEDs (a heading slip) and repeats the fuse under 'Safety devices' → merged in Unit 14 with a correction strip.

How to read the colour code

De-dup note	what was merged or moved, and from where (nothing is dropped silently).
Source correction	a typo or slip in the lecture text, fixed visibly with the original wording quoted.
ADDED BY EDITOR	material that is NOT in the lecture PDFs (extra data, a missing derivation, a topic BPSC asks but the lectures skipped).
PYQ lens	how BPSC has asked this exact point — exam and question number quoted; full text in Appendix A.

Suggested passes: (1) read the PYQ Topic Priority pages first and mark the A-priority units; (2) study each unit — the amber strips tell you which lines have already been examined; (3) finish with Appendix A, which lists every reviewed physics question with its answer and a one-line reason.

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UNIT ★

PYQ Topic Priority — what BPSC actually asks in physics

Consolidated from: all physics questions of 56–59th (2015), 60–62nd, 63rd … 71st (2025) BPSC Prelims — manually classified

All **1,623 questions** of the eleven papers were read one by one; **95** are physics or physics-adjacent (**63** of them in the last six papers, 66th–71st). Physics is a steady block: **≈10–11 questions per paper** since the 64th BPSC (64th 11, 65th 10, 66th 11, 67th 11, 68th 11, 69th 10, 70th 10, 71st 10) against only 3–4 in the 56–59th, 60–62nd and 63rd papers. Questions are counted against the unit of this book that answers them; 'recent' = 66th–71st. Score = all + recent + 0.5 × papers in which the topic appeared, so a topic that keeps returning ranks above a one-off cluster.

Topic ranking (all papers, recent papers weighted)

Rank	Unit / topic	All Qs	Recent Qs (66–71st)	Papers	Priority
1	U16 Modern & Nuclear Physics	19	9	10 (5 recent)	A
2	U10 Laws of Motion, Friction, Work-Power-Energy	9	9	3 (3 recent)	A
3	U14 Electrical Power, Devices & Domestic Electricity	11	5	5 (2 recent)	B
4	U11 Circular Motion & Gravitation	7	6	5 (4 recent)	A
5	U1 Units, Dimensions & Measuring Instruments	9	2	4 (2 recent)	B
6	U13 Electric Charge, Field & Current Electricity	5	5	3 (3 recent)	B
7	U7 Heat, Temperature & Thermometry	6	3	5 (2 recent)	B
8	U2 Light I: Nature of Light, Reflection & Mirrors	5	3	4 (2 recent)	B
9	U3 Light II: Refraction, Lenses & the Human Eye	4	4	3 (3 recent)	B
10	U6 Sound	4	3	3 (2 recent)	B
11	U15 Magnetism & Electromagnetism	3	3	3 (3 recent)	B
12	Space & Astronomy (extended only)	3	3	2 (2 recent)	B
13	U5 Waves & the Electromagnetic Spectrum	3	3	2 (2 recent)	B
14	U9 Kinematics	3	2	2 (1 recent)	C
15	U8 Thermal Expansion, Radiation Laws & Thermodynamics	2	2	2 (2 recent)	C
16	U12 Mechanical Properties of Fluids	1	1	1 (1 recent)	C
17	U4 Light III: Dispersion, Scattering, Wave Optics & Instruments	1	0	1 (0 recent)	C

Priority A = asked in ≥6 of the last-six-paper questions, or ≥9 questions overall with presence in ≥3 recent papers. **B** = ≥3 recent questions or ≥6 overall. **C** = occasional. Unit numbers refer to the units of this book.

What the pattern says — read this before the units

- **Modern & nuclear physics (Unit 16) is the single most examined block** — 19 questions across 10 of 11 papers: photoelectric effect (Einstein's Nobel 64th; photoelectric cell 69th; photodiode 68th), nucleus & nuclide arithmetic (65th, 66th, 67th), alpha/cosmic radiation, Higgs boson, fusion breakthrough, Manhattan Project, Thomson's atomic model (70th). Expect 1–2 questions every year.
- **Mechanics has become numerical since the 68th paper.** Laws of motion / friction / work-energy (Unit 10) drew 9 questions in the last four papers alone — the 71st paper set four short sums (F = ma, W = F·s, friction at constant velocity, energy conservation) plus a kinematics one ($s = ½at^2$). Practise the one-line formulas in Units 9–10.
- **Current electricity & household electricity (Units 13–14)** are asked every year: units of power (watt, kWh → joule), ammeter/voltmeter, motor & dynamo energy conversions, Ohm's law forms, parallel-resistance sum (66th), semiconductor resistance vs temperature, short circuit & fuse (71st), Faraday constant, tesla

(70th).

- **Units, dimensions & instruments (Unit 1)** gave a cluster of 9 questions in the 64th–65th papers (angstrom, hertz, pressure unit, strain, hygrometer, light-year, odometer). They are cheap marks — memorise the SI table, the 'no-unit' quantities and the instrument list.
- **Optics is asked in small, conceptual bites** — spectrum range (3900–7600 Å, 63rd), red vs violet wavelength (64th), speed of light in media (67th), frequency unchanged on refraction (66th), concave-mirror image at C (69th), colour mixing (69th), plane-mirror focal length (71st), black strips on a lens (68th). Wave optics/instruments (Unit 4) appears rarely — read once, keep the rainbow angles and scattering law.
- **Heat & sound**: Celsius→Fahrenheit conversion twice (64th, 65th), poorest heat conductor (66th), radiation needs no medium (70th), sound is longitudinal (65th), pitch ↔ frequency (68th), speed of sound in solids (71st). The thermodynamics laws, Stefan/Wien and thermal expansion (Unit 8) have appeared only through the solar constant (70th) and the Thomson effect (67th).
- **Gravitation (Unit 11)** keeps recurring in reasoning form: weight at the equator if the spin increases (67th), free fall on the Moon (69th), centripetal force (68th), centrifugation in a washing machine (67th), comets' orbits (70th).
- Physics-flavoured current affairs (LIGO-India, fusion ignition, space missions, pulsars) count as 'extended' — glance at them in Appendix A; they are not in the lectures and are marked as editor additions where mentioned.

Paper-by-paper physics load

Paper	56-59th	60-62nd	63rd	64th	65th	66th	67th	68th	69th	70th	71st
Physics Qs	4	3	4	11	10	11	11	11	10	10	10

Most-repeated question stems (learn these cold)

- ★ Unit of electric power → watt (64th Q11, 65th Q120); device to measure current → ammeter (64th Q14, 65th Q119).
- ★ °C → °F: 40 °C = 104 °F (64th Q10); 50 °C = 122 °F (65th Q116). Formula F = 9C/5 + 32.
- ★ Frequency is measured in hertz (64th Q4, 65th Q117); angstrom = unit of wavelength (64th Q3); light-year = distance (66th Q5).
- ★ Unit of pressure: N/m^2 (64th Q20), kg/cm^2 (65th Q111). Strain has no unit (65th Q114); pressure is a scalar (65th Q113); speed is not a vector (67th Q67).
- ★ Photoelectric effect: Einstein's Nobel (64th Q16); photoelectric cell converts light → electrical energy (69th Q16); photodiode for digital use (68th Q113).
- ★ Infrared radiation for muscle/body ache (66th Q1, 70th Q45) — the same fact twice in five papers.
- ★ Electric motor: electrical → mechanical (64th Q12); dynamo produces AC (69th Q26); transformer works only on AC; fuse = protective device (71st Q130).
- ★ Nucleus arithmetic: mass number = p + n (65th Q92); neutrons in ${}_{94}Pu^{242}$ = 148 (66th Q9); nucleus = protons + neutrons (67th Q75).

UNIT 1

Units, Dimensions & Measuring Instruments

Consolidated from: Lecture 19 (Unit & Dimension, Measurement) pp. 2–4 and p. 7; instrument list of Lecture 17 p. 3

★ **PYQ lens:** 9 questions across 64th–70th BPSC come straight from this unit: angstrom = unit of wavelength (64th Q3), frequency in hertz (64th Q4, 65th Q117), unit of pressure N/m² (64th Q20) & kg/cm² (65th Q111), strain has no unit (65th Q114), hygrometer measures humidity (64th Q19), light-year = distance (66th Q5), odometer = distance moved by a vehicle (70th Q6). Priority B — cheap, repeatable marks.

Units and measurement

- To measure a physical quantity like length, mass and time we require a **standard of measurement**. This standard is called the **unit** of that physical quantity — e.g. the unit of length is the metre, and a standard length of 1 metre has a precise definition.
- To measure the length of an object we determine how many times this standard metre is contained in the length of the object.
- **Measurement** = the comparison of a physical quantity with a standard quantity.

Physical quantities

- Those quantities which can describe the laws of physics are called physical quantities. A physical quantity is one that can be measured — length, mass, time, pressure, temperature, current and resistance are physical quantities.
- **Fundamental (base) quantities**: physical quantities that are independent of each other.
- **Derived quantities**: all other quantities, which can be expressed in terms of the fundamental quantities.

Units — properties, types and systems

Unit = the reference standard used to measure a physical quantity. A good unit:

- should be of suitable size and well-defined;
- should be easily reproducible (should not change with place) and must not change with time;
- should not change with physical conditions like temperature, pressure, etc.;
- must be easily comparable experimentally with similar physical quantities.
- **Fundamental units** — units defined for the fundamental quantities. **Derived units** — units of all other quantities, derived from the fundamental units.

System	Length / Mass / Time	Remark
FPS	foot / pound / second	British system
CGS	centimetre / gram / second	Gaussian system
MKS	metre / kilogram / second	Forerunner of SI
SI	metre / kilogram / second (+ K, A, mol, cd)	Used in all measurements throughout the world; based on **seven base units and two supplementary units**

SI base units

Quantity	Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Temperature	kelvin	K
Electric current	ampere	A
Amount of substance	mole	mol
Luminous intensity	candela	cd

Supplementary quantity	Unit	Symbol
Plane angle	radian	rad
Solid angle	steradian	sr

✎ **Source correction:** The source table prints the symbols as "M", "Kg", "S", "Mol" (capitalised). SI symbols are case-sensitive: m, kg, s, mol (K, A and cd are correct as printed).

Definitions of the base units (as given in the lecture)

- **Metre (m)**: the distance travelled by light in vacuum during a time interval of 1/299 792 458 second.
- **Kilogram (kg)**: the mass of a platinum-iridium cylinder kept at the International Bureau of Weights and Measures, Sèvres (Paris).
- **Second (s)**: the time taken by the light of a specified wavelength emitted by a caesium-133 atom to execute 9 192 631 770 vibrations.
- **Ampere (A)**: the current which, when passed through two straight parallel conductors of infinite length and negligible cross-section kept 1 metre apart in vacuum, produces between them a force of 2×10^{-7} newton per metre length.
- **Kelvin (K)**: the fraction 1/273.16 of the thermodynamic temperature of the triple point of water.
- **Candela (cd)**: 1/60th of the luminous intensity of 1 square centimetre of a perfect black body maintained at the freezing temperature of platinum (1773 °C).
- **Mole (mol)**: the amount of substance that contains as many elementary units as there are atoms in 0.012 kg of carbon-12.
- **Radian (rad)**: the angle subtended at the centre of a circle by an arc whose length equals the radius.
- **Steradian (sr)**: the solid angle subtended at the centre of a sphere by a surface of area equal to the square of its radius.

✎ **Source correction:** Source prints "kept at the National Bureau of Weights and Measurements, Paris" (it is the *International* Bureau, BIPM), "1/273.6 of the thermodynamic temperature" (correct fraction is **1/273.16**), and "kept at 1A. meter apart" (read: 1 metre apart).

▌ADDED BY EDITOR — not in source lectures▐ 2019 redefinition of the SI (context, not in the lectures)

Since 20 May 2019 all seven base units are defined through fixed values of natural constants — the kilogram via Planck's constant $h = 6.626\ 070\ 15 \times 10^{-34}$ J s (Kibble balance), the ampere via the elementary charge $e = 1.602\ 176\ 634 \times 10^{-19}$ C, the kelvin via Boltzmann's constant $k = 1.380\ 649 \times 10^{-23}$ J/K and the mole via Avogadro's number $6.022\ 140\ 76 \times 10^{23}$. The platinum-iridium cylinder ("Le Grand K") is therefore no longer the definition of the kilogram, though the lecture's older definitions are still what most objective papers quote.

Dimensional formula

The dimensional formula of a physical quantity tells which of the fundamental units have been used for its measurement, and to what power.

How is the dimensional formula of a physical quantity written?

- (1) Write the formula of the quantity, with the quantity on the left-hand side.
- (2) Express all quantities on the right-hand side in terms of fundamental quantities — mass, length and time.
- (3) Replace mass, length and time by M, L and T respectively.
- (4) Write the powers of the terms.

↻ **De-dup note:** Lecture 19 prints this introduction and the four-step question twice (bottom of p. 3 and top of p. 4). Kept once.

Characteristics of dimensions

- Dimensions do not depend on the system of units.
- Only quantities with the same dimensions can be added or subtracted.
- Dimensions can be obtained from the units of the physical quantity and vice versa.

- Two different quantities can have the same dimension (e.g. work and torque).
- When two quantities are multiplied or divided, their dimensions multiply/divide to give the dimension of the third quantity.

Uses of dimensional formulae

- (1) To check the correctness of an equation. (2) To convert a unit from one system to another. (3) To deduce the relation connecting physical quantities.

Worked dimensional formulae (Lecture 19 p. 3 — re-typed from the rendered page)

Physical quantity	Working	Unit	Dimension
Distance / displacement	—	m	$M^0L^1T^0$
Speed / velocity	distance ÷ time	m/s	$M^0L^1T^{-1}$
Acceleration (a)	Δv ÷ t = (m/s) ÷ s	m/s^2	$M^0L^1T^{-2}$
Momentum (p)	mass × velocity	kg m/s	$M^1L^1T^{-1}$
Force (F)	mass × acceleration (m × a)	$kg\ m/s^2$ = newton	$M^1L^1T^{-2}$
Work (W)	F · S (force × displacement) = $kg\ m/s^2 \times m$	joule	$M^1L^2T^{-2}$
Gravitational constant (G)	$G = F r^2/(m_1 m_2) = (kg\ m/s^2)\cdot m^2/kg^2 = m^3/(kg\ s^2)$	$N\ m^2/kg^2$	$M^{-1}L^3T^{-2}$
Electrostatic constant (k)	$F = k\, q_1q_2/r^2 \Rightarrow k = F r^2/(q_1q_2)$; q = It (C = A s) ⇒ $kg\ m^3/(s^4A^2)$	$N\ m^2/C^2$	$M^1L^3T^{-4}A^{-2}$
Torque (τ) — similar to work	r × F (length × force) = $m \times kg\ m/s^2$	N m	$M^1L^2T^{-2}$
Impulse (I) — change in momentum	Δp	kg m/s = N s	$M^1L^1T^{-1}$
Strain	change in length ÷ original length = m/m	unitless	dimensionless

✎ **Source correction:** Source prints impulse as "$M^1L^2T^{-1}$" and describes it as "change in maximum"; impulse = change in *momentum*, dimension $\mathbf{M^1L^1T^{-1}}$ (matches the second table on p. 4).

Units and dimensions of a few derived quantities (Lecture 19 p. 4)

Physical quantity	Unit	Dimensional formula
Displacement	m	$M^0L^1T^0$
Area	m^2	$M^0L^2T^0$
Volume	m^3	$M^0L^3T^0$
Velocity	$m\ s^{-1}$	$M^0L^1T^{-1}$
Acceleration	$m\ s^{-2}$	$M^0L^1T^{-2}$
Density	$kg\ m^{-3}$	$M^1L^{-3}T^0$
Momentum	$kg\ m\ s^{-1}$	$M^1L^1T^{-1}$
Work / Energy / Heat	joule ($kg\ m^2/s^2$)	$M^1L^2T^{-2}$
Power	watt (joule/s)	$M^1L^2T^{-3}$
Angular velocity	$rad\ s^{-1}$	$M^0L^0T^{-1}$
Angular acceleration	$rad\ s^{-2}$	$M^0L^0T^{-2}$
Resistance / impedance	ohm ($kg\ m^2/(s\ C^2)$)	$M\ L^2T^{-1}Q^{-2}$
EMF / voltage / potential	volt ($kg\ m^2/(s^2\ C)$)	$M\ L^2T^{-2}Q^{-1}$
Permeability	henry/m ($kg\ m/C^2$)	$M\ L\ Q^{-2}$
Permittivity	farad/m ($s^2C^2/(kg\ m^3)$)	$T^2Q^2M^{-1}L^{-3}$
Frequency	hertz (s^{-1})	T^{-1}
Wavelength	m	L^1
Moment of inertia	$kg\ m^2$	$M^1L^2T^0$
Force	newton ($kg\ m/s^2$)	$M^1L^1T^{-2}$
Pressure	newton/m^2 ($kg\ m^{-1}/s^2$)	$M^1L^{-1}T^{-2}$

Physical quantity	Unit	Dimensional formula
Impulse	newton s (kg m/s)	$M^1L^1T^{-1}$
Inertia (moment of)	kg m^2	$M^1L^2T^0$
Electric current	ampere (C/s)	$Q\ T^{-1}$

↻ **De-dup note:** The p. 4 table uses charge Q as a base dimension (an older convention); in SI the base is current A, so resistance = $M\ L^2T^{-3}A^{-2}$, voltage = $M\ L^2T^{-3}A^{-1}$, current = A. Both forms are kept because either may appear in an option.

★ **PYQ lens:** 65th Q114: "Which quantity does not have a unit?" → **Strain** (stress, force, pressure all have units). 65th Q113: "Which is a scalar?" → **Pressure** (force, velocity, acceleration are vectors). 67th Q67: "Which is not a vector?" → **Speed**.

▌ADDED BY EDITOR — not in source lectures ▌ Handy unit facts BPSC keeps asking (not in the lecture text)

1 Å (angstrom) = 10^{-10} m — a unit of **wavelength/atomic size** (64th Q3)

1 light-year = distance light travels in one year ≈ 9.46 × 10^{15} m — a unit of **distance**, not time (66th Q5)

1 parsec ≈ 3.26 light-years

1 astronomical unit (AU) ≈ 1.496 × 10^{11} m (Earth–Sun distance)

1 fermi = 10^{-15} m (nuclear size)

1 micron = 10^{-6} m

1 nautical mile = 1852 m

1 bar = 10^5 Pa ≈ 1 atm (1 atm = 1.013 × 10^5 Pa = 760 mm Hg)

1 kWh = 3.6 × 10^6 J

1 eV = 1.6 × 10^{-19} J

1 hp = 746 W

1 calorie = 4.186 J

1 dyne = 10^{-5} N

1 erg = 10^{-7} J.

▌ADDED BY EDITOR — not in source lectures ▌ Quantities with no unit / same dimensions (exam favourites)

Dimensionless: strain, refractive index, relative density (specific gravity), coefficient of friction, Poisson's ratio, angle (radian is a ratio), efficiency, magnification, dielectric constant (relative permittivity), relative permeability. **Same dimension pairs**: work–energy–torque–heat ($M\ L^2T^{-2}$); impulse–momentum ($M\ L\ T^{-1}$); pressure–stress–Young's modulus–energy density ($M\ L^{-1}T^{-2}$); frequency–angular velocity–velocity gradient–decay constant (T^{-1})

Planck's constant–angular momentum ($M\ L^2T^{-1}$); force constant–surface tension ($M\ T^{-2}$).

Devices and what they measure (Lecture 19 p. 7)

Device	Measures	Device	Measures
Microscope	Magnification (very small objects)	Hydrometer	Density of liquids
Telescope	Distant objects	Lactometer	Density (purity) of milk
Spectrometer	Wavelength	Ammeter	Electric current
Calorimeter	Heat energy	Voltmeter	Voltage (potential difference)
Thermometer	Temperature	Wattmeter	Electric power
Pyrometer	High temperature	Energy meter	Electrical energy (kWh)
Thermocouple	Temperature difference	Galvanometer	Small current (detects current)

Device	Measures	Device	Measures
Hygrometer	Humidity	Ohmmeter	Resistance
Anemometer	Wind speed	Multimeter	Voltage, current & resistance
Wind vane	Wind direction	Potentiometer	Potential difference / emf
Rain gauge	Rainfall	Clamp meter	Current (no contact)
Lux meter	Light intensity	Oscilloscope (CRO)	Voltage signal (waveform)
Sound level meter	Sound (loudness in dB)	Frequency meter	Frequency
Barometer	Atmospheric pressure	Megger	Insulation resistance
Manometer	Gas pressure	Speedometer	Speed (instantaneous)
Aneroid barometer	Pressure (without liquid)	Odometer	Distance travelled
Tachometer	Rotational speed (rpm)	Accelerometer	Acceleration

↻ **De-dup note:** The electrical instruments (galvanometer, ammeter, voltmeter, potentiometer, voltameter, wattmeter) are explained in detail in Unit 14; this table is the Lecture-19 summary list.

★ **PYQ lens:** 64th Q19 hygrometer → humidity (options: hydrometer, hygrometer, pyrometer, lactometer — a classic confusables set). 70th Q6 odometer → distance moved by a vehicle (vs speedometer = speed, chronometer = precise time, ammeter = current). 64th Q14 / 65th Q119: device to measure current → **ammeter** (not voltameter, which measures charge by electrolysis).

▌ADDED BY EDITOR — not in source lectures ▌ More instruments that have appeared in BPSC/UPSC-style objective papers

Seismograph — earthquake waves (Richter/ moment-magnitude scale)

Fathometer / echo-sounder (SONAR) — ocean depth

Altimeter — altitude (an aneroid barometer calibrated in height)

Audiometer — hearing / intensity of sound

Sphygmomanometer — blood pressure

Stethoscope — heart & lung sounds

Periscope — seeing over obstacles (submarines)

Radar — position/speed of distant objects by radio waves

Lidar — distance by laser

Bolometer — radiant heat (mentioned in Lecture 8)

Photometer — luminous intensity

Viscometer — viscosity

Dynamometer — force/torque/power of engines

Cryometer — very low temperatures

Chronometer — accurate time (navigation)

Sextant — angle between horizon and a star (latitude)

Geiger–Müller counter — radioactivity

Dosimeter — radiation dose

Pedometer — steps walked

Salinometer — salinity

Nephelometer — turbidity/cloudiness

Planimeter — area of a plane figure

Screw gauge (least count 0.01 mm) and vernier callipers (0.1 mm) — small lengths/diameters.

UNIT 2

Light I: Nature of Light, Reflection & Mirrors

Consolidated from: Lecture 1 pp. 3–4 (light intro, reflection), Lecture 2 (colour mixing, plane & spherical mirrors), Lecture 3 p. 2 (mirror formula, sign convention, magnification)

★ **PYQ lens:** Optics is asked in small conceptual bites: visible-spectrum range 3900–7600 Å (63rd Q85), time for sunlight to reach Earth (65th Q112), concave-mirror image of the same size → object at C (69th Q15), colour mixing — magenta/teal/mauve/cyan (69th Q40), focal length of a plane mirror = infinity (71st Q127). Priority B.

Light and optics

- **Optics** — the branch of physics which deals with the behaviour of light.
- **Light** is a form of energy; it is considered an **electromagnetic wave** — it does not require a medium to travel. It is made up of the visible spectrum, whose wavelength lies between **3800 Å and 7500 Å** (Lecture 1; the standard textbook range is 3900–7600 Å, which is the value BPSC accepted in 63rd Q85).
- **Dual nature of light**: it behaves as a **wave** (interference, diffraction) and as a **particle** (photoelectric effect).
- **Two components of optics** — (i) **Ray (geometrical) optics**: treats light as rays travelling in straight lines (reflection, refraction, mirrors, lenses); (ii) **Wave optics**: treats light as a wave (interference, diffraction, polarisation).

✎ **Source correction:** Lecture 1 prints "(i) Ray optics – Wave nature of light; (ii) Wave optics – Wave nature of light". Ray optics is the *particle/rectilinear-ray* model; wave optics is the wave model.

Rectilinear propagation

- Light travels along a straight-line path in a certain medium or in vacuum; the path changes only where the medium changes. This is **rectilinear (straight-line) propagation**. A bundle of light rays is called a **beam** of light.
- Apart from vacuum and gases, light can travel through some liquids and solids.
  - **Transparent** medium — light travels through it without attenuation over large distances (water, glycerine, glass, clear plastics).
  - **Opaque** medium — light cannot travel through it (wood, metals, bricks).
  - **Translucent** medium — light travels some distance but its intensity reduces rapidly (oiled paper, frosted glass).

Speed of light

$$c\ (\text{vacuum}) = 3 \times 10^8\ \text{m/s} \cdot \text{water} \approx 2.25 \times 10^8\ \text{m/s} \cdot \text{glass} \approx 2 \times 10^8\ \text{m/s}$$

- Light is slowest in the optically densest medium — of air, water, glass and vacuum, its speed is **minimum in glass** (67th Q66).

★ **PYQ lens:** 65th Q112 "Sunlight reaches the Earth in ...": key marked (E), because the exact value is **8 min 20 s (≈ 500 s)** for 1.5×10^{11} m ÷ 3×10^8 m/s — not a round "8 minutes". Moonlight reaches us in ≈ 1.3 s.

Visible spectrum and colour

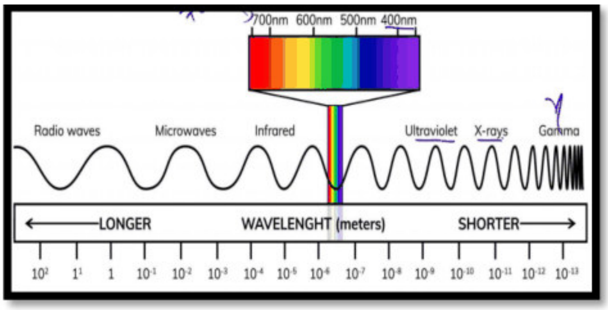

Fig. 2-1 · The electromagnetic spectrum with the visible band expanded (V–I–B–G–Y–O–R) [Lecture 1, p. 3]

- **VIBGYOR**: Violet (λ ≈ 4000 Å) → Indigo → Blue → Green → Yellow → Orange → Red (λ ≈ 7000 Å). Violet has the shortest wavelength and highest frequency; red the longest wavelength.
- (i) Least sensitive colour for the human eye: **violet and red**; (ii) most sensitive colour: **yellow(-green)**; (iii) soothing colour: **green**.

$c = \lambda \times \nu$ (c = speed of light, ν = frequency, λ = wavelength) · $E = h\nu$ (h = Planck's constant)

- Frequency ↑ ⇒ wavelength ↓ (inverse relation). **Frequency is the property that never changes** when light passes from one medium to another (66th Q6) — speed and wavelength change, refractive index differs.

★ **PYQ lens:** 64th Q9: "Wavelength of red light is more than that of violet light" is the correct statement.

↻ **De-dup note:** Lecture 1 also prints "Scattering of light ∝ $1/\lambda^4$ to spread visible spectrum" here. Scattering is treated fully in Unit 4 (Lecture 5) — the Lecture-1 line is merged there.

Colour mixing in light (Lecture 2 p. 2)

- In light, colour intermixing follows **additive mixing** — different coloured lights combine to produce new colours. The **primary colours of light** are **Red (R), Green (G), Blue (B)**.

Mixture	Result (additive)
Red + Green	Yellow
Red + Blue	Magenta
Green + Blue	Cyan
Red + Green + Blue	White light

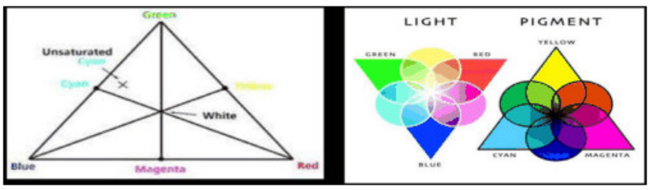

Fig. 2-2 · Additive colour triangle (light) vs subtractive (pigment) mixing [Lecture 2, p. 2]

★ **PYQ lens:** 69th Q40 matched: Magenta = red + blue; Teal = blue, green and white (a light blue-green); Mauve = blue, red and white (pale purple); Cyan = green + blue → answer (A).

▌ADDED BY EDITOR — not in source lectures ▌ Secondary & complementary colours (context for such match questions)

Secondary colours of light = yellow, magenta, cyan. A **complementary pair** adds to white: red–cyan, green–magenta, blue–yellow. **Pigment (subtractive) primaries** are cyan, magenta, yellow (painters call them blue, red, yellow); all three pigments together give black. An object's colour is the colour it *reflects* — a red rose in green light looks black; white reflects all, black absorbs all.

Reflection of light

Reflection = bouncing back of light from a surface when it strikes the boundary of two media (e.g. air and glass) — a part of the light returns into the same medium.

Laws of reflection

- (i) The incident ray, the reflected ray and the normal at the point of incidence all lie in the same plane.
- (ii) Angle of incidence = angle of reflection (∠i = ∠r). The angle between the ray and the *surface* is 90° – i (glancing angle).

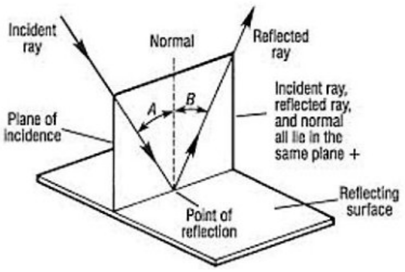

Fig. 2-3 · Incident ray, normal and reflected ray in one plane; ∠i = ∠r [Lecture 1, p. 4]

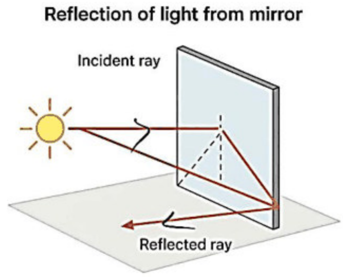

Fig. 2-4 · Reflection of light from a plane mirror — the reflected ray leaves at the same angle [Lecture 1, p. 4]

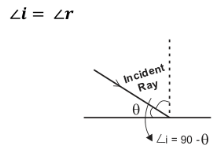

Fig. 2-5 · Lecture-1 sketch: ∠i = ∠r, and the glancing angle is 90° – θ [Lecture 1, p. 3]

Regular vs diffused reflection

- **Regular / specular reflection** — from a perfectly plane, polished surface; parallel incident rays remain parallel after reflection (mirror).
- **Irregular / diffused reflection** — from a rough surface; light is reflected by the bits of its plane surfaces in different directions. This is what lets us see a non-luminous object from any position.

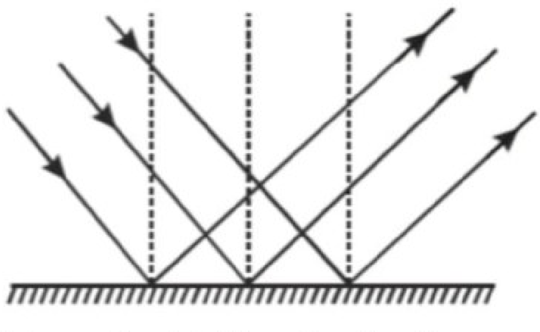

Fig. 2-6 · Regular reflection — parallel rays stay parallel [Lecture 1, p. 4]

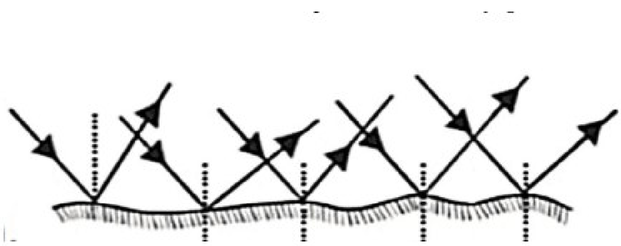

Fig. 2-7 · Diffused reflection from a rough surface [Lecture 1, p. 4]

↻ De-dup note: Lecture 1 prints "Reflection: bouncing back … laws (i)/(ii)" on p. 3 and again "Reflection of light … regular / irregular" on p. 4; Lecture 2 repeats "a mirror is a smooth, highly polished surface that reflects light according to the laws of reflection". All three are merged into this one sequence.

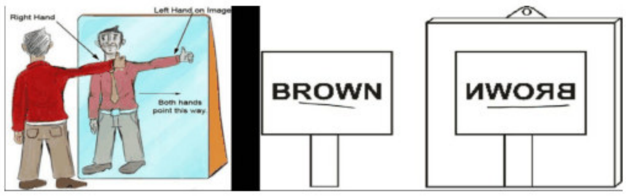

Fig. 2-8 · Everyday reflection: lateral inversion — BROWN reads NWORB in the mirror [Lecture 2, p. 2]

Mirrors

A **mirror** is a smooth, highly polished surface that reflects light according to the laws of reflection.

Plane mirror

- Image is **virtual, erect and of the same size**; image distance = object distance (image is as far behind the mirror as the object is in front); the image is **laterally inverted** (left ↔ right).
- **Focal length of a plane mirror is infinity** ($R = \infty$, $f = R/2 = \infty$); its power is zero (71st Q127).
- **Velocity of image**: if the object moves towards the mirror with velocity U, the image moves towards it with U, so the velocity of the image **with respect to the object** = U + U = **2U**.

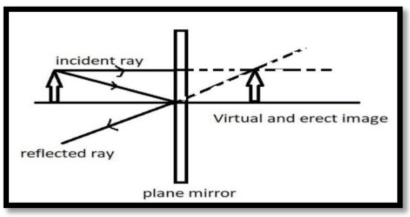

Fig. 2-9 · Image formation in a plane mirror — virtual, erect, same size [Lecture 2, p. 2]

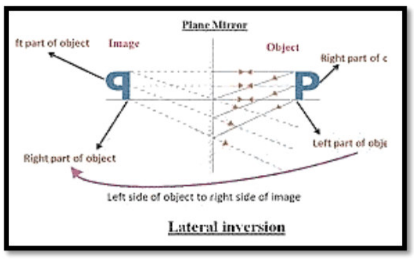

Fig. 2-10 · Lateral inversion — the right part of the object becomes the left part of the image [Lecture 2, p. 2]

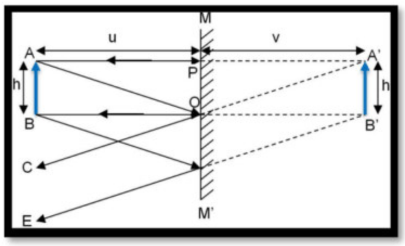

Fig. 2-11 · Ray construction for a plane mirror (Lecture 1): object AB at distance u, image A′B′ at v = u behind the mirror, same height h [Lecture 1, p. 4]

Number of images between two plane mirrors

An object placed between two plane mirrors inclined at angle θ forms multiple images:

$$m = 360° / \theta$$

- **Case I** — m is an **odd** number: number of images = m (irrespective of the object's position)… (as printed; strictly, odd m gives m images only when the object is *symmetric*, m – 1 otherwise — see editor note).
- **Case II** — m is an **even** number (2, 4, 6, 8 …): number of images N = **m – 1**.
- **Case III** — object at the bisector of the two inclined mirrors: N = m – 1 (e.g. θ = 60°: m = 6 → 5 images).
- **Case IV** — 360°/θ comes out as a decimal: take the integer part, e.g. m = 3.8 → n = 3.
- **Format m = n.xyz**: if n is odd → number of images = n; if n is even → n – 1 (e.g. m = 4.8, n = 4 → 3 images; m = 2.3, n = 2 → 1 image).
- **Case V** — mirrors parallel (θ = 0°): **infinite** images due to continuous multiple reflections.

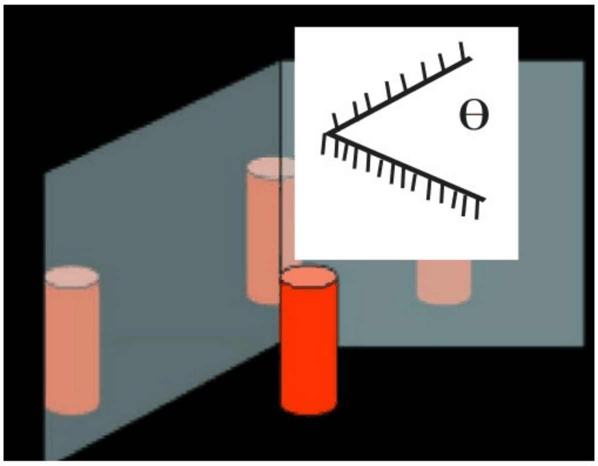

Fig. 2-12 · Two plane mirrors inclined at angle θ — multiple images [Lecture 2, p. 2]

▌ADDED BY EDITOR — not in source lectures ▌ Standard textbook rule (clarifies Cases I–III)

Let m = 360°/θ. If m is **even** → images = m – 1 (always). If m is **odd** → images = m when the object is placed *asymmetrically*, and m – 1 when the object lies on the *bisector* (symmetric position). Quick examples: 90° → 3 images

60° → 5

45° → 7

30° → 11

72° → 4 (m = 5, odd, symmetric → 4; asymmetric → 5).

Minimum size of a plane mirror

- To see one's **full image**, the minimum length of a plane mirror = **H/2** (H = height of the person). This does not depend on the distance from the mirror.
- To see the full image of a **wall behind** you, the minimum mirror length = **H/3** (H = height of the wall/building; person midway between wall and mirror). Example: L = 15/3 = 5 cm (as printed).

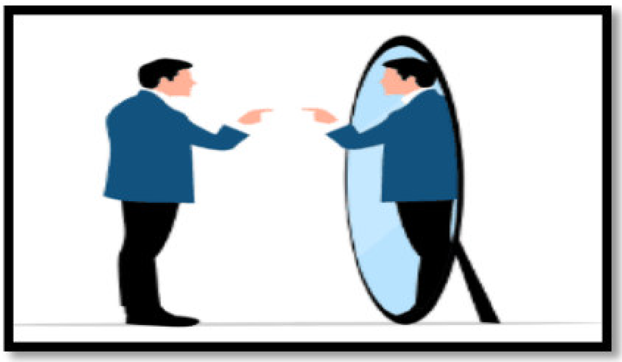

Fig. 2-13 · Full-length image needs a mirror only H/2 tall [Lecture 2, p. 3]

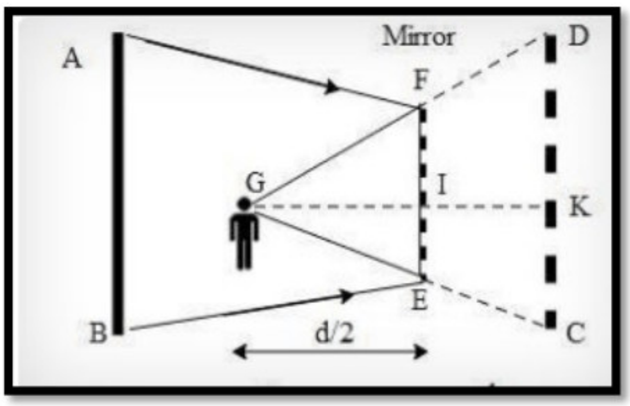

Fig. 2-14 · Seeing the whole wall behind: mirror = H/3 [Lecture 2, p. 3]

Spherical mirrors — definitions

Spherical mirrors are mirrors whose reflecting surface is a part of a sphere. A curved mirror is part of a hollow sphere: if reflection takes place from the **inner** surface the mirror is **concave**; if the **outer** surface reflects, it is **convex**.

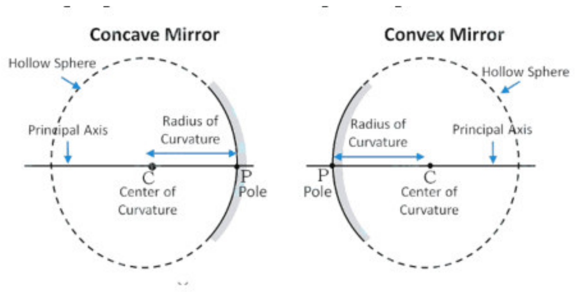

Fig. 2-15 · Concave and convex mirrors cut from a hollow sphere — pole, centre and radius of curvature, principal axis [Lecture 2, p. 4]

- **Pole (P)**: any point on the reflecting surface; for convenience the central point of the mirror.
- **Principal section**: any section of the mirror such as MM′ passing through the pole.
- **Centre of curvature (C)**: the centre of the sphere of which the mirror is a part. **Radius of curvature (R)**: the radius of that sphere.
- **Principal axis**: the line CP joining the pole and the centre of curvature.
- **Principal focus (F)**: the image point on the principal axis for an object at infinity. **Focal length (f)**: the distance PF between pole and focus. **f = R/2**.
- **Linear aperture**: the effective diameter of the light-reflecting area of the mirror. **Focal plane**: the plane through the focus perpendicular to the principal axis.

Concave mirror	Convex mirror
Part of a hollow sphere whose **inside** surface reflects	Part of a hollow sphere whose **outside** surface reflects
Converging in nature	**Diverging** in nature
Generally forms a **real and inverted** image (virtual, erect & magnified only when the object is between P and F)	**Always** forms a **virtual, erect and diminished** image

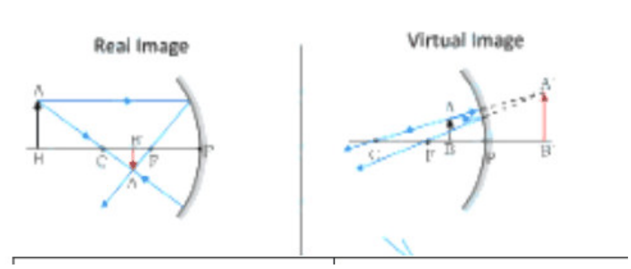

Fig. 2-16 · Real image (concave) vs virtual image (convex) [Lecture 2, p. 3]

Image formation by a concave mirror

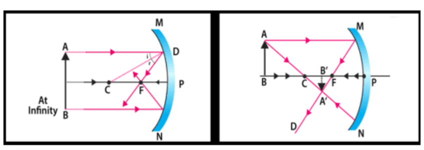

Fig. 2-17 · Object at infinity → image at F; object beyond C → between F and C [Lecture 2, p. 4]

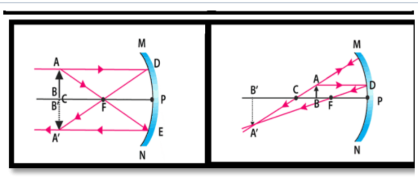

Fig. 2-18 · Object at C → at C, same size; object between C and F → beyond C, magnified [Lecture 2, p. 4]

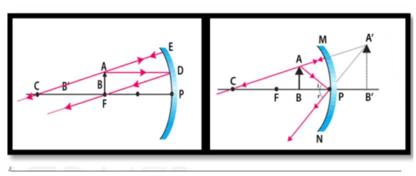

Fig. 2-19 · Object at F → image at infinity; object between P and F → virtual, erect, magnified image behind the mirror [Lecture 2, p. 4]

Sl.	Position of object	Position of image	Nature of image	Size of image
1	At infinity	At focus (F)	Real & inverted	Highly diminished (point-sized)
2	Beyond centre of curvature (beyond C)	Between F and C	Real & inverted	Diminished

Sl.	Position of object	Position of image	Nature of image	Size of image
3	At centre of curvature (at C)	At C	Real & inverted	Same size
4	Between C and F	Beyond C	Real & inverted	Magnified
5	At focus (F)	At infinity	Real & inverted	Infinitely large
6	Between F and pole (P)	Behind the mirror	Virtual & erect	Magnified

★ **PYQ lens:** 69th Q15: concave mirror, image real, inverted and same size as the object → object is **at the centre of curvature** (row 3).

Convex mirror

- Always forms a **virtual, erect, diminished** image between P and F behind the mirror (object at infinity → point image at F; object anywhere else → between P and F).

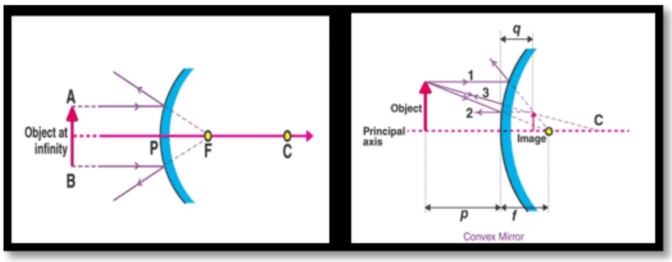

Fig. 2-20 · Image formation by a convex mirror — object at infinity and at a finite distance [Lecture 2, p. 4]

Uses of mirrors

- **Plane mirror**: looking/dressing mirror (same-size image); **periscope** (submarines, tanks); **kaleidoscope** (multiple reflection patterns); optical instruments (reflecting devices).
- **Concave mirror**: shaving/make-up mirror (enlarged image, object close); dentist's mirror (magnified view of teeth); reflectors of torches, search-lights and **car headlights** (bulb at focus → parallel beam); **solar cooker / furnace** (concentrates sunlight at the focus).
- **Convex mirror**: **rear-view mirror** of vehicles (wide field of view — erect, diminished image); security/surveillance mirrors in shops and parking areas; road-safety mirrors at blind turns; street-light reflectors (spread light over a wide area).

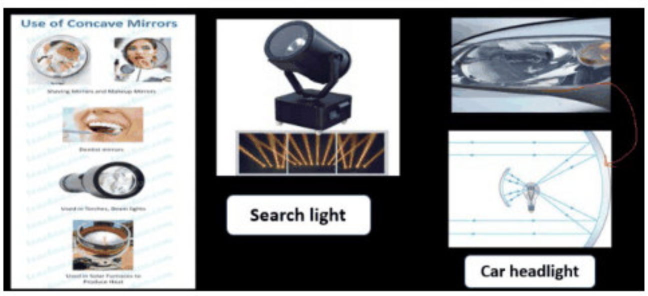

Fig. 2-21 · Uses of concave mirror — shaving mirror, search light, dentist's mirror, car headlight [Lecture 2, p. 4]

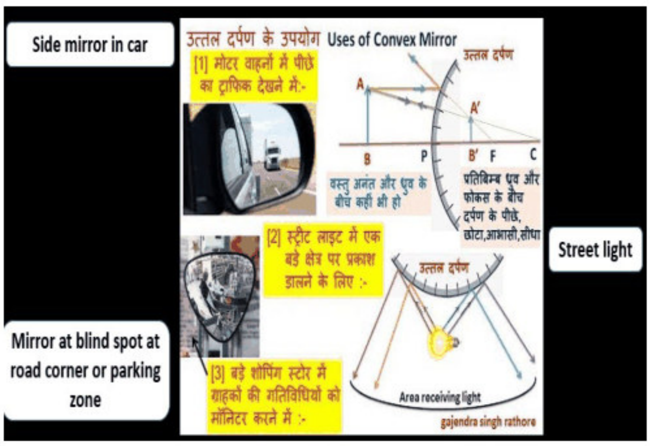

Fig. 2-22 · Uses of convex mirror — side mirror, blind-spot mirror at road corners, street light (source card is in Hindi) [Lecture 2, p. 4]

↻ **De-dup note:** Periscope and kaleidoscope are listed here (Lecture 2, uses of plane mirror) and described in detail in Lecture 5 (optical instruments). The descriptions and diagrams are kept once, in Unit 4.

Mirror formula, sign convention and magnification (Lecture 3 p. 2)

$$1/f = 1/v + 1/u$$

- **f** (focal length) → distance between pole P and focus F; **v** (image distance) → distance of image from the mirror; **u** (object distance) → distance of object from the mirror.
- Image distance is negative (real image, in front) for all concave-mirror cases except one — the object between P and F, where v is positive (virtual image).

Sign convention (New Cartesian system)

- All distances are measured from the **pole**.
- Distances to the **left** of the pole (direction of incident light) are **negative**; to the right **positive**. So u is always negative; f and R are negative for a concave mirror and positive for a convex mirror.
- Heights above the principal axis are positive (erect); heights below are negative (inverted).

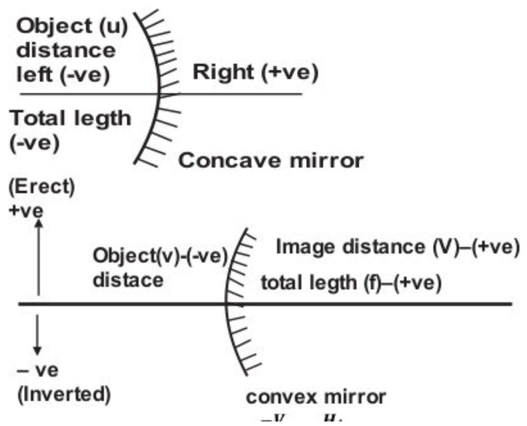

Fig. 2-23 · Sign convention for concave and convex mirrors (Lecture 3 sketch) [Lecture 3, p. 2]

Magnification

$$m = -v/u = h_i/h_o$$

- m > 1 → enlarged image; m < 1 → diminished image; m positive → virtual, erect image; m negative → real, inverted image.

↻ **De-dup note:** The mirror formula opens Lecture 3 (Light Part-02) in the source; it is moved here so the whole mirror story reads in one place. The lens versions of these formulas are in Unit 3.

▌ADDED BY EDITOR — not in source lectures ▌ Worked example (not in source)

An object 10 cm in front of a concave mirror of focal length 15 cm: u = –10, f = –15 ⇒ 1/v = 1/f – 1/u = –1/15 + 1/10 = 1/30 ⇒ v = +30 cm (behind the mirror), m = –v/u = +3 → virtual, erect, 3× magnified — exactly the shaving-mirror situation (object between P and F).

UNIT 3

Light II: Refraction, Total Internal Reflection, Lenses & the Human Eye

Consolidated from: Lecture 3 pp. 2–4 (refraction, Snell's law, examples, TIR), Lecture 4 (lenses, lens formula, power, human eye, defects)

★ **PYQ lens:** Frequency does not change on refraction (66th Q6); speed of light minimum in glass (67th Q66); magnifying glass = convex lens (67th Q73); black strips painted on a convex lens → full image, less bright (68th Q111). Priority B.

Refraction of light

Refraction is the bending of light when it passes obliquely from one medium to another (the speed changes, so the direction changes).

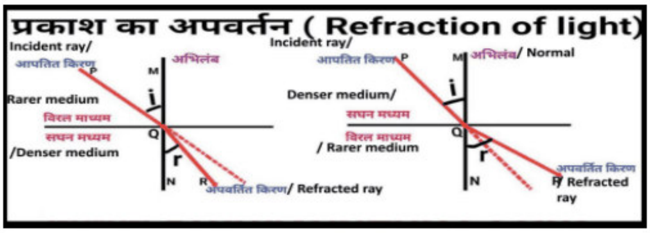

Fig. 3-1 · Refraction of light — incident, refracted rays and the normal (source card in Hindi/English) [Lecture 3, p. 2]

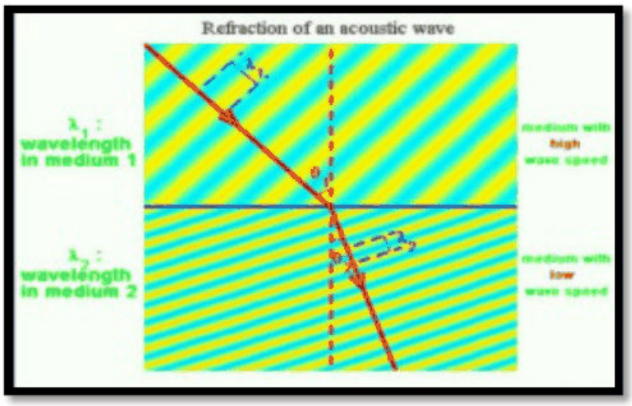

Fig. 3-2 · Refraction of a wave front: the wavelength shortens in the slower medium while the frequency stays the same [Lecture 3, p. 2]

Laws of refraction (Snell's law)

- (i) The incident ray, the refracted ray and the normal at the point of incidence lie in the same plane.
- (ii) The ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant for a given pair of media:

$$\sin i / \sin r = \mu_2/\mu_1 = {}_1\mu_2 \text{ (refractive index of medium 2 with respect to medium 1)}$$

Refractive index

$$\mu = \text{speed of light in vacuum / speed of light in medium} = c/v$$

- Light bends **towards the normal** when it enters a denser medium (slows down) and **away from the normal** when it enters a rarer medium (speeds up).
- Refractive index has no unit; μ (vacuum) = 1, air ≈ 1.0003, water = 1.33 (4/3), glass ≈ 1.5 (3/2), diamond = 2.42.

★ **PYQ lens:** 66th Q6 "Which does not change when light travels from one medium to another?" → **frequency** (velocity, wavelength change; refractive index is a property of the medium).

▌ADDED BY EDITOR — not in source lectures ▌ Useful relations (not spelled out in the lectures)

$\lambda_{medium} = \lambda_{vacuum}/\mu$ and $v = c/\mu$, while ν is fixed by the source. Real depth / apparent depth = μ (a pond of depth 4 m looks 3 m deep through water). Normal incidence (i = 0) → no bending, only a change of speed. Optical density ≠ mass density (kerosene is less dense than water but optically denser).

Everyday examples of refraction

- **1. Bending of an object** — a pencil in a glass of water appears bent at the surface (light from the immersed part bends away from the normal on leaving water).
- **2. Fish appears raised in water** — the apparent position of the fish is above its actual position, so a spear-fisher aims below the image.
- **3. Early sunrise and late sunset** — atmospheric refraction bends sunlight around the curvature of the Earth; the Sun is seen about **2 minutes before** actual sunrise and **2 minutes after** actual sunset, and appears flattened (oval) near the horizon.
- **4. Twinkling of stars** — the refractive index of atmospheric layers fluctuates (hot and cold air), so the refracted starlight keeps changing its path and intensity. Planets, being extended sources, do not twinkle appreciably.

Fig. 3-3 · A spoon in a glass of water looks bent at the surface [Lecture 3, p. 2]

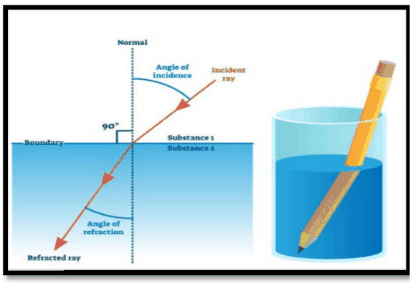

Fig. 3-4 · Refraction at the water surface — why a pencil looks bent [Lecture 3, p. 3]

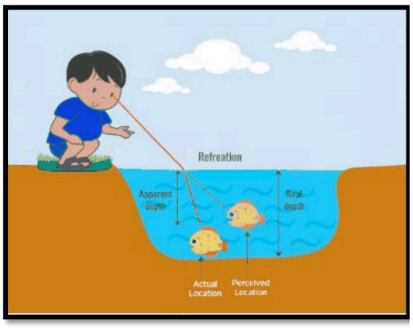

Fig. 3-5 · Fish appears raised — apparent vs actual position [Lecture 3, p. 3]

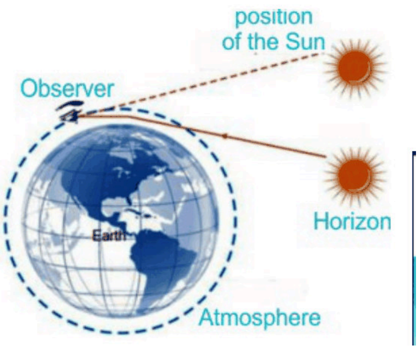

Fig. 3-6 · Early sunrise & late sunset — refraction through the atmosphere [Lecture 3, p. 3]

Fig. 3-7 · The setting Sun looks flattened and red [Lecture 3, p. 3]

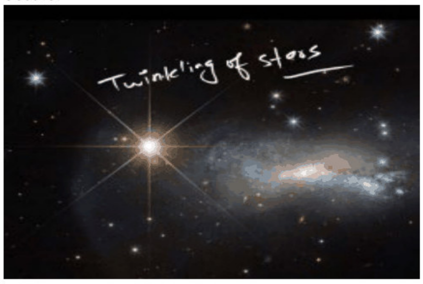

Fig. 3-8 · Twinkling of stars — fluctuating refractive index of air layers [Lecture 3, p. 3]

Total internal reflection (TIR)

- When light travels from a **denser to a rarer** medium and the angle of incidence exceeds the **critical angle (C)**, the light is completely reflected back into the denser medium instead of being refracted.
- Two conditions: (1) light must go from denser to rarer medium; (2) angle of incidence > critical angle.

$\sin C = 1/\mu$ (critical angle: the angle of incidence in the denser medium for which the angle of refraction is 90°)

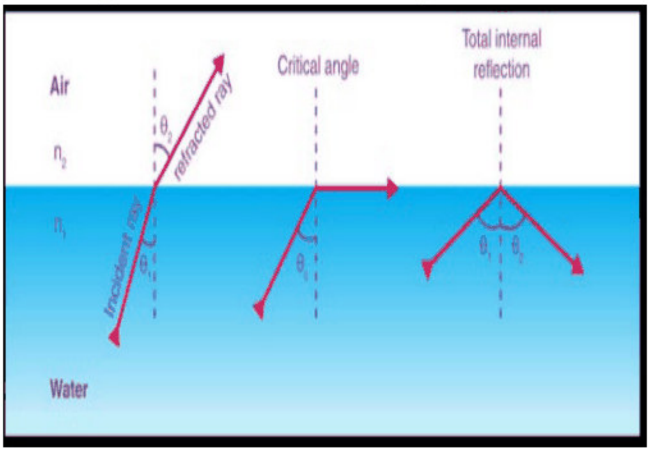

Fig. 3-9 · Refraction, critical angle and total internal reflection at a water–air surface [Lecture 3, p. 3]

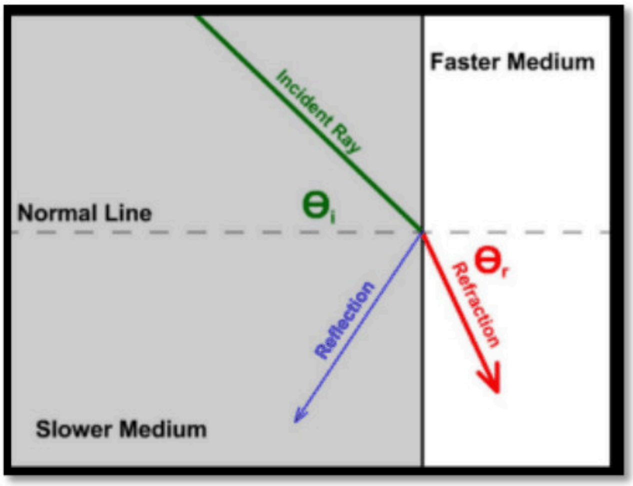

Fig. 3-10 · Reflection vs refraction when a ray meets a faster (rarer) medium [Lecture 3, p. 4]

Examples of total internal reflection

- **I. Sparkling of diamond** — refractive index of diamond is 2.42 (source: 2.5), so its critical angle is only **≈ 24°**. Diamond is cut so that light entering it meets the faces at i > C again and again, undergoes repeated TIR and finally beams out from a few faces into the observer's eye — hence the sparkle.
- **II. Optical fibre** — light propagates through a thin glass/quartz pipe (core, μ_1) surrounded by cladding of lower refractive index ($\mu_2 < \mu_1$) by multiple total internal reflections along the axis, even when the fibre is bent. Used in telecommunication and endoscopy.
- **III. Mirage and looming** — in deserts/hot regions the refractive index of air near the hot surface becomes smaller than that above; light from a distant object bends successively away from the normal until i > C and TIR takes place, so an **inverted image** appears on the ground like a pool of water (mirage). **Looming** is the reverse in cold regions (warm air above cold air) — objects like ships appear raised in the sky.

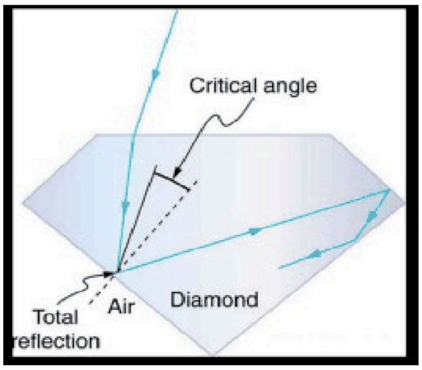

Fig. 3-11 · TIR inside a cut diamond [Lecture 3, p. 4]

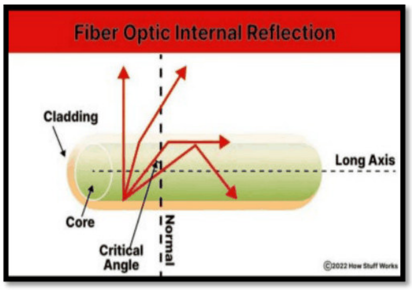

Fig. 3-12 · Optical fibre — core, cladding and TIR along the axis [Lecture 3, p. 4]

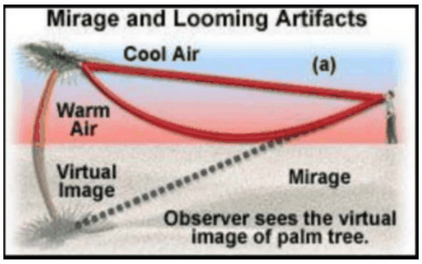

Fig. 3-13 · Mirage (a) — warm air near ground, virtual image of the palm tree [Lecture 3, p. 4]

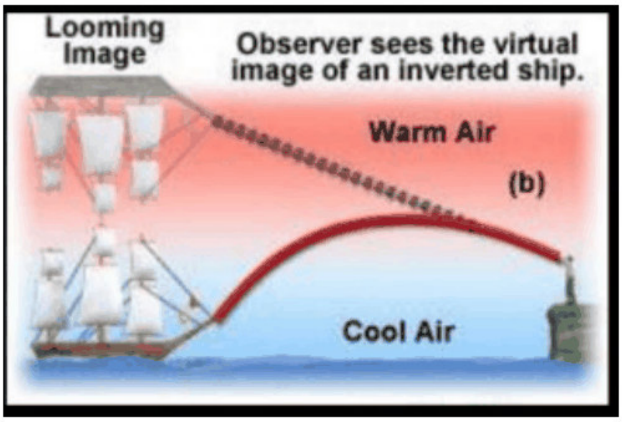

Fig. 3-14 · Looming (b) — cold air near the surface; the observer sees the virtual image of an inverted ship above it [Lecture 3, p. 4]

✎ **Source correction:** Source prints μ(diamond) = 2.5; the accepted value is **2.42** (C ≈ 24.4°). Glass μ = 1.5 gives C ≈ 42°; water μ = 1.33 gives C ≈ 49°.

▌ADDED BY EDITOR — not in source lectures ▌ Other TIR examples worth remembering

Totally reflecting prisms (45°–90°–45°) in binoculars and periscopes (Unit 4); brilliance of air bubbles in water; a test tube held obliquely in water looks silvery

"cat's-eye" road reflectors; endoscope (medical) and fibre-optic broadband. A rainbow involves refraction + *internal* (not total) reflection inside drops.

Lenses

A **lens** is a transparent optical medium bounded by two surfaces, at least one of which is curved. It works on the principle of **refraction** of light.

Convex lens (converging)	Concave lens (diverging)
Thicker at the centre, thinner at the edges	Thinner at the centre, thicker at the edges
Forms a **real, inverted** image except when the object is inside the focus (then virtual, erect, magnified)	**Always** forms a **virtual, erect, diminished** image
Used in camera, telescope, microscope, **magnifying glass**; corrects **hypermetropia**	Used in spectacles for **myopia**; peep-holes in doors

Image formation by a convex lens

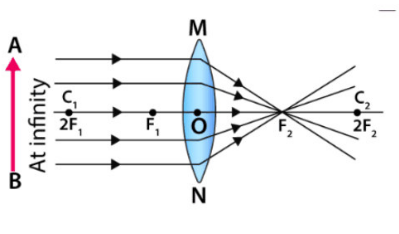

Fig. 3-15 · 1. Object at infinity → image at F_2, point-sized [Lecture 4, p. 2]

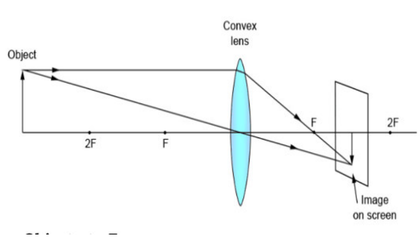

Fig. 3-16 · 2. Object beyond 2F → image between F and 2F, real, inverted, diminished [Lecture 4, p. 2]

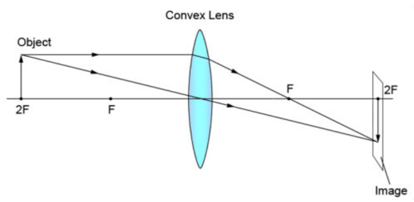

Fig. 3-17 · 3. Object at 2F → image at 2F, same size [Lecture 4, p. 2]

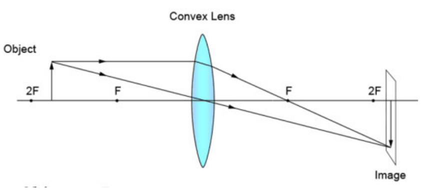

Fig. 3-18 · 4. Object between F and 2F → beyond 2F, magnified [Lecture 4, p. 2]

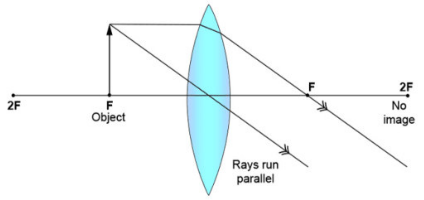

Fig. 3-19 · 5. Object at F → rays emerge parallel, image at infinity [Lecture 4, p. 2]

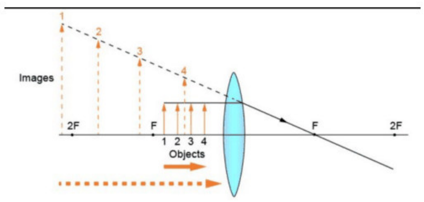

Fig. 3-20 · 6. Object between F and optical centre O → virtual, erect, magnified image on the same side (magnifying glass) [Lecture 4, p. 2]

Position of object	Position of image	Nature	Size
At infinity	At focus (F_2)	Real & inverted	Highly diminished (point image)
Beyond $2F_1$	Between F_2 and $2F_2$	Real & inverted	Diminished
At $2F_1$	At $2F_2$	Real & inverted	Same size
Between F_1 and $2F_1$	Beyond $2F_2$	Real & inverted	Magnified
At F_1	At infinity	Real & inverted	Highly enlarged
Between F_1 and optical centre (O)	On the same side of the lens as the object	Virtual & erect	Magnified

★ **PYQ lens:** 67th Q73: lens used in a magnifying glass → **convex lens** (object placed within the focal length). 68th Q111: black paint strips on a convex lens (f = 20 cm) → **every part** of a lens forms the complete image, so the horse is still whole, only **less bright** (option C).

Image formation by a concave lens

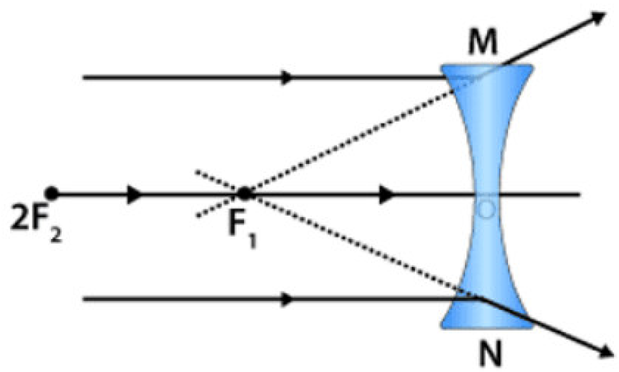

Fig. 3-21 · Case I — object at infinity: image at F_1, virtual, erect, highly diminished [Lecture 4, p. 3]

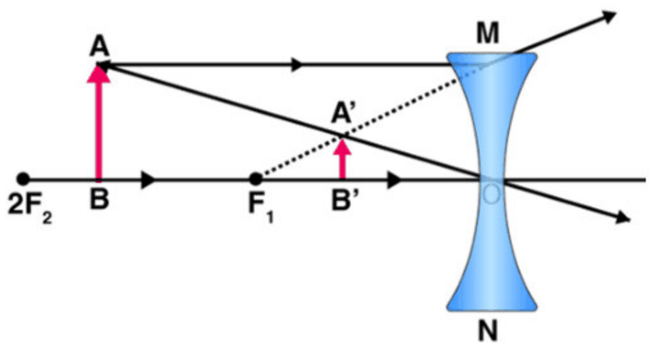

Fig. 3-22 · Case II — object between infinity and O: image between F_1 and O, virtual, erect, diminished [Lecture 4, p. 3]

Object location	Image location	Nature	Size
Infinity	At F_1	Virtual & erect	Highly diminished, point-sized
Between infinity and optical centre	Between focus (F_1) and optical centre (O)	Virtual & erect	Diminished

Lens formula, magnification and power

1/f = 1/v – 1/u (f = focal length, v = image distance, u = object distance)

$m = h_i/h_o = v/u$ (h_i = height of image, h_o = height of object)

Power of a lens P = 1 / f (in metres) unit: dioptre (D)

- Convex lens: **positive** power (f positive); concave lens: **negative** power (f negative). A lens of f = 50 cm has P = +2 D; f = –25 cm gives P = –4 D.

↻ **De-dup note:** Lecture 4 prints the lens formula and magnification twice (after the convex-lens table and again after the concave-lens table). Kept once here. Note the sign difference from the mirror formula (1/v + 1/u) and magnification (–v/u).

▌ADDED BY EDITOR — not in source lectures ▌ Combination and quick facts (not in source)

Powers of thin lenses in contact simply add: $P = P_1 + P_2$. A convex lens immersed in a liquid of higher refractive index becomes diverging (air bubble in water acts as a concave lens). Focal length of a lens depends on colour — least for violet, most for red (chromatic aberration). Sunlight focused by a convex lens can burn paper because the image of the Sun forms at F.

The human eye

The human eye is a natural optical instrument that works like a camera — it forms images using refraction of light and allows us to see objects.

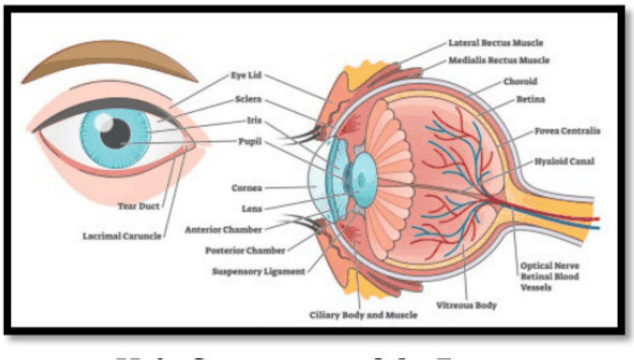

Fig. 3-23 · Anatomy of the human eye — cornea, iris, pupil, lens, retina, optic nerve [Lecture 4, p. 3]

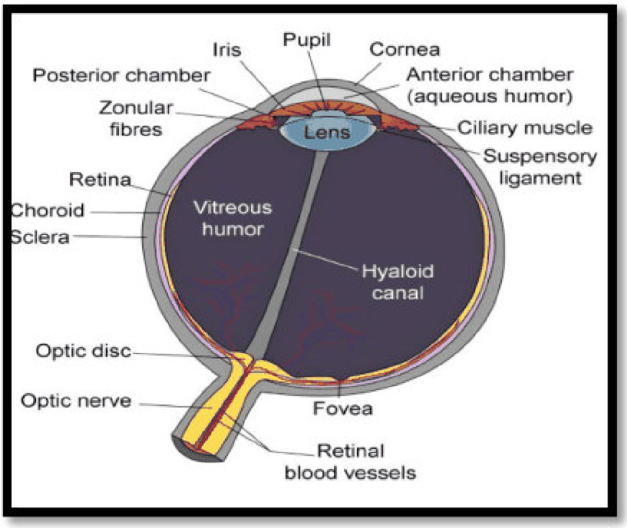

Fig. 3-24 · Sectional view — aqueous humour, vitreous humour, ciliary muscle, fovea, blind spot [Lecture 4, p. 3]

Main components of the eye

- **1. Cornea** — transparent, outermost layer; first surface where light enters; provides **maximum refraction** (bending) of light; acts like a fixed convex lens.
- **2. Aqueous humour** — clear fluid between cornea and lens; maintains intraocular pressure; nourishes cornea and lens.
- **3. Iris** — coloured part of the eye; controls the size of the pupil; works like the diaphragm of a camera.
- **4. Pupil** — central opening in the iris; regulates the amount of light entering the eye; expands in dim light, contracts in bright light.
- **5. Eye lens** — transparent, flexible **convex** lens; focuses light on the retina; can change its shape (**accommodation**): thick → near objects, thin → distant objects.
- **6. Ciliary muscles** — surround the lens; control its curvature; responsible for the power of accommodation.
- **7. Vitreous humour** — jelly-like substance behind the lens; maintains the shape of the eyeball; helps in light transmission.
- **8. Retina** — light-sensitive inner layer containing photoreceptors: **rods** → vision in dim light, **cones** → colour vision. The image formed here is **real, inverted and diminished**.
- **9. Fovea (yellow spot)** — central part of the retina; highest concentration of cones; provides sharpest vision.
- **10. Blind spot** — the point where the optic nerve leaves the eye; no photoreceptors → no vision.
- **11. Optic nerve** — carries visual signals to the brain; connects the eye to the visual cortex.

▌ADDED BY EDITOR — not in source lectures▐ Near point, far point and persistence of vision (standard facts the lecture skips)

Least distance of distinct vision (near point) for a normal eye = **25 cm**; far point = infinity. Persistence of vision ≈ 1/16 s (basis of cinema at 24 frames/s). The brain re-inverts the inverted retinal image.

Common eye defects

Defect	Problem	Cause	Correction
Myopia (near-sightedness)	Distant objects appear blurry (e.g. a student cannot read the board from the back of the class)	Eyeball too long / lens too converging — image forms **in front of** the retina; far point closer than infinity	**Concave** (diverging) lens of suitable negative power
Hypermetropia (far-sightedness)	Nearby objects appear blurry (cannot read a book without holding it at arm's length)	Eyeball too short / lens too weak — image forms **behind** the retina; near point farther than 25 cm	**Convex** (converging) lens of suitable positive power
Presbyopia	Age-related inability to focus on close objects, starts after about age 40	Ciliary muscles weaken, lens loses flexibility (power of accommodation falls)	Reading glasses; **bifocal** lenses (upper concave for distance + lower convex for reading)
Astigmatism	Blurry or distorted vision at all distances; lights at night seen with halos or streaks	Unevenly curved cornea (different curvature in different planes)	**Cylindrical** lens

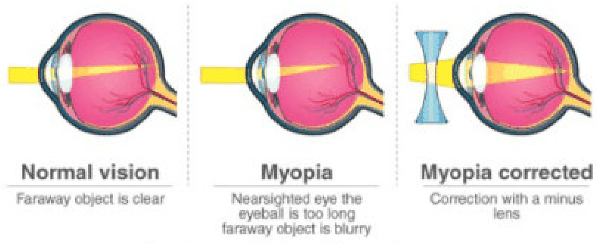

Fig. 3-25 · Myopia: image before the retina — corrected with a concave lens [Lecture 4, p. 4]

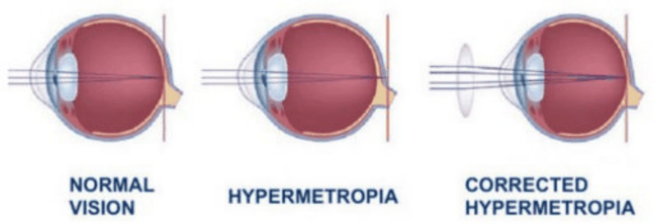

Fig. 3-26 · Hypermetropia: image behind the retina — corrected with a convex lens [Lecture 4, p. 4]

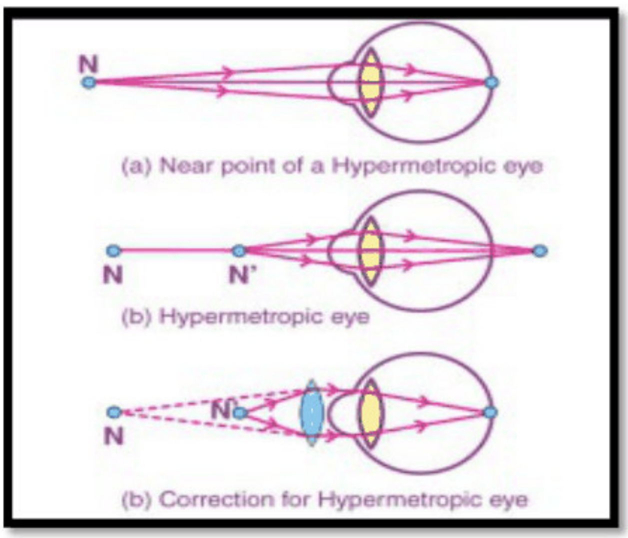

Fig. 3-27 · Hypermetropic eye — near point and correction with a convex lens [Lecture 4, p. 4]

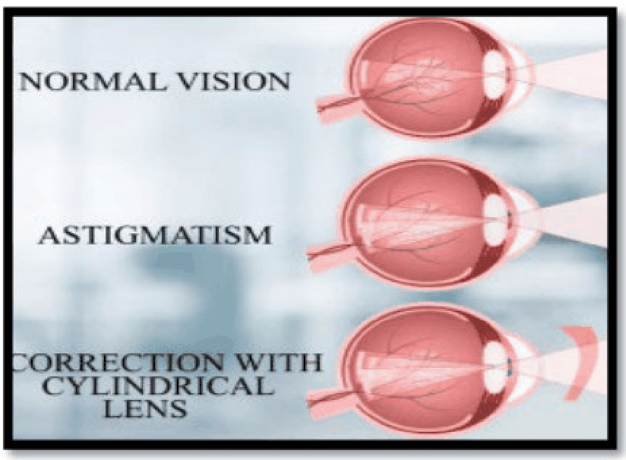

Fig. 3-28 · Astigmatism — corrected with a cylindrical lens [Lecture 4, p. 4]

▌ADDED BY EDITOR — not in source lectures ▌ Cataract & colour blindness (frequently asked, not in the lecture)

Cataract — the eye lens becomes cloudy/opaque (often with age); treated by surgically replacing the lens with an artificial intra-ocular lens. **Colour blindness** — genetic (X-linked, far commoner in males) deficiency of certain cones, usually red–green; it cannot be corrected by lenses. **Night blindness** — deficiency of vitamin A (rod pigment rhodopsin).

UNIT 4

Light III: Dispersion, Scattering, Wave Optics & Optical Instruments

Consolidated from: Lecture 5 (Light Part-04) pp. 2–5; scattering line of Lecture 1 p. 3; periscope & kaleidoscope entries of Lecture 2 p. 4

★ PYQ lens: This unit is examined rarely (only 64th Q9 "red λ > violet λ" maps here). Priority C — read once; keep the rainbow angles, Rayleigh's $1/\lambda^4$ law, the sky/sunset colours and which instrument uses what.

Dispersion of light

- Dispersion is the **splitting of white light into its constituent colours (VIBGYOR)** as it passes through a medium like a prism.
- It happens because the refractive index of the medium is different for each colour, so each colour bends by a different angle: **violet bends the most, red the least**.
- After rainfall, water droplets present in the atmosphere act as prisms.

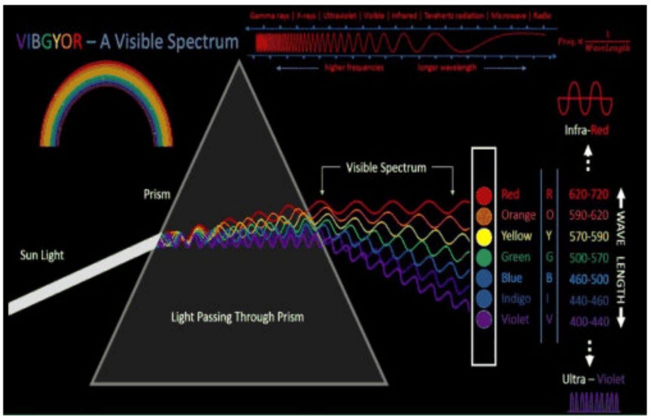

Fig. 4-1 · White light dispersed by a prism into the visible spectrum (VIBGYOR with wavelengths) [Lecture 5, p. 2]

Fig. 4-2 · Rainbow — nature's dispersion by raindrops [Lecture 5, p. 2]

Rainbows

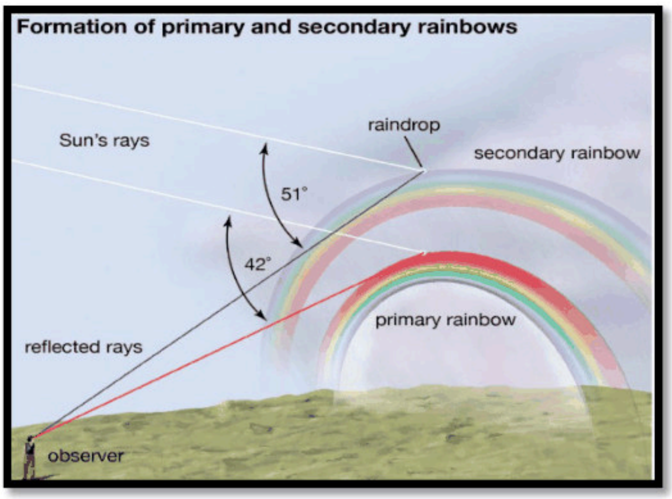

Fig. 4-3 · Formation of primary and secondary rainbows — angles of 42° and 51° with the observer's eye [Lecture 5, p. 2]

Primary rainbow	Secondary rainbow
Makes an angle of **42°** with the human eye	Makes an angle of about **52°** (source: 51°/52°) with the human eye
Inner band **violet**, outer band **red**	Inner band **red**, outer band **violet** (colours reversed)
One refraction–one internal reflection–one refraction inside each drop; brighter	Two internal reflections inside each drop; fainter

▌ADDED BY EDITOR — not in source lectures▐ Extra rainbow facts (not in the lecture)

A rainbow is seen only with the **Sun behind the observer**; the region between the two bows is darker (Alexander's dark band). Red of the primary bow is at 42°, violet at 40°; for the secondary bow red is at ≈ 50°, violet ≈ 54°. From an aeroplane a full circular rainbow can be seen.

Scattering of light

- Scattering is the phenomenon in which a beam of light is redirected in various directions after interacting with tiny particles such as air molecules or dust — the light is absorbed and then re-emitted in different directions.
- The strength of scattering depends on the wavelength of light and the size of the particles: **shorter wavelengths scatter more easily**.

Rayleigh scattering law: Intensity of scattered light ∝ $1/\lambda^4$

- **Blue colour of the sky** — the blue (short-λ) part of sunlight is scattered by gas molecules of the atmosphere much more than red (blue scatters about 5.5 times more than red — $(700/450)^4$).
- **Red colour of sunrise & sunset** — near the horizon light travels through a much thicker layer of atmosphere; most of the blue is scattered away and the transmitted light is red/orange.

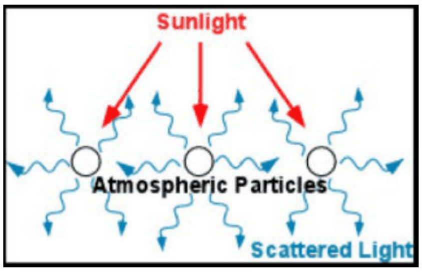

Fig. 4-4 · Sunlight scattered by atmospheric particles [Lecture 5, p. 2]

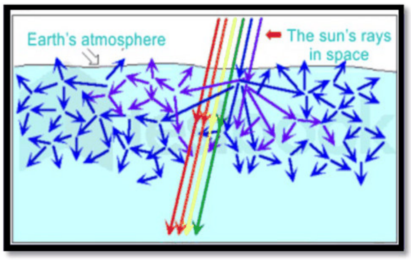

Fig. 4-5 · Blue light is scattered many times before reaching the eye [Lecture 5, p. 2]

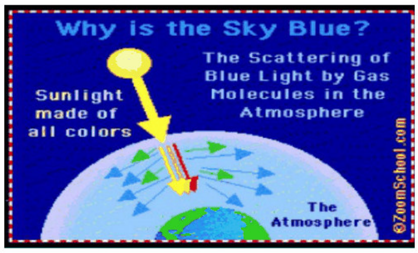

Fig. 4-6 · Why the sky is blue — scattering of blue light by gas molecules [Lecture 5, p. 3]

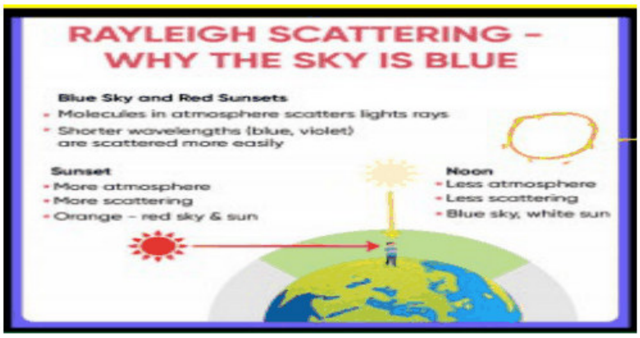

Fig. 4-7 · Rayleigh scattering — blue sky by day, red sky at sunset; more atmosphere → more scattering [Lecture 5, p. 3]

↻ **De-dup note:** Lecture 1 (p. 3) already printed "Scattering of light ∝ $1/\lambda^4$ to spread visible spectrum" and Lecture 5 (p. 3) repeats "Scattering of light: explains blue colour of sky, red colour of sunrise & sunset, Rayleigh scattering law". All merged here.

▌ADDED BY EDITOR — not in source lectures ▌ Related facts often asked (not in the lecture)

Danger signals are red because red scatters least and is visible farthest through fog/smoke. Clouds look white because water droplets are much larger than λ (Mie scattering — all colours equally). The sky looks **black** from the Moon or a spacecraft (no atmosphere). Rayleigh scattering applies when particle size ≪ λ

C. V. Raman's Nobel (1930) was for a different, inelastic scattering (Raman effect).

Interference of light

Interference is a phenomenon of **wave optics** in which two or more coherent light waves superpose (overlap) and produce a pattern of bright and dark fringes.

- **Constructive interference** → bright fringes (crest meets crest); **destructive interference** → dark fringes (crest meets trough).
- **Young's double-slit experiment (YDSE)** proves the wave nature of light; it produces alternate bright and dark bands called interference fringes and is responsible for effects like the colours on soap bubbles and oil slicks.

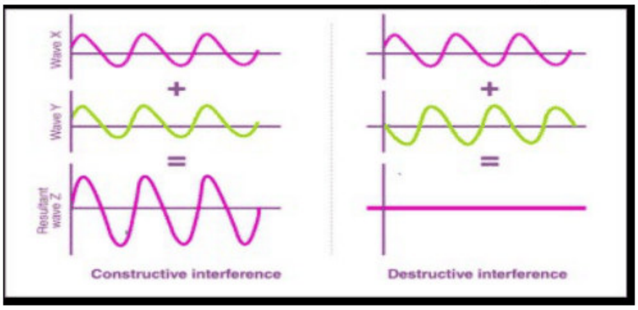

Fig. 4-8 · Constructive (+) and destructive (–) interference of two waves [Lecture 5, p. 3]

Conditions for sustained interference

- Sources must be **coherent** (same frequency and a constant phase difference).
- Light should be **monochromatic** (single wavelength).
- The waves should have **comparable amplitudes** (for good contrast).
- The sources should be **close** to each other (narrow fringes otherwise).

Examples of interference of light

- 1. Colours of a **soap bubble**. 2. Colours on a **CD/DVD surface**. 3. Colourful **oil film** on a water surface (thin-film interference).

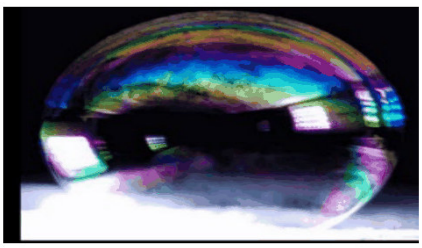

Fig. 4-9 · Colours in a soap bubble [Lecture 5, p. 3]

Fig. 4-10 · Colours on a CD-DVD surface [Lecture 5, p. 3]

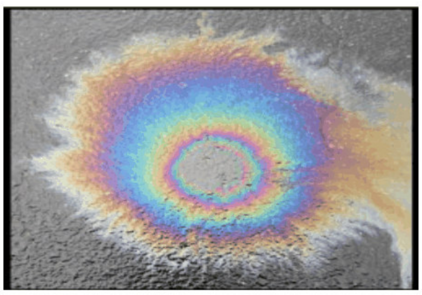

Fig. 4-11 · Colourful oil film on water [Lecture 5, p. 3]

Diffraction of light

- Diffraction is the **bending of light waves around the edges of an obstacle or aperture** when the size of the obstacle/aperture is comparable to the wavelength of light.
- A ray gets bent and spreads out, deviating from its path while passing through the edges of an obstacle or a hole; it also illuminates the area that would normally be in the shadow region.

- More pronounced for **longer wavelengths** (that is why sound, with metre-scale wavelengths, bends round corners easily while light, with $\lambda \sim 10^{-7}$ m, casts sharp shadows).

Polarisation of light

- Polarisation is the phenomenon in which the vibrations of light waves are restricted to a particular direction (plane) perpendicular to the direction of propagation.
- **Only transverse waves can be polarised** — polarisation proves that light is a transverse wave. (Sound, being longitudinal, cannot be polarised — Lecture 7.)
- After polarisation the vibrations are confined to a single plane, creating polarised light.
- Uses: sunglasses (cut glare), 3-D movies, stress analysis (photo-elasticity), LCD screens.

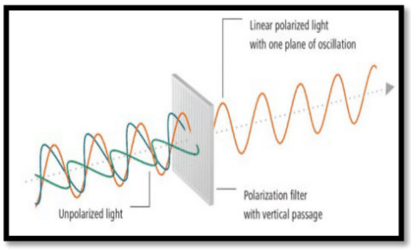

Fig. 4-12 · Unpolarised light → polarising filter → linearly polarised light [Lecture 5, p. 4]

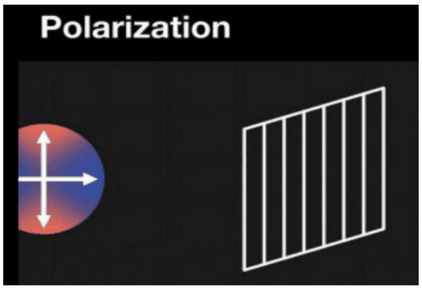

Fig. 4-13 · Polarisation: vibrations restricted to one plane [Lecture 5, p. 4]

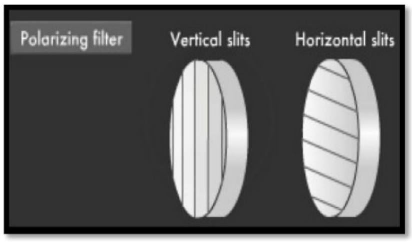

Fig. 4-14 · Polarising filter with vertical and horizontal slits — crossed filters block light [Lecture 5, p. 4]

Phenomenon	Shows light is…	Key condition	Everyday example
Interference	a wave	coherent sources	soap-bubble colours
Diffraction	a wave	obstacle size ≈ λ	coloured fringes round a street lamp seen through fine cloth
Polarisation	a **transverse** wave	polariser (Polaroid)	polarised sunglasses, 3-D films
Photoelectric effect	a particle (photon)	ν > threshold	photocells, solar panels (Unit 16)

↻ **De-dup note:** The comparison table above is an editor summary of the four phenomena as described in the lectures (no new facts).

Optical instruments

Compound microscope

- Uses **two convex lenses** — an objective of very short focal length and an eyepiece of short focal length; produces an **enlarged, inverted** (virtual) image.

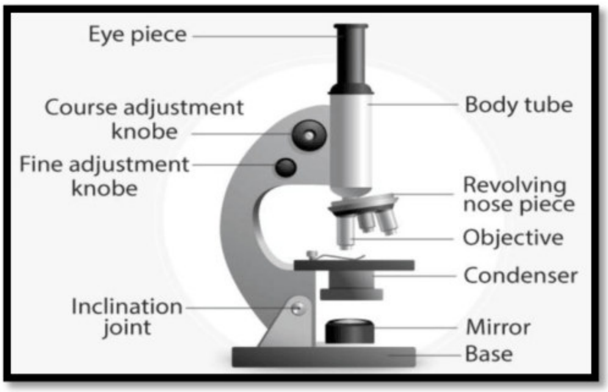

Fig. 4-15 · Parts of a compound microscope — eyepiece, coarse & fine adjustment, revolving nose piece, objective, condenser, mirror, base [Lecture 5, p. 4]

Astronomical telescope

- An optical instrument used to observe distant objects by collecting and magnifying light; used in astronomy to study stars, planets and galaxies.
- **Objective**: large focal length (and large aperture, to gather light); **eyepiece**: small focal length; produces an **inverted** image. (Magnifying power = f_o/f_e.)

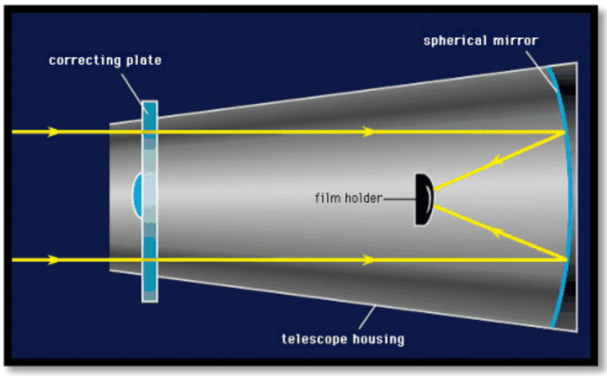

Fig. 4-16 · Reflecting telescope — correcting plate, spherical mirror, film holder [Lecture 5, p. 4]

▌ADDED BY EDITOR — not in source lectures ▌ Types of telescope (context)

Refracting telescopes use lenses (Galileo); reflecting telescopes use a concave mirror as objective (Newton) — cheaper for large apertures and free of chromatic aberration; most big observatories and the Hubble/James Webb space telescopes are reflectors. A terrestrial telescope adds an erecting lens; binoculars use totally-reflecting prisms to erect the image and shorten the tube.

Periscope

- An optical instrument used to see objects that are not in the direct line of sight, especially from a hidden or enclosed position (submarines, trenches, tanks).
- Two plane mirrors (or two totally-reflecting 45° prisms) are inclined at **45°** to the line of sight, parallel to each other.

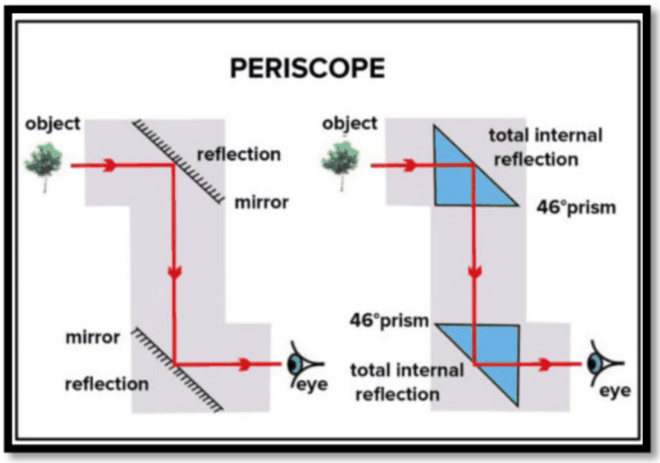

Fig. 4-17 · Periscope with two mirrors (left) and with two total-internal-reflection prisms (right) [Lecture 5, p. 5]

Kaleidoscope

- An optical instrument with two or more reflecting surfaces (mirrors) tilted to each other at an angle, so that one or more objects at one end are shown as a symmetrical pattern when viewed from the other end, due to **repeated (multiple) reflection**.

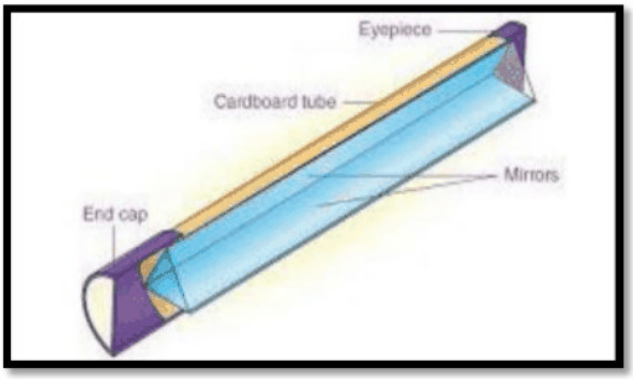

Fig. 4-18 · Kaleidoscope — cardboard tube with mirrors and an eyepiece [Lecture 5, p. 5]

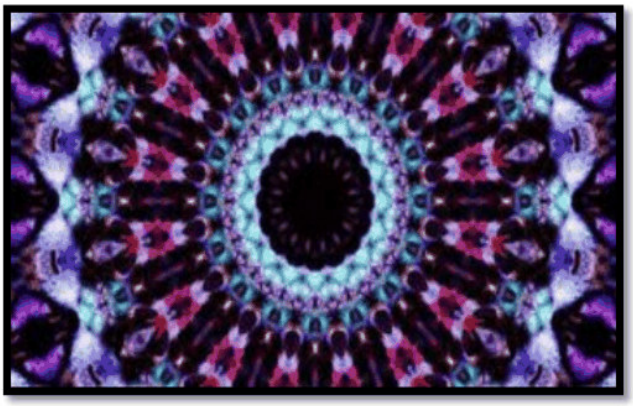

Fig. 4-19 · Typical kaleidoscope pattern [Lecture 5, p. 5]

↻ **De-dup note:** Periscope and kaleidoscope are also listed under "uses of plane mirror" in Lecture 2 — kept once here with the diagrams.

▌ADDED BY EDITOR — not in source lectures ▌ Simple microscope & camera (frequently asked, not detailed in the lectures)

A **simple microscope** (magnifying glass) is a single convex lens with the object inside F; magnifying power ≈ 1 + D/f (D = 25 cm). A **camera** uses a convex lens forming a real, inverted, diminished image on the film/sensor; the aperture (f-number) and shutter control exposure. **Spectrometer** uses a prism/grating to measure wavelengths; **binoculars** = two telescopes with prisms.

UNIT 5

Waves & the Electromagnetic Spectrum

Consolidated from: Lecture 6 (WAVE) pp. 2–3; Lecture 7 p. 2 (mechanical waves, transverse vs longitudinal) and p. 5 (list of wave phenomena)

★ **PYQ lens:** Infrared radiation is used for muscle/body aches (66th Q1 and again 70th Q45); speed of EM waves 3×10^8 m/s (66th Q7). Priority B — the same infrared fact has come twice in five papers.

What is a wave?

A **wave** is a disturbance that transfers energy from one place to another **without transfer of matter**. Examples: sound waves, water waves, light waves.

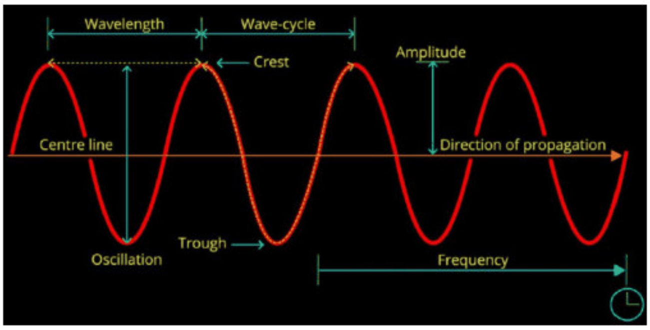

Fig. 5-1 · Anatomy of a wave — wavelength, crest, trough, amplitude, oscillation, frequency, direction of propagation [Lecture 6, p. 2]

Terms related to waves

- **Wavelength (λ)**: the distance between successive crests (or two compressions / two rarefactions) of a wave.
- **Frequency (ν or f)**: the number of wave cycles passing a given point in one second; unit hertz (Hz).
- **Time period (T)**: time for one complete cycle; $T = 1/f$.
- **Amplitude (A)**: maximum displacement measured between the crest and the mid-point (mean position) of the wave.
- **Velocity (v)**: speed and direction of propagation; $v = f\lambda$.
- **Intensity**: wave energy passing through unit area per unit time; $I = P/A$ (W/m^2).

↻ **De-dup note:** The wave terms are printed in Lecture 6 (p. 2) and again as "physical characteristics of a sound wave" in Lecture 7 (pp. 2–3). Generic definitions are kept here; the sound-specific meanings (loudness, pitch) are in Unit 6.

Classification of waves

Basis	Type	Description / examples
Need of medium	**Mechanical** waves	Require a material medium (solid, liquid or gas); transfer energy without transporting matter permanently — sound, water waves, seismic waves, waves on a string
	Electromagnetic (non-mechanical) waves	Need no medium; oscillating electric & magnetic fields — light, radio, X-rays
Direction of oscillation	**Transverse** waves	Particles of the medium oscillate **perpendicular** to the direction of propagation (crests & troughs) — light, waves on a string, ripples on water

Basis	Type	Description / examples
	Longitudinal waves	Particles oscillate **parallel** to the direction of propagation (compressions & rarefactions) — sound in air (formed by change in density/pressure)

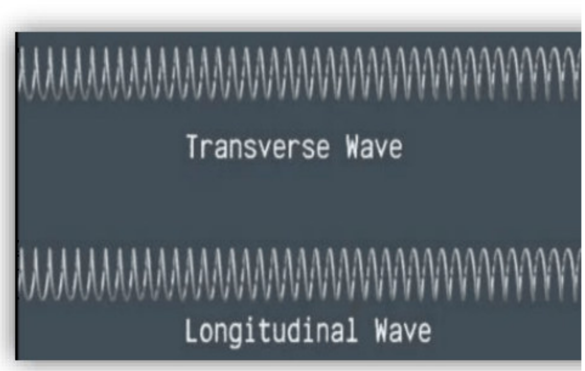

Fig. 5-2 · Transverse vs longitudinal wave on a spring [Lecture 7, p. 2]

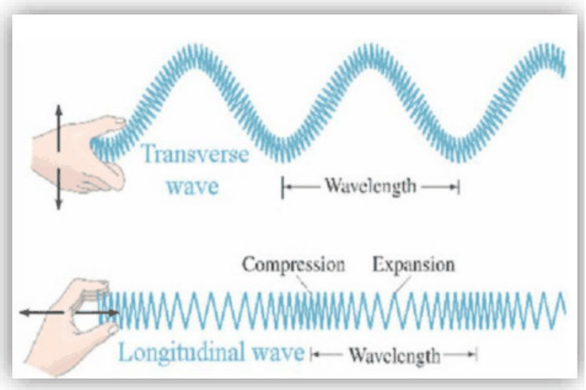

Fig. 5-3 · Transverse wave (wavelength = crest to crest) and longitudinal wave (compression–expansion) [Lecture 7, p. 2]

★ **PYQ lens:** 65th Q115: "Sound wave in air is ..." → **longitudinal** (not transverse, electromagnetic or polarised).

▌ADDED BY EDITOR — not in source lectures ▌ Matter waves & the three classes (standard classification, not in the lecture)

Physics classifies waves as (1) mechanical, (2) electromagnetic, (3) **matter (de Broglie) waves** associated with moving particles (Unit 16). Longitudinal waves need a medium with volume elasticity (so they travel in solids, liquids and gases); transverse mechanical waves need shear elasticity (so they travel only in solids and on liquid surfaces) — that is why earthquake S-waves cannot pass through the Earth's liquid outer core, proving it is liquid.

Electromagnetic waves

- Electromagnetic (EM) waves are a form of energy consisting of **oscillating electric and magnetic fields** that can travel through a vacuum at the speed of light.
- They are created by the **acceleration of charged particles** and travel as **transverse** waves: the electric field (E) and magnetic field (B) are perpendicular to each other and to the direction of propagation.
- EM waves are represented by a sinusoidal graph; the highest point is the **crest**, the lowest the **trough**. In vacuum all EM waves travel at **$c = 3 \times 10^8$ m/s**.

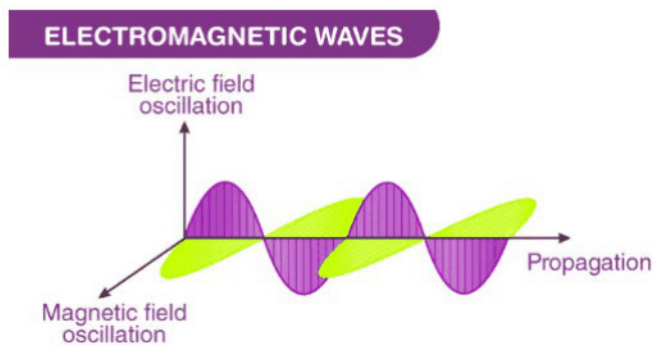

Fig. 5-4 · Electromagnetic wave — electric field oscillation ▯ magnetic field oscillation ▯ propagation [Lecture 6, p. 2]

Features of electromagnetic waves (Lecture 6)

- 1. **No medium required** — travel in vacuum (space), unlike sound waves.
- 2. **Transverse nature** — E ▯ B ▯ direction of propagation.
- 3. **Speed** — same for all EM waves in vacuum; independent of frequency.
- 4. **Produced by accelerating charges** — moving (oscillating) charged particles generate EM waves.
- 5. **Carry energy and momentum** — can exert pressure (radiation pressure); example: solar radiation.
- 6. **Obey wave phenomena** — reflection, refraction, diffraction, interference (and polarisation).
- 7. **Do not need a material medium** — can travel through empty space (space communication possible).
- 8. **Electric & magnetic fields oscillate** — fields vary sinusoidally and are always in phase.
- 9. **Relationship between speed, frequency and wavelength**: c = ν λ.
- 10. **Wide range of spectrum** — from radio waves to gamma rays; they differ only in wavelength and frequency.

Electromagnetic spectrum — different kinds of EM waves and their applications (Lecture 6 p. 3)

Radiation	Discoverer	How produced	Wavelength	Frequency (Hz)	Properties	Applications
γ-rays	Henri Becquerel & Marie Curie	Decay of radioactive nuclei	10^{-14} – 10^{-10} m	3×10^{22} – 3×10^{18}	(a) Highly penetrating; (b) uncharged; (c) low ionising power	(a) Information on nuclear structure; (b) medical treatment (radiotherapy) etc.
X-rays	Röntgen	Collisions of high-energy electrons with a heavy target	6×10^{-12} – 10^{-9} m	5×10^{19} – 3×10^{17}	(a) Low penetrating power (less than γ); (b) other properties similar to γ-rays except wavelength	(a) Medical diagnosis & treatment; (b) study of crystal structure; (c) industrial radiography
Ultraviolet	Ritter	Ionised gases, sun-lamp, spark etc.	6×10^{-10} – 3.8×10^{-7} m	5×10^{17} – 7×10^{14}	(a) All properties of light; (b) photoelectric effect	(a) To detect adulteration, writing & signature (forgery); (b) sterilisation of water — destructive action on bacteria
Visible light (sub-parts: violet, blue, green, yellow, orange, red)	Newton	Outer-orbit electron transitions in atoms, gas discharge tubes, incandescent solids & liquids	3.8×10^{-7} – 7.8×10^{-7} m (violet 3.9–4.55; blue 4.55–4.92; green 4.92–5.77; yellow 5.77–5.97; orange 5.97–6.22; red 6.22–7.8, all $\times 10^{-7}$ m)	8×10^{14} – 4×10^{14} (violet 7.69–6.59; blue 6.59–6.10; green 6.10–5.20; yellow 5.20–5.03; orange 5.03–4.82; red 4.82–3.84, all $\times 10^{14}$)	(a) Sensitive to human eye	(a) To see objects; (b) to study molecular structure
Infrared	William Herschel	(a) Rearrangement of outer-orbital electrons in atoms & molecules; (b) change of molecular vibrational & rotational energies; (c) bodies at high temperature	7.8×10^{-7} – 10^{-3} m	4×10^{14} – 3×10^{11}	(a) **Thermal effect**; (b) all properties similar to light except λ	(a) Industry, medicine (heat therapy) & astronomy; (b) fog/haze photography; (c) elucidating molecular structure
Microwaves	Hertz	Special electronic devices such as the klystron tube	10^{-3} – 0.3 m	3×10^{11} – 10^{9}	(a) Reflection, refraction and diffraction	(a) **Radar** and telecommunication; (b) analysis of fine details of molecular structure
Radio waves	Marconi	Oscillating circuits	0.3 m – few km	10^{9} – few Hz	(a) Exhibit wave-like properties more than particle-like	(a) Radio communication

Radio sub-band	Wavelength	Frequency (Hz)	Use
(A) SHF — Super High Frequency	0.01 – 0.1 m	3×10^{10} – 3×10^{9}	Radar, radio & satellite communication (microwaves)
UHF — Ultra High Frequency	0.1 – 1 m	3×10^{9} – 3×10^{8}	Radar, television broadcast, short-distance communication
VHF — Very High Frequency	1 – 10 m	3×10^{8} – 3×10^{7}	Television & FM radio
(B) HF — High Frequency	10 – 100 m	3×10^{7} – 3×10^{6}	Medium-distance communication
MF — Medium Frequency	100 – 1000 m	3×10^{6} – 3×10^{5}	Telephone communication, marine & navigation use
LF — Low Frequency	1000 – 10 000 m	3×10^{5} – 3×10^{4}	Long-range communication
VLF — Very Low Frequency	10 000 – 30 000 m	3×10^{4} – 10^{4}	Long-distance communication

✎ **Source correction:** The source table contains typesetting slips which are corrected above: γ-ray frequency "3×10^{22} Hz to 3×40^{15}" → 3×10^{22} – 3×10^{18} Hz; UV wavelength "6×19^{-10} m" → 6×10^{-10} m; the energy column (γ 10^{7}–10^{4} eV; X 2.4×10^{5}–1.2×10^{3} eV; UV 2×10^{3}–3 eV; visible 3.2–1.6 eV; IR 1.6–10^{-3} eV; microwave 10^{-3}–10^{-5} eV; radio 10^{-3} eV → 0) is kept here in text because the widths did not allow an eighth column.

★ **PYQ lens:** 66th Q1 & 70th Q45: radiation used in the treatment of muscle ache / body aches → **infrared** (thermal effect). 66th Q7: velocity of EM waves → **3×10^{8} m s^{-1}**. 64th Q3: angstrom is a unit of wavelength.

▌ADDED BY EDITOR — not in source lectures ▌ Memory aids & extra spectrum facts (not in the lecture)

Order of increasing wavelength: "**G**andhiji **X**-rayed **U**ma's **V**ery **I**ll **M**other **R**epeatedly" (γ, X, UV, visible, IR, microwave, radio) — frequency and photon energy fall in the same order. Microwave ovens use 2.45 GHz absorbed by water molecules; the ozone layer absorbs UV-B/UV-C; the greenhouse effect is trapping of terrestrial **infrared** by CO_2/water vapour; remote controls use IR; mobile phones use UHF (≈ 0.7–3 GHz)

5G uses 3.5 GHz and 26 GHz bands. UV causes tanning/vitamin-D synthesis and is detected by fluorescence

X-rays were discovered in 1895 and are produced by Coolidge tubes.

Wave phenomena — a summary (Lecture 7 p. 5)

Wave phenomena are the various behaviours shown by waves when they interact with media, obstacles or other waves.

- **1. Reflection** — bouncing back of waves from a surface; angle of incidence = angle of reflection; examples: echo (sound), mirror (light).
- **2. Refraction** — bending of waves when they enter a different medium, caused by change in speed; example: sound travelling farther at night (cool air near the ground bends sound waves back towards the surface).
- **3. Diffraction** — bending/spreading of waves around obstacles; more significant when the wavelength is large — sound shows strong diffraction (we hear round corners).
- **4. Interference** — superposition of two or more waves: constructive (increased amplitude) or destructive (decreased amplitude).
- **5. Beats** — produced by the interference of waves of slightly different frequencies; beat frequency = $|f_1 - f_2|$.
- **6. Doppler effect** — apparent change in frequency due to relative motion between source and observer; examples: ambulance siren, train horn (pitch rises as it approaches, falls as it recedes).
- **7. Standing (stationary) waves** — formed by superposition of two waves travelling in opposite directions; features: **nodes** (zero displacement) and **antinodes** (maximum displacement).
- **8. Polarisation** — restriction of vibrations to a particular direction; **only transverse waves** can be polarised; **sound waves cannot be polarised** (longitudinal).

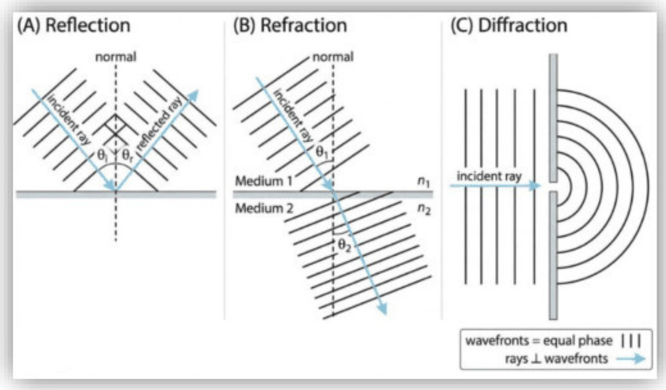

Fig. 5-5 · (A) Reflection, (B) refraction and (C) diffraction of wave fronts
[Lecture 7, p. 5]

ADDED BY EDITOR — not in source lectures **Doppler effect — the formula and its famous uses (announced in the Lecture-1 plan but never taught)**

Apparent frequency $f' = f\,(v \pm v_o)/(v \mp v_s)$ — upper signs for approach ($f' > f$), lower for recession. Uses: police speed radar, Doppler ultrasound of blood flow, weather radar, and **red-shift** of galaxies (light from receding galaxies shifts to longer wavelengths — Hubble's evidence for the expanding universe). The effect exists for light as well as sound; a star approaching us looks bluer.

ADDED BY EDITOR — not in source lectures **Mach number & shock waves (also in the Lecture-1 plan, not taught)**

Mach number = speed of object ÷ speed of sound in that medium (Mach 1 ≈ 340 m/s in air at sea level). Subsonic < 1, transonic ≈ 1, **supersonic** 1–5, hypersonic > 5. A supersonic object compresses the air ahead into a cone-shaped **shock wave**; the sudden pressure jump heard on the ground is the **sonic boom**. Bullets, Concorde, BrahMos (Mach 2.8–3) are supersonic; hypersonic glide vehicles exceed Mach 5.

UNIT 6

Sound

Consolidated from: Lecture 7 (SOUND WAVE) pp. 2–5

★ **PYQ lens:** Sound in air is longitudinal (65th Q115); shrillness/pitch is decided by frequency (68th Q112 — key "none of the above" because amplitude, wavelength, velocity were the options); a sitarist tightening the string adjusts its frequency (68th Q120); speed of sound is maximum in steel among steel/water/air/hydrogen (71st Q124). Priority B.

Sound — nature and origin

- **Sound** is a form of energy that travels through a medium (solid, liquid or gas) as vibrations or a disturbance which, on reaching the ear, causes the sensation of hearing.
- **Type of wave**: sound is a **mechanical, longitudinal** wave — particles of the medium vibrate parallel to the direction of travel, creating high-pressure **compressions** and low-pressure **rarefactions**.
- **Medium requirement**: sound needs a material medium; it **cannot travel in vacuum** (outer space).
- **Origin**: sound is always produced by a **vibrating** object.

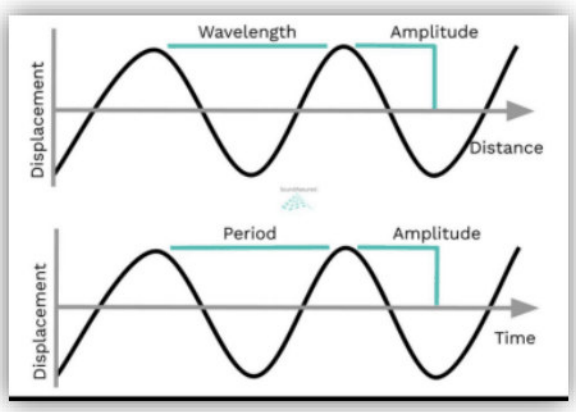

Fig. 6-1 · Displacement–distance and displacement–time graphs of a sound wave: wavelength, period, amplitude [Lecture 7, p. 2]

Physical characteristics of a sound wave

- **Amplitude** — maximum displacement or pressure variation from the equilibrium position; determines the **loudness** (higher amplitude → louder).
- **Frequency** — number of complete cycles (compressions and rarefactions) passing a point per second, in hertz; determines the **pitch** (higher frequency → higher pitch).
- **Wavelength** — distance between two consecutive identical points, e.g. two compressions or two rarefactions.
- **Time period** — time for one complete cycle; the inverse of frequency.
- **Speed (velocity)** — distance travelled per unit time; $v = \nu \lambda$.
- **Intensity** — wave energy passing through unit area per unit time, perpendicular to the direction of propagation:

$$I = P / A \quad I \text{ in W/m}^2,\ P = \text{power of source (W)},\ A = \text{area (m}^2)$$

Classification of sound by frequency

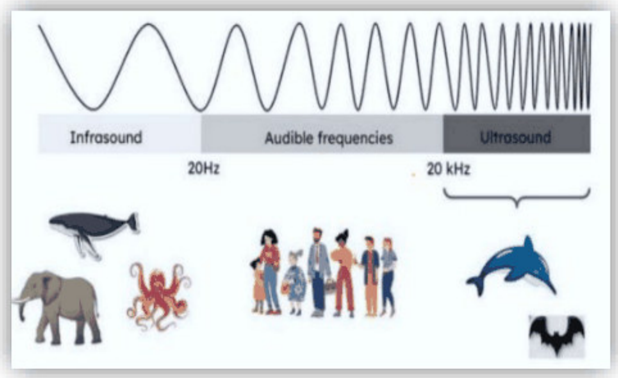

Fig. 6-2 · Infrasound (< 20 Hz) — audible range (20 Hz – 20 kHz) — ultrasound (> 20 kHz), with the animals that use them [Lecture 7, p. 3]

- **(i) Infrasonic** — below **20 Hz**; produced by landslides, volcanoes, earthquakes and meteorites; not audible to humans; also produced by large sources (elephants, whales communicate with it).
- **(ii) Sonic / audible** — **20 Hz to 20 kHz**; the sensitive audible range for humans.
- **(iii) Ultrasonic** — above **20 kHz**; frequency of sound higher than humans can hear; produced and heard by animals such as bats, cats, dogs, whales and dolphins.

Applications of ultrasonic sound waves

- 1. Ultrasound / ultrasonography in medicine; ECG and EEG (as listed in source — see correction).
- 2. To kill some animal-like pests (rats, frogs etc.).
- 3. To locate submarines and other objects under water — **SONAR**.
- 4. To clean clothes, machinery parts and jewellery (ultrasonic cleaning).

✎ **Source correction:** Source lists "ECG and EEG" under ultrasonic uses. ECG (electrocardiogram) and EEG (electroencephalogram) record *electrical* signals of the heart/brain — they do not use ultrasound. **Echocardiography** (heart ultrasound) is the ultrasonic test.

▌ADDED BY EDITOR — not in source lectures ▌ More ultrasonic uses (standard list, not in the lecture)

Detecting flaws/cracks in metal blocks, welding plastics, breaking kidney stones (lithotripsy), measuring depth of sea (echo-sounder), bats' echolocation, ultrasonic dog whistles, drilling holes in glass, and in industry for mixing/emulsifying liquids.

Velocity of sound

The speed of sound depends on the **elasticity** and **density** of the medium. Elasticity order: solid > liquid > gas, so the speed of sound also follows **solid > liquid > gas**.

Medium	Formula	Symbols
Solid (rod)	$v = \sqrt{Y/\rho}$	Y = Young's modulus of elasticity, ρ = density
Liquid	$v = \sqrt{B/\rho}$	B = bulk modulus of elasticity, ρ = density of liquid
Gas / air (Newton)	$v = \sqrt{P/\rho} \approx$ **319 m/s** (too low)	P = pressure, ρ = density (isothermal assumption)
Gas / air (Laplace correction)	$v = \sqrt{\gamma P/\rho} = \sqrt{\gamma RT/M} \approx$ **332 m/s** ✔	γ = ratio of specific heats (≈ 1.4 for air), R = gas constant, T = absolute temperature, M = molar mass

✎ **Source correction:** Source calls γ the "degree of freedom = 1.42". $\gamma = C_p/C_V$, the **adiabatic index (ratio of specific heats)** — about 1.4 for diatomic air (its value is *related* to the degrees of freedom by $\gamma = 1 + 2/f$, f = 5 for air, but it is not the degree of freedom itself).

↻ **De-dup note:** Newton's formula and the Laplace correction are each printed twice on Lecture 7 p. 3 (once in symbols, once with the symbol list). Kept once as the table above.

▌ADDED BY EDITOR — not in source lectures ▌ Typical values to remember (not tabulated in the lecture)

Air 0 °C ≈ 331–332 m/s, air 20 °C ≈ 343 m/s; hydrogen ≈ 1270 m/s (light gas → faster: $v \propto 1/\sqrt{M}$, so **among gases hydrogen is fastest**, but still slower than water or steel); water ≈ 1480 m/s; sea water ≈ 1530 m/s; iron/steel ≈ 5000–5900 m/s; aluminium ≈ 6400 m/s; glass ≈ 4500 m/s; rubber only ≈ 60 m/s (an exception — rubber is a solid but very in-elastic). Speed of sound in air ≈ 1/900 000 of the speed of light — hence lightning is seen before thunder is heard (count seconds ÷ 3 ≈ km to the strike).

★ **PYQ lens:** 71st Q124: "In which medium is the speed of sound maximum?" options steel / water / air / hydrogen → **steel**.

Factors affecting the velocity of sound in air

- **(i) Temperature**: on increasing the temperature the velocity of sound increases; for every 1 °C rise the velocity increases by about **0.6 m/s** ($v \propto \sqrt{T}$; $v_1/v_2 = \sqrt{T_1/T_2}$).
- **(ii) Humidity**: with increasing humidity the density of air decreases, so the velocity of sound **increases** (sound travels faster in moist air than in dry air).
- **(iii) Pressure**: at constant temperature a change of pressure changes the density in the same ratio, so **pressure has no effect** on the velocity of sound.

✎ **Source correction:** Source prints "velocity increases with 6 m/s per °C" — the accepted figure is ≈ **0.61 m/s per °C**.

▌ADDED BY EDITOR — not in source lectures ▌ Effect of wind and frequency (standard additions)

Wind blowing along the direction of sound adds its speed to the sound's speed (that is why sound carries farther downwind). The speed of sound in a given medium is **independent of frequency, wavelength and amplitude** — all notes of an orchestra reach you together.

Properties of sound waves

- **Reflection of sound**: (1) **echo**; (2) **reverberation** (multiple reflections). Sound also shows refraction, diffraction (strong, because λ is large) and interference (beats), but **not polarisation**.

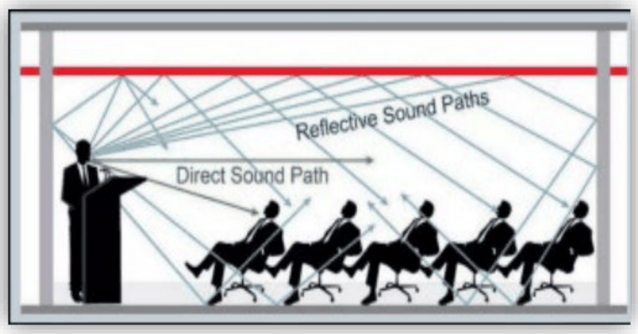

Fig. 6-3 · Direct sound path and reflective sound paths in a hall — the origin of reverberation [Lecture 7, p. 4]

Characteristics of musical sound

(i) Loudness

- Loudness decides how "high" or "low" (strong or weak) a sound is; it follows the **intensity** of sound (which depends on the **amplitude**) and is measured in **decibel (dB)**.

- Human threshold capacity: **120–140 dB**; loudness without pain for humans → up to about **85 dB**.

Sound	Approximate loudness
Jet engine (near)	140 dB
Normal talk	50–60 dB
Library / hospital	30–40 dB

▌ADDED BY EDITOR — not in source lectures ▌ Decibel scale details (not in the lecture)

0 dB = threshold of hearing (10^{-12} W/m²); every +10 dB means **10× the intensity** (+20 dB = 100×). Whisper ≈ 20–30 dB, busy traffic ≈ 70–80 dB, pain threshold ≈ 120–130 dB. WHO/CPCB norms: residential area day limit 55 dB, night 45 dB; industrial 75/70 dB; silence zones (hospitals, schools, courts) 50/40 dB. Prolonged exposure above 85 dB damages hearing.

(ii) Pitch

- Pitch is the sensation reserved by the ear that depends on the **frequency** of the sound wave: higher frequency → high pitch (shrill), lower frequency → low pitch (grave). It measures the sharpness/shrillness of a voice or sound.

★ PYQ lens: 68th Q112: shrillness of sound is determined by **frequency** — since the options were amplitude / wavelength / velocity, the key is "None of the above" (E). 68th Q120: a sitarist adjusting tension and plucking before a concert is tuning the **frequency** of the string to the accompanying instruments.

(iii) Timbre / quality

- Two sounds having the same frequency and wavelength can differ in their **overtones** (waveform); this is quality or timbre — it lets us distinguish a sitar from a flute playing the same note, or recognise a friend's voice.

▌ADDED BY EDITOR — not in source lectures ▌ Overtones and harmonics (one line of context)

A musical note = fundamental frequency + overtones (integral multiples = harmonics). Noise has no regular waveform; a musical sound has periodic vibrations.

Reverberation

- Reverberation is the persistence of sound in an enclosed space after the original sound has stopped, caused by multiple reflections of sound waves off surfaces.
- **Factors**: size of the space (larger rooms → more reverberation); materials (hard, non-absorbent surfaces like concrete or glass reflect more than soft, absorbent materials like carpet or curtains); absorption (more absorbing material → less reverberation).
- **Measurement**: **Reverberation time (RT60)** — the time for the sound level to decrease by **60 decibels** after the source has stopped; the standard quantity for describing reverberation.

▌ADDED BY EDITOR — not in source lectures ▌ Controlling reverberation & Sabine's work (context)

Auditorium walls are lined with fibre-board, rough plaster, curtains and false ceilings; seats are cushioned. W. C. Sabine's formula: $T = 0.161\, V / \Sigma \alpha S$ (V = volume, α = absorption coefficient, S = area). Good speech halls have RT ≈ 1 s; concert halls 1.5–2 s.

Echo

- An echo is the repetition of sound heard after it is reflected from a distant surface (hill, building, wall).
- **Condition for hearing an echo**: the minimum time gap between the original and the reflected sound must be **0.1 s** (persistence of hearing). Based on the speed of sound (≈ 343 m/s), the minimum distance of the reflecting surface ≈ **17 m** (source: distance ≥ 17 m; with 332 m/s the figure is 16.6 m).

Formula of echo: $t = 2d / v$ t = time for echo, d = distance of the reflecting surface, v = speed of sound (sound travels twice the distance — to and fro)

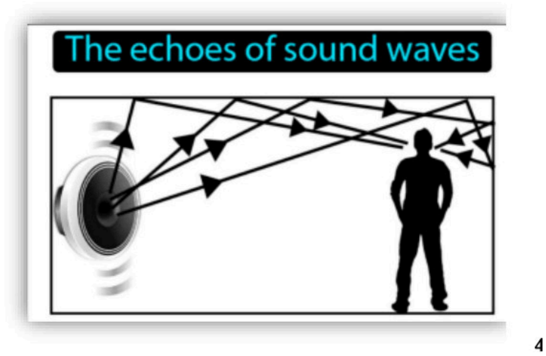

Fig. 6-4 · Echo — reflection of sound from a distant obstacle [Lecture 7, p. 5]

Applications of echo

- 1. **SONAR** (Sound Navigation And Ranging) — measures the depth of the sea, detects submarines, shipwrecks, shoals of fish (ultrasonic pulses; depth = v t / 2).
- 2. **Medical imaging** — ultrasonic waves used in ultrasound scanning (echo from tissues).
- 3. **Echolocation** — used by bats and dolphins to navigate and hunt.

▌ADDED BY EDITOR — not in source lectures▌ Echo vs reverberation (quick distinction, not in source)

An echo is a *distinct* repetition heard after ≥ 0.1 s; reverberation is the *overlapping* prolongation when reflections arrive within 0.1 s. Multiple echoes give "rolling thunder". Whispering galleries (Gol Gumbaz, Bijapur; St Paul's) work on repeated reflection of sound along curved walls.

UNIT 7

Heat, Temperature & Thermometry

Consolidated from: Lecture 8 (HEAT-01) pp. 2–4; Lecture 9 (TEMPERATURE) pp. 2–4

★ **PYQ lens:** °C → °F conversion (40 °C = 104 °F, 64th Q10; 50 °C = 122 °F, 65th Q116); poorest conductor of heat among copper/lead/mercury/zinc → lead (66th Q20); heat transfer that needs no medium → radiation (70th Q15); sublimation (63rd Q90); liquefaction of gases needs low T & high P (70th Q20). Priority B — 6 questions in 5 papers.

Heat and temperature

- **Heat** is a form of energy that flows from a body at higher temperature to a body at lower temperature because of the temperature difference. It is **energy in transit** — not a property of a body; it depends on mass, temperature change and material.
- **SI unit of heat**: **joule (J)**. Another unit: **calorie** — 1 calorie = **4.186 J** (heat needed to raise 1 g of water by 1 °C). 1 kcal = 1000 cal.
- **Temperature** is the **degree of hotness or coldness** of a body; it determines the *direction* of heat flow (from higher to lower temperature). SI unit **kelvin (K)**; other units °C and °F.

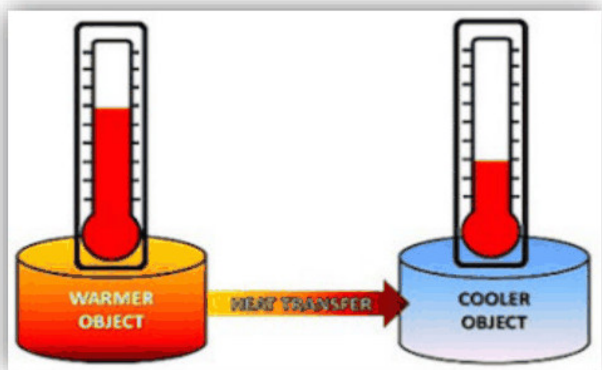

Fig. 7-1 · Heat flows from the warmer object to the cooler object until both reach the same temperature [Lecture 8, p. 2]

Heat	Temperature
Form of energy (energy in transit)	Measure of hotness (average kinetic energy of molecules)
Depends on mass (extensive)	Independent of mass (intensive)
Unit: joule (or calorie)	Unit: kelvin (or °C, °F)
Flows from one body to another	Indicates the thermal state; decides direction of heat flow
Measured with a calorimeter	Measured with a thermometer

↻ **De-dup note:** "Temperature — degree of hotness or coldness; determines direction of heat flow; SI unit kelvin; other units °C, °F" is printed in Lecture 8 p. 2 and again in Lecture 9 p. 2 (and the K/°C/°F fixed-point table three times in all). Kept once; the master scale table is below.

Modes of heat transfer

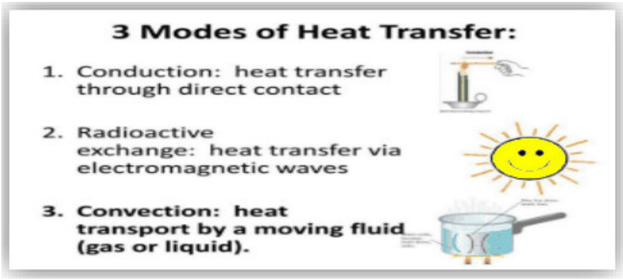

Fig. 7-2 · Three modes: conduction (direct contact), radiation (electromagnetic waves), convection (moving fluid) [Lecture 8, p. 2]

Conduction	Convection	Radiation
Depends upon temperature difference	Depends upon density difference	Heat transfer **without any medium**
Electrons/molecules transfer their vibration from one end to another without actual movement of matter	Transfer of heat occurs due to actual motion of particles (fluid)	Due to electromagnetic radiation (infrared)
Slow process	Slow process	Fast — at the speed of light, 3×10^8 m/s
Irregular path	Irregular path	Straight-line path, just like light
Mainly in solids	Liquids and gases	Any medium or vacuum

↻ **De-dup note:** The comparison table is from Lecture 9 p. 2; the detailed descriptions and applications below are from Lecture 8 pp. 2–3. Both kept together.

★ **PYQ lens:** 70th Q15: "Heat transfer that does not require a medium" → **radiation**.

I. Conduction

- Transfer of heat **without movement of particles** (from particle to particle); occurs mainly in **solids**.

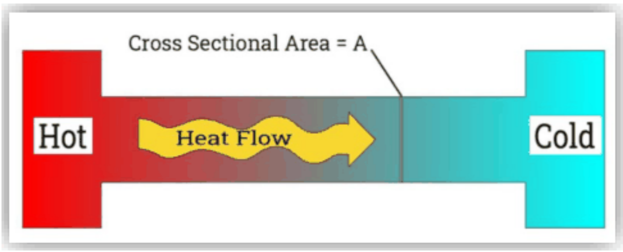

Fig. 7-3 · Heat flow along a bar of cross-section A from the hot end to the cold end [Lecture 8, p. 2]

Applications of thermal conduction

- In winter, iron chairs appear colder than wooden chairs (iron conducts heat away from the hand faster).
- Cooking utensils are made of aluminium and brass (good conductors) whereas their handles are made of wood (insulator).
- Ice is covered in gunny bags to prevent melting (trapped air is a bad conductor).
- We feel warm in woollen clothes and fur coats (air trapped between fibres).
- Two thin blankets are warmer than a single blanket of double thickness (air layer between them).
- Birds often swell their feathers in winter (trapped air).
- A new quilt is warmer than an old one (more trapped air).

★ **PYQ lens:** 66th Q20: poorest conductor of heat among copper, **lead**, mercury, zinc → lead (thermal conductivity: Cu 400 > Zn 116 > Pb 35 > Hg 8 W/m·K — strictly mercury is lower, but the key follows the standard "lead is a poor conductor" option; note both).

▌ADDED BY EDITOR — not in source lectures▐ Conductors vs insulators (numbers, not in the lecture)

Best thermal conductors: **silver > copper > gold > aluminium**; diamond conducts heat even better than silver (≈ 2200 W/m·K) though it is an electrical insulator. Poor conductors: glass, wood, water, air (0.025 W/m·K), wool, cork, styrofoam. Vacuum is the perfect insulator for conduction and convection (thermos flask).

II. Radiation

- The transfer of heat from one place to another **without heating the intervening medium**. When a body is heated and placed in vacuum it still loses heat, because heat radiation needs no medium (conduction and convection both require the presence of a material medium between source and surroundings).
- When radiation passes through any medium the medium absorbs it slightly according to its absorptive power, so the temperature of the medium rises slightly.
- Radiation intensity is measured with a **bolometer**.
- Heat radiations are always obtained in the **infrared** region of the electromagnetic spectrum, so they are called infra-red rays.

Applications of thermal radiation

- Heat from the Sun reaching the Earth; solar cookers and solar panels; infrared heaters and thermal imaging; black surfaces absorb more heat (used in solar devices); thermos flasks minimise heat loss via radiation (silvered walls).

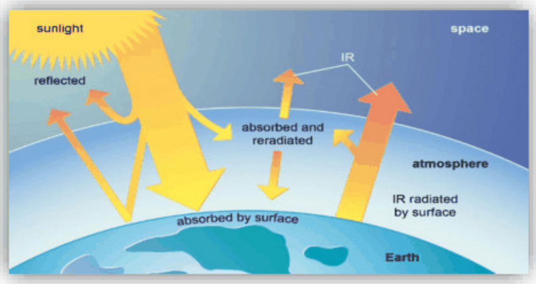

Fig. 7-4 · Earth's radiation balance — sunlight absorbed and re-radiated, infrared trapped by the atmosphere [Lecture 8, p. 3]

III. Convection

- Convection is the transfer of heat in **liquids and gases** due to the **actual movement of particles**; it occurs because of density differences created by temperature variation (hot fluid expands, becomes lighter and rises; cold fluid sinks).

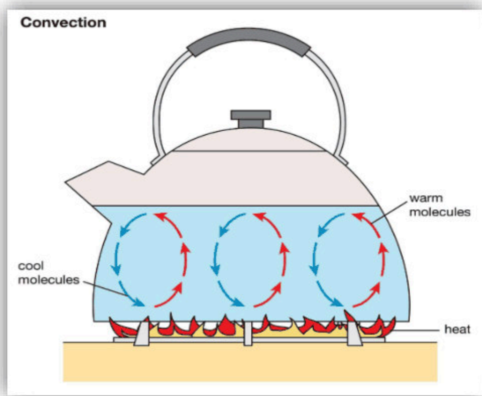

Fig. 7-5 · Convection currents in a kettle — warm molecules rise, cool molecules sink [Lecture 8, p. 3]

Applications of convection

- **Land & sea breezes** — **Day**: land heats faster → air over land rises → cool air moves from sea to land (**sea breeze**). **Night**: land cools faster → air moves from land to sea (**land breeze**).
- **Trade winds** — Equator: high temperature → low pressure (air rises); Poles: low temperature → high pressure. Air moves from poles to the equator and gets deflected due to the Earth's rotation, forming steady north-east trade winds.
- **Monsoons** — land heats more than the ocean in summer; moist air from the ocean moves to land, rises, cools and condenses → causes rainfall.
- **Ventilation** — warm air rises and exits through ventilators/exhaust; fresh air enters from outside (based on natural/forced convection).

▌ADDED BY EDITOR — not in source lectures ▌ Other convection examples (standard, not in the lecture)

Chimney draught, hot-air balloons (also in Unit 8), room heaters placed near the floor and air-conditioners near the ceiling, water heating in a geyser from below, the Earth's mantle convection (plate tectonics), the Sun's convective zone. In a vacuum or in orbit (weightlessness) convection stops — a candle flame in a spacecraft is spherical and blue.

Specific heat capacity and latent heat

Specific heat capacity (c)

- The amount of heat required to raise the temperature of **1 kg** of a substance by **1 °C** (or 1 K).

$Q = m c \Delta T$ Q = heat, m = mass, c = specific heat capacity, ΔT = change in temperature

▌ADDED BY EDITOR — not in source lectures ▌ Values & consequences (asked repeatedly in general-science papers)

Water has the **highest specific heat among common substances: 4186 J/kg·K (1 cal/g·°C)**; ice ≈ 2100, steam ≈ 2000, aluminium ≈ 900, iron ≈ 450, copper ≈ 385, mercury ≈ 140, lead ≈ 130 J/kg·K. Hence water is used as a coolant in car radiators and as a hot-water-bottle filler, sea breezes exist, coastal climates are moderate, and a watermelon stays cool longer. **Calorimetry / principle of mixtures** (Unit 8): heat lost = heat gained.

Latent heat

- Heat absorbed or released **without change in temperature** during a change of state.
- **I. Latent heat of fusion** — heat required to convert solid → liquid at constant temperature (ice → water). Temperature remains constant during melting; the energy is used to break intermolecular forces; unit **J/kg**.
- **II. Latent heat of vaporisation** — heat required to convert liquid → gas at constant temperature (water → steam). Requires **more heat than fusion**; used to overcome strong intermolecular forces; responsible for the cooling effect of evaporation.

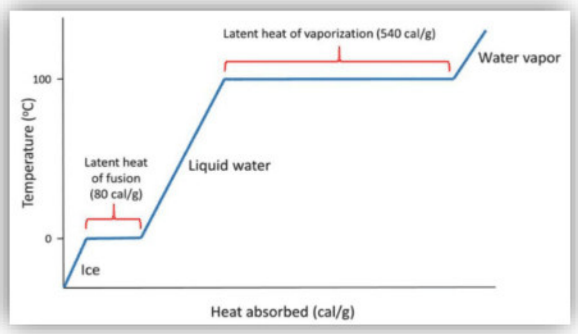

Fig. 7-6 · Heating curve of water: temperature vs heat absorbed — flat steps at 0 °C (latent heat of fusion 80 cal/g) and 100 °C (latent heat of vaporisation 540 cal/g) [Lecture 8, p. 4]

▌ADDED BY EDITOR — not in source lectures ▌ Latent-heat values & why steam burns worse (not in the lecture text, only in the graph)

L_{fusion}(ice) = **80 cal/g = 3.34 × 10^5 J/kg**

$L_{vaporisation}$(water) = **540 cal/g = 2.26 × 10^6 J/kg**. Steam at 100 °C causes a more severe burn than water at 100 °C because it first gives up 540 cal/g on condensing. Ice at 0 °C cools a drink better than water at 0 °C because it absorbs 80 cal/g while melting.

Effects of heat

- Increase in temperature; change in state (latent heat); expansion of matter (Unit 8); increase in molecular motion.

Change of state of matter

Phase conversion due to hidden (latent) heat represents change in the state of matter.

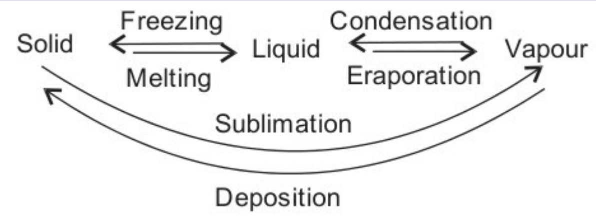

Fig. 7-7 · Solid ⇌ liquid ⇌ vapour: melting/freezing, evaporation/condensation, sublimation/deposition [Lecture 9, p. 2]

- **Melting** → conversion of a solid into its liquid form at constant temperature (**endothermic** — heat absorbed).
- **Freezing** → reverse of melting, with the release of energy (**exothermic** — heat released).
- **Evaporation** → conversion of surface liquid into vapour at **all temperatures**; the higher the temperature (and the larger the surface/wind, the lower the humidity), the higher the evaporation. It causes cooling.
- **Boiling** → evaporation within the whole mass at a constant temperature; the temperature at which the vapour pressure of the liquid equals the atmospheric pressure is the boiling point.
- **Sublimation** → conversion of a solid directly into vapour — e.g. solid CO_2 (dry ice), camphor, naphthalene, iodine, ammonium chloride.
- **Deposition** → reverse of sublimation (vapour → solid), e.g. frost.
- **Condensation** → conversion of vapour into liquid (dew, clouds).
- **Note**: when impurities are added to a liquid, its **boiling point rises** (and its freezing point falls — salt on icy roads, antifreeze).

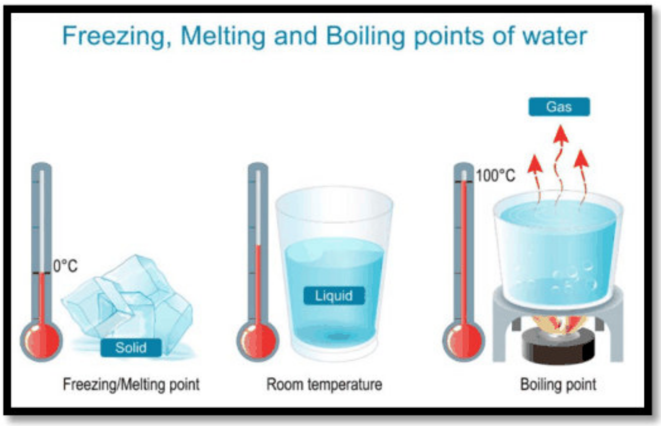

Fig. 7-8 · Freezing/melting point (0 °C), room temperature and boiling point (100 °C) of water [Lecture 9, p. 2]

✎ **Source correction:** Lecture 9 labels freezing as an "Endothermic process". Freezing **releases** heat — it is **exothermic**; melting, evaporation, boiling and sublimation are endothermic.

★ **PYQ lens:** 63rd Q90: conversion of a solid directly into gas → **sublimation**. 70th Q20: suitable condition for liquefaction of gases → **low temperature, high pressure** (below the critical temperature).

▌ADDED BY EDITOR — not in source lectures ▌ Pressure and boiling/melting points (pressure-cooker physics — not in the lecture)

Boiling point rises with pressure (pressure cooker ≈ 120 °C at 2 atm, food cooks faster) and falls with altitude (water boils near 70 °C on Everest; cooking is difficult in hills). The melting point of ice **falls** with pressure (regelation — skating on ice, glaciers flowing); for most other solids it rises. **Anomalous expansion of water**: water is densest at **4 °C**; it expands on cooling from 4 °C to 0 °C and on freezing (ice floats, pipes burst, lakes freeze from the top so fish survive) — this Lecture-1 plan topic is treated in Unit 8.

Temperature scales

Scale	Lower fixed point (freezing point of water)	Upper fixed point (boiling point of water)	Divisions / note
Celsius (°C)	0 °C	100 °C	100 divisions; SI-derived
Kelvin (K)	273.15 K	373.15 K	SI base unit; absolute scale (no negative values)
Fahrenheit (°F)	32 °F	212 °F	180 divisions
Réaumur (°R / °Re)	0°	80°	80 divisions
Rankine (°Ra)	491.67°	671.67°	Absolute Fahrenheit scale

↻ **De-dup note:** Lecture 9 gives this 5-scale table on p. 2 and a 3-scale (Celsius/Fahrenheit/Kelvin) "temperature scales comparison" again on p. 4; Lecture 8 lists the units once more. One master table is kept.

Conversion formulae

$$C/100 = (F - 32)/180 = (K - 273)/100 \Leftrightarrow C/5 = (F - 32)/9 = (K - 273)/5$$

$$C = (5/9)(F - 32) \quad F = (9/5)C + 32 \quad K = C + 273.15 \quad (C - 0)/100 = (F - 32)/(212 - 32) = (K - 273.16)/(373.16 - 273.16)$$

- **–40° is the same on the Celsius and Fahrenheit scales** (–40 °C = –40 °F). Absolute zero = 0 K = –273.15 °C = –459.67 °F. Normal human body temperature = 37 °C = 98.6 °F = 310 K.

★ **PYQ lens:** 64th Q10: 40 °C = (9/5 × 40) + 32 = **104 °F**. 65th Q116: 50 °C = 90 + 32 = **122 °F**. Practise 0 °C = 32 °F, 37 °C = 98.6 °F, 100 °C = 212 °F, 25 °C = 77 °F.

Measuring instruments — thermometers

Accurate measurement of temperature is essential in science, medicine, industry and daily life; thermometers operate on different physical principles such as thermal expansion, electrical resistance and radiation.

Types of thermometer

- **Mercury thermometer** — **Principle**: uniform expansion of mercury in a sealed glass tube. **Range**: wide, from **–38 °C to 356 °C** (can be extended by filling nitrogen above the mercury). **Pros**: opaque silver liquid (clear readings), doesn't wet glass, good conductor, accurate, uniform expansion. **Cons**: toxic and hazardous if broken; less useful clinically now (banned in many countries).
- **Alcohol thermometer** — **Principle**: coloured alcohol (ethyl alcohol etc.) expands in a glass tube. **Range**: very low temperatures (down to about **–115 °C / source: –200 °C**), limited above by alcohol's boiling point (≈ 78 °C). **Pros**: non-toxic, safe, great for very cold conditions (freezers, polar regions). **Cons**: transparent (needs dye), can evaporate, less durable, wets glass.
- **Clinical thermometer** — a mercury thermometer with a **constriction (narrowing) in the bore** so the mercury column does not fall back until shaken; range **35–42 °C** (95–108 °F); normal body temperature **36.9 °C / 98.4 °F** (source; 37 °C / 98.6 °F is the textbook value).

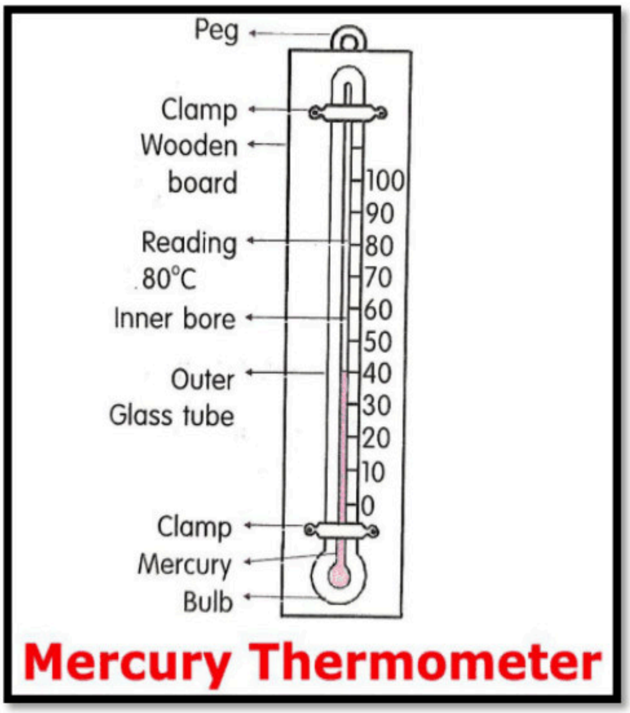

Fig. 7-9 · Mercury thermometer — bulb, capillary bore, scale [Lecture 9, p. 3]

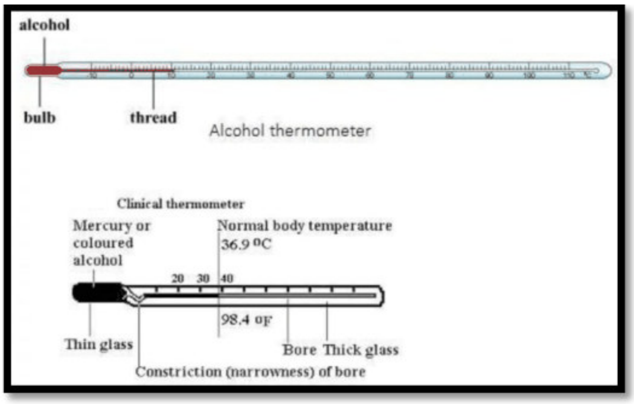

Fig. 7-10 · Alcohol thermometer (top) and clinical thermometer with constriction (bottom) [Lecture 9, p. 3]

- **Resistance thermometer (RTD)** — **Principle**: resistance of a conductor (usually platinum) changes with temperature; the metal's resistance increases with temperature. Common materials Pt100, Pt1000 (2-, 3- and 4-wire arrangements). **Use**: high accuracy, wide range, good linearity — common in laboratories and industry.

$R = R_0(1 + \alpha T)$ R = resistance at temperature T, R_0 = resistance at 0 °C, α = temperature coefficient of resistance

- **Platinum resistance thermometer (PRT)** — a highly accurate instrument using platinum wire whose resistance changes predictably with temperature; one of the most precise thermometers in science and industry (used to define the practical temperature scale).

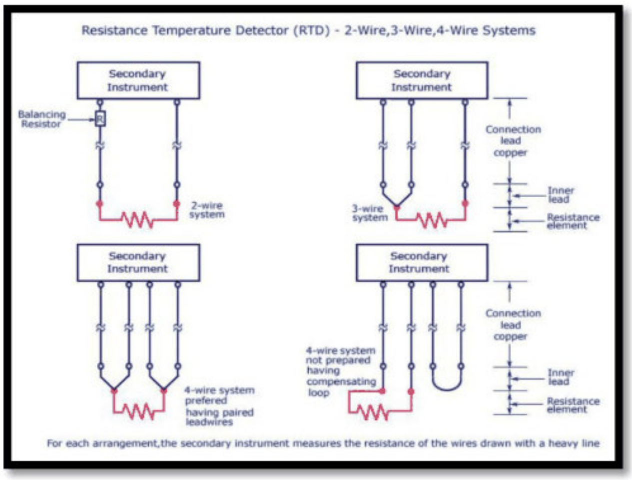

Fig. 7-11 · RTD — 2-wire, 3-wire and 4-wire measuring arrangements [Lecture 9, p. 3]

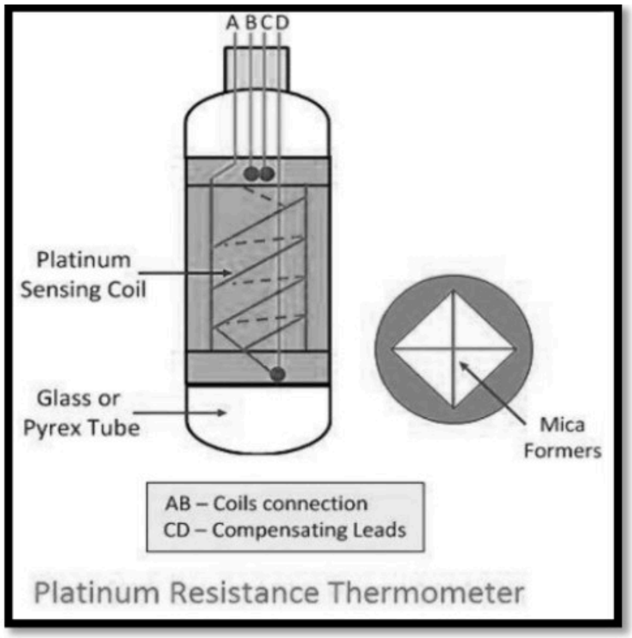

Fig. 7-12 · Platinum resistance thermometer — platinum sensing coil in a glass/Pyrex tube with compensating leads [Lecture 9, p. 4]

- **Pyrometer (infrared thermometer)** — **Principle**: detects the **infrared energy (heat) radiated** by objects; no contact needed. **Use**: measures very high temperatures (furnaces, molten metal, > 1000 °C) where ordinary thermometers cannot be used; non-contact, good for moving objects and body-temperature screening.

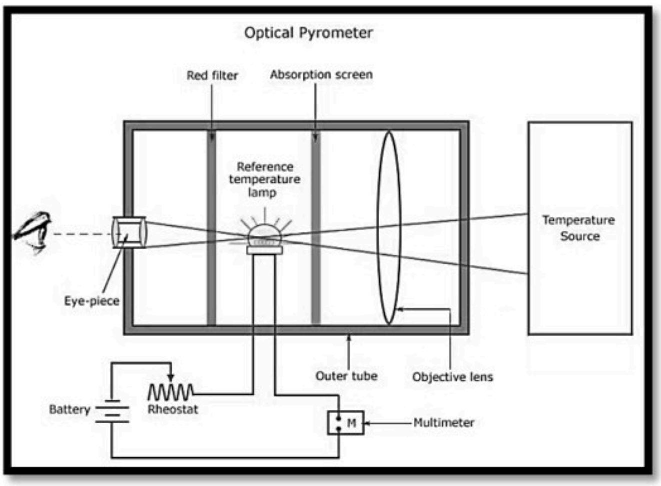

Fig. 7-13 · Optical pyrometer — the brightness of a reference lamp filament is matched with that of the hot source [Lecture 9, p. 4]

- **Digital thermometer** — uses electronic sensors (thermistors, thermocouples); displays digitally; easy for home/clinical use.
- **Bimetallic strip thermometer** — two metals bonded together bend with temperature change; used in ovens and thermostats (Unit 8).
- **Thermocouple** — two different metals joined; the voltage (Seebeck emf) changes with temperature; used for high/low industrial temperatures (Unit 8, Seebeck effect).
- **Gas thermometer** — measures temperature via gas pressure/volume (e.g. constant-pressure or constant-volume hydrogen/helium gas thermometer) — the most accurate primary standard.

Effects of temperature (summary from Lecture 9)

- Expansion and contraction of substances; change in physical state (solid ⇌ liquid ⇌ gas); increase in molecular kinetic energy.

▌ADDED BY EDITOR — not in source lectures ▌ Which thermometer for what (quick reference, not in the lecture)

Six's maximum–minimum thermometer (weather stations); thermistor (fast, small — digital fever thermometers); liquid-crystal strip thermometers; cryometer for < –100 °C (uses gas or resistance); the highest temperatures (stars) are found from Wien's law of the emitted spectrum (Unit 8). Mercury freezes at –39 °C, so alcohol (freezes –115 °C) is used in polar and upper-air thermometers.

UNIT 8

Thermal Expansion, Radiation Laws & Laws of Thermodynamics

Consolidated from: Lecture 10 (HEAT-PART-3) pp. 2–5

★ **PYQ lens:** Thomson (thermoelectric) effect — potential variation along a conductor due to temperature variation (67th Q72); solar constant 1.4 kW/m² (70th Q113). Priority C, but the thermodynamics laws and Stefan/Wien are standard general-science material.

Thermal expansion

Thermal expansion is the tendency of a substance to change its dimensions (length, area or volume) when its temperature changes — usually expanding on heating and contracting on cooling.

Types of thermal expansion

Type	What changes	Coefficient	Example
Linear	Length of a solid	$\alpha = \Delta l / (l_{original} \times \Delta T)$	Railway tracks expand in summer, so gaps are left between rails
Areal (superficial)	Area of a surface	$\beta = \Delta A / (A_{original} \times \Delta T)$	Metal plates expand when heated
Volume (cubical)	Volume of a solid, liquid or gas	$\gamma = \Delta V / (V_{original} \times \Delta T)$	Liquid expands in a thermometer when heated

▌ADDED BY EDITOR — not in source lectures ▌ Relation between the coefficients (standard result the lecture omits)

For an isotropic solid **$\beta = 2\alpha$ and $\gamma = 3\alpha$**. Liquids have only volume expansion ($\gamma_{real} = \gamma_{apparent} + \gamma_{vessel}$); gases expand most ($\gamma \approx 1/273$ per °C at constant pressure — Charles's law). Invar (Fe–Ni alloy) has $\alpha \approx 0$ and is used in pendulum clocks and measuring tapes

Pyrex glass expands little, so it does not crack on sudden heating; ordinary glass does.

Applications of thermal expansion (solids, liquids, gases)

- **1. Expansion joints in structures** — bridges, flyovers and buildings are provided with expansion joints that let materials expand and contract; prevents cracking, bending or structural failure, especially in regions with extreme temperature variation.
- **2. Railway tracks and roads** — small gaps are left between rails; in hot weather rails expand and the gaps prevent buckling or warping; concrete roads are also built with expansion gaps.
- **3. Thermometers (liquid expansion)** — mercury or alcohol thermometers work on the volume expansion of liquids; the liquid expands and moves up the capillary tube (medical, laboratory and environmental measurements).
- **4. Bimetallic strip (temperature-control devices)** — two metals with different expansion rates bonded together; on heating one expands more, causing bending. Used in electric irons, thermostats, circuit breakers — automatic temperature regulation.
- **5. Loosening tight lids and mechanical fittings** — metal lids expand when heated and open easily; heating a nut makes it expand and loosen from a bolt.
- **6. Overhead electric wires** — kept slightly loose (sagging) during installation: they expand in summer and contract in winter; prevents snapping due to contraction in cold weather.
- **7. Riveting** — rivets are heated before joining metal plates; on cooling they contract and form a tight, strong joint (bridges, boilers, shipbuilding).
- **8. Glass and metal fitting (thermal stress use)** — metal rims are heated to fit tightly around wooden wheels or glass objects; after cooling they contract and hold the object firmly.
- **9. Pipelines and machinery** — oil pipelines and steam pipes are designed with loops or bends to allow expansion and prevent damage due to thermal stress.

- **10. Hot-air balloons (gas expansion)** — air expands when heated, becomes less dense and provides lift.

Effect of temperature on the time period of a simple pendulum

- A pendulum clock keeps correct time at temperature θ. If the temperature rises to θ′ (> θ), the length of the pendulum increases by linear expansion, its time period increases ($T = 2\pi\sqrt{l/g}$) and the **clock loses time (runs slow)** in summer; in winter it gains time.

▌ADDED BY EDITOR — not in source lectures ▌ Anomalous expansion of water (Lecture-1 plan topic; only the graphic appears in Lecture 10)

Water **contracts** on heating from 0 °C to 4 °C and expands above 4 °C, so its density is **maximum at 4 °C (1000 kg/m³)**. Consequences: ice (density 917 kg/m³) floats; in winter a lake cools until the whole body is 4 °C, then the colder surface water (lighter) stays on top and freezes — ice insulates the water below, aquatic life survives at 4 °C; water pipes and rocks crack when water freezes (≈ 9 % expansion); ocean water is densest near 4 °C. Hope's experiment demonstrates it.

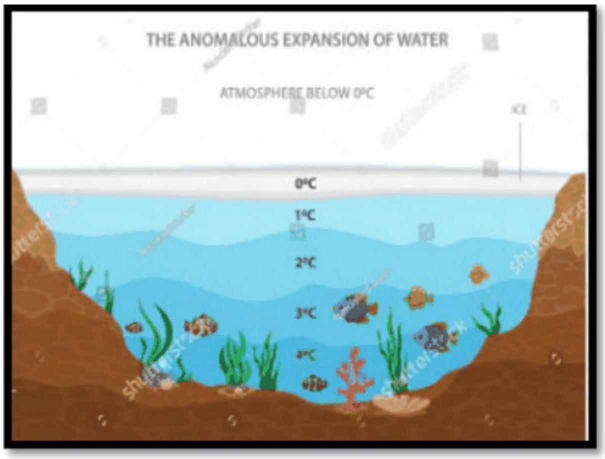

Fig. 8-1 · Anomalous expansion of water — a pond in winter: ice at 0 °C on top, water at 4 °C at the bottom [Lecture 10, p. 3]

Thermoelectric (Seebeck) effect

- The **Seebeck effect** is the phenomenon in which an electric voltage (emf) is generated in a circuit made of two dissimilar conductors when their junctions are kept at different temperatures.
- **Basic principle**: two different metals joined form two junctions — a **hot junction** (higher temperature) and a **cold junction** (lower temperature). Because of the temperature difference, charge carriers (electrons/holes) diffuse from the hot to the cold region, creating a potential difference (voltage) in the circuit. This is the working principle of the **thermocouple**.

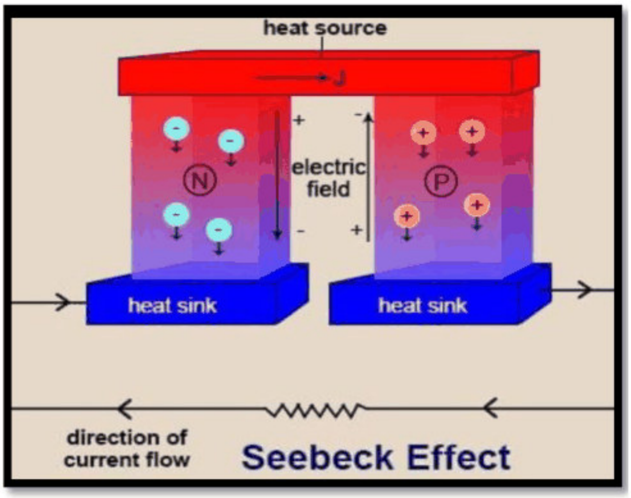

Fig. 8-2 · Seebeck effect — charge carriers diffuse from the heat source towards the heat sinks, driving a current [Lecture 10, p. 2]

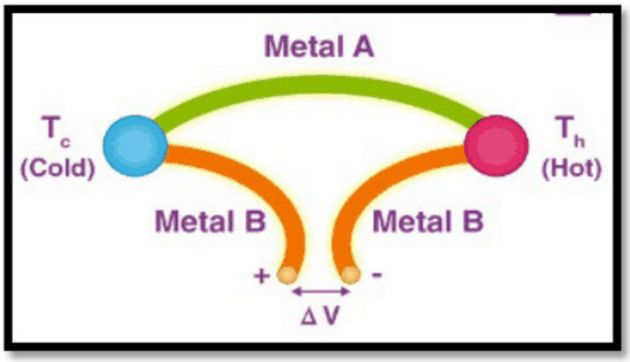

Fig. 8-3 · Thermocouple: metals A and B with hot (T_h) and cold (T_c) junctions giving ΔV [Lecture 10, p. 2]

▌ADDED BY EDITOR — not in source lectures ▌ Peltier & Thomson effects (67th Q72 asked exactly this — not in the lecture)

Seebeck (1821): temperature difference across two junctions → emf. **Peltier** (1834): the reverse — a current through the junction of two metals absorbs heat at one junction and releases it at the other (used in portable thermoelectric coolers). **Thomson** (Kelvin, 1851): when a current flows through a *single* conductor whose ends are at different temperatures, heat is absorbed or evolved along its length — equivalently, a temperature gradient along a conductor produces a potential gradient along it. Joule heating (I^2Rt) is not a thermoelectric effect.

★ **PYQ lens:** 67th Q72: "Due to temperature variation along a conductor, potential variation occurs along it. This is known as …" → **Thomson effect** (not Seebeck, which needs two dissimilar metals).

Law of mixtures — principle of calorimetry

- When two bodies at different temperatures are mixed, heat is transferred from the body at higher temperature to the body at lower temperature until both acquire the **same temperature**.
- The hotter body releases heat and the colder body absorbs it: **heat lost = heat gained** (the principle of calorimetry) — it represents the **law of conservation of heat energy** (no heat lost to the surroundings).

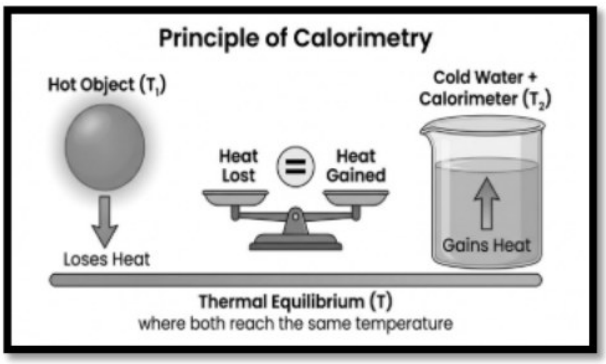

Fig. 8-4 · Principle of calorimetry — heat lost by the hot object equals heat gained by the cold water + calorimeter [Lecture 10, p. 4]

▌ADDED BY EDITOR — not in source lectures▐ Worked example (not in source)

100 g of water at 80 °C is mixed with 200 g at 20 °C: 100 × (80 – T) = 200 × (T – 20) ⇒ T = 40 °C. Mixing equal masses of ice at 0 °C and water at 80 °C: 80 cal/g is used to melt the ice, leaving 0 °C water — the final temperature is 0 °C with all ice just melted.

Thermal radiation laws

Ideal black body

- A body whose surface absorbs **all** incident thermal radiation at low temperature, irrespective of wavelength, and emits out all these absorbed radiations at high temperature, is an ideal black body.
- Key characteristics: perfect absorber (absorbs 100 % — no reflection or transmission); perfect emitter (emits maximum radiation); **emissivity e = 1**; radiation depends **only on temperature**, not on the material; appears perfectly black at all temperatures (a cavity with a small hole, lamp-black or platinum-black are close approximations).

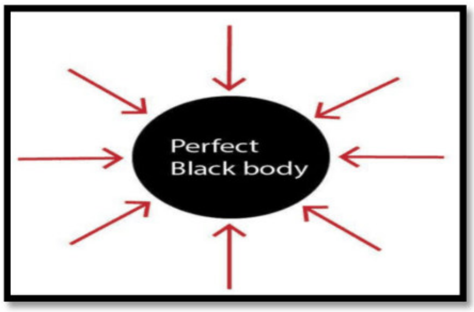

Fig. 8-5 · A perfect black body absorbs every ray that falls on it [Lecture 10, p. 4]

Stefan's law (Stefan–Boltzmann law)

- The amount of radiation emitted per second per unit area by an ideal black body is directly proportional to the **fourth power of its absolute temperature**.

$E \propto T^4$ $P = e\sigma A T^4$ σ (Stefan–Boltzmann constant) = 5.67×10^{-8} W m^{-2} K^{-4} (e = emissivity, A = surface area)

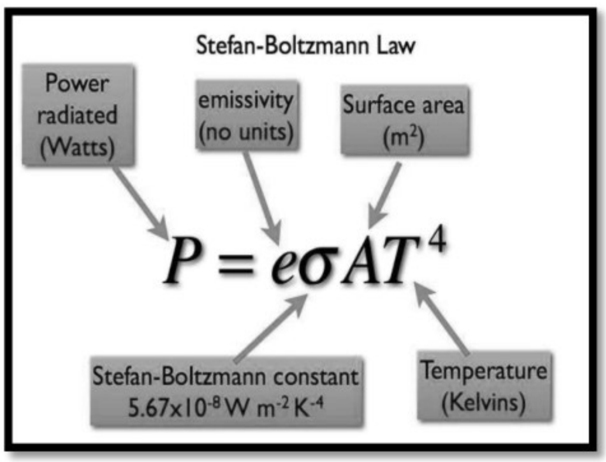

Fig. 8-6 · Stefan–Boltzmann law: $P = e\sigma AT^4$ — power radiated, emissivity, area, constant, temperature [Lecture 10, p. 4]

Wien's displacement law

- The wavelength corresponding to maximum emission of radiation **decreases with increasing temperature**: $\lambda_m \propto 1/T$.

$\lambda_m T = b$ b (Wien's constant) = 2.89×10^{-3} m K dimensions of b: $M^0 L^1 T^0 \theta^1$

▌ADDED BY EDITOR — not in source lectures ▌ What Wien's law explains (context, not in source)

A heated iron glows dull red → orange → yellow → white as T rises (λ_m shifts to shorter λ). Colour of stars gives their temperature: red stars (Betelgeuse ≈ 3500 K) are cooler than blue-white ones (Rigel ≈ 12 000 K); the Sun (5800 K) peaks at ≈ 500 nm (green-yellow). Human bodies (310 K) radiate mainly at ≈ 9.3 µm (infrared) — the basis of thermal imaging and IR thermometers.

Newton's law of cooling

- The rate of cooling (–dT/dt) of a body is directly proportional to the **excess of its temperature over the surroundings**, provided the excess is small ($T - T_0 \lesssim 35$ °C, as printed): $-dT/dt \propto (T - T_0)$.

$-dT/dt = K (T - T_0)$ T = temperature of the body, T_0 = temperature of the surroundings (all in °C), K = constant

- The negative sign indicates that the temperature falls with time — so the rate of cooling keeps decreasing as the body approaches the surrounding temperature. (Hence hot tea cools fast at first and slowly later; adding cold milk *later* leaves the tea hotter than adding it at once.)

Solar constant

▌ADDED BY EDITOR — not in source lectures ▌ Solar constant — asked in 70th Q113 (not in the lecture)

The **solar constant** is the solar energy received per second per unit area at the top of the Earth's atmosphere, at the mean Earth–Sun distance, perpendicular to the rays: ≈ **1.361–1.4 kW/m² (≈ 2 cal cm⁻² min⁻¹)**. Using Stefan's law with it gives the Sun's surface temperature ≈ 5800 K. About 30 % of it is reflected (albedo); only ≈ 1 kW/m² reaches the ground at noon on a clear day — the figure used for rating solar panels.

Laws of thermodynamics

The laws of thermodynamics are fundamental principles that govern how energy is transferred and transformed in physical systems; they are essential in physics, chemistry, engineering and even environmental science.

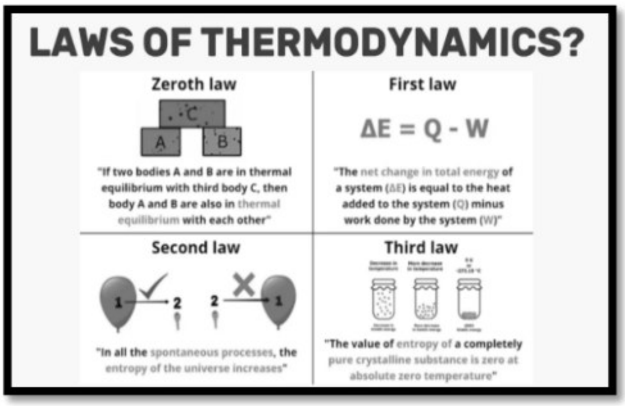

Fig. 8-7 · Zeroth, first, second and third laws of thermodynamics in one card [Lecture 10, p. 5]

I. Zeroth law

- If two systems are each in thermal equilibrium with a third system, then they are in thermal equilibrium with each other.
- **Meaning**: it defines the concept of **temperature**; it is the basis for using thermometers. **Example**: if body A = body B (same temperature) and body B = body C, then A = C.

II. First law (law of energy conservation)

- Energy can neither be created nor destroyed; it can only change from one form to another. Heat supplied to a system is used to increase its internal energy or to do work.

ΔQ = ΔU + ΔW ΔQ = heat supplied, ΔU = change in internal energy, ΔW = work done by the system

- **Examples**: a steam engine converts heat into mechanical work; the human body converts food into energy.

▌ADDED BY EDITOR — not in source lectures ▌ Sign convention & processes (standard additions)

Heat *given to* the system and work done *by* the system are positive. Isothermal (T constant, ΔU = 0 → Q = W), adiabatic (Q = 0 → W = –ΔU, e.g. bursting of a cycle tube, cloud formation), isochoric (V constant, W = 0), isobaric (P constant). A **perpetual-motion machine of the first kind** (work from nothing) is impossible by this law.

III. Second law

- It is impossible to convert all heat into work without any loss (Kelvin–Planck statement).
- **Clausius statement**: heat cannot flow spontaneously from a colder body to a hotter body (a refrigerator needs external work).
- **Key concept — Entropy (S)**: a measure of disorder/randomness. In an isolated system entropy always **increases** (never decreases). Natural processes are **irreversible**; it explains why the efficiency of a heat engine is always **less than 100 %**. **Example**: heat flows from hot tea to the surrounding air, not the reverse.

▌ADDED BY EDITOR — not in source lectures ▌ Carnot efficiency (announced in the Lecture-1 plan; not in the lecture)

Maximum possible efficiency of a heat engine working between source T_1 and sink T_2 (kelvin) is $\eta = 1 - T_2/T_1$

100 % would need $T_2 = 0$ K. Coefficient of performance of a refrigerator = $T_2/(T_1 - T_2)$. A **perpetual-motion machine of the second kind** (engine with 100 % efficiency) is impossible. Diesel engines (η ≈ 35–40 %) beat petrol engines (≈ 25–30 %) because of their higher compression ratio.

IV. Third law

- As the temperature approaches **absolute zero**, the entropy of a perfect crystal approaches zero.
- **Meaning**: absolute zero (0 K = –273.15 °C) is **unattainable** in a finite number of steps; the system becomes perfectly ordered.

▌ADDED BY EDITOR — not in source lectures ▌ Absolute zero in practice

The lowest laboratory temperatures (≈ 10^{-10} K) are reached by laser cooling and magnetic refrigeration; the cosmic microwave background is at 2.7 K; helium liquefies at 4.2 K and becomes a superfluid at 2.17 K; many metals become **superconductors** near absolute zero (mercury at 4.2 K — Kamerlingh Onnes, 1911).

UNIT 9

Kinematics — Motion in a Straight Line

Consolidated from: Lecture 11 (Kinematics) pp. 2–4; Lecture 12 p. 2 (graphs) and pp. 2–3 (types of motion)

★ **PYQ lens:** Scalar vs vector: pressure is a scalar (65th Q113), speed is not a vector (67th Q67); Galileo was the first to define speed (67th Q69). The 71st paper turned kinematics numerical: $s = \frac{1}{2}at^2$ gives $a = 2$ m/s² for a truck covering 400 m in 20 s from rest (71st Q122, continued in Unit 10). Priority C alone, A when combined with Unit 10.

Kinematics — basic ideas

- **Kinematics** is the branch of classical mechanics that describes the motion of points, objects and systems using displacement, velocity and acceleration, **without considering the forces** causing the motion (dynamics does that).
- **Rest and motion**: an object is at rest if its position does not change with time relative to its surroundings; otherwise it is in motion. Rest and motion are **relative** terms — they depend entirely on the chosen **frame of reference** (a passenger is at rest relative to the train, in motion relative to the platform).

↻ **De-dup note:** Lecture 11 states the definition of kinematics twice and the distance/displacement and speed/velocity definitions twice each (p. 2). Each is kept once below, with the comparison table.

Distance and displacement

- **Distance** — the actual path length covered by an object; a **scalar**.
- **Displacement** — the change in position: the shortest (straight-line) length from the initial to the final position, with direction; a **vector**.

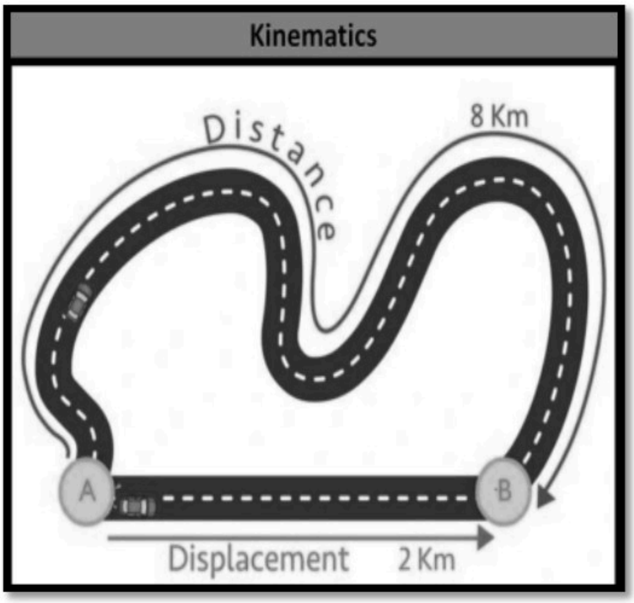

Fig. 9-1 · Distance (8 km along the road) vs displacement (2 km straight from A to B) [Lecture 11, p. 2]

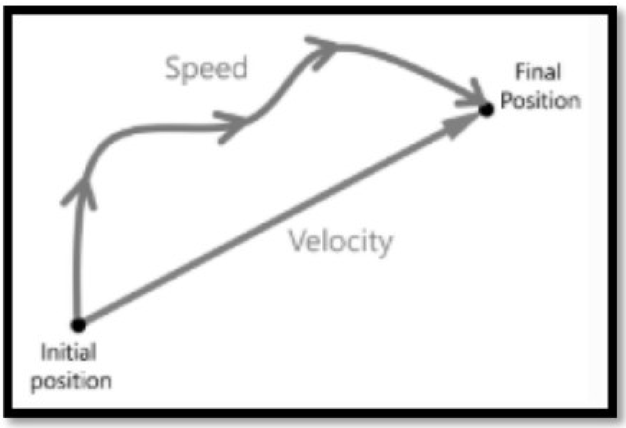

Fig. 9-2 · Speed is measured along the actual path; velocity along the displacement [Lecture 11, p. 2]

Distance	Displacement
Total length covered between the initial and final points	Shortest length between the initial and final points
Has only magnitude → **scalar** (e.g. "a person goes 10 m")	Has magnitude as well as direction → **vector** (e.g. "Mohan goes 10 m towards east")
Always **positive**	Can be positive, negative or **zero** (a round trip has zero displacement)
Distance = speed × time	Displacement = velocity × time

Distance	Displacement
Distance ≥ |displacement|; equal only for straight-line motion without reversal	Magnitude of displacement ≤ distance

Speed and velocity

- **Speed** — rate of change of distance; a scalar; SI unit m/s (1 km/h = 5/18 m/s; 1 m/s = 3.6 km/h).
- **Velocity** — rate of change of displacement; specifies magnitude *and* direction; a vector. A change in **either** speed **or** direction changes the velocity.
- **Uniform speed**: equal distances in equal intervals of time; **non-uniform speed**: unequal distances in equal intervals.

Uniform vs non-uniform velocity

- **Uniform velocity** — both speed and direction remain unchanged: equal displacements in equal intervals of time in a fixed direction; acceleration is zero. Example: a car moving on a straight road at a constant 60 km/h towards east. Rare in real life, because even a slight change in direction or speed makes velocity non-uniform.
- **Non-uniform velocity** — velocity changes with time, due to a change in speed, in direction, or both; unequal displacements in equal intervals of time; acceleration is present. Examples: a car in traffic (changing speed); a body moving on a circular path at constant speed (direction changing).

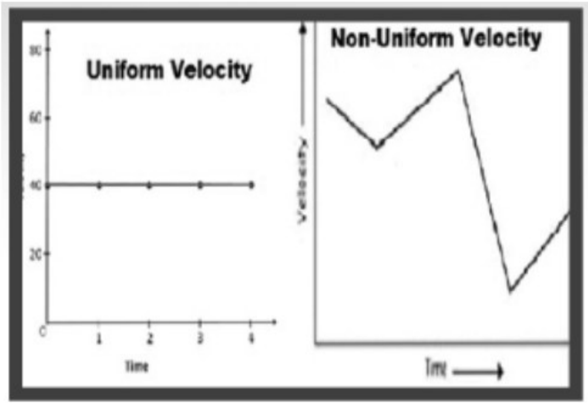

Fig. 9-3 · Uniform velocity (flat v–t graph) vs non-uniform velocity [Lecture 11, p. 3]

Average speed

- **Average speed** = total distance ÷ total time; **average velocity** = total displacement ÷ total time.
- **Case I** — a particle travels the first half *distance* with speed v_1 and the second half with speed v_2:

$$v_{av} = (s + s)/(t_1 + t_2) = 2s / (s/v_1 + s/v_2) = 2\, v_1 v_2 / (v_1 + v_2) \text{ (harmonic mean of the two speeds)}$$

- **Case II** — a particle travels with speed v_1 for the first half *time* and v_2 for the second half time: $s_1 = v_1 t$, $s_2 = v_2 t$; total distance = $(v_1 + v_2)t$; total time = $2t$:

$$v_{av} = (v_1 + v_2)\, t / 2t = (v_1 + v_2) / 2 \text{ (arithmetic mean of the two speeds)}$$

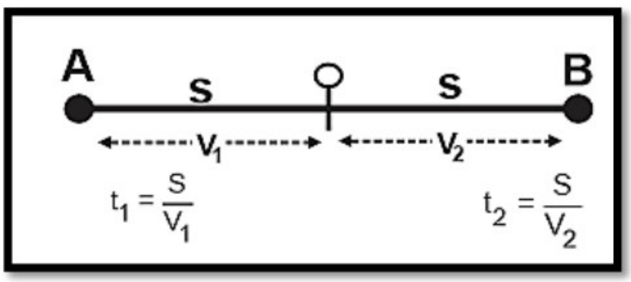

Fig. 9-4 · Case I — equal distances s at v_1 and v_2 [Lecture 11, p. 4]

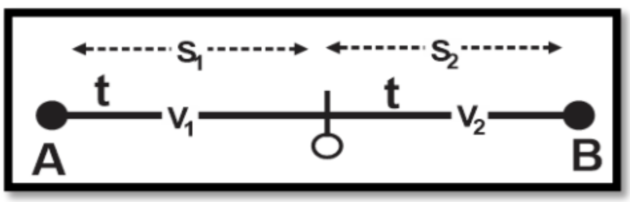

Fig. 9-5 · Case II — equal times t at v_1 and v_2 [Lecture 11, p. 4]

★ **PYQ lens:** 67th Q69: the first person to define speed → **Galileo** (distance per unit time). 67th Q67: not a vector → **speed** (torque, displacement, velocity are vectors).

▌ADDED BY EDITOR — not in source lectures ▌ Scalars vs vectors — the list BPSC draws its options from

Scalars: distance, speed, mass, time, temperature, work, energy, power, pressure, density, charge, electric potential, current (treated as scalar), frequency, volume. **Vectors**: displacement, velocity, acceleration, force, weight, momentum, impulse, torque, angular velocity, electric field, magnetic field, gravitational field, area (as a directed quantity). Current has direction but does not add by the parallelogram law, so it is a scalar.

Acceleration

- **Acceleration** = rate of change of velocity (a vector), unit m/s²; negative acceleration (slowing down) is **retardation/deceleration**.
- **Uniform acceleration** — velocity changes by **equal amounts in equal intervals of time**; acceleration is constant, the object gains or loses velocity at a steady rate. Example: a freely falling body near the Earth's surface — acceleration due to gravity **g ≈ 9.8 m/s²** (ignoring air resistance). The standard equations of motion apply.
- **Non-uniform acceleration** — velocity changes by unequal amounts in equal intervals; acceleration varies in magnitude, direction or both. Examples: a car in city traffic (accelerating and decelerating), a body on a circular path (direction of velocity changes continuously).

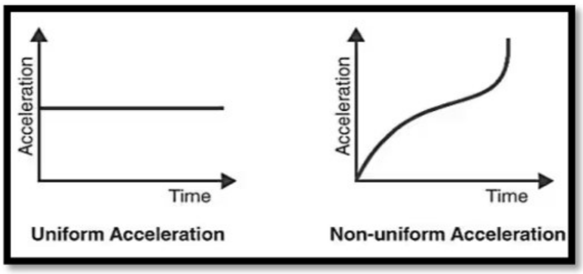

Fig. 9-6 · Uniform acceleration (flat a–t graph) vs non-uniform acceleration [Lecture 11, p. 3]

▌ADDED BY EDITOR — not in source lectures ▌ Acceleration facts that trip candidates (not in source)

A body can have zero velocity and non-zero acceleration (a ball at the top of its flight — v = 0, a = g downward). Uniform circular motion has constant speed but non-zero (centripetal) acceleration. A body moving at constant velocity has zero acceleration and zero net force. Acceleration is in the direction of the *change* of velocity, not necessarily of the velocity itself.

Equations of motion (uniform acceleration)

For an object moving with constant acceleration, the following relate initial velocity u, final velocity v, acceleration a, displacement s and time t:

$v = u + at$ $s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$ Displacement in the n^{th} second: $S_n = u + (a/2)(2n - 1)$

- We obtain following equations:

$$v = u + at$$
$$s = ut + \frac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$
$$\text{Displacement in the } n^{th} \text{ second}: S_{nth} = u + \frac{a}{2}(2n-1)$$

Fig. 9-7 · The four equations as printed in Lecture 11 [Lecture 11, p. 3]

▌ADDED BY EDITOR — not in source lectures ▌ Motion under gravity — the special case BPSC actually sets (not in the lecture)

Replace a by g = 9.8 m/s² (≈ 10 for quick sums), taking upward as negative. Body dropped from height h: $v = \sqrt{2gh}$, $t = \sqrt{2h/g}$; distances in successive seconds are in the ratio 1 : 3 : 5 : 7 (Galileo's odd-number rule). Body thrown up with u: maximum height $u^2/2g$, time to top u/g, total time of flight 2u/g, and it returns with the same speed u. **All bodies fall with the same g irrespective of mass** in vacuum (a feather and a coin dropped together on the Moon land together — 69th Q21, Unit 11). Projectile: range $R = u^2 \sin 2\theta / g$ is maximum at **45°**; the same range is obtained for θ and 90° – θ; the horizontal and vertical motions are independent.

▌ADDED BY EDITOR — not in source lectures ▌ Worked sums in the 71st-BPSC style

(1) Truck from rest covers 400 m in 20 s: $s = \frac{1}{2}at^2 \Rightarrow a = 2 \times 400/400 =$ **2 m/s²** (then F = ma, Unit 10). (2) A car at 20 m/s brakes uniformly to rest in 50 m: $v^2 = u^2 + 2as \Rightarrow a = -400/100 = -4$ m/s²; time = 5 s. (3) Stone dropped from a 45 m tower: $t = \sqrt{2 \times 45/10} = 3$ s, hits at 30 m/s.

Graphical analysis of motion (Lecture 12 p. 2)

Position–time (s–t) graph

- The **slope** of the graph gives the **velocity**; a straight line indicates constant velocity (a horizontal line = rest); a curve of increasing slope = acceleration.

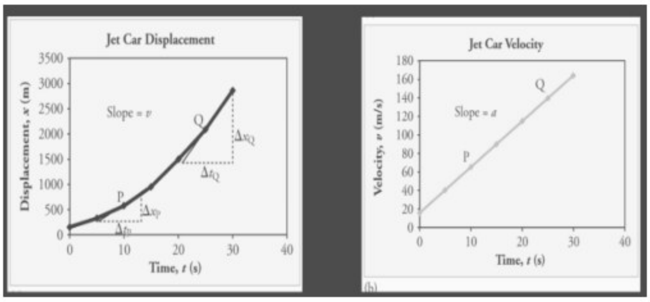

Fig. 9-8 · Displacement–time graph: slope Δx/Δt = velocity (jet-car example) [Lecture 12, p. 2]

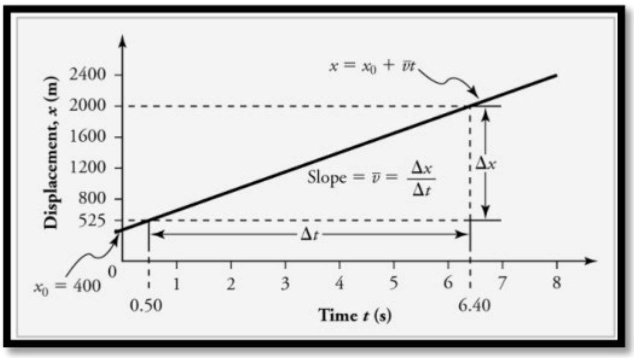

Fig. 9-9 · Straight-line s–t graph → constant velocity ($x = x_0 + vt$) [Lecture 12, p. 2]

Velocity–time (v–t) graph

- The **slope** gives the **acceleration**; the **area under the graph** gives the **displacement**; a horizontal line indicates constant velocity (zero acceleration); a straight sloping line = uniform acceleration.

Acceleration–time (a–t) graph

- The **area under the graph** gives the **change in velocity**.

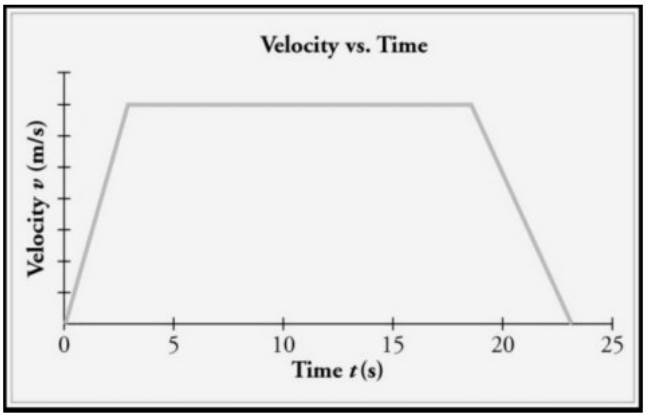

Fig. 9-10 · Velocity–time graph (trapezium): area = displacement [Lecture 12, p. 2]

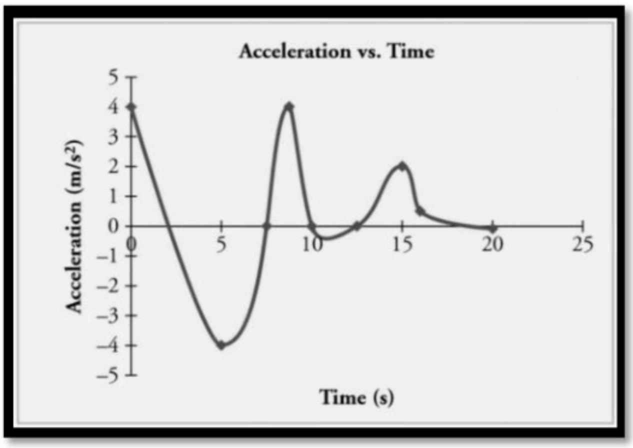

Fig. 9-11 · Acceleration–time graph: area = change in velocity [Lecture 12, p. 2]

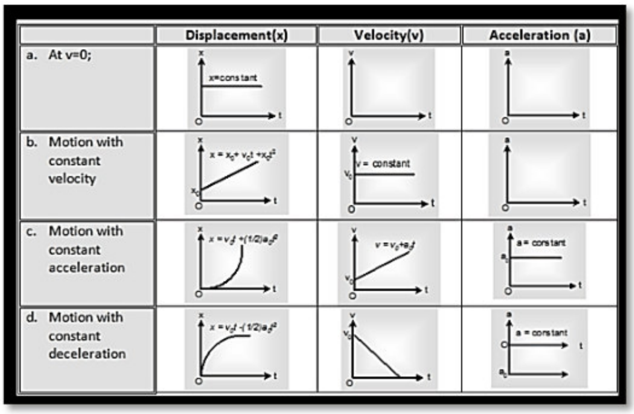

Fig. 9-12 · Shapes of s–t, v–t and a–t graphs for rest, constant velocity, constant acceleration and constant deceleration [Lecture 12, p. 2]

Types of motion (Lecture 12 pp. 2–3)

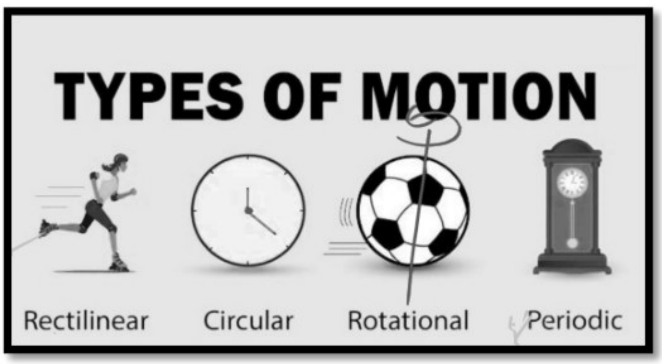

Fig. 9-13 · Rectilinear, circular, rotational and periodic motion [Lecture 12, p. 2]

- **(1) Translational motion** — all points of the body move through the same displacement in the same time. Two types: **(i) Rectilinear** — motion along a straight line (a person walking straight); **(ii) Curvilinear/circular** — motion along a curved or circular path.
- **(2) Rotational motion** — a body spins about an axis passing through it, e.g. rotation of the Earth on its own axis, a spinning top, a wheel (Fig. shows v = rω).
- **(3) Periodic motion** — motion that repeats itself at fixed intervals of time, e.g. beating of the heart, the Earth's revolution around the Sun, a clock's hands.

- **(4) Oscillatory / simple harmonic motion (SHM)** — a to-and-fro repeated motion about a mean position, e.g. a simple pendulum, a loaded spring, a swing. SHM is a special periodic motion in which the restoring force ∝ displacement.

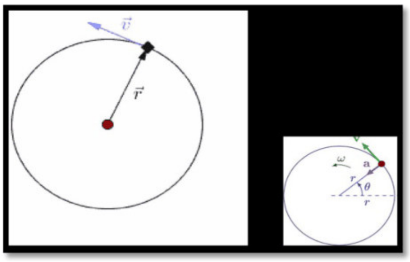

Fig. 9-14 · Circular motion — position vector r and tangential velocity v; inset: ω, a and θ (Lecture 12 p. 3) [Lecture 12, p. 3]

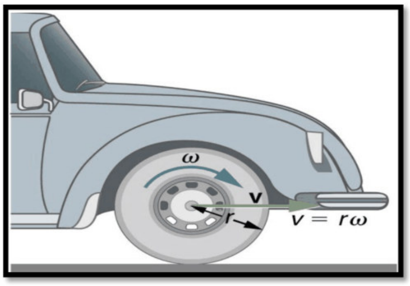

Fig. 9-15 · Rotational motion of a car wheel — v = rω [Lecture 12, p. 3]

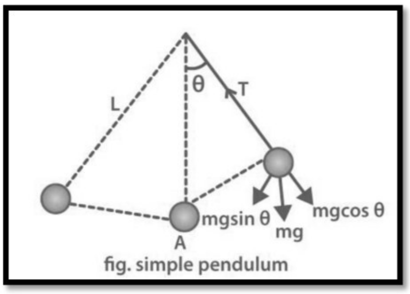

Fig. 9-16 · Simple pendulum — restoring component mg sin θ [Lecture 12, p. 3]

▌ADDED BY EDITOR — not in source lectures ▌ Simple pendulum facts (not in the lecture text)

Time period T = 2π√(l/g) — independent of the mass and (for small angles) of the amplitude; l is measured to the centre of the bob. A **seconds pendulum** has T = 2 s (l ≈ 99.4 cm ≈ 1 m). T increases at higher altitude / deeper mines / at the equator (g smaller) and in a lift accelerating downward; it is infinite in a freely falling lift or an orbiting satellite (g_{eff} = 0). Frequency of SHM of a spring: f = (1/2π)√(k/m).

UNIT 10

Laws of Motion, Momentum, Friction & Work–Power–Energy

Consolidated from: Lecture 12 (Motion Part-02) pp. 3–6 (Newton's laws, conservation of momentum, friction); Lecture 19 p. 6 (Work, Power and Energy)

★ **PYQ lens: Priority A — 9 questions in the last four papers, all answerable from this unit.** 68th: ball bearings convert sliding into rolling friction (Q114), a goalkeeper pulls his hands back to reduce the rate of change of momentum (Q115), friction turns kinetic energy into heat (Q118), a bus turning right throws passengers to the left (Q119); 70th Q85 magnetic force is a non-contact force; 71st: friction on a box pushed at constant velocity = the 200 N push (Q121), F = ma = 7000 kg × 2 m/s² = 14 000 N (Q122), a falling body does not violate energy conservation (Q125), W = 140 N × 15 m = 2100 J (Q126). Since the 71st paper this unit is **numerical** — practise the one-line formulas.

Force — the idea behind the laws

▌ADDED BY EDITOR — not in source lectures▐ What a force is, and the contact / non-contact split (announced in the Lecture-1 plan, never taught — needed for 70th Q85)

A **force** is a push or pull that can change the state of rest or of uniform motion of a body, its direction, or its shape. It is a **vector**

SI unit **newton (N)** — 1 N gives a 1 kg mass an acceleration of 1 m/s² (1 N = 10^5 dyne; 1 kgf = 9.8 N)

Contact forces act only when bodies touch: muscular force, normal reaction, tension, friction, air resistance / drag, spring force, impact force

Non-contact (field) forces act at a distance through a field: **gravitational**, **magnetic**, **electrostatic**, (nuclear forces at the sub-atomic scale) — 70th Q85 "Which is a non-contact force?" → **magnetic force** (friction and impact are contact forces)

Balanced forces (net force zero) leave the velocity unchanged — the body may still be moving; **unbalanced** forces produce acceleration. A body acted on by several forces behaves as if only their vector sum (**resultant**) acted on it.

Newton's laws of motion

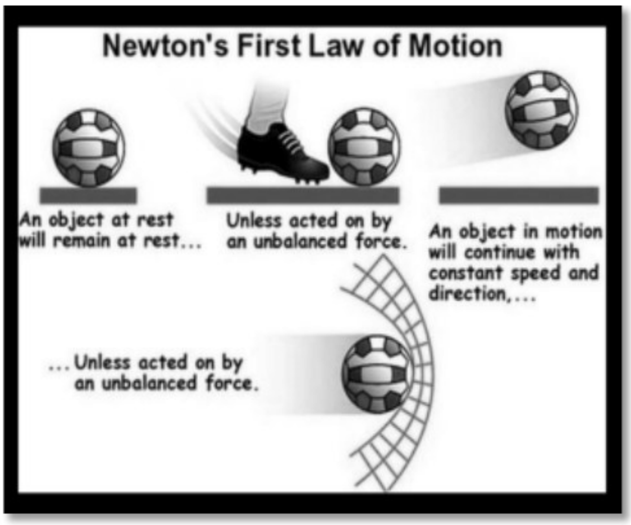

Fig. 10-1 · Newton's first law — an object at rest stays at rest, an object in motion continues with constant speed and direction, unless acted on by an unbalanced force [Lecture 12, p. 4]

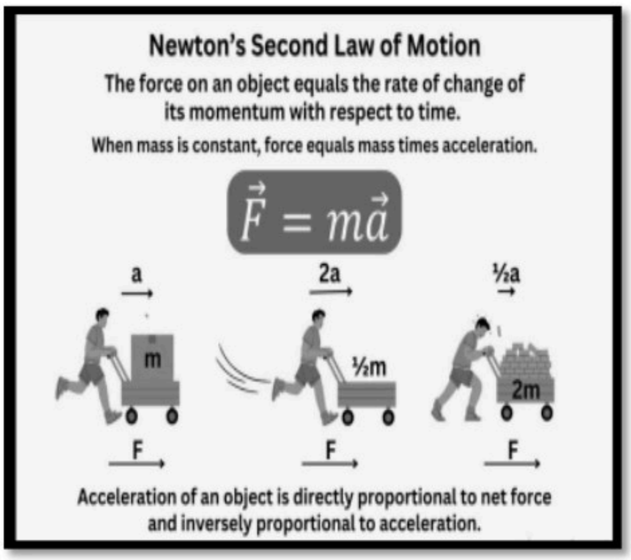

Fig. 10-2 · Newton's second law — F = ma: acceleration is directly proportional to the net force and inversely proportional to the mass [Lecture 12, p. 4]

First law (law of inertia / Galileo's law of inertia)

Statement: a body at rest remains at rest, and a body in motion continues in uniform motion in a straight line, unless acted upon by an external unbalanced force.

- Objects **resist change** in their state of motion — this property is called **inertia**. The mass of a body is the measure of its inertia (a loaded truck is harder to start and harder to stop than an empty one).
- If no external force acts, the velocity remains constant (zero acceleration).
- The first law defines force qualitatively: force is that which changes (or tends to change) the state of rest or uniform motion.

↻ **De-dup note:** Lecture 12 states the first law three times — as "Statement ...", again as "(Law of Inertia / Galileo's law of inertia): a body continues its state of motion or rest until it is not compelled by any external force", and the inertia-of-rest bus example is printed twice on p. 3. One statement and the three inertia cases are kept.

The three kinds of inertia

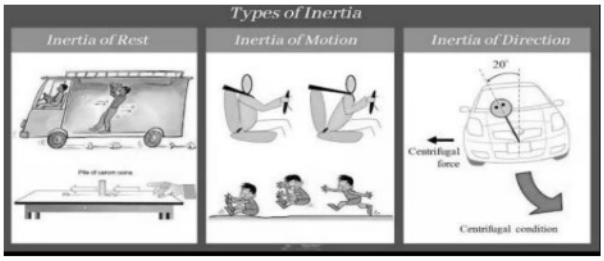

Fig. 10-3 · Types of inertia — inertia of rest (bus starts, passenger jerks back), inertia of motion (bus brakes, passenger lurches forward), inertia of direction (car turns, passenger is thrown outward) [Lecture 12, p. 3]

Kind	Inertia opposes a change in...	Everyday examples (lecture + common exam cases)
Inertia of rest	the state of **rest**	When a bus **suddenly starts**, standing passengers **fall backward**; dust comes out of a carpet when it is beaten; the fruit falls when a branch is shaken; a coin on a card over a glass drops into the glass when the card is flicked.
Inertia of motion	the state of **motion**	When a moving bus **suddenly brakes**, passengers **fall forward**; a **bowler runs up before delivering** the ball so that his own velocity is added to the ball — hence fast bowling; an athlete runs before a long jump; a passenger jumping off a moving bus must run forward a few steps.
Inertia of direction	the **direction** of motion	A bus **turning sharply to the right** throws passengers to the **left** (they tend to keep moving straight); mud flies off tangentially from a rotating wheel; a stone whirled on a string flies off along the tangent when the string breaks.

★ **PYQ lens:** 68th Q119: a bus moving straight suddenly turns right → passengers **bend towards the left** (inertia of direction). This is the same physics as "sparks fly off tangentially from a grinding wheel".

Second law (law of acceleration)

Statement: the rate of change of momentum of an object is directly proportional to the applied (net) force and takes place in the direction of the force.

$$F = ma \text{ (F in newton, m in kg, a in m/s}^2\text{)} \cdot \text{momentum } p = mv \cdot F = \Delta p / \Delta t = (mv - mu)/t$$

- If the applied force increases, the acceleration increases proportionally; for the same force a heavier body accelerates less ($a \propto 1/m$).

- Example: kicking a football harder increases its acceleration and speed.

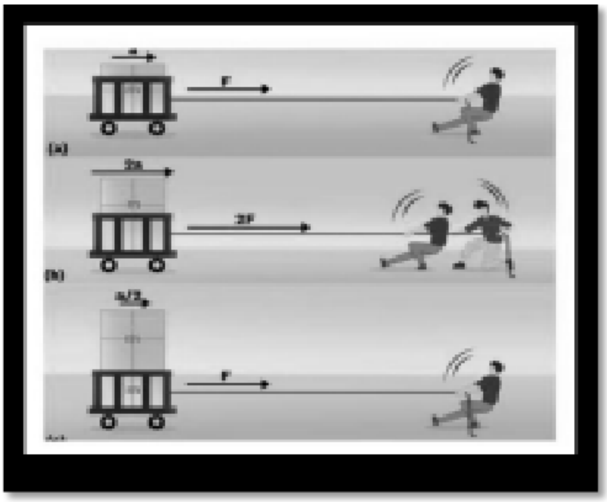

Fig. 10-4 · Same mass, double force → double acceleration; same force, double mass → half acceleration [Lecture 12, p. 4]

↻ **De-dup note:** The heading "Second Law of Motion (Law of Acceleration)" and the momentum line p = mv are printed twice on Lecture 12 p. 4; kept once.

▌ADDED BY EDITOR — not in source lectures ▌ Momentum and impulse — the part of the second law BPSC actually sets

Momentum $p = mv$ is a **vector** (kg m/s or N s); a bullet and a truck can have the same momentum. **Impulse** = force × time = change in momentum ($J = F\Delta t = \Delta p$); it is the *area under a force–time graph*

Why we "give" with the ball — for the *same* change of momentum, increasing the time of contact (Δt) reduces the force ($F = \Delta p/\Delta t$): a goalkeeper pulls his hands back, a cricketer lowers his hands while catching, athletes land on sand/foam, vehicles have crumple zones and air-bags, china is packed in straw, shock absorbers and padded gloves work the same way

Why hammers and karate chops hurt — the reverse: a short contact time makes the force large for the same Δp.

★ **PYQ lens:** 68th Q115: goalkeeper pulls his hands backwards after holding the ball → it **decreases the rate of change of momentum** (so the force on the hands is smaller). 71st Q122: truck from rest covers 400 m in 20 s ⇒ $a = 2s/t^2 = 2$ m/s²; weight 7 tons = 7000 kg ⇒ $F = ma$ = **14 000 N** (the "hill" is a distractor — use the given kinematic data).

Third law (action–reaction law)

Statement: to every action there is an equal and opposite reaction.

- Forces always occur in **pairs**, but the two forces act on **different bodies** — so they never cancel each other out for a single body.
- Action and reaction are simultaneous, equal in magnitude, opposite in direction and of the same kind (both gravitational, both contact, etc.).

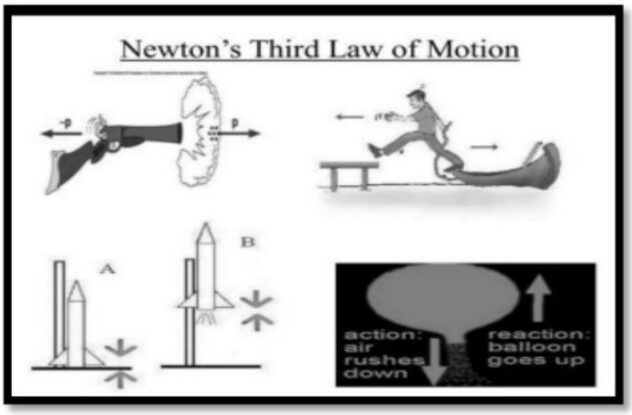

Fig. 10-5 · Third-law pairs — gun recoils as the bullet leaves; gases pushed down lift the rocket; air rushing out pushes the balloon forward [Lecture 12, p. 4]

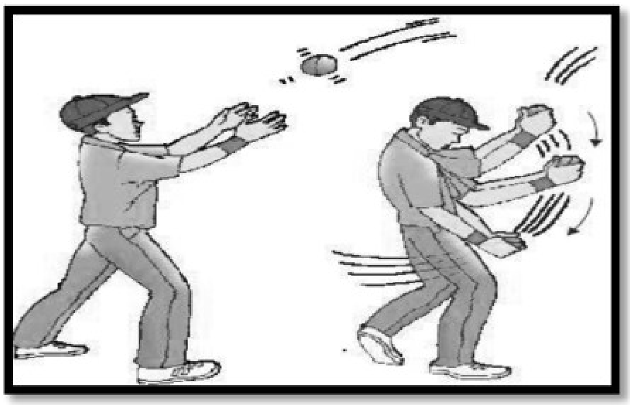

Fig. 10-6 · Two boys pushing each other apart — each feels an equal and opposite force [Lecture 12, p. 4]

- **Examples** — rocket propulsion (gases pushed downward → rocket moves upward); rowing a boat (oar pushes water backward → boat moves forward); walking (foot pushes the ground backward → the ground pushes the foot forward); recoil of a gun; a swimmer pushing water backward; a book on a table presses the table down and the table pushes the book up (normal reaction).

▌ADDED BY EDITOR — not in source lectures ▌ Common traps on the third law (not in source)

The pair for the weight of a book on a table is *not* the table's normal reaction — it is the pull of the book on the Earth. A horse can pull a cart even though action = reaction because the *friction of the ground on the horse's hooves* is the external force that moves the horse-cart system. Rockets work in vacuum — they push on their own exhaust, not on the air.

Law of conservation of linear momentum

If the net external force on a system is zero, its total momentum remains constant — the initial momentum equals the final momentum.

$$p_{final} = p_{initial} \cdot m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

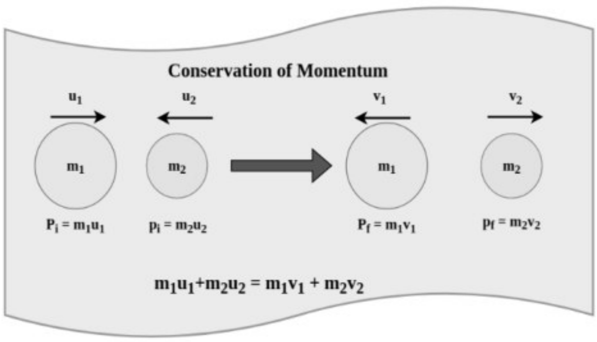

Fig. 10-7 · Conservation of momentum — before and after a collision the total m×v of the system is unchanged [Lecture 12, p. 5]

✎ **Source correction:** Lecture 12 p. 5 prints the law as "P_f = P_i, $mv_1 = mv_2$". The general two-body form is $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ (exactly as in the source figure); "$mv_1 = mv_2$" holds only for a single body of constant mass, in which case it just says the velocity is unchanged.

▌ADDED BY EDITOR — not in source lectures ▌ Applications of momentum conservation (standard examples)

Recoil of a gun: bullet momentum forward = gun momentum backward, so recoil velocity = $(m_{bullet}/M_{gun}) \times v_{bullet}$ — a heavy gun recoils gently

Rocket / jet: the mass ejected backward per second × its exhaust speed = thrust; the rocket accelerates faster as fuel burns because its mass falls

Two skaters pushing apart move off with velocities inversely proportional to their masses

Explosions: a bomb at rest bursting into fragments has zero total momentum after the burst

Collisions: momentum is conserved in every collision; kinetic energy is conserved only in a perfectly *elastic* one (billiard balls ≈ elastic; a bullet embedding in a block = perfectly inelastic). Angular momentum ($L = I\omega$) is likewise conserved — a spinning skater speeds up when she pulls in her arms; the Earth's rotation, a diver's somersault and a helicopter's tail rotor are all angular-momentum stories.

Friction

Friction is a force that opposes the relative motion (or the tendency of motion) between two surfaces in contact. It acts along the surface, opposite to the (attempted) motion.

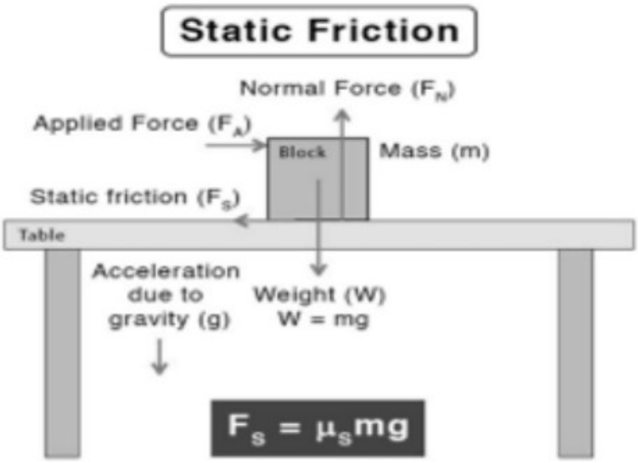

Fig. 10-8 · Static friction — the block does not move until the applied force exceeds F_s(max) = μ_s mg [Lecture 12, p. 5]

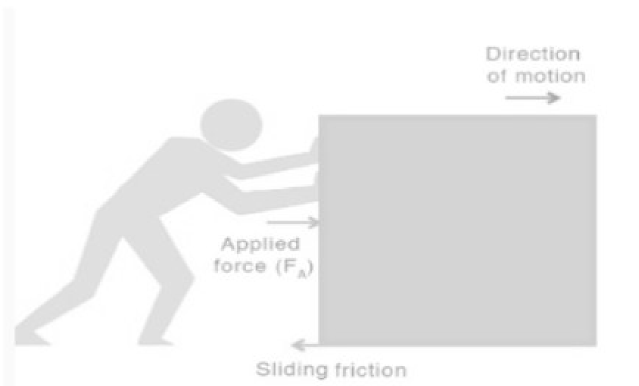

Fig. 10-9 · Sliding (kinetic) friction acts backward on a box being pushed along the floor [Lecture 12, p. 5]

Types of friction (Lecture 12 pp. 5–6)

- **I. Static friction** — acts when the object is **at rest**; prevents motion from starting; self-adjusting (equal to the applied force) up to a maximum value:

$$f_{s(max)} = \mu_s N \ (\mu_s = \text{coefficient of static friction, } N = \text{normal reaction})$$

- **II. Kinetic (sliding) friction** — acts when the object is **moving**; **always less than the maximum static friction** (it is easier to keep a box sliding than to start it):

$$f_k = \mu_k N \text{ with } \mu_k < \mu_s$$

- **III. Rolling friction** — acts in rolling motion (wheels, balls); the **smallest** of all — which is why wheels and ball bearings are used. $\mu_{rolling} \ll \mu_{kinetic} < \mu_{static}$.

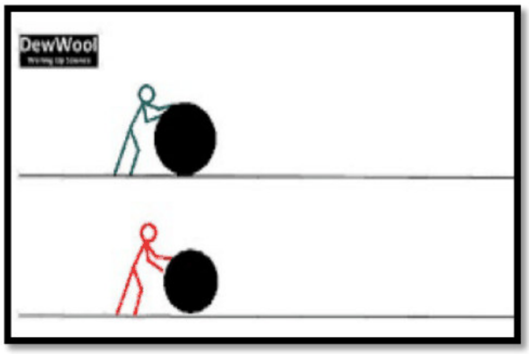

Fig. 10-10 · Rolling a heavy roller is far easier than sliding it — rolling friction is the least [Lecture 12, p. 6]

✎ **Source correction:** Under "Kinetic (sliding) friction" Lecture 12 re-prints the *static* formula "f_s(max) = μ_s N". The kinetic-friction formula is f_k = μ_k N, and μ_k is smaller than μ_s.

★ **PYQ lens:** 68th Q114: ball bearings are used to convert sliding friction into **rolling friction**. 71st Q121: a box pushed at **constant velocity** with 200 N — net force is zero, so the friction force is exactly **200 N** (any other value would accelerate or decelerate the box). 68th Q118: energy change that involves friction → **kinetic energy → heat**.

▌ADDED BY EDITOR — not in source lectures▐ Friction — facts the lecture leaves out

Friction is caused by interlocking of surface irregularities and by molecular adhesion at the points of real contact; it is **independent of the area of contact** (for the same normal force) and, for kinetic friction, roughly independent of speed; the **coefficient μ has no unit** (ratio of two forces) and is usually < 1 (rubber on dry concrete ≈ 1, steel on ice ≈ 0.03, Teflon ≈ 0.04)

Angle of friction θ: $\tan\theta = \mu_s$; a body just starts sliding down an incline at the *angle of repose* α where $\tan\alpha = \mu_s$

Friction is a necessary evil — we could not walk, write, hold a pen, light a match, tie knots, use brakes or drive on roads without it

Reducing friction: lubricants (oil, grease, graphite), polishing, ball bearings, streamlining (drag), air cushion (hovercraft), Teflon coating

Increasing friction: treaded tyres, sand on icy roads, chalk on gymnasts' hands, spikes on shoes, kabaddi players rubbing soil on palms. **Fluid friction (drag)** grows with speed and depends on shape — hence streamlined cars, aircraft, fish and birds.

Work, power and energy (Lecture 19 p. 6)

Work

Work is done when a force applied to a body causes a displacement in the direction of the force.

W = F · S = F S cos θ (θ = angle between the force and the displacement)

- **SI unit**: joule (J) — 1 J = 1 N × 1 m; **CGS unit**: erg; **1 joule = 10^7 erg**.
- Work is a **scalar** quantity and can be **positive, negative or zero**: θ < 90° → positive work (force helps the motion); θ = 90° → **zero** work (cos 90° = 0); θ = 180° → negative work, e.g. work done by friction, since cos 180° = –1.

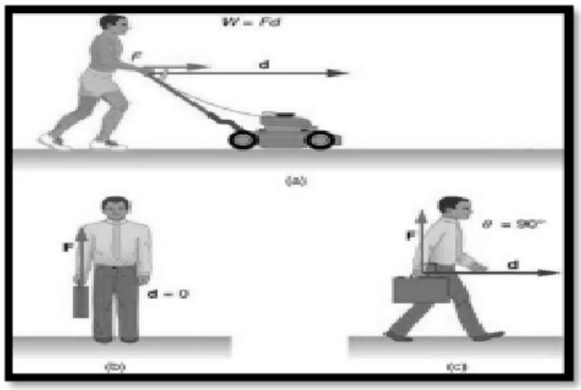

Fig. 10-11 · (a) Pushing a mower: W = F·d; (b) holding a suitcase still: d = 0, no work; (c) carrying it horizontally: θ = 90°, no work by the lifting force [Lecture 19, p. 6]

★ **PYQ lens:** 71st Q126: a pair of oxen exerts 140 N over a 15 m furrow → W = F × s = 140 × 15 = **2100 J**.

▌ADDED BY EDITOR — not in source lectures ▌ Zero-work situations examiners love (not in source)

A coolie carrying a load on his head along a level platform does no work *against gravity* (θ = 90°); a satellite in circular orbit — gravity does no work (force ⊥ velocity); a person pushing a wall that does not move; the centripetal force in uniform circular motion; the magnetic force on a moving charge (always ⊥ v). Work is *done* by a person lifting a load (positive), by gravity on a falling body (positive) and by gravity on a rising body (negative). **Work–energy theorem**: net work done on a body = change in its kinetic energy ($W_{net} = ½mv^2 – ½mu^2$).

Energy

- **Energy** is the capacity to do work; it has the **same unit as work** (joule) and is a scalar.
- Forms: **kinetic energy** (energy of motion) and **potential energy** (energy of position or configuration), and many others — heat, light, sound, chemical, electrical, nuclear, magnetic.

Kinetic energy KE = ½ m v² Potential energy (gravitational) PE = m g h

- **Law of conservation of energy**: energy can neither be created nor destroyed; it can only be transformed from one form to another. The total energy of an isolated system is constant.

★ **PYQ lens:** 71st Q125: "The potential energy of a freely falling body continuously decreases" → the principle of conservation of energy is **not violated** — the lost PE appears as KE (at every instant PE + KE = mgh_{top}), and on hitting the ground it becomes heat, sound and deformation.

▌ADDED BY EDITOR — not in source lectures ▌ Energy — relations, forms and conversions that keep appearing

KE and momentum: KE = $p^2/2m$; doubling the speed quadruples the KE (stopping distance ∝ v^2); at the same momentum the *lighter* body has more KE; at the same KE the *heavier* body has more momentum

Elastic PE of a spring = $½kx^2$

Free fall from height h: KE at the ground = mgh, speed √(2gh), independent of mass; at half height KE = PE

Other units: 1 kWh = 3.6 × 10^6 J (the "unit" of the electricity bill — Unit 14), 1 eV = 1.6 × 10^{-19} J, 1 calorie = 4.186 J, 1 erg = 10^{-7} J

Conversions (device → change): electric motor: electrical → mechanical; generator/dynamo: mechanical → electrical; battery/cell: chemical → electrical; electric heater/iron/bulb: electrical → heat (+ light); photoelectric / solar cell: light → electrical; loudspeaker: electrical → sound; microphone: sound → electrical

LED / bulb: electrical → light; plants (photosynthesis): light → chemical; friction/brakes: kinetic → heat (68th Q118); a bow: elastic PE → KE of the arrow; a candle: chemical → heat + light; the Sun: nuclear (fusion) → radiant

Mass–energy: $E = mc^2$ — mass itself is a form of energy (Unit 16). **Conservative forces** (gravity, spring, electrostatic): work done is independent of path, so mechanical energy is conserved; **non-conservative** forces (friction, air drag) dissipate mechanical energy as heat.

Power

- **Power** is the rate of doing work or of transferring energy.

$P = W / t$ or $P = F \times v$ (force × velocity, for a constant force along the motion)

- **SI unit**: watt (W) — 1 W = 1 joule per second; larger units kW (10^3 W), MW (10^6 W); another common unit is the **horsepower**, 1 hp ≈ **746 W**.
- Power is a **scalar** quantity.

ADDED BY EDITOR — not in source lectures **Power — quick numbers (not in source)**

Energy = power × time, so a 100 W bulb for 10 h uses 1 kWh = 1 "unit"; a healthy adult can sustain ≈ 100 W and produce ≈ 1 hp for short bursts; a car engine ≈ 50–100 kW; a nuclear power unit ≈ 700–1000 MW; the **watt** is named after James Watt, the **joule** after James Prescott Joule, the **newton** after Isaac Newton. In electricity $P = VI = I^2R = V^2/R$ (Unit 14; 71st Q128).

ADDED BY EDITOR — not in source lectures **Worked sums in the 71st-BPSC style**

(1) A 50 kg boy climbs 20 stairs of 15 cm each in 6 s: $W = mgh = 50 \times 10 \times 3 = 1500$ J, $P = 250$ W

(2) A 2 kg ball dropped from 5 m hits the floor at $v = \sqrt{2gh} = 10$ m/s with KE = 100 J

(3) A car of 1000 kg going at 20 m/s is stopped in 4 s: retardation 5 m/s², braking force 5000 N, work done by brakes = ½ × 1000 × 400 = 2×10^5 J (all of it becomes heat)

(4) A bullet of 10 g at 400 m/s has KE = ½ × 0.01 × 160 000 = 800 J and momentum 4 kg m/s

(5) A crane lifts 500 kg by 10 m in 25 s: $P = 500 \times 10 \times 10 / 25 = 2000$ W ≈ 2.7 hp.

UNIT 11

Circular Motion & Gravitation

Consolidated from: Lecture 13 (Circular Motion & Gravitation) pp. 2–4; Lecture 14 p. 2 (escape velocity, factors affecting g, gravitational field) and p. 3 (mass vs weight table); Kepler's laws from Lecture 12 p. 5

★ **PYQ lens: Priority A — 7 questions in 5 papers, 6 of them recent, always in reasoning form.** Copernicus first said the Earth goes round the Sun (56–59th Q139); washing machine works by centrifugation (67th Q65); if the Earth spins faster, weight at the equator **decreases** (67th Q68); the Sun is the chief body of the solar system (67th Q141); centripetal force keeps a body on its circular path (68th Q117); two masses falling freely on the Moon have the **same velocity at any instant** (69th Q21); comets have highly elongated elliptical orbits (70th Q70).

Circular motion

If a particle moves in a plane such that its distance from a fixed (or moving) point remains constant, its motion is **circular motion** with respect to that point. The fixed point is the **centre** and the constant distance is the **radius** of the circular path.

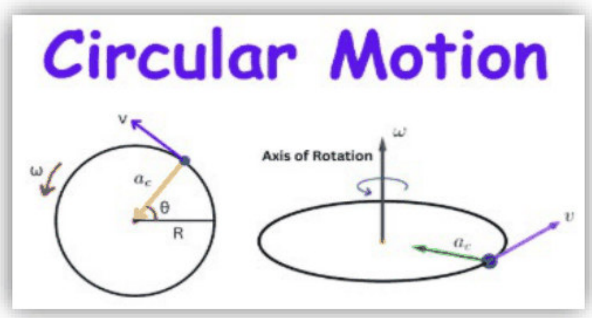

Fig. 11-1 · Circular motion — velocity v along the tangent, centripetal acceleration a_c toward the centre, angular velocity ω about the axis [Lecture 13, p. 2]

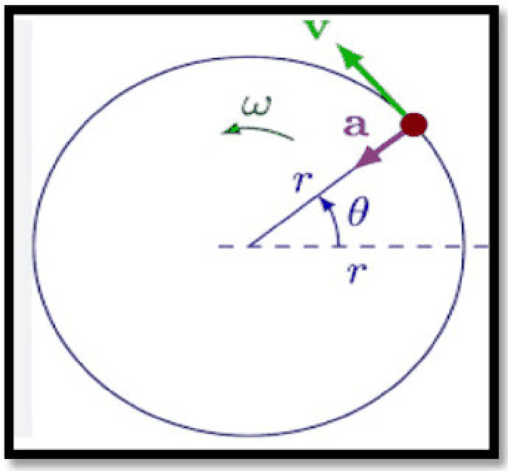

Fig. 11-2 · Uniform circular motion: constant speed, continuously changing direction — a always points to the centre [Lecture 13, p. 2]

Terms related to circular motion (Lecture 13 p. 3)

- **Angular displacement (θ)** — the angle traced by the position vector of the particle about the fixed point (unit radian).
- **Time period (T)** — the time taken by the particle to complete one revolution.
- **Frequency (n)** — the number of revolutions per second; unit rps or rpm. **n = 1/T**.
- **Angular velocity (ω)** — the rate of change of angular displacement with respect to time (rad/s).

$$\omega = \theta/t = 2\pi/T = 2\pi n \cdot \text{linear speed } v = r\omega \cdot \text{centripetal acceleration } a_c = v^2/r = r\omega^2$$

Uniform circular motion

- The particle moves with **constant speed**; the **tangential acceleration is zero** — so the net force has no component along the motion, but the *direction* of velocity changes continuously, so there **is** an acceleration (centripetal), directed towards the centre.
- The essential force required for circular motion is the **centripetal force**, $F = mv^2/r = mr\omega^2$, always directed towards the centre and perpendicular to the velocity — it does **no work** and changes only the direction of motion.

✎ **Source correction:** Lecture 13 p. 2 writes "If a particle moves with constant speed. It means net external force applied is zero." The *net force is not zero* in uniform circular motion — a centripetal force mv^2/r must act; only the *tangential* component of the force (and hence the tangential acceleration, $\Delta v/\Delta t = 0$) is zero, which is what the following line in the source says.

Non-uniform circular motion

- Motion along a circular path in which the **speed (magnitude of velocity) changes with time**, so the velocity changes both in magnitude and direction. Example: a car accelerating or slowing down on a circular track; circular motion in a vertical plane (speed is greatest at the bottom, least at the top).

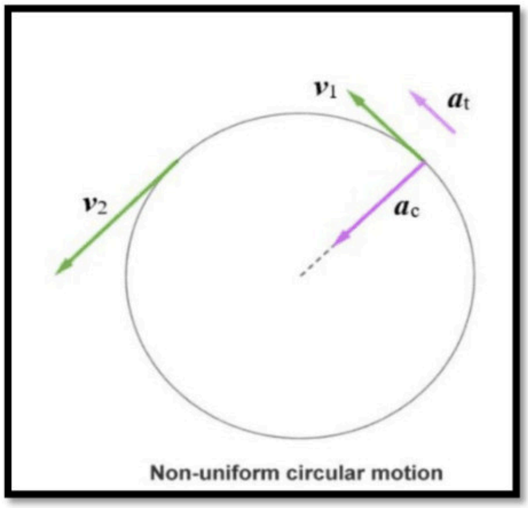

Fig. 11-3 · Non-uniform circular motion — the acceleration has a radial (centripetal) part a_c and a tangential part a_t [Lecture 13, p. 2]

- **Centripetal (radial) acceleration** — acts toward the centre, responsible for the change in *direction*, always perpendicular to the velocity: $\mathbf{a_c = v^2/r}$.
- **Tangential acceleration** — acts along the tangent, responsible for the change in *speed*; positive when the speed increases, negative when it decreases: $\mathbf{a_t = dv/dt}$.

Total acceleration $a = \sqrt{a_c^2 + a_t^2}$ (the two components are perpendicular)

★ **PYQ lens:** 68th Q117: centripetal force is responsible to **keep the body moving along the circular path**. 67th Q65: a washing machine (spin-dryer) works on **centrifugation** — the drum's wall supplies the centripetal force to the clothes, but water can escape through the holes and flies off along the tangent.

▌ADDED BY EDITOR — not in source lectures ▌ Centripetal vs "centrifugal", and the everyday cases BPSC draws from

Centripetal force is a real force supplied by something — tension in a string (stone whirled round), gravity (Moon, satellites, planets), friction (car on a level bend — hence skidding on wet roads), the normal reaction of a banked road or track (roads and railway tracks are **banked**, cyclists **lean inward** on a turn), the wall of the drum (washing machine, centrifuge, cream separator, the "rotor" ride). **Centrifugal force** is a *pseudo-force* felt only in the rotating frame — it is the "outward push" a passenger feels in a turning bus (really the inertia of direction, Unit 10). **Centrifuge**: heavier particles move to the outside — used to separate cream from milk, plasma from blood cells, uranium isotopes and honey from the comb. Turning **faster** or on a **tighter** curve needs a bigger centripetal force ($F \propto v^2/r$): that is why vehicles slow down on sharp bends and why the outer rail is raised. A **satellite** is a body in permanent free fall — gravity provides exactly the centripetal force needed.

Gravitation

Gravitation is the universal force of attraction that exists between any two bodies in the universe having mass, acting along the line joining their centres.

The discovery of the law of gravitation (Lecture 13 p. 3)

The way the law of universal gravitation was discovered is often considered the paradigm of modern scientific technique. The major steps were:

- the hypothesis about planetary motion given by **Nicolaus Copernicus** (1473–1543) — the Sun, not the Earth, at the centre (heliocentric model);

- the careful experimental measurements of the positions of the planets and the Sun by **Tycho Brahe** (1546–1601);
- analysis of the data and formulation of the empirical laws by **Johannes Kepler** (1571–1630);
- development of the general theory by **Isaac Newton** (1642–1727).

↻ **De-dup note:** The heading "The discovery of the law of gravitation" is printed twice on Lecture 13 p. 3 (once with only the first sentence). Merged into one list.

★ **PYQ lens:** 56–59th Q139: the scientist who first discovered that the Earth revolves round the Sun → **Copernicus** (Newton, Dalton, Einstein were the distractors). 67th Q141: the chief heavenly body of the solar system → the **Sun** (≈ 99.8 % of the system's mass).

▌ADDED BY EDITOR — not in source lectures ▌ Galileo and the falling bodies (context for 69th Q21)

Galileo Galilei (1564–1642) showed — against Aristotle — that in the absence of air all bodies fall with the *same* acceleration regardless of mass (the Leaning-Tower story; the feather-and-hammer drop performed on the Moon by Apollo 15 in 1971). He also formulated the law of inertia (Unit 10) and first defined speed (67th Q69). Newton's *Principia* (1687) contained the three laws of motion and the law of gravitation; the story of the falling apple dates from 1666.

Kepler's laws of planetary motion (Lecture 12 p. 5)

Kepler's laws describe how planets move around the Sun; they were formulated by Johannes Kepler in the early 17th century (1609–1619) from Tycho Brahe's observations.

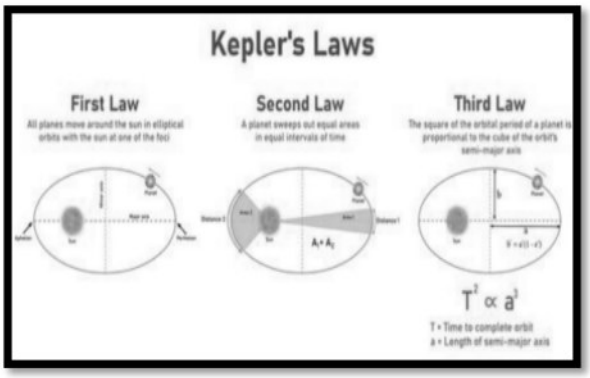

Fig. 11-4 · Kepler's three laws — elliptical orbits with the Sun at a focus, equal areas in equal times, $T^2 \propto a^3$ [Lecture 12, p. 5]

- **1. Law of orbits (elliptical law)** — planets move in **elliptical orbits** around the Sun, with the **Sun at one focus** of the ellipse (not at the centre). This is why the distance between a planet and the Sun keeps changing.
- **2. Law of areas (equal-area law)** — the line joining the planet and the Sun sweeps **equal areas in equal intervals of time**. Hence a planet moves **faster when closer** to the Sun (**perihelion**) and **slower when farther** (**aphelion**) — the orbital speed varies. (This is conservation of angular momentum.)
- **3. Law of periods (harmonic law)** — the **square of the orbital period is proportional to the cube of the mean distance** (semi-major axis) from the Sun. Farther planets take much longer to revolve — Mars takes more time than the Earth.

$$T^2 \propto R^3 \Rightarrow (T_1/T_2)^2 = (R_1/R_2)^3$$

↻ **De-dup note:** Kepler's laws are printed in Lecture 12 between conservation of momentum and friction; they belong with gravitation, so they are moved here (the Lecture-13 history list refers to them anyway).

★ **PYQ lens:** 70th Q70: highly elongated elliptical orbit → a **comet** (e.g. Halley's comet, period ≈ 76 years; next return 2061). Asteroids have nearly circular orbits between Mars and Jupiter; a meteor is the streak of light, a meteorite the piece that reaches the ground.

▌ADDED BY EDITOR — not in source lectures▐ Kepler's laws in numbers (context)

Earth: perihelion ≈ 147 million km (around 3 January), aphelion ≈ 152 million km (around 4 July) — so the Earth moves fastest in January; the seasons are due to the axial tilt (23.5°), *not* the varying distance. Third-law check: Mars at 1.52 AU → $T = 1.52^{1.5} \approx 1.88$ years ✓. The same laws hold for satellites round the Earth and for the Moon (T ≈ 27.3 days at 384 400 km). Kepler's laws are empirical

Newton *derived* them from the inverse-square law.

Newton's law of gravitation (Lecture 13 p. 3)

Statement: every particle in the universe attracts every other particle with a force that is **directly proportional to the product of their masses** and **inversely proportional to the square of the distance** between them, acting along the line joining them.

$$F = G\, m_1\, m_2 / r^2 \quad G = 6.67 \times 10^{-11}\ \text{N m}^2\ \text{kg}^{-2}$$

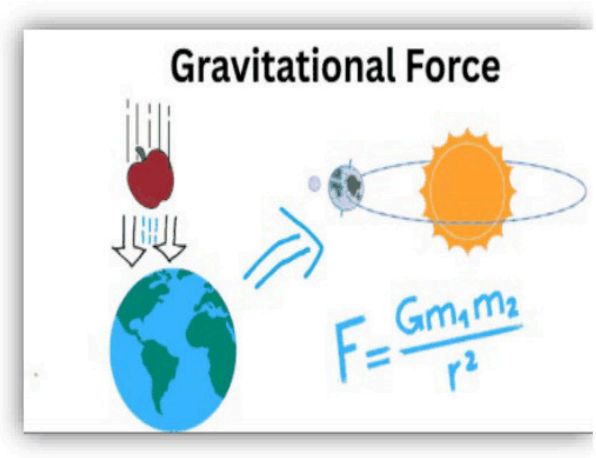

Fig. 11-5 · Gravitational force — the same law governs the falling apple and the Earth's orbit round the Sun [Lecture 13, p. 3]

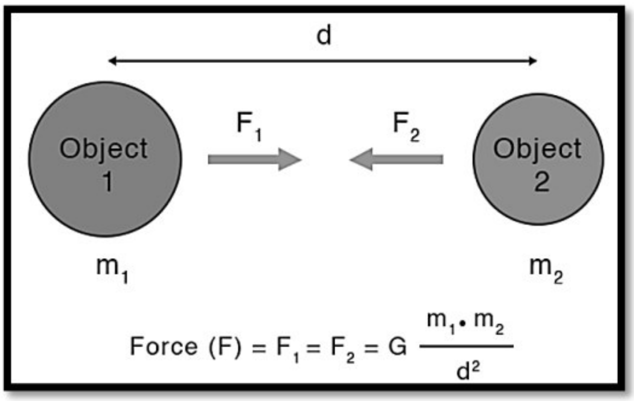

Fig. 11-6 · Two masses m_1 and m_2 a distance d apart attract each other with equal and opposite forces $F_1 = F_2 = G\, m_1 m_2/d^2$ [Lecture 13, p. 3]

Universal gravitational constant G

- G is a **scalar**; its value is the same throughout the universe; it does not depend on the nature or size of the bodies, nor on the nature of the medium between them.
- Its value was first measured by **Henry Cavendish** (1798) with the **torsion-balance** experiment ("weighing the Earth").
- SI unit N m²/kg²; dimensional formula $M^{-1}L^3T^{-2}$ (Unit 1).

Properties of the gravitational force (Lecture 13 p. 3 + Lecture 14 p. 2)

- It is **always attractive**, developed as **action-reaction pairs** — so it obeys Newton's third law (the Earth pulls the apple; the apple pulls the Earth equally).
- It is **independent of the medium** between the two masses (cannot be shielded).
- It is a **central force** — acts along the line joining the centres of gravity of the two bodies.
- It is a **conservative force** — work done does not depend on the path; the work done round any closed path is zero.
- It is the **weakest force in nature** — about 10^{36} times smaller than the electrostatic force and 10^{38} times smaller than the (strong) nuclear force. It dominates at astronomical scales only because it is always attractive and has infinite range.
- The law applies to **all bodies** irrespective of size, shape and position. Force between any two masses = *gravitational force*; force between the Earth and a body = *force of gravity*.

✎ **Source correction:** Lecture 14 p. 2 prints "10^{36} times smaller than electrostatic force and 10^{-18} times smaller than nuclear force" — the nuclear ratio should read about 10^{38} (a "10^{-18} times smaller" force would be *stronger*, and the strong force is stronger than electrostatic, not weaker).

↻ **De-dup note:** The properties list appears in Lecture 13 ("Gravitational force is always attractive ...") and again in Lecture 14 ("Gravitation Force — central as well as conservative, weakest force ..."). Merged above.

Acceleration due to gravity (g)

- In Newton's law of gravitation the attraction between any two bodies is *gravitation*; if one of the bodies is the **Earth**, the attraction is called **gravity** — the force by which the Earth attracts a body towards its centre. It is a special case of gravitation.

For the Earth taken as a uniform sphere of mass M and radius R, the force on a particle of mass m at distance r from the centre (outside the Earth) is $F = GMm/r^2$. By Newton's second law $F = mg$, so:

$$g = G\,M / R^2 \text{ (at the surface)} \approx 9.8\ \text{m/s}^2 \cdot \text{at height } r > R\text{: } g' = G\,M / r^2$$

- The value of g **does not depend on** (i) the shape and size of the object, (ii) the **mass of the object** — a feather and a stone fall together in vacuum.
- The value of g **depends on** (i) the mass of the Earth/planet, (ii) the radius of the Earth/planet.
- Any value of g measured at a given location differs from the calculated value for three reasons: **(1)** the Earth's mass is not distributed uniformly; **(2)** the Earth is not a perfect sphere; **(3)** the Earth **rotates**.

↻ **De-dup note:** The heading "ACCELERATION DUE TO GRAVITY" is printed twice on Lecture 13 p. 4. Merged.

★ **PYQ lens:** 69th Q21: two objects of different masses falling freely near the surface of the Moon → they **have the same velocity at any instant** (same $g_{Moon} \approx 1.6$ m/s² for both; the forces differ, the inertia is unchanged, and there is no air on the Moon to separate them).

Factors affecting g (Lecture 14 p. 2)

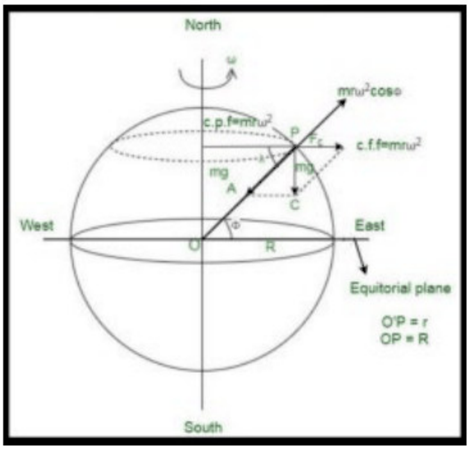

Fig. 11-7 · Rotating Earth — at latitude λ the centrifugal term $m\omega^2 r \cos\lambda$ reduces the effective weight; it is largest at the equator (r = R) and zero at the poles [Lecture 14, p. 2]

- **Shape of the Earth** — $g \propto 1/R^2$. The Earth is not a perfect sphere but an oblate spheroid (elliptical): its **polar diameter is about 42 km less** than the equatorial diameter. Therefore **g is minimum at the equator and maximum at the poles** (≈ 9.78 vs 9.83 m/s²).

▌ADDED BY EDITOR — not in source lectures ▌ Rotation, altitude and depth — the rest of the standard list (only the shape effect is in the lecture)

Rotation of the Earth: part of gravity supplies the centripetal force needed to carry a body round with the Earth, so the effective g at latitude λ is $g_{eff} = g - \omega^2 R\cos^2\lambda$ — the reduction is greatest at the **equator** (≈ 0.034 m/s², about 0.3 %) and zero at the poles. **If the Earth spun faster, g and hence the weight at the equator would decrease** (67th Q68); at about 17 times the present spin (T ≈ 84 min) objects at the equator would become weightless. **Altitude**: g decreases with height, $g_h \approx g(1 - 2h/R)$ for h ≪ R (about 0.3 % less at Everest; 8.7 m/s² on the ISS at 400 km — astronauts float not because g ≈ 0 but because they are in free fall). **Depth**: g also *decreases* going down a mine, $g_d = g(1 - d/R)$, and is **zero at the centre** of the Earth. So g is **maximum at the surface at the poles**. **Other planets**: $g \propto M/R^2$ — Moon ≈ 1.62 m/s² (1/6 of Earth), Mars ≈ 3.7, Jupiter ≈ 24.8, Sun ≈ 274 m/s².

★ **PYQ lens:** 67th Q68: if the spinning speed of the Earth increases, the weight of a body at the equator will **decrease** (a larger share of gravity is used up as centripetal force). Note the two independent reasons why equatorial weight is least: the bulge (larger R) *and* the spin.

Gravitational field (Lecture 14 p. 2)

- The space around a body in which its gravitational pull can be experienced by other bodies is its **gravitational field**.
- **Intensity of gravitational field** (E_g or I) — the gravitational force acting per unit mass at a point in the field:

$E_g = I = F/m = GM/r^2$ — a vector directed towards the centre of gravity of the body; SI unit N/kg (= m/s²); dimensional formula $[L\,T^{-2}]$

✎ **Source correction:** Lecture 14 prints the SI unit as "N/m"; force per unit *mass* is **N/kg**, which is the same as m/s² (at the Earth's surface the field intensity is just g).

▌ADDED BY EDITOR — not in source lectures ▌ Gravitational potential & potential energy (needed for the escape-velocity derivation)

Gravitational potential V = –GM/r (work done per unit mass in bringing a body from infinity; zero at infinity, negative everywhere else). Potential energy of a mass m at distance r: U = –GMm/r; near the surface this reduces to the familiar ΔU = mgh. A body needs kinetic energy ≥ |U| = GMm/R to reach infinity — that is the escape condition below.

Escape velocity

Escape velocity is the **minimum velocity** with which a body has to be projected (vertically upwards) from a planet's surface so that it just crosses the planet's gravitational field and never returns. When the projectile just escapes to infinity it has neither kinetic energy nor potential energy left: $\frac{1}{2}mv_e^2 - GMm/R = 0$.

$$v_e = \sqrt{2GM/R} = \sqrt{2gR} = \sqrt{8\pi\rho G R^2/3} \quad (\rho = \text{mean density of the planet})$$

- **Escape speed depends on**: the mass M and radius R of the planet, and the position (height) from which the particle is projected.
- **Escape speed does not depend on**: (i) the **mass m of the body** projected, (ii) the **angle (direction) of projection**, nor on the shape or size of the body.
- If a body is thrown from the Earth's surface with the escape speed it leaves the Earth's gravitational field and never returns. **Escape velocity on the Earth = 11.2 km/s** (≈ 40 000 km/h).

↻ **De-dup note:** Lecture 14 introduces escape velocity twice on p. 2 (once at the top of the page, once under "Gravitational Field"), and a stray line "...scape velocity does not depend upon the mass or shape..." is repeated under Pascal's law on p. 3. All merged into this one section.

✎ **Source correction:** The source formula strip reads "E_g = V_e = √2GM/R..." and "E_g = V_e = √2 V_0" — the "E_g =" prefix is a typesetting carry-over from the field-intensity lines above it; escape velocity is v_e, not the field intensity.

Heavenly body	Escape velocity	Remark
Moon	2.3 km/s	too low to hold an atmosphere
Mercury	4.28 km/s	almost no atmosphere
Earth	11.2 km/s	retains N_2, O_2 but not free H_2, He
Jupiter	60 km/s	retains even hydrogen and helium
Sun	618 km/s	—
Neutron star	2×10^5 km/s	≈ two-thirds of the speed of light

Relation between escape velocity and orbital velocity

$v_e = \sqrt{2} \times v_o$ (v_o = orbital velocity of a satellite close to the surface = $\sqrt{gR} \approx 7.9$ km/s)

▌ADDED BY EDITOR — not in source lectures ▌ Satellites — orbital velocity, geostationary orbit and weightlessness (in the Lecture-1 plan; the lectures only give v_e = √2 v_o)

Orbital velocity $v_o = \sqrt{GM/(R+h)}$ — for a near-Earth orbit ≈ **7.9 km/s** (the "first cosmic velocity"; 11.2 km/s is the "second"); it does *not* depend on the satellite's mass and *decreases* with height. **Time period** $T = 2\pi(R+h)/v_o$: ≈ 84 min near the surface, 90–93 min for the ISS (≈ 400 km). **Geostationary satellite**: T = 24 h, height ≈ **35 786 km** (≈ 36 000 km), circular orbit in the **equatorial plane**, moving west→east — appears fixed in the sky; used for communication (INSAT/GSAT), DTH and weather; three such satellites cover the whole globe (Arthur C. Clarke, 1945). **Polar / Sun-synchronous satellites** (≈ 500–900 km, T ≈ 100 min) pass over the poles and image the whole Earth strip by strip — remote sensing (IRS, Cartosat), reconnaissance. **Weightlessness** in orbit is the *free-fall* condition — the satellite and the astronaut fall towards the Earth with the same acceleration, so the astronaut presses on nothing (a pendulum clock stops, water floats as spheres, candles burn as dim blue spheres). **A satellite needs no fuel to stay in orbit** — gravity supplies the centripetal force; fuel is used only to correct the orbit and to counter the thin-air drag in low orbits. If a satellite's speed becomes < v_o it spirals down; if it exceeds v_e it escapes on a hyperbola. The **Moon** is the Earth's natural satellite (v ≈ 1 km/s, T = 27.3 days sidereal / 29.5 days synodic).

▌ADDED BY EDITOR — not in source lectures ▌ Why the Moon has no atmosphere, and why hydrogen is rare in ours (the classic application)

A gas escapes when the typical molecular speed ($\propto \sqrt{T/M}$) approaches a sizeable fraction of the escape velocity. The Moon's 2.3 km/s cannot hold any gas; the Earth's 11.2 km/s holds N_2, O_2, CO_2, Ar but loses the light H_2 and He

Jupiter's 60 km/s keeps everything. A **black hole** is the limiting case — its escape velocity exceeds the speed of light (Schwarzschild radius $r_s = 2GM/c^2$; for the Sun ≈ 3 km). Gravitational waves (ripples in space-time from accelerating masses) were first detected by **LIGO** in 2015

LIGO-India is being built at Aundha (Hingoli, **Maharashtra**) — 60–62nd Q71.

Mass vs weight (Lecture 14 p. 3)

Basis	Mass	Weight
Definition	Amount of matter in a body	Force exerted by gravity on a body
Nature	Scalar quantity	Vector quantity
SI unit	kilogram (kg)	newton (N) (also kgf / kg-wt: 1 kgf = 9.8 N)
Formula	—	W = mg
Dependence on gravity	Independent of gravity	Depends on gravity (on g)
Value	Same everywhere (constant)	Changes with location (Earth, Moon, poles, equator, altitude)
Instrument	Beam (physical / pan) balance	Spring balance

Basis	Mass	Weight
Direction	No direction	Always towards the centre of the Earth (vertically down)
Dimension	[M]	$[M\ L\ T^{-2}]$
Zero possibility	Cannot be zero (except with no matter)	Can be zero — in space, in free fall, at the centre of the Earth

ADDED BY EDITOR — not in source lectures **Weight puzzles that recur in objective papers**

A body of mass 60 kg weighs 588 N on Earth but ≈ 98 N (1/6) on the Moon — its mass is still 60 kg; a **spring balance** would read 10 kg-wt on the Moon, a **beam balance** still 60 kg (both pans are affected alike); a body weighs *most at the poles* and *least at the equator* — and slightly less at the top of a mountain or down a mine; in a **lift** the apparent weight is m(g + a) accelerating up, m(g – a) accelerating down, **zero** when the cable snaps (free fall); a body is *lightest* in a lift accelerating downward. Buoyancy of air makes true weight slightly more than measured weight — a balloon "weighs" negative. Weight is what a bathroom scale measures — the *normal reaction*.

UNIT 12

Mechanical Properties of Fluids — Pressure, Buoyancy, Viscosity, Surface Tension

Consolidated from: Lecture 14 (Gravitation & Mechanical Properties of Fluids) pp. 3–5

★ **PYQ lens:** Only one direct question so far — the highest viscosity among water / air / blood / honey → **honey** (66th Q10) — but the Lecture-1 plan lists the whole block (Pascal's law, surface tension, capillarity, Bernoulli, Archimedes) and every other state PSC asks it. Priority C; read once, keep the "why does it float / rise / fly" reasons.

Fluids and their mechanical properties

- The **mechanical properties of fluids** (liquids and gases) encompass their behaviour at rest (**hydrostatics**) and in motion (**hydrodynamics**), defined by pressure, density, viscosity and surface tension.
- **Fluids** are substances — liquids and gases — that lack a fixed shape and continuously deform (**flow**) under an applied shear stress. Their behaviour revolves around their ability to flow, deform under shear and exert pressure.

↻ **De-dup note:** Lecture 14 p. 3 opens with two near-identical sentences on "mechanical properties of fluids"; merged.

▌ADDED BY EDITOR — not in source lectures ▌ Density & relative density — the missing first step

Density ρ = mass/volume (kg/m³; water = 1000 kg/m³ = 1 g/cm³ at 4 °C; mercury 13 600; air ≈ 1.29; ice 917 — which is why ice floats with ≈ 9/10 of its volume submerged). **Relative density (specific gravity)** = density of substance ÷ density of water — a pure number; measured with a **hydrometer** (density of liquids) or **lactometer** (milk). Density of a gas rises with pressure and falls with temperature; that of water is *maximum at 4 °C* (Unit 8).

Pressure

Pressure P = Force / Area = F / A SI unit N/m² = pascal (Pa); 1 atm = 1.013 × 10^5 Pa = 760 mm Hg = 1.013 bar

- Pressure is a **scalar** (65th Q113) although force is a vector — it acts equally in all directions at a point in a fluid.
- For the same force, a **smaller area gives a larger pressure**: knives and needles are sharpened, nails are pointed; camels' broad feet, snow-shoes, wide tractor tyres, skis and broad foundations *reduce* the pressure by spreading the force.

Pressure in a liquid column

P = h ρ g (h = depth, ρ = density of the liquid, g = acceleration due to gravity)

- Liquid pressure **increases with depth** (dams are thicker at the base; deep-sea divers need armoured suits) and depends on **density and g** — but **not** on the shape or width of the container or the total amount of liquid (hydrostatic paradox).
- Total pressure at depth h in an open vessel = atmospheric pressure + hρg. A liquid finds its own level; pressure is the same at all points on the same horizontal level in a liquid at rest.

▌ADDED BY EDITOR — not in source lectures ▌ Atmospheric pressure — measured and felt

The atmosphere presses with ≈ 10^5 N on every square metre (≈ 1 kg-wt per cm²), balanced from inside our bodies

Torricelli (1643) measured it with the mercury **barometer** — 76 cm (760 mm) of mercury at sea level; a water barometer would need ≈ 10.3 m. **Aneroid barometer** (no liquid; the *altimeter* is one calibrated in height). Pressure **falls with altitude** (≈ 1 cm Hg per 100 m near the ground) — that is why water boils below 100 °C on hills and pressure cookers are used, why ears "pop", ink pens leak and biscuit packets puff up in aircraft. A sudden fall of the barometer reading forecasts a storm; a slow rise, fair weather. Straws, syringes, suction pads, siphons, vacuum cleaners and "sucking" of water by a hand-pump all work because atmospheric pressure *pushes* the liquid in — a suction pump can therefore lift water at most ≈ 10 m. **Manometer** measures gas pressure; the **sphygmomanometer** measures blood pressure (normal ≈ 120/80 mm Hg).

Pascal's law

Pascal's law: pressure applied to a **confined** (enclosed) fluid is transmitted **equally and undiminished** in all directions to every point of the fluid and to the walls of the container. Stated in the source in three ways:

- the pressure in a fluid at rest is the same at all points if gravity is ignored;
- a liquid exerts equal pressures in all directions;
- if the pressure in an enclosed fluid is changed at a particular point, the change is transmitted to every point of the fluid and to the walls without being diminished in magnitude.

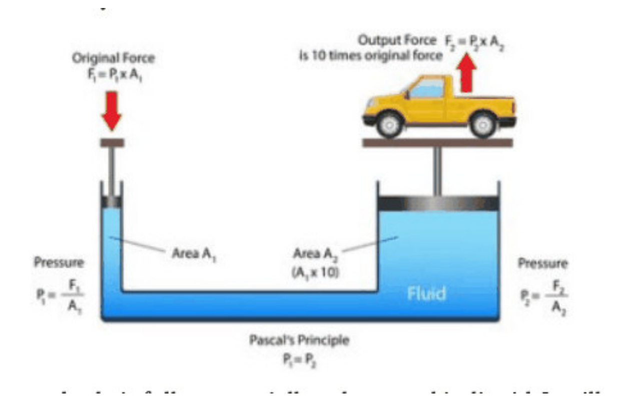

Fig. 12-1 · Pascal's principle — $P_1 = P_2$, so $F_2/A_2 = F_1/A_1$: a small force on the small piston lifts a car on the piston of 10× the area with 10× the force [Lecture 14, p. 3]

$$F_1/A_1 = F_2/A_2 \Rightarrow F_2 = F_1 \times (A_2/A_1) \text{ (force multiplication of a hydraulic machine)}$$

- **Applications**: hydraulic **jacks**, hydraulic **lifts** (garages), hydraulic **presses** (cotton bales, oil extraction), hydraulic **brakes** of cars (equal braking pressure reaches all four wheels simultaneously), power steering, dentist's/barber's chairs, excavator arms.

↻ **De-dup note:** Lecture 14 states Pascal's law twice on p. 3 (a short bullet version and a three-way statement with applications); a stray escape-velocity line printed between them has been moved to Unit 11. Merged here.

▌ADDED BY EDITOR — not in source lectures ▌ Why liquids and not gases in hydraulics (context)

Liquids are practically **incompressible**, so the pressure change is transmitted instantly and completely; gases would first compress. A hydraulic machine multiplies *force* but not *work* — the small piston must move proportionately farther (energy conservation). Blaise Pascal (1623–1662) also gave the *Pascal's vases* / hydrostatic paradox demonstration; the SI unit of pressure is named after him.

Buoyancy and Archimedes' principle

When a body is fully or partially submerged in a liquid it experiences an upward force known as the **buoyant force (upthrust)**. It arises because the pressure on the lower surface of the body (deeper) exceeds that on the upper surface.

Buoyant force $F_b = \rho g V$ (ρ = density of the fluid, g = acceleration due to gravity, V = volume of fluid displaced = $A \cdot h$ for a submerged block)

Archimedes' principle

A body wholly or partially immersed in a fluid experiences an upward buoyant force **equal to the weight of the fluid displaced** by it.

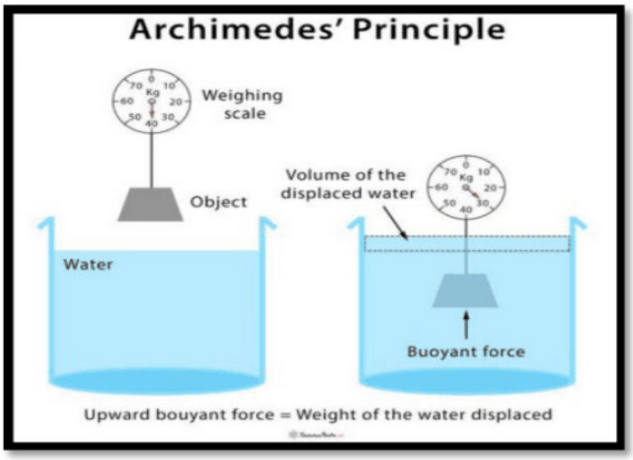

Fig. 12-2 · Archimedes' principle — the scale reads less in water; the loss of weight equals the weight of the water displaced [Lecture 14, p. 4]

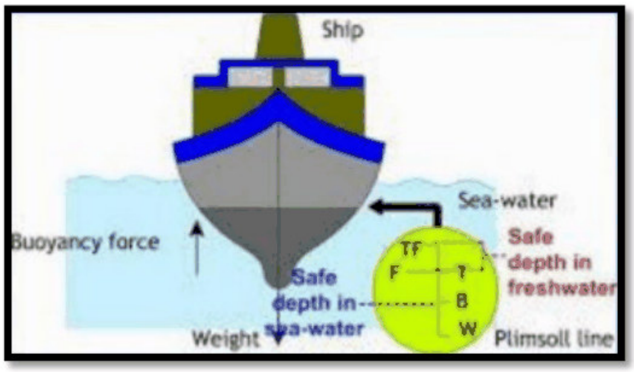

Fig. 12-3 · A ship floats when buoyancy = weight; the Plimsoll line marks the safe loading depth in sea water and in less-dense fresh water [Lecture 14, p. 4]

- **Displaced fluid**: the magnitude of the buoyant force equals the weight of the fluid that the object pushes aside. Apparent weight in the fluid = true weight – buoyant force.
- **Density and floating**: an object **floats if its average density is less** than that of the fluid, because it displaces a weight of fluid equal to its own weight before becoming fully submerged; it **sinks** if denser; it stays suspended if equal. A floating body displaces fluid equal to its own *weight*; a sunk body displaces fluid equal to its own *volume*.
- **Applications** — floating **ships** (a steel hull encloses air, so its *average* density is less than water's); **submarines** controlling depth (ballast tanks take in water to dive, blow it out with compressed air to surface); **hydrometers** (measuring the density of liquids — floats higher in denser liquid); **hot-air balloons** and airships (hot air / helium is less dense than the surrounding air); life-jackets, icebergs, swimming.

↻ **De-dup note:** Lecture 14 gives the buoyancy formula first as "Buoyancy force = ρgv" and again as "F_b = ρgV" with the symbol list; kept once.

▌ADDED BY EDITOR — not in source lectures ▌ Floating facts examiners like

A ship rises when it moves from fresh water into sea water (sea water is denser — hence the **Plimsoll line** in the source figure) and sinks slightly when it enters a river from the sea; it is easier to swim in the sea (and one floats effortlessly in the very dense Dead Sea). An **iceberg** floats with about 90 % of its volume under water ($\rho_{ice}/\rho_{sea\ water}$ = 0.917/1.025). When ice floating in a glass melts the **water level does not change** (the melt exactly fills the volume displaced). A **lactometer** floats lower in watered-down milk (milk 1.03 g/cm³). A balloon filled with hydrogen or helium rises because the buoyant force of air exceeds its weight; a body appears lighter in water — this is how Archimedes tested King Hiero's crown ("Eureka!"). The **centre of buoyancy** must lie above the centre of gravity for stable floating — ships carry ballast low down. Egg sinks in fresh water but floats in strong brine.

Viscosity (η)

- A property of **liquids and gases** — the internal **resistance a fluid offers to flow** (fluid friction between adjacent layers moving at different speeds). Honey and tar are highly viscous; water and petrol are not; gases have very small viscosity.

Fig. 12-4 · Low- vs high-viscosity liquids — the thick liquid flows off the spoon slowly [Lecture 14, p. 4]

Viscous force $F = \eta A\ (dv/dx)$ (A = area of the layer, dv/dx = velocity gradient between layers; η = coefficient of viscosity, SI unit Pa·s = N s/m²; CGS unit poise, 1 Pa·s = 10 poise)

- **Effect of temperature** — for a **liquid**, raising the temperature **decreases** viscosity and the flow increases (warm honey pours easily; engine oil thins when hot); for a **gas**, raising the temperature **increases** viscosity (faster molecular exchange between layers), so the flow resistance rises.

✎ **Source correction:** Lecture 14 p. 4 says "Velocity of liquid decreases on increasing temperature liquid" and "Velocity of gas increases..." — read **viscosity**, not velocity (its own summary line "Temp ↑ = Viscosity ↓ = Flow increase" has it right). Viscosity is also written with the symbol η (eta), not "c".

★ **PYQ lens:** 66th Q10: highest viscosity among water, air, blood, honey → **honey** (order: air < water < blood < honey; blood is ≈ 3–4 times as viscous as water).

▌ADDED BY EDITOR — not in source lectures▐ Viscosity in action (not in the lecture)

Stokes' law $F = 6\pi\eta r v$ gives the drag on a small sphere — a falling raindrop, a dust particle or a parachutist soon reaches a constant **terminal velocity** (weight = drag + buoyancy), which is why rain does not hurt and why the smallest drops (fog) stay suspended

Motor oils are graded by viscosity (SAE 10W-40) — multigrade oils flow when cold yet stay thick when hot

Blood viscosity rises with dehydration and red-cell count

Lubricants are chosen with viscosity high enough to keep the surfaces apart but low enough to flow

Viscometer measures viscosity. A perfectly non-viscous fluid is a *superfluid* (liquid helium below 2.17 K). Streamlined (laminar) flow changes to turbulent flow above a critical speed — described by the **Reynolds number** (< 2000 laminar, > 3000 turbulent).

Ideal fluid and Bernoulli's equation

Ideal fluid

- **Steady (streamline) flow** — the velocity of the fluid at any point is constant with respect to time; every particle follows the path (streamline) of the one before it; streamlines never cross.
- **Incompressible** — density constant; and **non-viscous** — no internal friction.

▌ADDED BY EDITOR — not in source lectures▐ Equation of continuity (used by Bernoulli, not printed in the lecture)

For steady flow of an incompressible fluid the volume passing every cross-section per second is the same: $A_1 v_1 = A_2 v_2$. So a liquid speeds up where the pipe narrows — a river runs fast in a gorge and slow where it widens; you narrow the mouth of a hose-pipe with your thumb to make the jet faster; the falling stream from a tap gets thinner as it speeds up.

Bernoulli's equation

Based on the **law of conservation of energy** applied to a flowing fluid; valid for ideal-fluid flow (incompressible, non-viscous, streamline). Along a streamline the sum of the pressure energy, kinetic energy and potential energy per unit volume is constant:

$$P + \tfrac{1}{2}\rho v^2 + \rho g h = \text{constant}$$

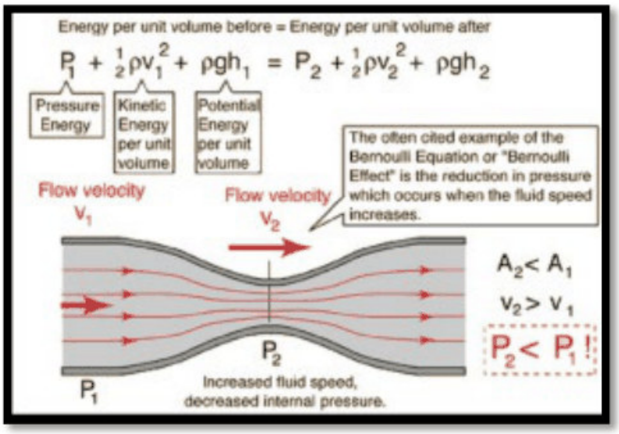

Fig. 12-5 · Bernoulli's principle — in the narrow section the flow speed rises and the internal pressure falls ($P_2 < P_1$)
[Lecture 14, p. 4]

- **Consequence**: where the **speed of a fluid is high, its pressure is low**, and vice versa ("Bernoulli effect" / Venturi effect).

▌ADDED BY EDITOR — not in source lectures ▌ Everyday applications of Bernoulli's theorem (standard list — not in the lecture)

Aerofoil lift of an aeroplane wing — air moves faster over the curved upper surface, lower pressure above, net upward lift; the **spin (Magnus effect)** of a cricket ball (swing), tennis ball (top-spin), football (banana kick)

Atomiser / sprayer / carburettor / Bunsen burner — a fast air jet lowers the pressure and draws up the liquid or gas

Venturimeter measures flow rate in pipes, the **pitot tube** measures aircraft air-speed

Roofs are blown off (not in) in a storm — fast wind over the roof reduces the pressure above it; **two boats or lorries moving close and parallel are pulled together**; a person standing near a fast train is pulled towards it; a **paper strip rises** when you blow over it; a ping-pong ball stays in an upward jet of air; blowing between two hanging balloons brings them together; the **chimney draught** and the smell drawn out of a hut by wind over it; the non-contact **Bernoulli gripper** that lifts silicon wafers in industry. **Torricelli's law**: liquid leaves a hole at depth h with speed $\sqrt{2gh}$, the same as a freely falling body. **Daniel Bernoulli** (Swiss), 1738.

Surface tension (T or γ)

Surface tension is the force acting per unit length along the surface of a liquid (perpendicular to an imaginary line drawn on the surface): **$T = F/L$**, SI unit **N/m** (also J/m^2 — it equals the surface energy per unit area). It arises from the unbalanced inward cohesive pull on the molecules of the surface layer.

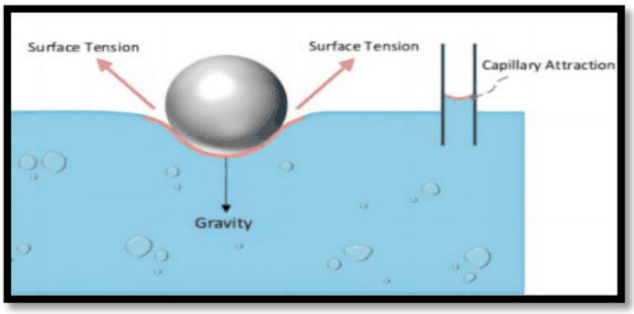

Fig. 12-6 · Surface tension holds up a steel ball / needle; capillary attraction at the wall [Lecture 14, p. 4]

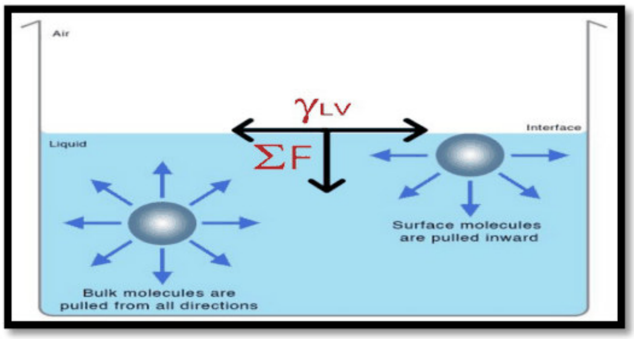

Fig. 12-7 · Molecules in the bulk are pulled equally in all directions; surface molecules are pulled inward — the origin of surface tension [Lecture 14, p. 4]

- It is responsible for the **spherical shape of droplets** (a sphere has the least surface area for a given volume) and for **insects floating (walking) on the water surface**.
- **Applications / examples** (Lecture 14 p. 5): the shape of drops of water and of mercury; insects (water-striders) remain floating on a liquid surface; a greased needle floats on water.

✎ **Source correction:** Lecture 14 p. 5 heads this list "Application of surface vision — Prolate of water and mercury" — read "surface *tension*" and "*droplets* of water and mercury".

▌ADDED BY EDITOR — not in source lectures ▌ Surface tension — factors and the classic examples

Decreases with temperature (hot water and soap solutions wet and clean better; it vanishes at the critical temperature); **soaps and detergents lower it** — the water then spreads into cloth fibres and lifts grease; **impurities** such as salt raise it slightly; **oil poured on rough sea** calms it (oil film lowers surface tension); **camphor bits dance** on water (dissolving camphor lowers T unevenly); **mosquito larvae** breathe at the surface — spraying kerosene/oil lowers the surface tension and they drown; a rain-coat or umbrella repels water because the fine pores are bridged by surface tension (pressing the cloth from inside lets water through); a **small drop is spherical**, a large one flattens under gravity; rain-drops and molten-lead shot become spheres while falling; **soap bubbles**: excess pressure inside = $4T/r$ (two surfaces), inside a drop = $2T/r$ — a *smaller* bubble has *higher* pressure and would blow up a larger one connected to it; **antiseptics** (Dettol) have low surface tension to spread over wounds; **hair of a paint-brush** cling together when taken out of water. **Cohesion** = attraction between like molecules (water–water); **adhesion** = between unlike molecules (water–glass). Water wets glass (adhesion > cohesion, angle of contact acute, **concave** meniscus); mercury does not (cohesion > adhesion, obtuse angle, **convex** meniscus).

Capillarity

Capillarity (capillary action) is the rise or fall of a liquid in a narrow tube (capillary) due to **surface tension and the angle of contact** between the liquid and the tube.

- **Water rises** in a glass capillary (concave meniscus; the narrower the tube, the higher the rise — $h \propto 1/r$); **mercury falls** (is depressed) in a glass capillary (convex meniscus).

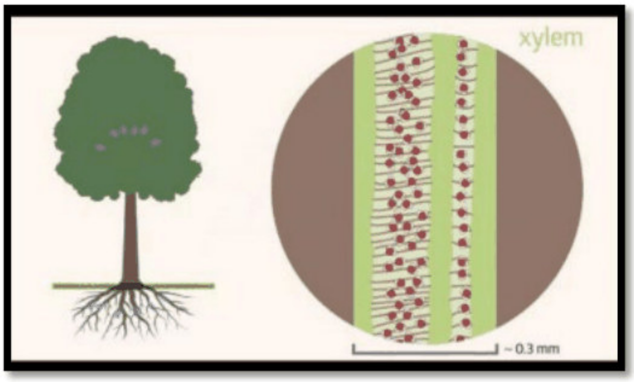

Fig. 12-8 · Capillary action in the xylem vessels carries water up from the roots of a tree [Lecture 14, p. 5]

Fig. 12-9 · A straw (narrow tube) lowered into water — the liquid rises inside it [Lecture 14, p. 5]

Capillary rise h = 2 T cos θ / (r ρ g) (θ = angle of contact, r = radius of the tube, ρ = density)

▌ADDED BY EDITOR — not in source lectures▐ Capillarity in daily life (standard list)

The wick of a lamp or candle draws up oil/wax; a towel, sponge or blotting paper soaks up water; ink rises in a nib; **sap and water rise in the xylem** of plants (with transpiration pull); water rises through the pores of soil and bricks — ploughing breaks the capillaries and *conserves* soil moisture, damp-proof courses stop the rise in walls; a chalk or sugar cube dipped at one corner gets wet all over; **in a freely falling lift or an orbiting spacecraft** (g = 0) the liquid rises to fill the whole capillary; a **paint brush** picks up paint. If the capillary is too short the liquid does *not* overflow — the meniscus merely flattens. The rise is greater in a narrower tube and in a liquid of higher surface tension; hot water rises less than cold.

UNIT 13

Electric Charge, Field, Potential & Current Electricity

Consolidated from: Lecture 15 (Electricity Part-01) pp. 2–3; Lecture 16 (Electricity Part-02) pp. 2–3

★ **PYQ lens: Priority B — 5 questions, all in the last six papers.** Two resistors in parallel giving 1.403 kΩ with one of 2 kΩ → the other is 4.70 kΩ (66th Q2); resistance of a semiconductor **decreases** on heating (66th Q3); salted water, orange juice and lemon juice are all conductors (69th Q19); current density is a **vector** (69th Q27); resistance of a wire = **V/I** (70th Q104). Expect one Ohm's-law / resistance question every year.

Electric charge (Lecture 15 p. 2)

Electric charge is the fundamental property of matter responsible for electrical phenomena. It can be **positive (+)** or **negative (–)** and is conserved. It is the property by which matter produces and experiences electrical and magnetic effects; the **excess or deficiency of electrons** in a body is the cause of its net charge (deficiency of electrons → positive; excess → negative).

- **Unit**: coulomb (C); **nature**: scalar; **charge on one electron** = -1.6×10^{-19} C (proton $+1.6 \times 10^{-19}$ C; 1 C = charge of 6.25×10^{18} electrons).
- The charge of fundamental particles (electron, proton) is their internal characteristic, while the charge on a body depends on the number of protons and electrons inside it.

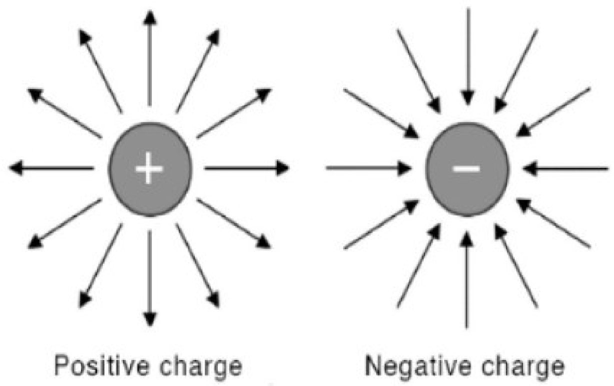

Fig. 13-1 · Field lines of a positive charge (outward) and a negative charge (inward) [Lecture 15, p. 2]

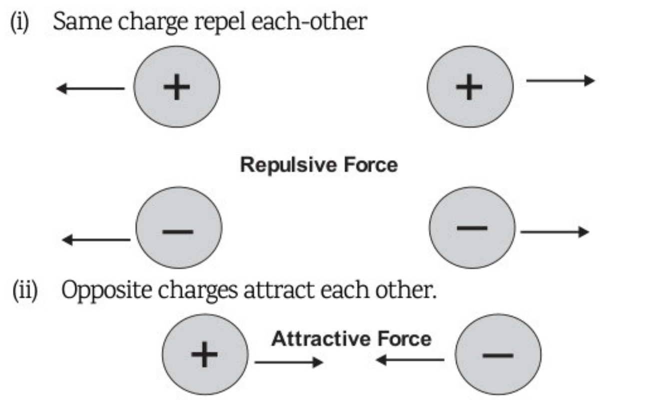

Fig. 13-2 · (i) Like charges repel — (ii) unlike charges attract [Lecture 15, p. 2]

Specific properties of charge

- **Charge is a scalar** — it represents an excess or deficiency of electrons.
- **Charge is transferable** — if a charged body is put in contact with another body, charge can be transferred to it (only electrons move; protons stay in the nucleus).
- **Charge is always associated with mass** — charge cannot exist without mass, though mass can exist without charge (neutron). The presence of charge is itself proof of the existence of mass. **The mass of a body changes after being charged**: a body given a **positive** charge loses electrons and its mass **decreases** (slightly); a body given a **negative** charge gains electrons and its mass **increases**.
- **Charge is quantised** — all free charges are integral multiples of a basic unit e: **q = ne**, n = ±1, ±2, … The quantum of charge is the charge of an electron or proton (Millikan's oil-drop experiment).
- **Charge is conserved** — in an isolated system the total charge does not change with time, though individual charges may change: charge can neither be created nor destroyed. Conservation of charge holds in all chemical (atomic) and nuclear reactions; no exception has ever been found.
- **Charge is invariant** — independent of the frame of reference: the charge on a body does not change whatever its speed (unlike mass).

▌ADDED BY EDITOR — not in source lectures▐ Methods of charging & electroscope (context frequently asked)

Friction (glass rod rubbed with silk → glass +, silk –; ebonite/plastic with fur → plastic –), **conduction** (contact) and **induction** (charging without contact; the induced charge is opposite). A **gold-leaf electroscope** detects charge (leaves diverge). Charge resides on the **outer surface** of a conductor and concentrates at sharp points (lightning conductor, Unit 14); inside a hollow conductor the field is zero (**Faraday cage** — why a car is safe in lightning). Quarks carry ±e/3 and ±2e/3 but are never free. **Electrostatics in life**: crackling when a nylon sweater is pulled off, hair standing on end with a comb, lightning, photocopiers and laser printers, electrostatic paint spraying, electrostatic precipitators in chimneys, fuel tankers trailing chains to earth static charge.

Coulomb's law (Lecture 15 p. 3)

The electrostatic force of interaction between two static point charges is **directly proportional to the product of the charges**, **inversely proportional to the square of the distance** between them, and acts along the straight line joining the two charges. Like charges repel, unlike charges attract; the force decreases with distance.

$F = k\, q_1 q_2 / r^2$ $k = 1/(4\pi\varepsilon_0) = 9 \times 10^9$ N m² C^{-2} ($\varepsilon_0 = 8.85 \times 10^{-12}$ C² N^{-1} m^{-2}, permittivity of free space)

▌ADDED BY EDITOR — not in source lectures▐ Coulomb vs Newton, and the medium (context)

The law has the same inverse-square form as gravitation, but the electric force is ~10^{36} times stronger (Unit 11) and can be attractive *or* repulsive. In a medium the force is *reduced* by the **dielectric constant** K (relative permittivity): $F_{medium} = F_{vacuum}/K$ (water K ≈ 80 — which is why ionic salts dissolve in water; mica ≈ 6, glass ≈ 5–10, metals → ∞). 1 coulomb is an enormous charge — two 1 C charges 1 m apart would repel with 9×10^9 N. The unit is named after **Charles-Augustin de Coulomb** (torsion balance, 1785).

Electric field and field intensity

An **electric field** is the region around a charge in which another charge experiences a force.

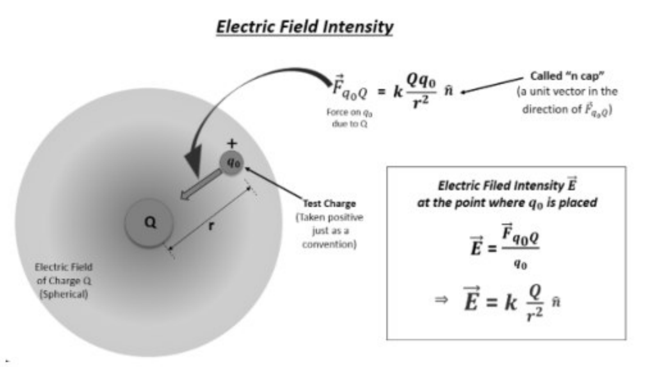

Fig. 13-3 · Electric field intensity — the force on a small positive test charge q_0 placed at distance r from Q: $E = F/q_0 = kQ/r^2$ [Lecture 15, p. 3]

- **Electric field intensity (E)** — the force per unit (positive test) charge: **E = F/q**; unit **N/C** (= V/m); a **vector** in the direction of the force on a positive charge. For a point charge Q at distance r: $E = kQ/r^2$.
- **Electric field lines** start on positive charges and end on negative ones, never cross, are crowded where the field is strong, and are perpendicular to a conductor's surface.

▌ADDED BY EDITOR — not in source lectures ▌ Capacitance & capacitors (in the Lecture-1 plan, never taught)

A **capacitor** stores charge and electrical energy on two conductors separated by an insulator (dielectric). **Capacitance** C = Q/V; unit **farad (F)** (1 F is huge — practical values are μF, nF, pF). Parallel-plate: $C = \varepsilon_0 KA/d$ — larger plates, smaller gap, and a dielectric all *increase* C. Energy stored $U = ½CV^2 = ½QV$. Capacitors in **parallel add** ($C = C_1 + C_2$), in **series** $1/C = 1/C_1 + 1/C_2$ (the opposite of resistors). Uses: camera flash, tuning circuits in radios, smoothing rectified DC, power-factor correction, defibrillators, keyboards and touch-screens (capacitive sensing), the **Leyden jar** was the first capacitor (1745). A charged capacitor blocks DC but passes AC.

Electric potential and potential difference

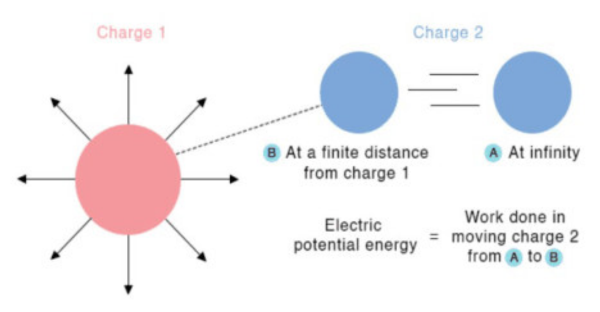

Fig. 13-4 · Electric potential energy = work done in moving charge 2 from infinity (A) to a finite distance (B) from charge 1 [Lecture 15, p. 3]

- **Electric potential (V)** — the work done per unit charge in bringing a positive test charge from infinity to that point: **V = W/q**; unit **volt (V)** — 1 V = 1 joule per coulomb. It is a **scalar**. For a point charge Q: V = kQ/r.
- **Potential difference (voltage)** — the difference in electrical potential energy per unit charge between two points in a circuit; it is what **drives the current**: $V_{AB} = W/q$. Positive charge flows from higher to lower potential (electrons the other way).
- **Current does not flow between two charged bodies at the same electric potential** — a potential difference is required (Lecture 17 p. 5). This is like heat flowing only between bodies at different temperatures, and water only between different levels.

▌ADDED BY EDITOR — not in source lectures ▌ Potential vs charge — the "big body, small potential" idea (context)

The Earth is taken as zero potential (earthing, Unit 14). A body at high potential need not have a large charge (a Van de Graaff dome at 10^6 V holds only microcoulombs), and a cloud with a huge charge may be at modest potential. Birds sit safely on a single high-tension wire because both feet are at the *same* potential — no potential difference, no current. Lightning is a discharge across ~10^8–10^9 V. The **electron-volt** (1 eV = 1.6×10^{-19} J) is the energy gained by an electron falling through 1 V. **Equipotential surfaces** are perpendicular to field lines; no work is done in moving a charge along them.

Electric current (Lecture 15 p. 3)

An **electric current** is a flow of charged particles — electrons or ions — through a conductor or through space; it is the **net rate of flow of electric charge** through a surface.

I = Q / t unit: ampere (A) — 1 A = 1 coulomb per second (1 A = 6.25×10^{18} electrons per second)

- **Direction**: the *conventional* current flows from **+ to –** (the direction positive charge would move); electrons actually drift from – to +.
- Electric current is treated as a **scalar** — it has a direction along the wire but does not obey the vector (parallelogram) law of addition. **Current density** J = I/A (A/m²), however, is a **vector** (69th Q27).

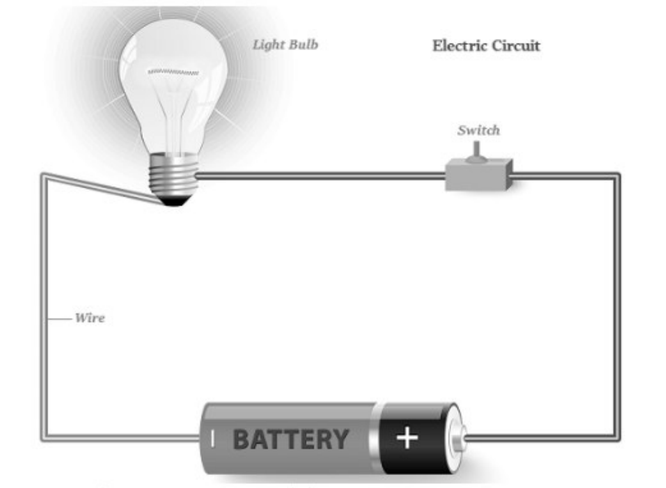

Fig. 13-5 · A simple circuit — battery, switch, wire and bulb: a closed conducting path is needed for a current [Lecture 15, p. 2]

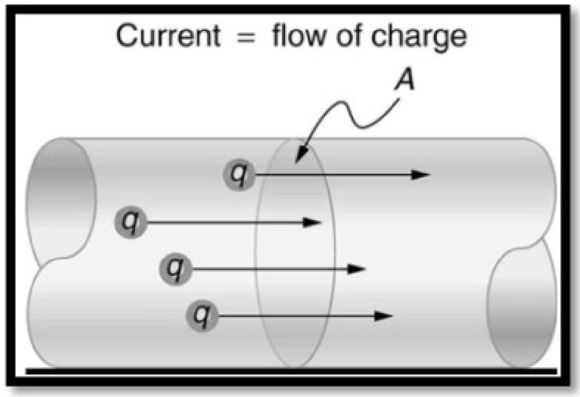

Fig. 13-6 · Current = flow of charge q through a cross-section A of the conductor [Lecture 15, p. 3]

★ **PYQ lens:** 69th Q27: current density → **a vector quantity**. 69th Q19: "Which liquid is a bad conductor of electricity?" — salted water, orange juice and lemon juice are all **electrolytes** (ions carry the current), so the answer is **none of the above**; pure/distilled water, oils and alcohol are the poor conductors.

▌ADDED BY EDITOR — not in source lectures ▌ Drift velocity, and what carries the current (context)

In a metal the free electrons drift at only ≈ 10^{-4} m/s (a fraction of a millimetre per second), yet the *signal* travels at nearly the speed of light because the field is set up along the whole wire almost instantly — the bulb lights immediately. Charge carriers: **electrons** in metals; **ions (both signs)** in electrolytes and gases (neon tubes, lightning); **electrons and holes** in semiconductors; **protons/ions** in plasma. A current of ≈ 10 mA through the body is painful, ≈ 100 mA across the heart can be fatal — it is the *current*, not the voltage alone, that kills, which is why a 12 V car battery is safe to touch but a 230 V mains socket is not. **Ammeter** measures current (series, low resistance), **voltmeter** measures potential difference (parallel, high resistance) — Unit 14.

Ohm's law (Lecture 16 p. 2)

At **constant temperature** (and other physical conditions), the current flowing through a conductor is **directly proportional to the potential difference** across its ends.

V = I R ⇒ I = V/R ⇒ R = V/I (R = resistance, unit ohm Ω; 1 Ω = 1 V/A)

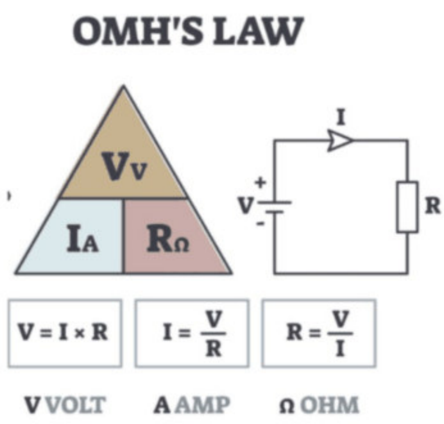

Fig. 13-7 · Ohm's-law triangle — cover the quantity you want: V = IR, I = V/R, R = V/I [Lecture 16, p. 2]

- Valid for **ohmic conductors** (metals, alloys at constant temperature) whose V–I graph is a straight line through the origin; **non-ohmic** devices (diodes, transistors, filament lamps whose temperature changes, electrolytes, thermistors) do not obey it.

★ **PYQ lens:** 70th Q104: the resistance of a wire → **R = V/I**. (The other options IV, 1/2V and I/V are traps; learn all three rearrangements from the triangle.)

▌ADDED BY EDITOR — not in source lectures ▌ Georg Simon Ohm (1827) and the graphs (context)

On a V–I graph the *slope* V/I is the resistance — a steeper line means a larger resistance; on an I–V graph the slope is the *conductance* G = 1/R (unit **siemens**, S, formerly "mho"). A **superconductor** (mercury below 4.2 K — Kamerlingh Onnes, 1911; YBCO ceramics up to ≈ 90 K) has *zero* resistance and expels magnetic fields (Meissner effect) — used in MRI magnets, maglev trains and the LHC.

Classification of materials by conductivity (Lecture 16 p. 2)

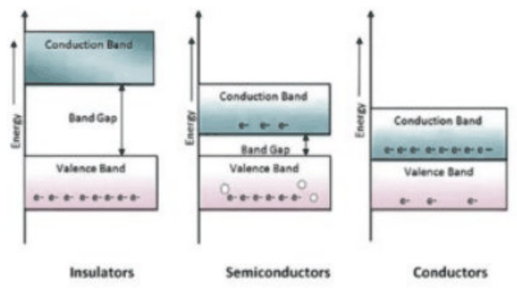

Fig. 13-8 · Energy bands — insulators (large gap), semiconductors (small gap, few electrons cross at room temperature), conductors (bands overlap) [Lecture 16, p. 2]

Conductors	Insulators	Semiconductors
Free electrons are present, hence electric current can flow easily (metals — silver, copper, aluminium, gold; also graphite, salt solutions, human body, earth)	Free electrons are **not available**, hence current cannot pass through them (rubber, glass, mica, dry wood, plastics, porcelain, pure/distilled water, dry air, paper, ebonite)	**Few** free electrons are available at room temperature, hence they conduct only slightly (silicon, germanium, gallium arsenide, selenium)
On increasing temperature the flow of current **decreases** (resistance rises — more collisions of electrons with vibrating ions)	Resistance is very high at all ordinary temperatures (breaks down only at very high voltage)	On increasing temperature more free electrons are released, hence the flow of current **increases** (resistance **falls**)
Resistivity 10^{-8} – 10^{-6} Ω m	Resistivity 10^{8} – 10^{16} Ω m	Resistivity 10^{-5} – 10^{3} Ω m (in between)

★ **PYQ lens:** 66th Q3: on heating, the resistance of a semiconductor **decreases** (negative temperature coefficient). Contrast: metals — resistance increases on heating; alloys such as manganin and constantan — almost no change (hence used for standard resistors).

▌ADDED BY EDITOR — not in source lectures ▌ Semiconductor extras BPSC touches (photodiode 68th Q113, LEDs, solar cells)

Doping a pure (intrinsic) semiconductor with pentavalent atoms (P, As, Sb) gives an **n-type** (extra electrons); with trivalent atoms (B, Al, Ga, In) a **p-type** (holes). A **p–n junction diode** conducts in one direction only → **rectifier** (AC → DC). **LED** — emits light when forward-biased (Unit 14); **photodiode** — reverse-biased, its current changes with light → light sensors, optical fibre receivers, digital/logic applications (68th Q113); **solar (photovoltaic) cell** — a junction that converts light directly to electricity (silicon); **transistor** (Bardeen, Brattain, Shockley, 1947) — amplifier/switch, the building block of all integrated circuits; **thermistor** — resistance falls steeply with temperature (fever thermometers, fire alarms). **Silver** is the best conductor, then copper, gold, aluminium; **nichrome** is used in heater coils (high resistivity, high melting point), **tungsten** in bulb filaments.

Resistance (Lecture 16 p. 2)

Resistance is the property of a material that opposes the flow of electric current — how difficult it is for electrons to move through a conductor. Unit **ohm (Ω)**.

Factors on which resistance depends

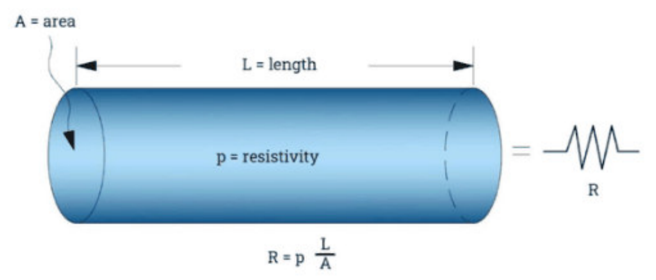

Fig. 13-9 · R = ρ L/A — a conductor of length L, cross-section A and resistivity ρ [Lecture 16, p. 2]

- **1. Length (L)** — a longer wire has more resistance: **R ∝ L**.
- **2. Area of cross-section (A)** — a thicker wire has less resistance: **R ∝ 1/A**.
- **3. Material (resistivity ρ)** — copper: low resistance; nichrome: high; rubber: very high.
- **4. Temperature** — **metals: resistance increases** with temperature; **semiconductors (and insulators, electrolytes): resistance decreases** with temperature.

R = ρ L / A (ρ = resistivity / specific resistance, unit Ω m — a property of the material and temperature only, not of the dimensions)

✎ **Source correction:** Lecture 16 prints the resistivity symbol as "δ" in "R = δℓ/A" and as "p" in the text; the standard symbol is ρ (rho). Conductivity σ = 1/ρ (unit S/m).

▌ADDED BY EDITOR — not in source lectures ▌ Stretching and folding a wire (the standard numerical)

The **volume** of a wire stays constant, so if it is stretched to n times its length, A becomes 1/n of the original and R becomes **n²** times (doubling the length by stretching → 4R); if a wire is folded/cut into n equal parts and bundled in parallel, R becomes R/n². Doubling the *diameter* (same length) makes R one-quarter. Resistivity order (room temperature): silver 1.6×10^{-8} Ω m < copper 1.7×10^{-8} < gold < aluminium 2.8×10^{-8} < tungsten 5.6×10^{-8} < iron ≈ 10^{-7} < nichrome ≈ 10^{-6} < germanium ≈ 0.5 < silicon ≈ 2000 < glass ≈ 10^{12}. **Rheostat** = variable resistance (used to control current in a circuit, e.g. fan regulator); colour code of carbon resistors: B B ROY of Great Britain had a Very Good Wife (0–9).

Combination of resistors (Lecture 16 pp. 2–3)

Two or more resistors may be connected in **series**, in **parallel**, or in a combination of both, to provide a specific total resistance, control the current and distribute voltage appropriately across the parts of a circuit.

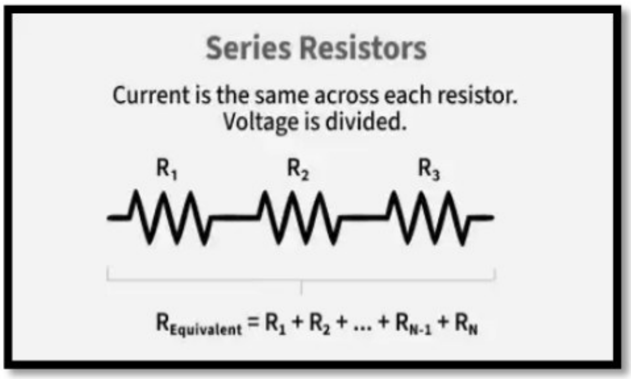

Fig. 13-10 · Series resistors — the same current flows through each; the voltage divides; $R = R_1 + R_2 + \ldots + R_n$ [Lecture 16, p. 3]

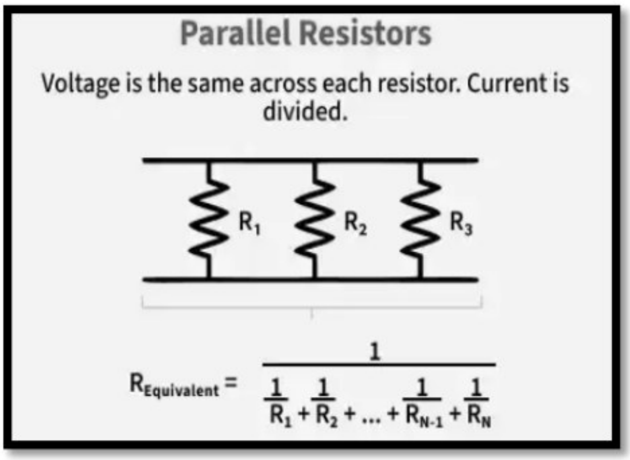

Fig. 13-11 · Parallel resistors — the same voltage across each; the current divides; $1/R = 1/R_1 + 1/R_2 + \ldots + 1/R_n$ [Lecture 16, p. 3]

I. Series combination

- Resistors connected one after another in a single path, so that the current has **only one path**; the **same current flows through each** resistor (charge leaving one enters the next without branching).
- The total potential difference is shared: $V = V_1 + V_2 + V_3$ (larger R takes the larger share, $V \propto R$).

$R_s = R_1 + R_2 + R_3 + \ldots$ (equivalent resistance is greater than the greatest individual resistance)

- If one element fails the whole circuit is broken — that is why old fairy lights all went out together, and why fuses and switches are connected in series.

II. Parallel combination

- Resistors connected between the **same two points**, giving **multiple paths**; the **potential difference across each is the same**, while the total current **divides** among the branches: $I = I_1 + I_2 + \ldots$ (the smaller R takes the larger current, $I \propto 1/R$).

$1/R_p = 1/R_1 + 1/R_2 + 1/R_3 + \ldots$ for two resistors $R_p = R_1R_2/(R_1 + R_2)$ (equivalent resistance is less than the smallest individual resistance)

- n equal resistors R: series → nR; parallel → R/n. Domestic appliances are wired in **parallel** so that each gets the full mains voltage and works independently (Unit 14).

★ **PYQ lens:** 66th Q2: two parallel resistors give 1.403 kΩ, one is 2.0 kΩ → $1/R_2 = 1/1.403 - 1/2 = 0.7128 - 0.5 = 0.2128 \Rightarrow R_2 =$ **4.70 kΩ** (check: 2 × 4.7/6.7 = 1.403 ✓).

▌ADDED BY EDITOR — not in source lectures ▌ Cells in series and parallel, EMF and internal resistance (standard follow-up)

EMF (E) of a cell = the potential difference across its terminals when no current is drawn; **terminal voltage** $V = E - Ir$, where r is the cell's **internal resistance** (a new dry cell ≈ 1.5 V, r ≈ 0.1–1 Ω; a car battery 12 V, r ≈ 0.01 Ω — hence it can deliver hundreds of amperes). Cells in **series** add their EMFs (a torch with two 1.5 V cells gives 3 V); identical cells in **parallel** give the same EMF but can supply more current for longer. **Kirchhoff's laws**: the sum of currents at a junction is zero (charge conservation); the sum of EMFs round a loop equals the sum of IR drops (energy conservation). **Wheatstone bridge** (P/Q = R/S when the galvanometer shows zero) measures an unknown resistance — the **metre bridge** is its laboratory form; the **potentiometer** compares EMFs without drawing current (Unit 14).

▌ADDED BY EDITOR — not in source lectures▐ Worked examples in the recent-BPSC style

(1) 2 Ω, 3 Ω and 6 Ω in parallel: 1/R = 1/2 + 1/3 + 1/6 = 1 ⇒ R = **1 Ω**; in series 11 Ω. (2) A 230 V heater draws 4.6 A: R = 50 Ω, P = 1058 W. (3) Three 60 Ω bulbs in parallel across 12 V: each draws 0.2 A, total 0.6 A, equivalent 20 Ω. (4) A wire of 8 Ω is stretched to double its length → 32 Ω; cut into four equal pieces connected in parallel → 0.5 Ω. (5) A 12 V battery of internal resistance 0.5 Ω across 5.5 Ω: I = 12/6 = 2 A, terminal voltage 11 V.

UNIT 14

Electrical Power, Effects of Current, Instruments, Devices, Lamps & Domestic Electricity

Consolidated from: Lecture 17 (Electricity Part-03) pp. 2–5; the electrical-instrument entries of Lecture 19 p. 7

★ **PYQ lens: Priority B — 11 questions in 5 papers (the largest count after Unit 16, though only two of them recent); the most repeated stems in the whole physics syllabus.** Unit of electric power = **watt** (64th Q11 *and* 65th Q120); device to measure current = **ammeter** (64th Q14 *and* 65th Q119); electric motor: electrical → mechanical energy (64th Q12); car-battery acid = **sulphuric acid** (63rd Q99); Faraday constant is a universal constant (66th Q4); 71st: 250 units = **9×10^8 J** (Q123), **IR^2** is not a power formula (Q128), current in a short circuit rises very high (Q129), fuse protects equipment (Q130).

Electrical power (Lecture 17 p. 2)

Electrical power is the rate at which electrical energy is transferred or used in a circuit — how much electricity is being consumed or produced at any moment.

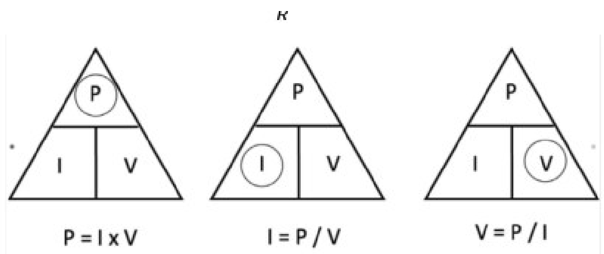

Fig. 14-1 · Power triangle — P = I × V, I = P/V, V = P/I [Lecture 17, p. 2]

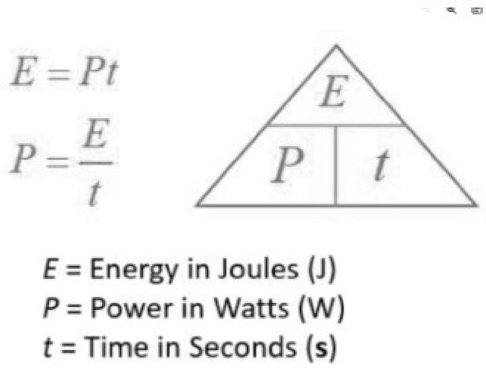

Fig. 14-2 · Energy triangle — E = P t, P = E/t: E in joules, P in watts, t in seconds [Lecture 17, p. 2]

$P = VI = I^2R = V^2/R$ (V = potential difference, I = current, R = resistance; unit watt (W) = 1 J/s = 1 V × 1 A; 1 kW = 1000 W; 1 hp = 746 W)

- For a **given resistance R** (constant): $P \propto V^2$ and $P \propto I^2$ — doubling the voltage across a heater quadruples its power.
- If the **voltage is constant** (as in the mains): $P \propto I$ — a device drawing twice the current uses twice the power; $P_1/P_2 = I_1/I_2$.

✎ **Source correction:** Lecture 17 p. 2 prints "$P \propto I^2$ … $P \propto I$" together under "If voltage remains constant"; $P \propto I^2$ holds for constant *R*, $P \propto I$ for constant *V* (as separated above).

★ **PYQ lens:** 64th Q11 / 65th Q120: unit of electric power → **watt** (ampere = current, volt = potential, coulomb = charge). 71st Q128: which does **not** denote electric power → **IR^2** (VI, I^2R and V^2/R all do).

▌ADDED BY EDITOR — not in source lectures ▌ Power ratings and the series/parallel bulb trick (standard objective material)

A device marked "100 W, 220 V" has $R = V^2/P = 484\ \Omega$ and draws $I = P/V \approx 0.45$ A at its rated voltage. **In parallel (normal house wiring)** the higher-wattage bulb glows brighter ($P \propto 1/R$ at fixed V); **in series** the *lower*-wattage bulb (higher R) glows brighter ($P = I^2R$, same I) — a 40 W and a 100 W bulb in series: the 40 W one is brighter. Two heaters of resistance R in series give ¼ of the power that both in parallel give. Typical ratings: LED bulb 9 W, CFL 15 W, fan 60–75 W, TV 100 W, refrigerator 150–250 W, electric iron 750–1000 W, geyser/kettle 1500–2000 W, AC 1500–2500 W (Lecture 17: fan < TV < iron < kettle).

Electrical energy and the commercial unit

Electrical energy is the total work done by an electric current in a circuit over a given time — the amount of power consumed or supplied during that time.

E = P × t (joule = watt × second) · commercial unit: kilowatt-hour (kWh) = 1 "unit" · 1 kWh = 1000 W × 3600 s = 3.6 × 10^6 J

- Energy meters at home (**kWh meter / "electricity meter"**) count units: a 1 kW appliance running for 1 hour = 1 unit; a 100 W bulb for 10 hours = 1 unit; a 2 kW geyser for 30 min = 1 unit.

✎ **Source correction:** Lecture 17 prints "1 kilowatt-hour (kWh) = 3.6 x 10 Joules" — the exponent dropped out in typesetting; it is **3.6 × 10^6 J**.

↻ **De-dup note:** "Electrical energy = power × time" and the kWh definition are printed twice in different forms on Lecture 17 p. 2; merged.

★ **PYQ lens:** 71st Q123: 250 units consumed → 250 × 3.6 × 10^6 = **9 × 10^8 J**. This is the same 1 kWh = 3.6 MJ conversion as in Unit 1 — learn it cold.

Effects of electric current (Lecture 17 p. 2)

1. Heating effect (Joule's law of heating)

When current passes through a conductor, electrical energy is converted into heat because the drifting electrons collide with the ions of the lattice.

Heat produced $H = I^2 R t$ (joule) = $V I t$ = $(V^2/R) t$ · in calories: $H = I^2Rt / 4.18$

- Heat produced is proportional to the **square of the current**, to the **resistance** and to the **time** (James Prescott Joule, 1841).
- **Applications**: electric heater, electric iron, geyser/immersion rod, toaster, electric kettle, **filament bulb**, and the **electric fuse** — used wherever heat is required (heating elements are of **nichrome**: high resistivity, high melting point, does not oxidise when red-hot).
- Undesirable heating: losses in transmission lines (minimised by transmitting at high voltage and low current, since loss ∝ I^2), heating of appliances and of the Earth's atmosphere by lightning.

2. Magnetic effect

- A current-carrying conductor produces a **magnetic field** around it — **Oersted's experiment** (1820; Unit 15).
- **Applications**: electromagnets, electric bell, electric motor, relay, loudspeaker, telephone receiver, magnetic cranes, MRI, maglev — the basis of modern electrical machines.

3. Chemical effect

- Electric current causes chemical changes in a conducting liquid (**electrolyte**) — studied in **electrolysis** (Faraday's laws).
- **Applications**: **electroplating** (chromium on taps, silver on cutlery, gold on jewellery, zinc galvanising), battery charging, extraction and refining of metals (aluminium, copper, sodium), electrotyping, production of hydrogen and oxygen, chlor-alkali industry — important in industry and chemistry.

▌ADDED BY EDITOR — not in source lectures ▌ Faraday's laws of electrolysis & the Faraday constant (asked in 66th Q4; only the constant is in the lecture)

First law: mass deposited m = Z I t = Z Q (Z = electrochemical equivalent). **Second law**: for the same charge, masses of different substances deposited are proportional to their chemical equivalents. **Faraday constant F = charge of one mole of electrons = N_A × e ≈ 96 485 C/mol** (≈ 96 500) — a **universal constant**; it does not depend on the amount of electrolyte, the current or the solvent. One faraday deposits one gram-equivalent of any substance (108 g Ag, 31.75 g Cu). A **voltameter** (Lecture 17 p. 3) is the electrolytic cell used to measure charge by the mass deposited — do not confuse it with the *voltmeter*.

▌ADDED BY EDITOR — not in source lectures ▌ Other effects worth a line (not in the lecture)

Physiological effect — nerves and muscles respond to current (electric shock, ECG/EEG, pacemakers, electric fish); **luminous effect** — gas-discharge tubes, LEDs; **thermoelectric (Seebeck/Peltier)** effects — Unit 8. The three classic effects plus their instruments: heating → hot-wire ammeter; magnetic → moving-coil galvanometer; chemical → voltameter.

Applications of the heating effect

I. Electric fuse

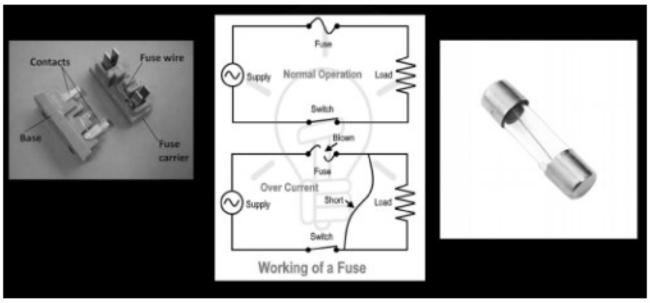

Fig. 14-3 · Fuse — contacts, fuse wire and carrier; normal operation vs over-current when the wire melts and opens the circuit [Lecture 17, p. 5]

- A **safety device** that works on the **heating effect** of current: a short wire of **high resistance and low melting point** — an alloy of **tin and lead (Sn + Pb)** in the ideal fuse; also tin–lead–zinc alloys, or copper/aluminium for heavy ratings.
- It is connected in **series** with the live wire at the main switchboard (and in plugs); when the current exceeds the rated value (overload or short circuit) the wire **melts and breaks the circuit**, protecting the appliances and wiring from damage and fire.
- Fuse ratings: 5 A for lighting circuits, 15 A for power circuits (heaters, geysers). A fuse should never be replaced by a thicker wire or a coin.

↻ **De-dup note:** Lecture 17 lists the fuse under "Application of heating effect" (p. 2) and again under "Safety devices" (p. 5, with its characteristics). Merged here; the other safety devices follow in the domestic-electricity section.

★ **PYQ lens:** 71st Q129: in a short circuit the current **increases very high** (the live and neutral touch directly → R ≈ 0 → I = V/R very large → heat, sparks, fire — unless the fuse melts). 71st Q130: device to protect equipment from electric shock/overload → **fuse** (strictly, a fuse/MCB protects the *equipment* from over-current; *earthing* and the *ELCB/RCCB* protect *people* from shock).

▌ADDED BY EDITOR — not in source lectures ▌ MCB, ELCB and the short-circuit story (modern equivalents, not in the lecture)

MCB (miniature circuit breaker) — an electromagnetic + thermal switch that trips on over-current and can be reset (replaces fuses in modern boards)

ELCB / RCCB (residual-current device) — trips within milliseconds when the live and neutral currents differ by ≈ 30 mA, i.e. when current is leaking to earth through a person — the real anti-shock device

Overloading — too many appliances on one socket exceed the wire's rating

Short circuit — insulation failure lets live touch neutral; both are cured by the fuse/MCB, not by earthing. **Surge protectors / stabilisers** guard against voltage spikes and fluctuations.

II. Electric bulb (incandescent lamp)

- Works on the **heating effect**: the filament is heated to incandescence (white heat) and emits light. Details of lamps below.

Electrical instruments (Lecture 17 p. 3 + Lecture 19 p. 7)

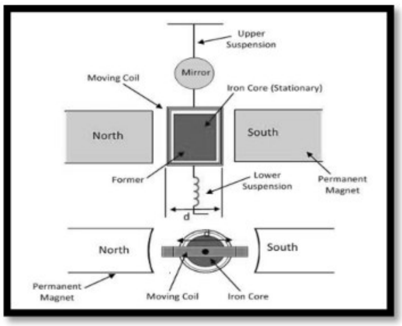

Fig. 14-4 · Moving-coil galvanometer — coil on a former between the poles of a permanent magnet, suspension and mirror [Lecture 17, p. 3]

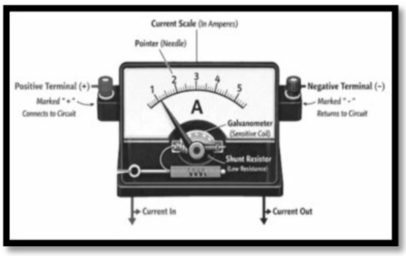

Fig. 14-5 · Ammeter = galvanometer + low-resistance shunt in parallel; the current scale reads in amperes [Lecture 17, p. 3]

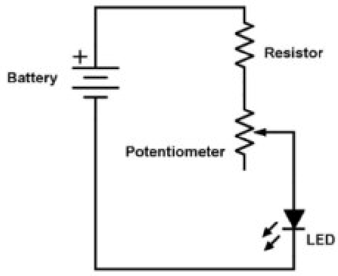

Fig. 14-6 · Potentiometer wired as a variable resistor (voltage divider) to control an LED — the source card [Lecture 17, p. 3]

Instrument	What it does / measures	Connection & construction
Galvanometer	**Detects** the presence and direction of a small current in a circuit	Moving coil in a magnetic field; very sensitive; zero at the centre in a "centre-zero" type
Ammeter	Measures the **amount of current** (amperes)	A galvanometer with a **low-resistance shunt in parallel**; connected in **series** with the circuit; ideal ammeter has **zero** resistance
Voltmeter	Measures the **potential difference** (volts) across a component	A galvanometer with a **high resistance in series**; connected in **parallel** across the component; ideal voltmeter has **infinite** resistance
Potentiometer	Highly sensitive instrument to **measure emf** / compare emfs and measure internal resistance (emf = electromotive force, the "voltage" of a source)	Long uniform wire + driver cell; null method — draws **no current** from the cell at balance, hence more accurate than a voltmeter
Voltameter	Measures the **charge** that has flowed (by the mass deposited in electrolysis)	Electrolytic cell (copper / silver voltameter) in series
Wattmeter	Measures **electrical power**; used in AC and DC circuits	Has **two coils** — a **current coil** (series) and a **potential (voltage) coil** (parallel)
Energy meter	Measures **electrical energy** consumed (kWh, "units")	Induction-disc (old) or digital; in series with the supply at the meter board
Ohmmeter / Multimeter	Resistance / voltage + current + resistance in one	Battery-powered; multimeter = "AVO" meter (amperes, volts, ohms)
Clamp meter	Current **without breaking the circuit**	Clamps round one conductor and senses its magnetic field
Oscilloscope (CRO)	Displays a voltage signal against time (waveform)	Cathode-ray tube or digital screen
Megger	**Insulation resistance** (megohms) of cables and windings	Hand-cranked or electronic high-voltage source

✎ **Source correction:** Lecture 17 expands "Emf" as "Electrometer force" — it is the **electromotive force**.

★ **PYQ lens:** 64th Q14 / 65th Q119 (identical): the device to measure electric current → **ammeter** (voltmeter — voltage; voltameter — charge; potentiometer — emf).

▌ADDED BY EDITOR — not in source lectures ▌ Converting a galvanometer (the classic reasoning question)

To make an **ammeter**, a very small resistance (**shunt**) is joined in **parallel** so most of the current bypasses the delicate coil — this also makes the ammeter's total resistance tiny, so inserting it in series does not disturb the circuit. To make a **voltmeter**, a large resistance (**multiplier**) is joined in **series** so that the meter draws negligible current when placed across a component. An ammeter wrongly connected in parallel burns out (short circuit); a voltmeter wrongly connected in series just reads the source voltage and the circuit stops working. **Range**: a shunt of $R_g/(n-1)$ extends the range n times.

Electrical devices and energy conversion (Lecture 17 p. 3)

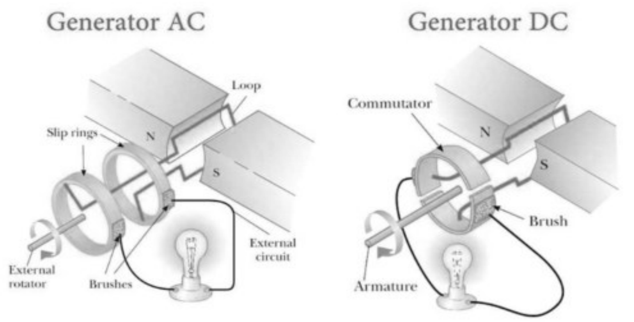

Fig. 14-7 · AC generator (slip rings) and DC generator (commutator) — a coil rotated in a magnetic field [Lecture 17, p. 3]

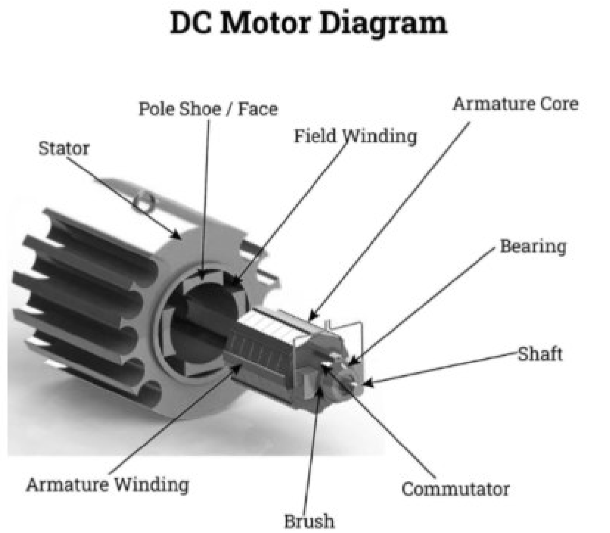

Fig. 14-8 · DC motor — stator with field winding, armature, commutator and brushes [Lecture 17, p. 3]

Dynamo / generator

- Converts **mechanical energy → electrical energy**; works on **electromagnetic induction** (Faraday, 1831 — a changing magnetic flux through a coil induces an emf).
- An **AC generator** uses **slip rings**; a **DC generator (dynamo)** uses a **split-ring commutator** (which reverses the connection every half turn so the output current always flows one way).

> ★ **PYQ lens:** 69th Q26: an AC current can be produced by a **dynamo** (a choke coil only limits AC; a transformer changes the voltage of an *existing* AC but does not generate it). 64th Q12: in an electric motor, **electrical energy is converted into mechanical energy**.

Electric motor

- Converts **electrical energy → mechanical energy** (motion); works on the **magnetic effect of current** — a current-carrying coil in a magnetic field experiences a torque (Fleming's left-hand rule, Unit 15). Used in fans, pumps, mixers, washing machines, electric vehicles, drills, lifts.
- **Low-voltage motors tend to burn out** because at reduced voltage they draw more current to deliver the same power ($P = VI \Rightarrow I = P/V$), overheating the windings (I^2R).

> ✎ **Source correction:** Lecture 17 says the motor "also works on electromagnetic induction". A motor works on the *magnetic (motor) effect* — force on a current-carrying conductor in a magnetic field; **electromagnetic induction** is the generator/transformer principle (a running motor does develop a *back emf* by induction, which is presumably what was meant).

Energy-conversion efficiency

- In an *ideal* motor converting electrical energy directly to motion there is no heat loss; **real** machines always lose some energy as heat (I^2R in the windings), friction and eddy currents — efficiencies are typically 75–95 % for motors, up to 99 % for large transformers.

Batteries (Lecture 17 p. 4)

- **General principle**: a cell converts stored **chemical energy → electrical energy**; a **battery** is a group of cells. **Components**: a positive electrode (cathode on discharge), a negative electrode (anode) and an **electrolyte**.

Feature	Lead-acid battery (car battery)	Nickel-cadmium (Ni-Cd) battery
Type	Secondary (rechargeable) — the "accumulator"	Secondary (rechargeable)
Voltage	**12 V** from **six 2 V cells** in series	1.2 V per cell

Feature	Lead-acid battery (car battery)	Nickel-cadmium (Ni-Cd) battery
Electrodes	**Lead** (–) and **lead dioxide, PbO_2** (+)	**Nickel oxide-hydroxide** (+) and **metallic cadmium** (–)
Electrolyte	**Dilute sulphuric acid** (≈ 35 % acid, 65 % water; relative density ≈ 1.25 when charged)	Alkaline **potassium hydroxide**
Capacity / uses	Expressed in **ampere-hours (Ah)**; cars, inverters, UPS	Calculators, cordless appliances, transistors, portable tools, emergency lights

★ **PYQ lens:** 63rd Q99: the acid used in a car battery → **sulphuric acid** (H_2SO_4).

▌ADDED BY EDITOR — not in source lectures ▌ Primary vs secondary cells and the cells around us (context)

Primary cells (not rechargeable): Volta's cell (1800 — first battery), Daniell cell (Zn/Cu, 1.1 V), Leclanché / **dry cell** (Zn can (–), carbon rod (+), NH_4Cl paste + MnO_2 depolariser, **1.5 V**), alkaline cell (1.5 V, longer life), **mercury and silver-oxide button cells** (watches, hearing aids), lithium coin cell (3 V). **Secondary cells**: lead–acid, Ni–Cd, Ni–MH, and **lithium-ion** (3.6–3.7 V per cell; mobiles, laptops, EVs — 2019 Nobel: Goodenough, Whittingham, Yoshino). **Fuel cell** — converts H_2 + O_2 continuously into electricity and water (spacecraft, buses). **Solar cell** — light → electricity. A battery's **Ah** rating × voltage = energy in Wh (a 12 V 100 Ah battery holds 1.2 kWh). Discharged lead–acid batteries show a fall in electrolyte density (checked with a hydrometer). **Inverter** batteries are deep-cycle lead–acid or tubular.

Transformers and AC/DC conversion (Lecture 17 p. 4)

Transformer

- **Purpose**: steps **up** or steps **down** an **AC voltage** (a step-up transformer increases voltage and decreases current; a step-down does the reverse — power is nearly conserved, $V_pI_p \approx V_sI_s$).
- **Principle**: **electromagnetic induction** (mutual induction between the primary and secondary coils wound on a common soft-iron core); it works **only with AC, not with DC** (DC gives no changing flux, so no induced emf).
- **Application**: a **mobile-phone charger** is a step-down transformer (e.g. 220 V AC → 9 V, then rectified to DC); power grids step up to 132–765 kV for transmission (to cut I^2R losses) and step down to 230 V for homes.

$$V_s/V_p = N_s/N_p \text{ (turns ratio; } N_s > N_p \rightarrow \text{step-up)}$$

AC/DC conversion

- **Rectifier**: converts **AC → DC** (needed for charging batteries and running all electronics; uses p–n junction diodes).
- **Inverter**: converts **DC → AC** (home inverters run mains appliances from a battery; solar inverters).
- **Frequency in India**: the AC mains operates at **50 Hz** (230 V single-phase; 415 V three-phase); USA/Canada use 60 Hz, 120 V.

▌ADDED BY EDITOR — not in source lectures ▌ AC vs DC — the facts behind the options (context)

AC reverses direction 100 times a second at 50 Hz (50 complete cycles); its **peak** value is √2 × the stated **rms** value (230 V rms ≈ 325 V peak). AC won the "war of the currents" (Tesla/Westinghouse vs Edison) because it can be transformed to high voltage for cheap long-distance transmission

DC is used in batteries, electronics, electroplating, electric trains (25 kV AC overhead lines feed DC traction motors) and now **HVDC** links for very long lines. A **choke coil** limits AC without wasting power as heat (tube-light ballast). **Nikola Tesla** — AC induction motor

Michael Faraday — induction, dynamo, laws of electrolysis

Thomas Edison — practical filament lamp (1879), DC supply.

Lighting technologies (Lecture 17 pp. 4–5)

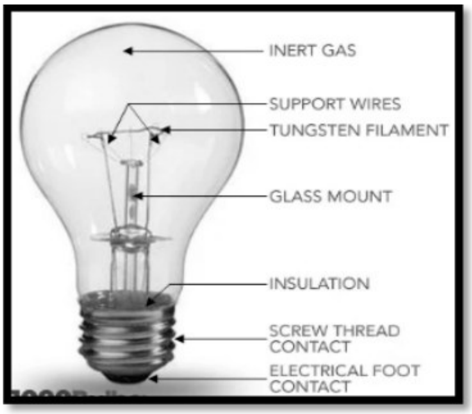

Fig. 14-9 · Incandescent bulb — inert gas, support wires, tungsten filament, glass mount, insulation, screw-thread and foot contact [Lecture 17, p. 3]

Incandescent (filament) bulbs

- **Filament**: made of **tungsten** (highest melting point of any metal, **3422 °C**); **operating temperature 2000–2500 °C**.
- **Gas filling**: inert gases such as **nitrogen or argon** (sometimes krypton) prevent oxidation and slow the evaporation of the filament; the earliest bulbs were evacuated.
- **Reasons for the short lifespan**: the filament wire is not perfectly uniform (thin spots overheat and break); the bulb cannot be completely evacuated of air; the supporting wires can melt.
- **Very inefficient**: only **5–10 %** of the energy is converted to light — the rest is heat (hence the phase-out in favour of LEDs).

Halogen lamps

- The filament works at a higher temperature inside a small quartz envelope containing a halogen (iodine/bromine) that returns evaporated tungsten to the filament — whiter light and longer life; used in car headlamps, projectors and floodlights.

✎ **Source correction:** Lecture 17 says the halogen-lamp filament "is an alloy of tungsten and sodium". The filament is **tungsten**; the *gas* contains a **halogen** (iodine or bromine). Sodium is the metal in *sodium-vapour* street lamps.

Fluorescent lamps (tube lights and CFLs)

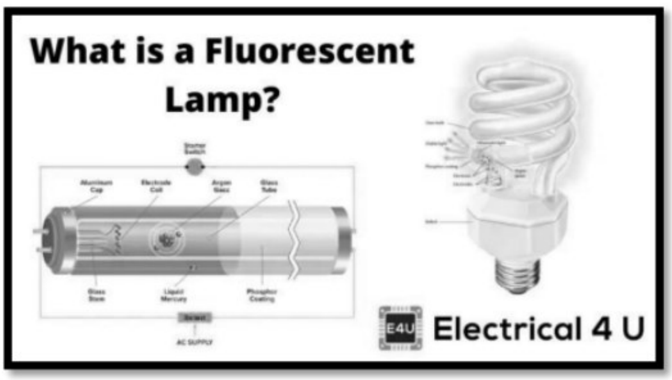

Fig. 14-10 · Fluorescent lamp / CFL — electrodes, mercury vapour and inert gas inside a phosphor-coated tube [Lecture 17, p. 4]

- **CFL** = **Compact Fluorescent Lamp**. **Gas**: **mercury vapour** plus inert gases (**neon, argon**).
- **Principle**: accelerated electrons excite mercury atoms, which emit **ultraviolet** radiation; a **phosphor coating** on the inside of the glass **fluoresces** and converts the UV into visible white light.

- **Component**: a **choke (inductor/ballast)** starts the lamp (with the starter) and then limits the current; electronic ballasts do the same in CFLs.
- **Efficiency**: about **22 %** of the electrical energy becomes light (≈ 4× a filament bulb); life 6 000–10 000 h; contain mercury, so must be disposed of carefully.

↻ **De-dup note:** The heading "Fluorescent Lamps" is printed twice on Lecture 17 p. 4; merged.

Light-emitting diodes (LEDs)

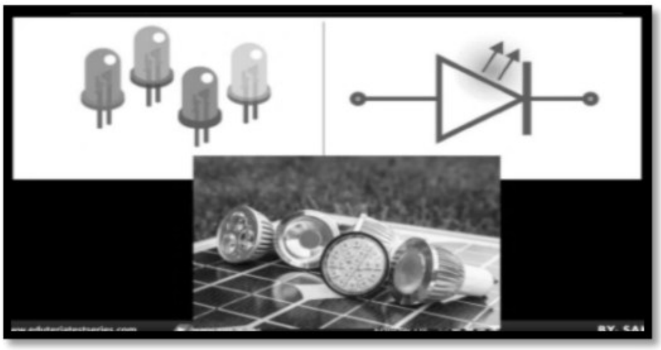

Fig. 14-11 · LEDs — the diode symbol and lamps; the solar-panel/photodiode panel of the source card [Lecture 17, p. 5]

- **Principle**: **electroluminescence** — light is emitted when electrons recombine with holes in a forward-biased **semiconductor** p–n junction (gallium arsenide, gallium phosphide, gallium nitride for blue/white — 2014 Nobel Prize: Akasaki, Amano, Nakamura).
- **Advantages over CFLs**: more energy-efficient (≈ 40–50 % of energy to light; 80–100 lumen/W); **much longer lifespan (~50 000 h vs 6 000–10 000 h for CFLs)**; no mercury; instant start; rugged; work on low DC voltage.
- **Advantages over sodium-vapour lamps (street lighting)**: **directional** light emission (~180°) so less light is wasted; longer lifespan; **better colour rendering** (white, not monochromatic yellow); dimmable.

✎ **Source correction:** Lecture 17 prints "Advantages of CFLs" as a sub-heading *under* LEDs, followed by "More energy efficient; longer lifespan (~50 000 hours vs CFL's 6 000–10 000 hours)". The bullets describe the advantages of **LEDs over CFLs**; the heading is a slip.

Sodium-vapour lamps

- Produce light in **all directions (360°)**; emit an almost **monochromatic yellow light** (589 nm sodium D-lines) — very efficient and penetrates fog well, hence the classic orange street light and highway lamp; colours of objects cannot be judged under it.

▌ADDED BY EDITOR — not in source lectures ▌ Other lamps in one line each (not in the lecture)

Neon sign — orange-red glow discharge in neon (other colours use other gases or phosphors); **mercury-vapour lamp** — bluish-white, used in factories and for UV sterilisation; **metal-halide** — stadium floodlights; **arc lamp** — carbon arc, cinema projectors, welding; **laser** — coherent, monochromatic, unidirectional light (Unit 4). Efficiency ranking: LED > sodium ≈ metal-halide > fluorescent/CFL > halogen > incandescent. India's UJALA scheme (2015) distributed LED bulbs to cut lighting load.

Safety devices and domestic electrical systems (Lecture 17 p. 5)

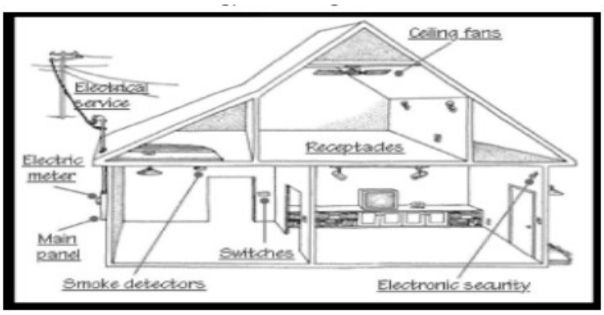

Fig. 14-12 · A house wiring layout — service entrance, electric meter, main panel, switches, receptacles (sockets), ceiling fans, smoke detectors and security system [Lecture 17, p. 5]

Safety devices

- **Fuse** — principle: heating effect; function: melts and breaks the circuit during over-current; characteristics: **high resistance, low melting point**, made of tin, lead or zinc alloys (details above).
- **Earthing (grounding)** — a thick copper wire connects the metal body of an appliance to a copper plate buried deep in moist earth; it provides a **safe, low-resistance path for fault (leakage) current to flow into the ground** so that a person touching the body is not shocked and the fuse blows. In a **three-pin plug the longest and thickest pin is the earth** terminal — it **connects first and disconnects last** (and its length opens the shutters of the socket).
- **Lightning conductor** — a pointed **metal rod** fixed at the top of a tall building and connected by a thick copper strip to a plate buried in the earth; it safely conducts the lightning's electric energy to the ground (and, by discharging the cloud through its sharp points, often prevents the strike altogether — Benjamin Franklin, 1752).

Domestic electrical systems

- **Wiring configuration**: basically a **parallel connection** — every appliance gets the same 230 V, can be switched independently, and a fault in one does not stop the others; the total current adds up (hence separate 5 A lighting and 15 A power circuits).
- **Wire colour coding (old Indian three-wire system)**: **Live** wire — **red** insulation; **Neutral** — **black**; **Earth** — **green**. (New IEC code used in modern wiring: live **brown**, neutral **blue**, earth **green-and-yellow**.)
- The **live** wire is at 230 V (rms) with respect to earth and carries the current in; the **neutral** is at ≈ 0 V (earthed at the substation) and returns it; switches and fuses are always fitted in the **live** wire so that an appliance is dead when switched off.
- **Power consumption of appliances (ascending order)**: **Fan < Television < Electric iron < Electric kettle** (≈ 75 W < 100 W < 1000 W < 2000 W).
- **Current flow**: current does not flow between two charged bodies at the same electric potential — a potential difference is required.

↻ **De-dup note:** The Faraday-constant paragraph printed here in Lecture 17 p. 5 has been merged into the chemical-effect section above; the "Current flow" line is also kept in Unit 13 (potential difference).

▌ADDED BY EDITOR — not in source lectures ▌ Electrical safety — why each rule works (not in the lecture)

Never touch a switch or appliance with wet hands (water with dissolved salts conducts, and wet skin has far lower resistance — dry skin ≈ 10^5 Ω, wet ≈ 10^3 Ω); wear rubber-soled shoes / stand on a wooden stool when working on wiring (insulation from earth); use a **tester** (neon lamp in series with a high resistance) to identify the live wire; switch off the mains before rescuing a shock victim, or push the person away with a dry wooden stick; a bird on one wire is safe, but touching two wires (or a wire and the pole) completes a circuit; in a lightning storm stay away from tall trees and out of open fields and water, and stay inside a car (Faraday cage). **Three-pin plugs** are used for high-power appliances with metal bodies (geyser, iron, fridge, washing machine); the thin two-pin type is enough for double-insulated plastic-bodied devices. The **Indian standard supply** is 230 V ± 10 %, 50 Hz; the "unit" price on the bill is per kWh.

UNIT 15

Magnetism & Electromagnetism

Consolidated from: Lecture 18 (Magnetism) pp. 2–4

★ **PYQ lens:** **Priority B — 3 questions in three of the last five papers (67th, 69th, 70th).** Paramagnetism applies to **platinum** among iron / platinum / nickel / mercury (67th Q74); AC is produced by a **dynamo** (69th Q26 — the device itself is described in Unit 14); **tesla = newton per ampere-metre** (70th Q123). Learn the three-material table and the unit of B.

Magnetism and the two kinds of current

Magnetism is a physical phenomenon produced by the motion of electric charges, resulting in attractive and repulsive forces between objects. (All magnetism — even that of a bar magnet — comes from moving charges: orbiting and spinning electrons.)

Types of current (Lecture 18 p. 2)

Electric current is classified into two types by the direction of charge flow: **direct current (DC)**, which flows in a single constant direction, and **alternating current (AC)**, which reverses direction periodically.

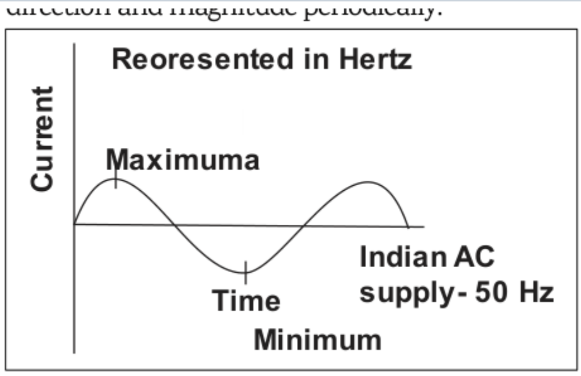

Fig. 15-1 · Alternating current — magnitude and direction change periodically; the Indian supply completes 50 cycles per second (50 Hz) [Lecture 18, p. 2]

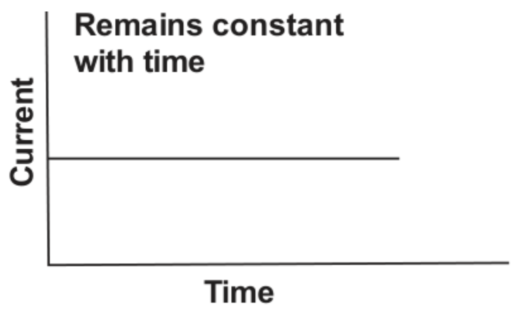

Fig. 15-2 · Direct current — constant magnitude and one direction with time [Lecture 18, p. 2]

- **Alternating current (AC)** — changes its direction and magnitude periodically (a sine wave with maxima and minima); frequency in hertz; **Indian AC supply: 50 Hz**, 230 V. Produced by AC generators (alternators); transmitted by the grid; easily transformed.
- **Direct current (DC)** — flows in only one direction with constant polarity; remains constant with time. Sources: cells and batteries, DC generators (dynamos), solar cells, rectifiers.

↻ **De-dup note:** AC/DC appear here in Lecture 18 and again (generator types, rectifier/inverter, 50 Hz) in Lecture 17 — the devices are kept in Unit 14, the definitions and graphs here.

Electromagnetism and the magnetic field

- **Electromagnetism** — the branch of physics that deals with the magnetism due to electric current or moving charge (an electric current is equivalent to charges/electrons in motion).
- **Magnetic field** — a region of space near a magnet, electric current or moving charged particle in which magnetic effects are exerted on any other magnet, current or moving charge. The field vector **B** is also called **magnetic flux density** or **magnetic induction**.

SI unit of B: tesla (T) = weber/metre² (Wb/m²) = newton per ampere-metre (N A^{-1} m^{-1}) · CGS unit: gauss (G); 1 tesla = 10^4 gauss

★ **PYQ lens:** 70th Q123: which is equivalent to tesla → **newton per ampere-metre** (from F = BIL ⇒ B = F/IL). Newton per coulomb is the unit of *electric* field; N/(A·s) = N/C.

▌ADDED BY EDITOR — not in source lectures ▌ Magnetic flux, and how strong fields are (in the Lecture-1 plan; not taught)

Magnetic flux $\Phi = B A \cos\theta$ — the number of field lines through an area; unit **weber (Wb)**

1 T = 1 Wb/m². **Typical fields**: Earth's surface ≈ 0.3–0.6 G (≈ 5×10^{-5} T), fridge magnet ≈ 0.01 T, strong bar magnet ≈ 0.1–1 T, MRI scanner 1.5–3 T, laboratory superconducting magnets ≈ 20–45 T, neutron star ≈ 10^8 T. **Magnetic permeability μ** = how easily a material lets flux pass ($\mu_0 = 4\pi \times 10^{-7}$ T m/A for vacuum; relative permeability $\mu_r \approx 1$ for air, thousands for soft iron). **Magnetic susceptibility χ** = M/H — negative for diamagnets, small positive for paramagnets, large positive for ferromagnets (the table below).

Oersted's discovery (Lecture 18 pp. 2–3)

- The relation between electricity and magnetism was discovered by **Hans Christian Oersted in 1820** — he found a magnetic field around a conductor carrying electric current.
- (1) An electric current through a conducting wire **deflects a magnetic needle** (compass) held near the wire.
- (2) When the **direction of the current is reversed**, the deflection of the needle is **also reversed**.
- (3) On **increasing the current**, or bringing the needle **closer** to the conductor, the deflection **increases** (field ∝ I, ∝ 1/distance for a straight wire).
- Related facts: a magnet at rest produces a magnetic field around it; an electric charge **at rest** produces only an **electric** field; a charge moving with **uniform velocity** (a steady current) has an **electric as well as a magnetic** field around it; an **oscillating or accelerated charge** produces **electromagnetic waves** in addition to the electric and magnetic fields (Unit 5).

↻ **De-dup note:** Lecture 18 narrates Oersted's discovery twice on pp. 2–3 ("Oersted showed that..." and again "Oersted discovered a magnetic field around a conductor... Other related facts are as follows") with the same "related facts" list printed in two garbled halves; consolidated into the numbered list above.

▌ADDED BY EDITOR — not in source lectures ▌ Direction rules and the fields of simple conductors (needed to use Oersted's result)

Right-hand thumb rule (Maxwell's cork-screw rule): grip the straight wire with the right hand, thumb along the current — the curled fingers give the direction of the circular field lines. **Straight wire**: concentric circles, $B = \mu_0 I/(2\pi r)$. **Circular loop**: field at the centre $B = \mu_0 I/(2r)$, the face where current is anticlockwise behaves as a north pole (**clock rule**). **Solenoid**: a long coil — its field is like that of a bar magnet, uniform inside, $B = \mu_0 n I$ (n = turns per metre); with a **soft-iron core** it becomes an **electromagnet**. **Ampère** (1820) gave the force between two parallel currents — like currents attract, unlike repel — which is how the ampere was defined (Unit 1). **Biot–Savart law** gives the field of any current element.

Magnetic field lines and magnets

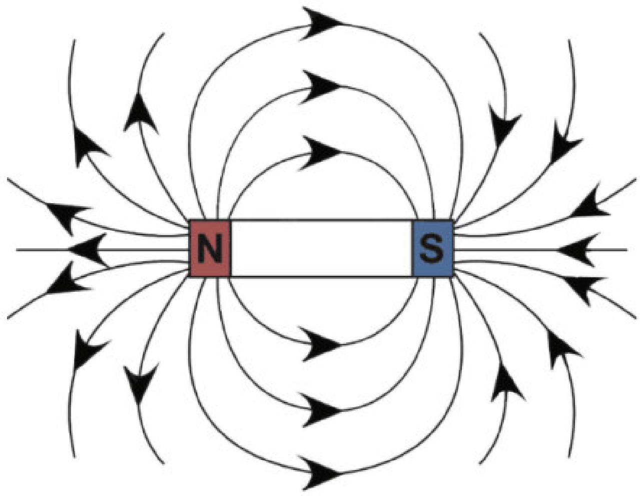

Fig. 15-3 · Field lines of a bar magnet — emerging from N, entering S, forming closed loops through the magnet [Lecture 18, p. 2]

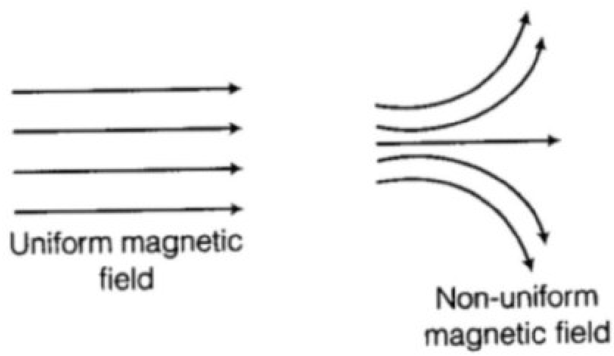

Fig. 15-4 · Uniform (parallel, equally spaced) and non-uniform magnetic fields [Lecture 18, p. 3]

- **Magnetic field lines** are imaginary lines that represent the direction and strength of the magnetic field around a magnet: they **emerge from the north pole and enter the south pole** outside the magnet and **form closed loops** inside it (S → N).
- Properties: the tangent at any point gives the direction of B; the **density (closeness) of lines gives the strength**; lines **never intersect** (the field cannot have two directions at a point); they are crowded at the poles.

Types of magnets (Lecture 18 pp. 3–4)

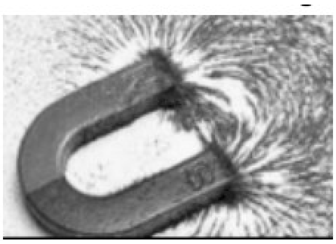

Fig. 15-5 · Horseshoe (U-shaped) magnet — iron filings map the strong field between the poles [Lecture 18, p. 4]

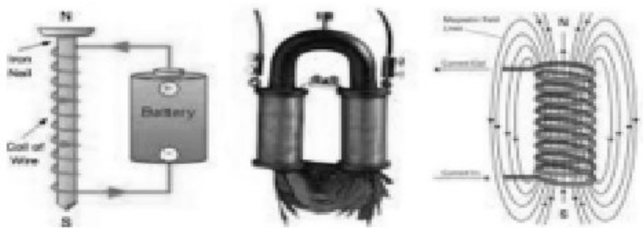

Fig. 15-6 · Electromagnets — nail wound with wire and a battery, a U-core electromagnet, and the field of a current-carrying coil (solenoid) [Lecture 18, p. 4]

- **Bar magnet** — the common laboratory magnet; a **permanent** magnet, typically **ferromagnetic**; a freely suspended bar magnet points **north–south** (the compass principle); like poles repel, unlike attract; **poles always exist in pairs** — cutting a magnet gives two smaller magnets (no magnetic monopole).
- **Horseshoe magnet** — a U-shaped permanent magnet designed to produce a **strong field between its poles** (the poles are close together); ferromagnetic material; used in loudspeakers, moving-coil meters, magnetos.
- **Electromagnet** — a coil of wire wound around a **ferromagnetic (soft-iron) core**; its magnetism is **temporary** and exists **only while current flows**; its strength increases with the current, the number of turns and the core. Uses: electric bell, relay, magnetic crane (scrap yards), telephone, MRI, maglev, separating iron from ore.

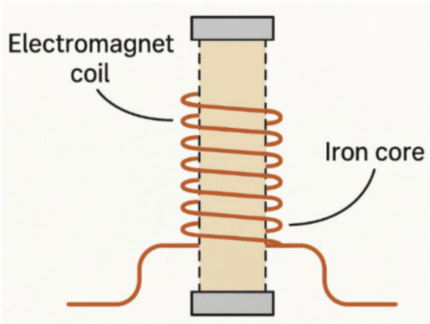

Fig. 15-7 · Electromagnet coil wound on an iron core [Lecture 18, p. 2]

▌ADDED BY EDITOR — not in source lectures ▌ Permanent vs temporary magnets — materials (context)

Permanent magnets need a material with high **retentivity** (retains magnetism) and high **coercivity** (hard to demagnetise): **steel**, alnico (Al–Ni–Co), ferrites, and the strongest — **neodymium (Nd–Fe–B)** and samarium–cobalt. **Temporary magnets / electromagnet cores** need **soft iron** — high permeability, low retentivity, so they magnetise strongly and lose it instantly when the current stops (also used in transformer cores and motor armatures, as laminated sheets to cut eddy currents). **Lodestone (magnetite, Fe_3O_4)** is the natural magnet, known to the Greeks of *Magnesia*. A magnet is **demagnetised** by heating above the Curie point, hammering, or an AC field of decreasing amplitude. **Magnetic keepers** (soft-iron bars) preserve bar magnets in storage.

Earth's magnetism (Lecture 18 p. 3)

- The **Earth behaves like a giant magnet** and its magnetic field is spread all around it (out to the magnetosphere, which shields us from the solar wind and produces the auroras).

▌ADDED BY EDITOR — not in source lectures ▌ Geomagnetism — the details the lecture skips (in the Lecture-1 plan)

The field is like that of a huge bar magnet **tilted ≈ 11° to the rotation axis**, whose *magnetic south* pole lies near the **geographic North** (in the Canadian Arctic) — that is why the compass's north pole points north. The field is generated by convection of molten iron in the outer core (**dynamo theory**), not by a solid magnet (the core is far above the Curie point). **Elements of the Earth's field**: (1) **declination** — angle between geographic and magnetic meridians

(2) **dip / inclination** — angle the field makes with the horizontal (0° at the magnetic equator, 90° at the magnetic poles — a dip needle stands vertical there)

(3) **horizontal component B_H** (≈ 0.3–0.4 G in India). The **magnetic equator (aclinic line)** passes through Thumba (Kerala) — used for sounding-rocket launches. The poles **wander** and the field **reverses** every few 10^5 years (recorded in sea-floor rocks). Uses: **mariner's compass** (Chinese, ~11th century), navigation of migratory birds and turtles, magnetic surveys for ore. **Neutral points**: where the field of a magnet cancels the Earth's B_H.

Curie temperature

- The temperature at which a **ferromagnetic** substance changes into a **paramagnetic** substance (thermal agitation destroys the alignment of the domains): iron ≈ 770 °C (1043 K), nickel ≈ 358 °C, cobalt ≈ 1121 °C, gadolinium ≈ 20 °C (just below room temperature). A red-hot iron nail is not attracted by a magnet. Named after Pierre Curie.

Classification of magnetic materials (Lecture 18 p. 3)

Materials are classified into three types according to their interaction with a magnetic field: **diamagnetic** (weakly repelled), **paramagnetic** (weakly attracted) and **ferromagnetic** (strongly attracted and able to retain magnetism).

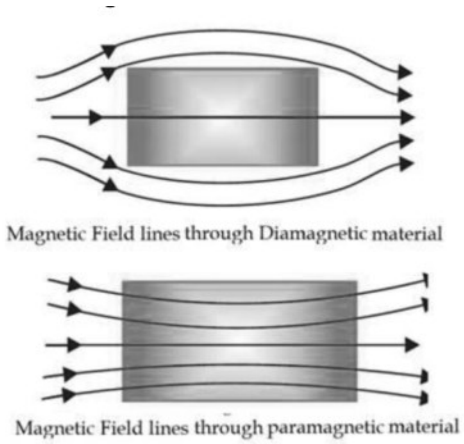

Fig. 15-8 · Field lines are pushed out of a diamagnetic material and drawn (slightly) into a paramagnetic one [Lecture 18, p. 3]

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Effect of a magnetic field	Weakly magnetised in the direction **opposite** to the field	Weakly magnetised **in the direction** of the field	**Strongly** magnetised in the direction of the field
Behaviour toward a magnet	**Weakly repelled** (moves from stronger to weaker field; sets perpendicular to the field)	**Weakly attracted** (moves toward the stronger field; sets parallel)	**Strongly attracted**
Magnetic susceptibility (χ)	Small and **negative** ($\chi < 0$; e.g. -10^{-5})	Small and **positive** ($\chi > 0$; $\approx 10^{-5}$ – 10^{-3})	Very large and positive ($\chi \gg 0$; 10^{2} – 10^{5})
Relative permeability μ_r	Slightly less than 1	Slightly more than 1	Very much greater than 1 (soft iron $\approx 10^{3}$–10^{5})
Effect of temperature	Independent of temperature	$\chi \propto 1/T$ (Curie's law) — weaker when hot	Becomes paramagnetic above the **Curie temperature**
Examples	**Bismuth** (strongest), copper, gold, silver, **water**, salt (NaCl), zinc, lead, mercury, hydrogen, nitrogen, diamond, most organic materials	**Manganese, aluminium, platinum**, sodium, chromium, **oxygen (O_2)** (liquid oxygen clings to magnet poles), rare-earth salts	**Iron, nickel, cobalt, steel, magnetite (Fe_3O_4)**, gadolinium, alnico, ferrites

★ **PYQ lens:** 67th Q74: the paramagnetic theory of magnetism applies to → **platinum** (iron and nickel are ferromagnetic; mercury is diamagnetic). Same table, other angles: "which is diamagnetic?" → bismuth/copper/water; "strongly attracted" → iron/nickel/cobalt; "repelled by a magnet" → diamagnetic.

▌ADDED BY EDITOR — not in source lectures ▌ Why materials behave this way, and superconductors (context)

In **diamagnets** the atoms have no net magnetic moment; the applied field induces tiny opposing currents (Lenz's law) — present in *all* materials but masked in the other two classes; a frog has been levitated in a 16 T field by its diamagnetism, and a **superconductor is a perfect diamagnet** ($\chi = -1$, Meissner effect) — the basis of magnetic levitation demonstrations. **Paramagnets** have permanent atomic moments (unpaired electrons) that are partly aligned by the field against thermal disorder. **Ferromagnets** have moments already aligned within microscopic **domains**; the field grows the favourably-oriented domains — hence the large effect, saturation and **hysteresis** (retentivity, coercivity; the B–H loop). **Antiferromagnets** (MnO, Cr) and **ferrimagnets** (ferrites — magnetite, used in transformer cores, magnetic tape, microwave devices) are the other two classes.

Fleming's rules (Lecture 18 p. 4)

Fleming's rules — the **left-hand rule for motors**, the **right-hand rule for generators** — give the mutually perpendicular directions of force (motion), magnetic field and current in electromagnetic machines. Hold the **thumb, forefinger and middle finger perpendicular** to each other; the mnemonic **F-B-I** (Force/motion, B-field, Induced/Injected current) fixes the assignment thumb–forefinger–middle finger.

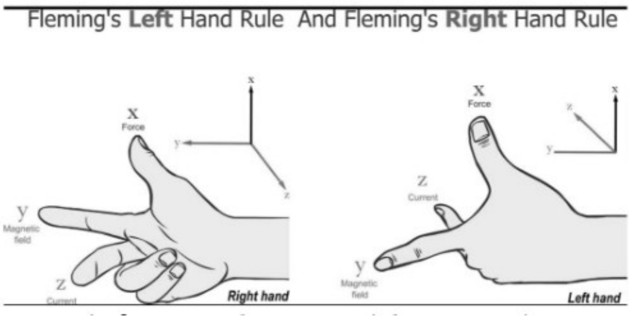

Fig. 15-9 · Fleming's left-hand rule (motor) and right-hand rule (generator) — thumb = force/motion, forefinger = field, middle finger = current [Lecture 18, p. 4]

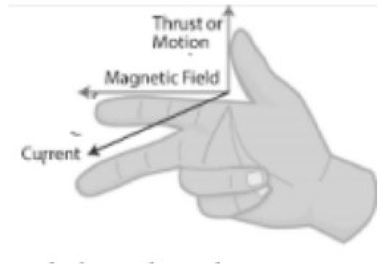

Fig. 15-10 · Right-hand rule as printed in the lecture — thumb: thrust/motion, forefinger: magnetic field, middle finger: current [Lecture 18, p. 4]

Finger	Fleming's left-hand rule (electric motor)	Fleming's right-hand rule (generator / dynamo)
Gives	Direction of the **magnetic force** on a current-carrying conductor placed in a field	Direction of the **induced current** when a conductor is moved in a field
Thumb	Direction of **force / motion** of the conductor	Direction of **motion** of the conductor
Forefinger	Direction of the **magnetic field** (N → S)	Direction of the **magnetic field** (N → S)
Middle finger	Direction of the **current** in the conductor	Direction of the **induced current**
Used in	Motors, loudspeakers, moving-coil galvanometer/ammeter/voltmeter, force between parallel wires	Generators, dynamos, induction cooktops, microphones, bicycle dynamo

✎ **Source correction:** Lecture 18 lists the middle finger of the *left-hand* rule as "direction of **induced** current" — in the motor rule the middle finger is simply the direction of the (applied) current; *induced* current belongs to the right-hand (generator) rule. Also, the heading "Fleming right hand thumb Pule" reads "Rule".

↻ **De-dup note:** Lecture 18 p. 4 gives the rules three times — the FBI paragraph, "Fleming left hand thumb Rule" list and "Fleming right hand thumb Rule" list. Merged into the table.

Magnetic force on a moving charge and on a current-carrying wire (Lecture 18 p. 4)

Force on a charge: $F = q v B \sin\theta$ (vector form $F = q (v \times B)$) q = charge, v = velocity, B = magnetic field, θ = angle between v and B

- The force is **maximum when the charge moves perpendicular** to the field (θ = 90°) and **zero when it moves parallel** to the field (θ = 0° or 180°) or is at rest (v = 0). It is always **perpendicular to both v and B**, so it does **no work** — it changes the *direction* of motion, not the speed (the charge describes a circle of radius $r = mv/qB$; this is the principle of the cyclotron, mass spectrometer and the trapping of cosmic-ray particles in the Van Allen belts). This is the **Lorentz force** (with the electric part, $F = qE + q v \times B$).

Force on a current-carrying wire: $F = B I L \sin\theta$ B = magnetic field, I = current, L = length of the wire in the field, θ = angle between the current and the field

- Maximum ($F = BIL$) when the wire is perpendicular to the field, zero when parallel. Direction by **Fleming's left-hand rule**. Two parallel wires with currents in the same direction **attract**, in opposite directions **repel**. This force drives every **electric motor** and defines the **tesla** (70th Q123) and the **ampere** (Unit 1).

✎ **Source correction:** In the wire formula Lecture 18 labels I as "induced current"; here I is the current *flowing in* the wire (the force is on a current-carrying conductor), not an induced one.

▌ADDED BY EDITOR — not in source lectures ▌ Electromagnetic induction — Faraday, Lenz and the devices (in the Lecture-1 plan; announced but never taught, yet it underlies the dynamo of 69th Q26 and the transformer of Unit 14)

Faraday's laws (1831): whenever the magnetic flux linked with a closed coil **changes**, an emf is induced in it (and a current if the circuit is closed); the emf is proportional to the **rate of change of flux**: $\varepsilon = -N\, d\Phi/dt$. Flux can be changed by moving a magnet in or out of a coil, moving the coil, rotating a coil in a field (generator), or switching the current in a neighbouring coil (transformer, induction coil). **Lenz's law**: the induced current opposes the change that produced it (energy conservation) — the minus sign; a magnet dropped through a copper tube falls slowly; a coin is "braked" in a field. **Motional emf** $\varepsilon = B\, l\, v$ for a rod of length l moving at speed v perpendicular to B. **Applications**: AC/DC generators, transformers, induction motors (Tesla), induction cooktops (eddy currents heat the pan), electromagnetic brakes on trains and roller-coasters, metal detectors, magnetic-stripe card readers, dynamo torches, wireless phone charging, microphones and guitar pick-ups, speedometer (eddy-current type), induction furnace, and the *back emf* of a running motor. **Eddy currents** are induced in bulk metal — useful in the above, wasteful in transformer cores (hence laminated cores). **Self-inductance** L (unit **henry, H**) opposes changes of current in a coil — the *choke*; **mutual inductance** links two coils (transformer). Faraday, Henry (self-induction), Lenz, Maxwell (unified electromagnetism, predicted EM waves — Unit 5).

UNIT 16

Modern Physics — Photoelectric Effect, Matter Waves, Radioactivity, Nuclear Fission & Fusion, Thermionic Emission

Consolidated from: Lecture 19 pp. 4–6 (Photoelectric effect, photon, wave–particle duality, de Broglie, atomic nucleus & decay law, fission/fusion, nuclear reactor, thermionic emission)

★ **PYQ lens: Priority A — the single most examined block: 19 questions in 10 of the 11 papers.** Nagasaki bomb = **plutonium** (56–59th Q138); cosmic rays are charged as well as uncharged (Q140); ionising radiation both causes and cures cancer (Q142); "God particle" = **Higgs boson** (60–62nd Q49); radio-isotope for leukaemia (Q50); LIGO-India in Maharashtra (Q71); nucleus = positively charged centre (63rd Q89); hydrogen has no neutron (64th Q2); Einstein's Nobel = **photoelectric effect** (64th Q16); mass number = 18 + 20 = 38 (65th Q92); neutrons in ${}_{94}Pu^{242}$ = **148** (66th Q9); theory of relativity — Einstein (67th Q71); nucleus = protons + neutrons (67th Q75); fusion breakthrough Dec 2022 = **Lawrence Livermore** (68th Q75); **photodiode** for digital applications (68th Q113); positively charged emission = **alpha** (68th Q116); Manhattan Project (69th Q14); photoelectric cell converts light → electricity (69th Q16); first atomic model — **J. J. Thomson** (70th Q150). Expect 1–2 questions every year.

Photoelectric effect (Lecture 19 p. 4)

Photoelectric effect — the emission of electrons from a metal surface when electromagnetic radiation of **sufficient frequency** falls on it. The emitted electrons are **photoelectrons**. (Discovered by **Heinrich Hertz** in 1887, investigated by Hallwachs and Lenard, explained by **Einstein in 1905** — his 1921 Nobel Prize.)

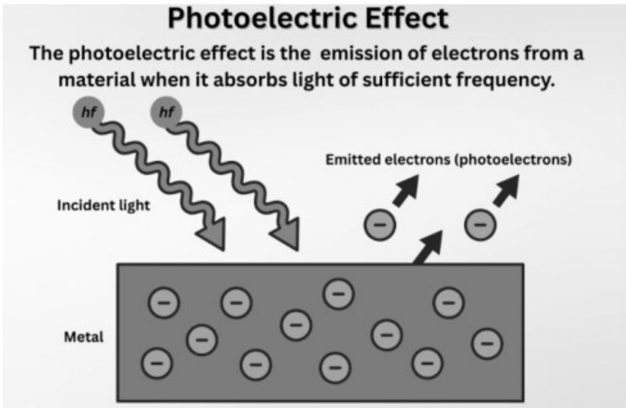

Fig. 16-1 · Photoelectric effect — photons of energy hf strike the metal and electrons are emitted [Lecture 19, p. 4]

Experimental observations

- **Threshold frequency (ν_0)** — below ν_0 **no photoemission occurs**, however intense the light (red light on sodium does nothing; faint violet/UV light ejects electrons at once). Every metal has its own ν_0 (alkali metals — Cs, K, Na — respond to visible light; zinc, copper need UV).
- **Instantaneous emission** — there is **no time lag** ($< 10^{-9}$ s) between the incidence of light and the emission of electrons, even for very weak light.
- The **kinetic energy** of the photoelectrons depends on the **frequency** of the light, **not on its intensity**.
- The **photoelectric current** (number of electrons per second) depends on the **intensity**, **not on the frequency**.
- (Wave theory could explain none of the first three — hence the need for photons.)

Einstein's photoelectric equation

$$h\nu = \varphi + K_{max} \text{ i.e. } K_{max} = h\nu - h\nu_0 \cdot \varphi = h\nu_0 = \text{work function} \cdot K_{max} = \tfrac{1}{2} m v_{max}^2 = eV_0 \ (V_0 = \text{stopping potential})$$

- **hν** = energy of the incident photon; **φ = hν$_0$** = **work function** of the metal — the minimum energy needed to pull an electron out of the surface (a few eV: Cs 2.1 eV, Na 2.3, Zn 4.3, Pt 5.6); the surplus appears as the electron's maximum kinetic energy.
- **Significance**: confirms the **particle nature of light** — light is absorbed in discrete quanta (photons), each of energy **E = hν**, one photon ejecting one electron; a graph of K_{max} against ν is a straight line of slope **h** (Millikan verified this and measured Planck's constant).

✎ **Source correction:** The source prints the equation as "hv = ⌀ + K_max" with the symbols φ and K partly dropped in typesetting; re-typed above with the standard symbols. Note the two different things called "photoelectric": the *effect* (this section) and the *cell* (device, below).

★ **PYQ lens:** 64th Q16: Einstein got the Nobel Prize (1921, awarded 1922) for the **photoelectric effect** — *not* for relativity (67th Q71 asks who *presented* the theory of relativity → **Einstein**, 1905 special / 1915 general).

★ **PYQ lens:** 69th Q16: a **photoelectric cell** is a device that **converts light energy into electric energy**. 68th Q113: the photoelectric device most suitable for **digital** applications → **photodiode** (fast on/off switching; a photovoltaic/solar cell generates power, a photo-emitter is the old vacuum phototube).

ADDED BY EDITOR — not in source lectures **Photoelectric devices and uses (the source stops at the equation)**

Photo-emissive cell (vacuum phototube — caesium cathode) — light meters, old sound-track readers in cinema, burglar alarms and automatic door openers (beam interrupted → current stops → relay)

Photovoltaic / solar cell — a silicon p–n junction that produces a voltage under light (≈ 0.5 V per cell, 15–25 % efficiency; solar panels, calculators, satellites — India's solar capacity > 100 GW in 2025)

Photodiode / phototransistor — reverse-biased junction whose current follows the light — fibre-optic receivers, TV remote sensors, barcode scanners, smoke detectors, digital cameras (CCD/CMOS sensors count photons)

Photoconductive cell (LDR, CdS) — resistance falls with light — street lights that switch on at dusk

Photomultiplier tube — detects single photons (scintillation counters, night-vision). **Solar constant** ≈ 1.36 kW/m² (Unit 8; 70th Q113). **Photon momentum** p = h/λ gives **radiation pressure** — solar sails, comet tails pointing away from the Sun.

The photon (Lecture 19 pp. 4–5)

A **photon** is the **quantum (packet) of electromagnetic radiation**. Properties:

- **Energy** E = hν = hc/λ (h = Planck's constant = 6.63 × 10^{-34} J s; a visible photon carries ≈ 2–3 eV, an X-ray photon keV, a gamma photon MeV);
- **Momentum** p = h/λ = E/c (photons have momentum although they have no mass — light exerts pressure);
- **Zero rest mass**; **electrically neutral**; not deflected by electric or magnetic fields;
- travels at the **speed of light** c in vacuum (in every frame); its energy is not changed by refraction (the frequency stays the same — Unit 3), but its speed and wavelength do; photons can be created (emission) and destroyed (absorption); the number of photons is not conserved; in a collision with an electron (Compton effect) the photon loses energy and its wavelength increases.

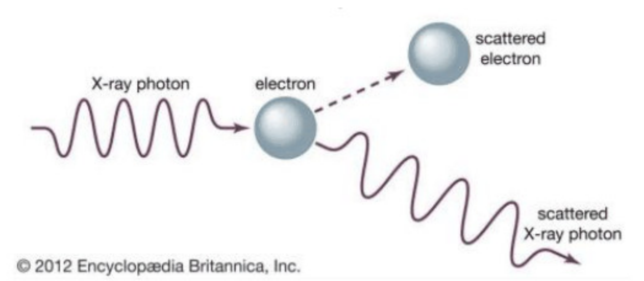

Fig. 16-2 · Compton scattering — an X-ray photon bounces off an electron and emerges with lower energy (longer wavelength) [Lecture 19, p. 5]

✎ **Source correction:** The source prints "Momentum = p = h/" with the λ missing; it is $p = h/\lambda$.

▌ADDED BY EDITOR — not in source lectures ▌ Quantum theory in one paragraph (context BPSC has begun to ask — Thomson, Bohr, Higgs)

Max Planck (1900) — energy is emitted and absorbed in quanta $E = h\nu$ (black-body radiation; Nobel 1918)

Einstein (1905) — light itself consists of photons

Niels Bohr (1913) — quantised electron orbits explain the hydrogen spectrum; **de Broglie (1924)** — matter waves

Heisenberg (1927) — uncertainty principle ($\Delta x \cdot \Delta p \geq h/4\pi$)

Schrödinger — wave equation

Dirac — predicted antimatter (positron found by Anderson, 1932)

Pauli — exclusion principle

S. N. Bose with Einstein — Bose–Einstein statistics (bosons: photon, gluon, W, Z, **Higgs**), and **Fermi**/Dirac — fermions (electron, proton, neutron, quarks, neutrinos). The **Higgs boson** ("God particle", proposed 1964 by Peter Higgs and others) gives mass to fundamental particles; discovered at CERN's **Large Hadron Collider** in **July 2012** (Nobel 2013 — Higgs & Englert) — 60–62nd Q49. **Standard Model**: 6 quarks (up, down, charm, strange, top, bottom), 6 leptons (electron, muon, tau and their neutrinos), force carriers (photon — electromagnetic, gluon — strong, W/Z — weak, graviton — hypothetical for gravity) and the Higgs. Protons and neutrons are each made of three quarks (uud, udd).

Wave–particle duality (Lecture 19 p. 5)

Light behaves both as a **wave** — interference, diffraction, polarisation (Unit 4) — and as a **particle** — photoelectric effect, Compton effect, black-body radiation. Neither picture alone is complete; which aspect shows up depends on the experiment (Bohr's complementarity).

de Broglie hypothesis (matter waves, 1924)

Every moving particle has a wave associated with it — the *matter wave* — with wavelength:

$$\lambda = h / p = h / (m v) \cdot \text{for an electron accelerated through a potential V: } \lambda = h / \sqrt{2 m e V} \approx 12.27 / \sqrt{V} \text{ Å}$$

- The wavelength is significant only for very light particles (electrons, neutrons — $\lambda \approx 1$ Å for a 150 V electron, comparable to atomic spacing, hence **electron diffraction** by crystals — Davisson & Germer 1927, G. P. Thomson); for a cricket ball it is ~10^{-34} m, utterly unobservable. de Broglie's Nobel: 1929.

✎ **Source correction:** The source drops λ from both formulas ("h/p = h/mv", "h/√2meV"); restored above.

▌ADDED BY EDITOR — not in source lectures ▌ Where matter waves matter (context)

The **electron microscope** (Ruska, 1931) uses electron waves ~10^5 times shorter than visible light to resolve viruses, DNA and atoms (TEM/SEM; up to ~10^6 ×, versus ~2000 × for a light microscope); **neutron diffraction** maps crystal and magnetic structures

Bohr's orbits are those in which a whole number of electron wavelengths fit round the circumference ($2\pi r = n\lambda$); electron waves are the basis of the quantum tunnelling used in flash memory and the scanning tunnelling microscope.

The atomic nucleus

▌ADDED BY EDITOR — not in source lectures ▌ The nucleus — composition, notation and the arithmetic BPSC sets every second year (the lecture jumps straight to the decay law)

Discovery: Rutherford's gold-foil (alpha-scattering) experiment (1911, with Geiger & Marsden) — most alphas pass straight through, a few bounce back ⇒ the atom has a tiny, dense, **positively charged nucleus** (63rd Q89) with electrons far outside

Thomson (1897, discovered the electron; 1904 "plum-pudding" model — the **first atomic model**, 70th Q150)

Bohr (1913) put the electrons in quantised orbits

Chadwick (1932) discovered the **neutron**. **Composition**: protons (+e, mass 1.0073 u) and neutrons (neutral, 1.0087 u) — together **nucleons** — held by the short-range **strong nuclear force** (67th Q75); the electron's mass is only 1/1836 of the proton's. **Notation** $_ZX^A$: **Z = atomic number = number of protons** (= electrons in the neutral atom)

A = mass number = protons + neutrons; **neutrons N = A – Z**. 65th Q92: 18 electrons (so Z = 18) and 20 neutrons → **A = 38** (argon-38). 66th Q9: $_{94}Pu^{242}$ → N = 242 – 94 = **148**. 64th Q2: ordinary **hydrogen (protium, $_1H^1$)** is the only atom with **no neutron** (deuterium has 1, tritium 2). **Size**: nuclear radius R ≈ 1.2 × 10^{-15} $A^{1/3}$ m (**fermi**), about 10^{-5} of the atom; nuclear density ≈ 2 × 10^{17} kg/m³ — the same for all nuclei (a teaspoon of nuclear matter ≈ 10^9 tonnes — neutron-star density). **Isotopes** — same Z, different A ($_1H^1$, $_1H^2$, $_1H^3$; C-12, C-14; U-235, U-238) — identical chemistry; **isobars** — same A, different Z (Ar-40, Ca-40); **isotones** — same N. **Mass defect & binding energy**: the nucleus weighs less than its separated nucleons; the difference Δm appears as binding energy **E = Δm c²** (1 u = 931.5 MeV) — the binding energy per nucleon is greatest (~8.8 MeV) around **iron-56**, which is why fusion of light nuclei and fission of heavy ones both release energy. **Atomic mass unit** u = 1/12 of the mass of a C-12 atom = 1.66 × 10^{-27} kg. **Cosmic rays** (56–59th Q140) are high-energy particles from space — mostly **protons and alpha particles (charged)** plus gamma photons and neutrinos (**uncharged**) — "charged as well as uncharged"; discovered by Victor Hess (1912) in balloon flights.

Radioactive decay law (Lecture 19 p. 5)

$$N = N_0 e^{-\lambda t} \cdot \text{half-life } T_{1/2} = \ln 2 / \lambda = 0.693 / \lambda \cdot \text{mean life } \tau = 1/\lambda \ (\lambda = \text{decay constant}; T_{1/2} = 0.693\,\tau)$$

- After n half-lives the undecayed fraction is $(½)^n$: 1 → ½ → ¼ → ⅛ …; after 10 half-lives ≈ 0.1 % remains. Half-lives range from microseconds to billions of years (U-238: 4.5 × 10^9 yr; C-14: 5730 yr; I-131: 8 days; Rn-222: 3.8 days). Decay is random and spontaneous — unaffected by temperature, pressure or chemical state.

✎ **Source correction:** The source prints "N = N_0 e^{-t}", "$T_{1/2}$ = ln2/" and "mean life = 1/" with the decay constant λ dropped; restored above.

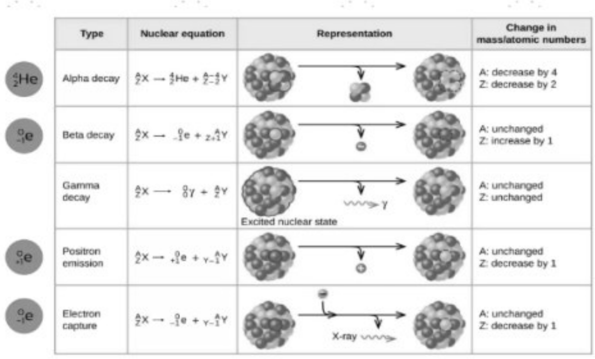

Fig. 16-3 · Types of decay — alpha, beta, gamma, positron emission and electron capture, with the change in mass/atomic numbers [Lecture 19, p. 5]

▌ADDED BY EDITOR — not in source lectures▐ Radioactivity — discovery, the three radiations and their uses (the decay law is all the lecture gives, yet BPSC asked alpha rays, ionising radiation and medical isotopes)

Discovery: **Henri Becquerel** (1896, uranium salts fogging a photographic plate)

Marie & Pierre Curie isolated polonium and radium (1898) — the Curies and Becquerel shared the 1903 Nobel

Marie Curie also won Chemistry 1911 (the only person with Nobels in two sciences)

Rutherford named α and β (1899), Villard found γ (1900). **Alpha (α)** = helium nucleus $_2He^4$, **positive** charge +2e (68th Q116), mass 4 u, speed ~10^7 m/s, **highest ionising power**, **least penetrating** (stopped by paper / skin / a few cm of air), deflected by E and B fields; the parent loses 4 in A and 2 in Z. **Beta (β^-)** = fast **electron** from the nucleus (a neutron → proton + electron + antineutrino), charge –e, speed up to 0.99c, moderate ionisation, stopped by a few mm of aluminium

Z rises by 1, A unchanged; β^+ (positron) emission lowers Z by 1. **Gamma (γ)** = high-energy **photon**, no charge, no mass, travels at c, **least ionising**, **most penetrating** (needs cm of lead / metres of concrete), not deflected by fields

A and Z unchanged — emitted when a nucleus de-excites after α/β decay. **Ionising radiation** (α, β, γ, X-rays, neutrons, UV-C) damages DNA — it can **cause cancer** (Hiroshima, Chernobyl, radon in mines) and, precisely targeted, **cure it** (radiotherapy) — 56–59th Q142. **Units**: activity — becquerel (Bq, 1 decay/s) and curie (1 Ci = 3.7 × 10^{10} Bq); absorbed dose — gray (Gy); biological dose — sievert (Sv) / rem. **Detectors**: Geiger–Müller counter, scintillation counter, cloud chamber (C. T. R. Wilson), photographic badge (dosimeter). **Uses of radio-isotopes**: **Co-60** — external γ-source for cancer radiotherapy and food/instrument sterilisation

I-131 — thyroid disorders (uptake test and therapy)

P-32 — leukaemia and polycythaemia (β-emitter; 60–62nd Q50 — the workbook key said Co-60, standard references give P-32)

Tc-99m — the commonest diagnostic imaging tracer

C-14 — radiocarbon dating of once-living material up to ~50 000 yr (Libby, Nobel 1960)

U-238/Pb-206 and K-40/Ar-40 — dating rocks and the Earth's age (4.5 × 10^9 yr)

Na-24 — blood-circulation studies and leak detection

Am-241 — smoke detectors

Sr-90 — thickness gauges

Cs-137 — industrial radiography

Pu-238 — radioisotope thermoelectric generators on Voyager, Curiosity and Perseverance. **Natural background** ≈ 2–3 mSv/yr (radon, cosmic rays, K-40 in bananas and in us).

Nuclear fission and fusion (Lecture 19 p. 5)

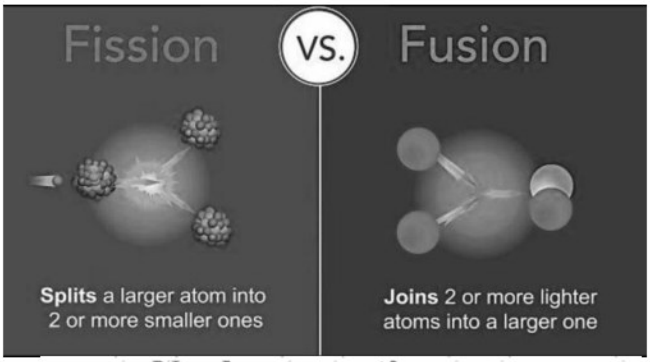

Fig. 16-4 · Fission splits a heavy nucleus into two lighter ones; fusion joins light nuclei into a heavier one [Lecture 19, p. 5]

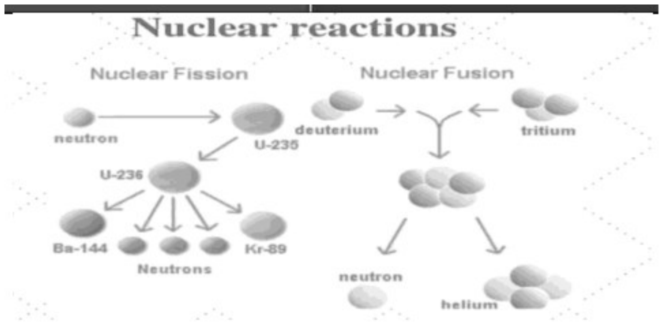

Fig. 16-5 · Nuclear reactions — a neutron splits U-235 into Ba-144 + Kr-89 + 3 neutrons; deuterium + tritium fuse into helium + a neutron [Lecture 19, p. 5]

Nuclear fission

Nuclear fission is the process in which a **heavy nucleus splits** into two or more lighter nuclei, releasing a large amount of energy along with 2–3 neutrons. Splitting of a heavy nucleus (U-235); used in nuclear reactors (controlled) and the atom bomb (uncontrolled).

$$_{92}U^{235} + {_0}n^1 \rightarrow {_{56}}Ba^{144} + {_{36}}Kr^{89} + 3\,{_0}n^1 + \approx 200 \text{ MeV}$$ **(energy comes from the mass defect, $E = \Delta m c^2$)**

▌ADDED BY EDITOR — not in source lectures ▌ Fission — the facts behind the questions

Discovered by **Otto Hahn and Fritz Strassmann** (Dec 1938), explained by **Lise Meitner and Otto Frisch** (1939); first self-sustaining **chain reaction** — **Enrico Fermi**, Chicago Pile-1, 2 Dec 1942. **Chain reaction**: each fission's neutrons cause further fissions — sustained only above the **critical mass** (≈ 52 kg of bare U-235, ≈ 10 kg of Pu-239). **Fissile materials**: **U-235** (only 0.7 % of natural uranium — hence *enrichment*), **Pu-239** (bred from U-238 in reactors) and **U-233** (bred from **thorium-232** — India's three-stage programme, because India has the world's largest thorium reserves in the Kerala/Odisha monazite sands). **Fission per kg of U-235** ≈ 8×10^{13} J ≈ energy of 2500 tonnes of coal. **The Manhattan Project** (1942–45, USA/UK/Canada; scientific director J. Robert Oppenheimer, Los Alamos; General Leslie Groves) produced the first nuclear weapons — 69th Q14

Trinity test, 16 July 1945 (Alamogordo, New Mexico)

Hiroshima, 6 Aug 1945 — "Little Boy", uranium-235

Nagasaki, 9 Aug 1945 — "Fat Man", plutonium-239 (56–59th Q138). **India**: Pokhran-I "Smiling Buddha" 18 May 1974 (Pu device), Pokhran-II 11–13 May 1998 (five tests, incl. a thermonuclear device — 11 May is *National Technology Day*)

Homi J. Bhabha — father of India's nuclear programme (BARC, Trombay; Apsara, 1956 — Asia's first research reactor)

Tarapur (1969) — first power station

Kudankulam (Russian VVER) — largest

NPCIL operates ≈ 8 GW (target 100 GW by 2047)

Kalpakkam PFBR — India's first 500 MW prototype fast-breeder reactor (fuel loading began in 2024), the second stage of the three-stage programme. **Nuclear waste** (spent fuel, high-level waste) stays radioactive for thousands of years — vitrified and stored in deep geological repositories.

Nuclear fusion

Nuclear fusion is the process in which **two light nuclei combine** to form a heavier nucleus, releasing enormous energy (per kg, about 4× that of fission). Combining light nuclei; the **source of energy in stars** (the Sun fuses ~600 million tonnes of hydrogen into helium every second via the proton–proton chain; heavier stars use the CNO cycle — Hans Bethe, Nobel 1967).

$$_1H^2 + {}_1H^3 \rightarrow {}_2He^4 + {}_0n^1 + 17.6\text{ MeV}$$ **(deuterium + tritium — the easiest reaction)**

- Fusion needs temperatures of ~10^7–10^8 K (a **plasma**) so that the positively charged nuclei overcome their Coulomb repulsion — hence "**thermonuclear**" reaction. It cannot yet be sustained on Earth for power, though the uncontrolled version is the **hydrogen bomb** (Ivy Mike 1952; triggered by a fission bomb).

★ **PYQ lens:** 68th Q75: the laboratory that claimed a historic nuclear fusion breakthrough in **December 2022 → Lawrence Livermore National Laboratory** (National Ignition Facility, California) — the first laser-driven fusion shot to produce more energy (3.15 MJ) than the laser energy delivered to the target (2.05 MJ): "**ignition**" / scientific break-even (repeated in 2023–24 with higher yields).

▌ADDED BY EDITOR — not in source lectures ▌ Fusion on Earth — the projects (context)

Tokamak — a doughnut-shaped magnetic bottle (Soviet design; magnetic confinement): **ITER** at Cadarache, France — 35-nation project including **India** (India builds the cryostat; first plasma now planned for the mid-2030s)

JET (UK) set the 69 MJ record in 2024

China's EAST "artificial sun" held plasma for 1066 s in 2025

India's own **Aditya** and **SST-1** tokamaks are at the Institute for Plasma Research, Gandhinagar. **Inertial confinement** (lasers) — NIF (USA), Laser Mégajoule (France). **Stellarator** — Wendelstein 7-X (Germany). Fusion fuel: deuterium from sea-water and tritium bred from lithium; no long-lived waste, no chain-reaction runaway — the "holy grail" of clean energy.

Nuclear reactor (Lecture 19 pp. 5–6)

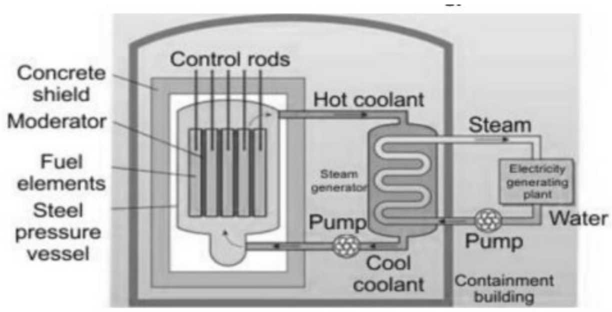

Fig. 16-6 · Nuclear power plant — concrete shield, control rods, moderator, fuel elements, steel pressure vessel, hot coolant → steam generator → turbine, cool coolant returned by pump; containment building [Lecture 19, p. 6]

Part	Function	Materials / examples
Reactor core	Contains the fuel rods (uranium pellets) where fission occurs; surrounded by the moderator	Uranium oxide pellets in zircaloy tubes
Fuel	Undergoes fission	**Enriched uranium (U-235, 3–5 %)**; natural uranium (PHWR/CANDU — India's main fleet); Pu-239 (MOX, fast breeders); **thorium** (can be converted to fissile U-233)
Moderator	**Slows down fast neutrons** to thermal speeds so that they are captured by U-235 and the reaction is sustained	**Heavy water (D_2O)**, **graphite**, **light (ordinary) water**; beryllium
Control rods	**Absorb neutrons** to control the reaction rate — pushed in to slow/shut down, pulled out to speed up	**Boron** (boron carbide / boron steel), **cadmium**; hafnium
Coolant	**Removes heat** from the core and carries it to the steam generator	**Water**, **heavy water**, liquid sodium (fast breeders), CO_2/helium gas, molten salt
Steam generator (heat exchanger)	Transfers the coolant's heat to a secondary water circuit, making steam	Primary/secondary loops keep the radioactive coolant away from the turbine

Part	Function	Materials / examples
Turbine & generator	Steam turns the turbine; the generator converts **mechanical → electrical** energy	Same as in a coal plant
Shielding / containment	Concrete shield stops neutrons and gamma rays; the steel pressure vessel and containment building prevent release of radioactivity	Thick concrete + lead + steel
Reflector	Bounces escaping neutrons back into the core	Graphite, beryllium

↻ **De-dup note:** Lecture 19 splits the reactor list across pp. 5–6 with stray words ("Control", "Steam") left by the page break; re-assembled as one table.

▍ADDED BY EDITOR — not in source lectures▍ Reactor types and accidents (context)

PHWR (pressurised heavy-water, natural uranium — India's 220/700 MW units), **PWR/BWR** (light water, enriched uranium — most of the world, Kudankulam), **Fast Breeder Reactor** (no moderator; breeds Pu-239 from U-238 or U-233 from Th-232; sodium coolant — Kalpakkam), **RBMK** (graphite, Chernobyl), **SMRs** (small modular reactors — India's Bharat Small Reactors plan). **Accidents**: Three Mile Island (USA, 1979), **Chernobyl** (USSR/Ukraine, 26 April 1986 — worst), **Fukushima Daiichi** (Japan, 11 March 2011 — tsunami). A reactor cannot explode like a bomb (fuel is far below weapons-grade); the danger is meltdown and release of I-131, Cs-137, Sr-90. India's regulator: **AERB**; fuel: UCIL mines (Jaduguda, Jharkhand); heavy water: HWB plants (Manuguru, Kota, Talcher…). Nuclear supplies ≈ 3 % of India's electricity, ≈ 9 % of the world's.

Thermionic emission (Lecture 19 p. 6)

Thermionic emission is the emission of electrons from the surface of a metal when it is **heated sufficiently** (Edison effect, 1883; explained by Richardson — Nobel 1928).

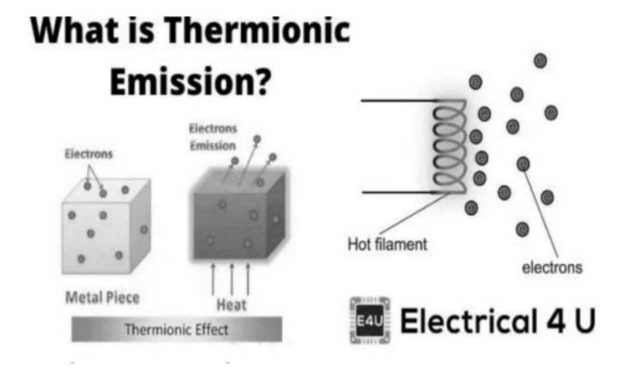

Fig. 16-7 · Thermionic emission — a heated metal / hot filament boils off electrons [Lecture 19, p. 6]

- **Basic principle**: metals contain free electrons; when heated, these electrons gain (thermal) energy; if their energy exceeds the **work function** — the minimum energy required to escape — they leave the metal surface. (The same work function φ as in the photoelectric effect; the energy is supplied by heat instead of light.)
- The emission rises steeply with temperature; cathodes are made of **tungsten** (high melting point) or tungsten coated with **thorium / barium oxide** (lower work function → emission at lower temperature).

▍ADDED BY EDITOR — not in source lectures▍ Where thermionic emission is used, and the four ways to free an electron (context)

Vacuum tubes / valves — the diode (Fleming, 1904) and triode (de Forest, 1906) that ran radio, radar and the first computers (ENIAC had 18 000 valves) before transistors; the **cathode-ray tube** of old TVs and oscilloscopes; the **X-ray tube** (electrons from a hot filament hit a tungsten target — Röntgen 1895, Nobel 1901); the **electron gun** of electron microscopes and CRTs; magnetrons in **microwave ovens** and radar. **Electron emission** can be caused by (1) heat — *thermionic*

(2) light — *photoelectric*

(3) a strong electric field — *field emission* (cold cathode)

(4) bombardment by other electrons — *secondary emission* (photomultipliers).

Work, power and energy — see Unit 10

↻ **De-dup note:** Lecture 19 p. 6 prints "Work, Power and Energy" between thermionic emission and the instrument table. That block (work = F·s cos θ, joule/erg, KE = ½mv², PE = mgh, conservation of energy, P = W/t = Fv, watt/horsepower) has been moved to Unit 10 to sit with Newton's laws and friction, where the 71st-BPSC numericals combine them; the instrument table of p. 7 is in Unit 1.

Relativity and the big picture

▌ADDED BY EDITOR — not in source lectures ▌ Einstein's relativity & mass–energy (asked directly in 67th Q71; the lectures only use E = mc² implicitly)

Special relativity (1905) — the laws of physics are the same in all inertial frames and the **speed of light in vacuum is the same for every observer** ($c = 3 \times 10^8$ m/s is the universal speed limit); consequences: **time dilation** (moving clocks run slow — GPS satellites' clocks must be corrected; muons from cosmic rays reach the ground), **length contraction**, relativity of simultaneity, and **$E = mc^2$** — mass is a form of energy (1 g of matter ≡ 9×10^{13} J ≈ 21 kilotons of TNT; the Sun loses 4 million tonnes per second as light). **General relativity (1915)** — gravity is the curvature of space-time by mass-energy; predicted the bending of starlight (confirmed by Eddington's 1919 eclipse expedition), the precession of Mercury's orbit, gravitational red-shift (clocks run *faster* at altitude — also in GPS), **black holes** (first image, M87-star by the Event Horizon Telescope in 2019; Sagittarius A-star in 2022) and **gravitational waves** (detected by **LIGO**, 14 Sept 2015, announced Feb 2016; Nobel 2017 — Weiss, Barish, Thorne). **LIGO-India** (Hingoli district, **Maharashtra**; approved 2016, sanctioned ₹2600 crore in 2023, target ≈ 2030) — 60–62nd Q71. Einstein's own Nobel (1921) was for the photoelectric effect (64th Q16), not relativity.

▌ADDED BY EDITOR — not in source lectures ▌ Space & astronomy facts that appeared as "physics" questions (extended scope — Appendix A)

First man on the Moon — Neil Armstrong (Apollo 11, 20/21 July 1969, with Buzz Aldrin; Michael Collins stayed in orbit) — 66th Q8; **first human in space** — Yuri Gagarin (12 April 1961); first woman — Valentina Tereshkova (1963)

Rakesh Sharma — first Indian in space (1984, Soyuz T-11)

Kalpana Chawla (1997, 2003 Columbia), Sunita Williams

Shubhanshu Shukla — first Indian on the ISS (Axiom-4, June–July 2025)

Gaganyaan — India's crewed mission (planned 2027). **Pulsars** = rapidly **rotating neutron stars** emitting beamed radio pulses (Jocelyn Bell Burnell, 1967) — 69th Q9; **neutron star** — collapsed core of a massive star (≈ 1.4 solar masses in ~20 km; escape velocity ≈ 2×10^5 km/s — Unit 11 table); **black hole** — escape velocity > c; **supernova** — the explosion; **white dwarf** — the Sun's eventual fate (Chandrasekhar limit 1.4 $M_\odot$ — S. Chandrasekhar, Nobel 1983); **quasar** — an active galactic nucleus. **Space missions** (69th Q2): Cassini–Huygens → **Saturn** and its rings/Titan (1997–2017), Juno → **Jupiter** (2011–), **Artemis** → the **Moon** (Artemis-I uncrewed test flight 2022; Artemis-II the first crewed lunar fly-by of the programme), VERITAS → **Venus** (NASA, ~2031)

ISRO: **Chandrayaan-3** (soft landing near the lunar south pole, 23 Aug 2023 — National Space Day), **Aditya-L1** (solar observatory at L1, Sept 2023), Mangalyaan (2014), **NISAR** (NASA–ISRO radar satellite, launched 30 July 2025), Shukrayaan (Venus, ~2028), Chandrayaan-4 (sample return, ~2027). **Comets** have highly elongated elliptical orbits (70th Q70)

Sun — the chief body of the solar system (67th Q141) — a G-type main-sequence star, 4.6 billion years old, surface 5800 K, core 1.5×10^7 K, energy by fusion; light takes 8 min 20 s to reach us (Unit 2). **James Webb Space Telescope** (2021, infrared, at L2) — deepest views of the early universe

Hubble (1990). The universe is 13.8 billion years old (Big Bang; cosmic microwave background at 2.7 K).

UNIT A

Appendix A — every reviewed physics PYQ (56–59th → 71st BPSC)

Consolidated from: verbatim question text from the supplied workbooks; study answer, status and one-line reason

Status codes: **source-key** = answer follows the verified key in the workbook; **added-answer** = the workbook has no key for that paper (71st BPSC), the answer was worked out by the editor and the working is shown; **editor-corrected** = the workbook key conflicts with standard references and both are shown. Scope **core** = taught in the lectures; **extended** = physics-adjacent context.

56-59th BPSC (2015) — 4 physics questions

Q138 · U16 Modern & Nuclear Physics · **source-key** · core

What was the fissionable material used in the bombs dropped at Nagasaki (Japan) in the year 1945?

(A) Sodium | (B) Potassium | (C) Plutonium | (D) Uranium

Answer: Plutonium — Answer follows the verified answer field of the supplied workbook.

Q139 · U11 Circular Motion & Gravitation · **source-key** · core

The scientist who first discovered that the Earth revolve round the Sun was

(A) Newton | (B) Dalton | (C) Copernicus | (D) Einstein

Answer: Copernicus — Answer follows the verified answer field of the supplied workbook.

Q140 · U16 Modern & Nuclear Physics · **source-key** · core

Cosmic rays

(A) are charged particles | (B) are uncharged particles | (C) can be charged as well as uncharged | (D) None of the above

Answer: can be charged as well as uncharged — Answer follows the verified answer field of the supplied workbook.

Q142 · U16 Modern & Nuclear Physics · **source-key** · extended

Which of the following can cause cancer as well as cure it depending upon its intensity and use?

(A) Tobacco | (B) Alcohol | (C) Ionised radiation | (D) Ultraviolet rays

Answer: Ionised radiation — Answer follows the verified answer field of the supplied workbook.

60-62nd BPSC (Combined) — 3 physics questions

Q49 · U16 Modern & Nuclear Physics · **source-key** · core

God Particle' is

(A) Neutrino | (B) Higgs Boson | (C) Meson | (D) Positron | (E) None of the above/More than one of the above

Answer: Higgs Boson — Answer follows the verified answer field of the supplied workbook.

Q50 · U16 Modern & Nuclear Physics · **editor-corrected** · extended

Which of the following radio isotopes is used in the treatment of blood cancer [leukemia]?

(A) Iodine -131 | (B) Sodium-24 | (C) Phosphorus-32 | (D) Cobalt-60 | (E) None of the above/More than one of the above

Answer: Phosphorus-32 (workbook key: Cobalt-60) — Workbook key says (d) Cobalt-60. P-32 (β-emitter) is the isotope used in leukaemia/polycythaemia therapy; Co-60 is an external γ source. Study answer follows the standard reference.

Q71 · U16 Modern & Nuclear Physics · **source-key** · extended

India's first Laser Interferometer Gravitational-Wave Observatory (LIGO) laboratory will be setup in which State ?

(A) Uttar Pradesh | (B) Maharashtra | (C) Andhra Pradesh | (D) Bihar | (E) None of the above/More than one of the above

Answer: Maharashtra — Answer follows the verified answer field of the supplied workbook.

63rd BPSC — 4 physics questions

Q85 · U2 Light I: Nature of Light, Reflection & Mirrors · **source-key** · core

The wavelength of visible spectrum is in the range

(A) 1300 Å-3900 Å | (B) 3900 Å-7600 Å | (C) 7800 Å-8200 Å | (D) 8500 Å-9800 Å | (E) None of the above/More than one of the above

Answer: 3900 Å-7600 Å — Answer follows the verified answer field of the supplied workbook.

Q89 · U16 Modern & Nuclear Physics · **source-key** · extended

The positively charged part at the centre of an atom is called as

(A) proton | (B) neutron | (C) electron | (D) nucleus | (E) None of the above/More than one of the above

Answer: nucleus — Answer follows the verified answer field of the supplied workbook.

Q90 · U7 Heat, Temperature & Thermometry · **source-key** · core

The conversion of a solid directly into gas is called as

(A) sublimation | (B) condensation | (C) evaporation | (D) boiling | (E) None of the above/More than one of the above

Answer: sublimation — Answer follows the verified answer field of the supplied workbook.

Q99 · U14 Electrical Power, Devices & Domestic Electricity · **source-key** · extended

The acid used in a car battery is

(A) acetic acid | (B) hydrochloric acid | (C) nitric acid | (D) sulfuric acid | (E) None of the above/More than one of the above

Answer: sulfuric acid — Answer follows the verified answer field of the supplied workbook.

64th BPSC — 11 physics questions

Q2 · U16 Modern & Nuclear Physics · **source-key** · extended

Which of the following elements does not contain neutrons?

(A) Oxygen | (B) Nitrogen | (C) Hydrogen | (D) Copper | (E) None of the above/More than one of the above

Answer: Hydrogen — Answer follows the verified answer field of the supplied workbook.

Q3 · U1 Units, Dimensions & Measuring Instruments · **source-key** · core

Angstrom is a unit of

(A) wavelength | (B) energy | (C) frequency | (D) velocity | (E) None of the above/More than one of the above

Answer: wavelength — Answer follows the verified answer field of the supplied workbook.

Q4 · U1 Units, Dimensions & Measuring Instruments · **source-key** · core

Frequency is measured in

(A) hertz | (B) meter/second | (C) radian | (D) watt | (E) None of the above/More than one of the above

Answer: hertz — Answer follows the verified answer field of the supplied workbook.

Q9 · U4 Light III: Dispersion, Scattering, Wave Optics & Instruments · **source-key** · core

Choose the correct statement.

(A) Wavelength of red light is less than violet light. | (B) Wavelength of red light is more than violet light. | (C) Wavelength of violet light is more than green light. | (D) Wavelength of violet light is more than yellow light. | (E) None of the above/More than one of the above

Answer: Wavelength of red light is more than violet light. — Answer follows the verified answer field of the supplied workbook.

Q10 · U7 Heat, Temperature & Thermometry · **source-key** · core

The value of 40 degrees Celsius in Fahrenheit scale is

(A) 104 °F | (B) 100 °F | (C) 102 °F | (D) 75 °F | (E) None of the above/More than one of the above

Answer: 104 °F — Answer follows the verified answer field of the supplied workbook.

Q11 · U14 Electrical Power, Devices & Domestic Electricity · **source-key** · core

The unit of electric power is

(A) ampere | (B) volt | (C) coulomb | (D) watt | (E) None of the above/More than one of the above

Answer: watt — Answer follows the verified answer field of the supplied workbook.

Q12 · U14 Electrical Power, Devices & Domestic Electricity · **source-key** · core

In electrical motor

(A) heat is converted into electrical energy | (B) electrical energy is converted into heat | (C) electrical energy is converted into mechanical energy | (D) mechanical energy is converted into electrical energy | (E) None of the above/More than one of the above

Answer: electrical energy is converted into mechanical energy — Answer follows the verified answer field of the supplied workbook.

Q14 · U14 Electrical Power, Devices & Domestic Electricity · **source-key** · core

The device to measure electric current is

(A) voltmeter | (B) ammeter | (C) voltameter | (D) potentiometer | (E) None of the above/More than one of the above

Answer: ammeter — Answer follows the verified answer field of the supplied workbook.

Q16 · U16 Modern & Nuclear Physics · **source-key** · core

Einstein got the Nobel Prize for

(A) relativity | (B) Bose-Einstein condensation | (C) mass-energy equivalence | (D) photoelectric effect | (E) None of the above/More than one of the above

Answer: photoelectric effect — Answer follows the verified answer field of the supplied workbook.

Q19 · U1 Units, Dimensions & Measuring Instruments · **source-key** · core

Which instrument is used to measure humidity?

(A) Hydrometer | (B) Hygrometer | (C) Pyrometer | (D) Lactometer | (E) None of the above/More than one of the above

Answer: Hygrometer — Answer follows the verified answer field of the supplied workbook.

Q20 · U1 Units, Dimensions & Measuring Instruments · **source-key** · core

What is the unit of pressure?

(A) Newton/sq. meter | (B) Newton-meter | (C) Newton | (D) Newton/meter | (E) None of the above/More than one of the above

Answer: Newton/sq. meter — Answer follows the verified answer field of the supplied workbook.

65th BPSC — 10 physics questions

Q92 · U16 Modern & Nuclear Physics · **source-key** · extended

The number of electrons and neutrons in an element is 18 and 20 respectively. Its mass number is

(A) 22 | (B) 2 | (C) 38 | (D) 20 | (E) None of the above/More than one of the above

Answer: 38 — Answer follows the verified answer field of the supplied workbook.

Q111 · U1 Units, Dimensions & Measuring Instruments · **source-key** · core

The unit of pressure is

(A) kg/cm^2 | (B) kg/cm | (C) kg/mm | (D) kg/cm^3 | (E) None of the above/More than one of the above

Answer: kg/cm^2 — Answer follows the verified answer field of the supplied workbook.

Q112 · U2 Light I: Nature of Light, Reflection & Mirrors · **source-key** · core

The sunlight from the sun to the earth reaches in

(A) 5 minutes | (B) 6 minutes | (C) 8 minutes | (D) 10 minutes | (E) None of the above/More than one of the above

Answer: None of the above/More than one of the above — Workbook key is (E). Light actually takes ≈ 8 min 20 s, which is why the bare '8 minutes' option was not accepted by the key; remember the value as 8 min 20 s (≈ 500 s).

Q113 · U9 Kinematics · **source-key** · core

Which one of the following is a scalar quantity?

(A) Force | (B) Pressure | (C) Velocity | (D) Acceleration | (E) None of the above/More than one of the above

Answer: Pressure — Answer follows the verified answer field of the supplied workbook.

Q114 · U1 Units, Dimensions & Measuring Instruments · **source-key** · core

Which one of the following quantities does not have unit?

(A) Stress | (B) Force | (C) Strain | (D) Pressure | (E) None of the above/More than one of the above

Answer: Strain — Answer follows the verified answer field of the supplied workbook.

Q115 · U6 Sound · **source-key** · core

Sound wave in air is

(A) transverse | (B) longitudinal | (C) electromagnetic | (D) polarized | (E) None of the above/More than one of the above

Answer: longitudinal — Answer follows the verified answer field of the supplied workbook.

Q116 · U7 Heat, Temperature & Thermometry · **source-key** · core

The value of 50 °C in Fahrenheit scale is

(A) 104 °F | (B) 122 °F | (C) 100 °F | (D) 75 °F | (E) None of the above/More than one of the above

Answer: 122 °F — Answer follows the verified answer field of the supplied workbook.

Q117 · U1 Units, Dimensions & Measuring Instruments · **source-key** · core

What is measured in hertz?

(A) Frequency | (B) Energy | (C) Heat | (D) Quality | (E) None of the above/More than one of the above

Answer: Frequency — Answer follows the verified answer field of the supplied workbook.

Q119 · U14 Electrical Power, Devices & Domestic Electricity · **source-key** · core

The device to measure electric current is

(A) voltmeter | (B) ammeter | (C) voltameter | (D) potentiometer | (E) None of the above/More than one of the above

Answer: ammeter — Answer follows the verified answer field of the supplied workbook.

Q120 · U14 Electrical Power, Devices & Domestic Electricity · **source-key** · core

The unit of electric power is

(A) ampere | (B) volt | (C) coulomb | (D) watt | (E) None of the above/More than one of the above

Answer: watt — Answer follows the verified answer field of the supplied workbook.

66th BPSC — 11 physics questions

Q1 · U5 Waves & the Electromagnetic Spectrum · **source-key** · core

The radiations used in the treatment of muscle ache are

(A) infrared | (B) microwave | (C) UV | (D) X-ray | (E) None of the above/More than one of the above

Answer: infrared — Answer follows the verified answer field of the supplied workbook.

Q2 · U13 Electric Charge, Field & Current Electricity · **source-key** · core

The total resistance of a circuit having two parallel resistors is 1·403 kilo-ohm. If one of the resistors is 2·0 kilo-ohm, then the other resistor will be

(A) 1·403 kilo-ohm | (B) 2·0 kilo-ohm | (C) 3·403 kilo-ohm | (D) 4·70 kilo-ohm | (E) None of the above/More than one of the above

Answer: 4·70 kilo-ohm — Answer follows the verified answer field of the supplied workbook.

Q3 · U13 Electric Charge, Field & Current Electricity · **source-key** · core

On heating, the resistance of a semiconductor

(A) increases | (B) decreases | (C) remains same | (D) first increases and then decreases | (E) None of the above/More than one of the above

Answer: decreases — Answer follows the verified answer field of the supplied workbook.

Q4 · U14 Electrical Power, Devices & Domestic Electricity · **source-key** · core

Faraday constant

(A) depends on the amount of the electrolyte | (B) depends on the current passed in the electrolyte | (C) depends on the volume of the solvent in which the electrolyte is dissolved | (D) is a universal constant | (E) None of the above/More than one of the above

Answer: is a universal constant — Answer follows the verified answer field of the supplied workbook.

Q5 · U1 Units, Dimensions & Measuring Instruments · **source-key** · core

Light-year' is a unit of

(A) time | (B) speed | (C) distance | (D) intensity of light | (E) None of the above/More than one of the above

Answer: distance — Answer follows the verified answer field of the supplied workbook.

Q6 · U3 Light II: Refraction, Lenses & the Human Eye · **source-key** · core

Which of the following does not change when light travels from one medium to another?

(A) Velocity | (B) Wavelength | (C) Frequency | (D) Refractive index | (E) None of the above/More than one of the above

Answer: Frequency — Answer follows the verified answer field of the supplied workbook.

Q7 · U5 Waves & the Electromagnetic Spectrum · **source-key** · core

The velocity of electromagnetic waves is

(A) 3×10^8 ms^{-1} | (B) 3×10^7 ms^{-1} | (C) 3×10^6 ms^{-1} | (D) 3×10^5 ms^{-1} | (E) None of the above/More than one of the above

Answer: 3×10^8 ms^{-1} — Answer follows the verified answer field of the supplied workbook.

Q8 · Space & Astronomy (extended only) · **source-key** · extended

The first man who placed his foot on the moon is

(A) Leonov | (B) Neil Armstrong | (C) Michael Collins | (D) James Van Allen | (E) None of the above/More than one of the above

Answer: Neil Armstrong — Answer follows the verified answer field of the supplied workbook.

Q9 · U16 Modern & Nuclear Physics · **source-key** · core

The number of neutrons in the nucleus of plutonium nuclide ($_{94}Pu^{242}$) is

(A) 94 | (B) 148 | (C) 242 | (D) 336 | (E) None of the above/More than one of the above

Answer: 148 — Answer follows the verified answer field of the supplied workbook.

Q10 · U12 Mechanical Properties of Fluids · **source-key** · core

The highest viscosity among the following is of

(A) water | (B) air | (C) blood | (D) honey | (E) None of the above/More than one of the above

Answer: honey — Answer follows the verified answer field of the supplied workbook.

Q20 · U7 Heat, Temperature & Thermometry · **source-key** · core

The poorest conductor of heat among the following is

(A) copper | (B) lead | (C) mercury | (D) zinc | (E) None of the above/More than one of the above

Answer: lead — Answer follows the verified answer field of the supplied workbook.

67th BPSC — 11 physics questions

Q65 · U11 Circular Motion & Gravitation · **source-key** · core

The working principle of a washing machine is

(A) centrifugation | (B) dialysis | (C) reverse osmosis | (D) diffusion | (E) None of the above/More than one of the above

Answer: centrifugation — Answer follows the verified answer field of the supplied workbook.

Q66 · U3 Light II: Refraction, Lenses & the Human Eye · **source-key** · core

The speed of light will be minimum while passing through

(A) air | (B) glass | (C) water | (D) vacuum | (E) None of the above/More than one of the above

Answer: glass — Answer follows the verified answer field of the supplied workbook.

Q67 · U9 Kinematics · **source-key** · core

Which of the following is not a vector quantity?

(A) Torque | (B) Displacement | (C) Speed | (D) Velocity | (E) None of the above/More than one of the above

Answer: Speed — Answer follows the verified answer field of the supplied workbook.

Q68 · U11 Circular Motion & Gravitation · **source-key** · core

If the spinning speed of the earth increases, then the weight of the body at the equator will

(A) remain same | (B) be doubled | (C) increase | (D) decrease | (E) None of the above/More than one of the above

Answer: decrease — Answer follows the verified answer field of the supplied workbook.

Q69 · U9 Kinematics · **source-key** · extended

Who is the first person to define speed?

(A) Kepler | (B) Ptolemy | (C) Galileo | (D) Newton | (E) None of the above/More than one of the above

Answer: Galileo — Answer follows the verified answer field of the supplied workbook.

Q71 · U16 Modern & Nuclear Physics · **source-key** · extended

The 'theory of relativity' is presented by which scientist?

(A) Stephen Hawking | (B) Marie Curie | (C) Albert Einstein | (D) Isaac Newton | (E) None of the above/More than one of the above

Answer: Albert Einstein — Answer follows the verified answer field of the supplied workbook.

Q72 · U8 Thermal Expansion, Radiation Laws & Thermodynamics · **source-key** · core

Due to temperature variation along conductor, potential variation occurs along it. This phenomenon is known as

(A) Seebeck effect | (B) Peltier effect | (C) Thomson effect | (D) Joule effect | (E) None of the above/More than one of the above

Answer: Thomson effect — Answer follows the verified answer field of the supplied workbook.

Q73 · U3 Light II: Refraction, Lenses & the Human Eye · **source-key** · core

What type of lens is used in magnifying glass?

(A) Convex lens | (B) Convex mirror | (C) Concave lens | (D) Plano-concave lens | (E) None of the above/More than one of the above

Answer: Convex lens — Answer follows the verified answer field of the supplied workbook.

Q74 · U15 Magnetism & Electromagnetism · **source-key** · core

The paramagnetic theory of magnetism applies to

(A) iron | (B) platinum | (C) nickel | (D) mercury | (E) None of the above/More than one of the above

Answer: platinum — Answer follows the verified answer field of the supplied workbook.

Q75 · U16 Modern & Nuclear Physics · **source-key** · extended

The nucleus of an atom consists of

(A) protons and neutrons | (B) electrons only | (C) electrons and neutrons | (D) electrons and protons | (E) None of the above/More than one of the above

Answer: protons and neutrons — Answer follows the verified answer field of the supplied workbook.

Q141 · U11 Circular Motion & Gravitation · **source-key** · extended

Which is the chief heavenly body of solar system?

(A) Saturn | (B) Sun | (C) Earth | (D) Jupiter | (E) None of the above/More than one of the above

Answer: Sun — Answer follows the verified answer field of the supplied workbook.

68th BPSC — 11 physics questions

Q75 · U16 Modern & Nuclear Physics · **source-key** · extended

Which of the following laboratories claimed a historic nuclear fusion breakthrough in December 2022?

(A) Lawrence Livermore National Laboratory | (B) Oak Ridge National Laboratory | (C) Los Alamos National Laboratory | (D) More than one of the above | (E) None of the above

Answer: Lawrence Livermore National Laboratory — Answer follows the verified answer field of the supplied workbook.

Q111 · U3 Light II: Refraction, Lenses & the Human Eye · **source-key** · core

Large number of thin strips of black paint are made on the surface of a convex lens of focal length 20 cm to catch the image of a white horse. The image will be

(A) a zebra of black stripes | (B) a horse of black stripes | (C) a horse of less brightness | (D) More than one of the above | (E) None of the above

Answer: a horse of less brightness — Answer follows the verified answer field of the supplied workbook.

Q112 · U6 Sound · **source-key** · core

Shrillness of sound is determined by

(A) amplitude of sound | (B) wavelength of sound | (C) velocity of sound | (D) More than one of the above | (E) None of the above

Answer: None of the above — Key is (E): shrillness (pitch) is decided by frequency, which is not among options A–C.

Q113 · U16 Modern & Nuclear Physics · **source-key** · core

Which of the following photoelectric devices is most suitable for digital applications?

(A) Photovoltaic cell | (B) Photoemitter | (C) Photodiode | (D) More than one of the above | (E) None of the above

Answer: Photodiode — Answer follows the verified answer field of the supplied workbook.

Q114 · U10 Laws of Motion, Friction, Work-Power-Energy · **source-key** · core

Ball bearings are used to convert static friction into

(A) drag | (B) sliding friction | (C) rolling friction | (D) More than one of the above | (E) None of the above

Answer: rolling friction — Answer follows the verified answer field of the supplied workbook.

Q115 · U10 Laws of Motion, Friction, Work-Power-Energy · **source-key** · core

A goalkeeper in a game of football pulls his hands backwards after holding the ball shot at the goal. This enables the goalkeeper to

(A) exert large force on the ball | (B) increase the force exerted by the ball on hands | (C) decrease the rate of change of momentum | (D) More than one of the above | (E) None of the above

Answer: decrease the rate of change of momentum — Answer follows the verified answer field of the supplied workbook.

Q116 · U16 Modern & Nuclear Physics · **source-key** · core

Which among the following is a positively charged particle emitted by a radioactive element?

(A) Beta ray | (B) Alpha ray | (C) Cathode ray | (D) More than one of the above | (E) None of the above

Answer: Alpha ray — Answer follows the verified answer field of the supplied workbook.

Q117 · U11 Circular Motion & Gravitation · **source-key** · core

Centripetal force is responsible to

(A) keep the body moving along the circular path | (B) fly the object along a straight line | (C) independent motion of the object in space | (D) More than one of the above | (E) None of the above

Answer: keep the body moving along the circular path — Answer follows the verified answer field of the supplied workbook.

Q118 · U10 Laws of Motion, Friction, Work-Power-Energy · **source-key** · core

Which of the following energy changes involves frictional force?

(A) Potential energy to sound energy | (B) Chemical energy to heat energy | (C) Kinetic energy to heat energy | (D) More than one of the above | (E) None of the above

Answer: Kinetic energy to heat energy — Answer follows the verified answer field of the supplied workbook.

Q119 · U10 Laws of Motion, Friction, Work-Power-Energy · **source-key** · core

A bus is moving along a straight path and takes a sharp turn to the right side suddenly. The passengers sitting in the bus will

(A) fall in the forward direction | (B) bent towards left side | (C) bent towards right side | (D) More than one of the above | (E) None of the above

Answer: bent towards left side — Answer follows the verified answer field of the supplied workbook.

Q120 · U6 Sound · **source-key** · core

Before playing the orchestra in a musical concert, a sitarist tries to adjust the tension and pluck the string suitably. By doing so, he/she is adjusting

(A) amplitude of sound | (B) intensity of sound | (C) frequency of the sitar string with the frequency of other musical instruments | (D) More than one of the above | (E) None of the above

Answer: frequency of the sitar string with the frequency of other musical instruments — Answer follows the verified answer field of the supplied workbook.

69th BPSC — 10 physics questions

Q2 · Space & Astronomy (extended only) · **source-key** · extended

Match List–I with List–II : (Space Mission) a. Cassini-Huygens b. Juno c. Artemis d. VERITAS (Exploration) 1. Jupiter 2. Saturn and its rings 3. Venus 4. Human Spaceflight—Moon to Mars

(A) a-2, b-1, c-4, d-3 | (B) a-3, b-1, c-4, d-2 | (C) a-2, b-3, c-4, d-1 | (D) a-3, b-1, c-2, d-4

Answer: a-2, b-1, c-4, d-3 — Answer follows the verified answer field of the supplied workbook.

Q9 · Space & Astronomy (extended only) · **source-key** · extended

In the universe, what are pulsars?

(A) A group of stars | (B) Rotating neutron stars | (C) Explosion of a star | (D) Radio waves emitted by a star

Answer: Rotating neutron stars — Answer follows the verified answer field of the supplied workbook.

Q14 · U16 Modern & Nuclear Physics · **source-key** · extended

What is the ‘Manhattan Project’?

(A) A research and development undertaking that produced the first nuclear weapons | (B) One of the largest art auctions of the world | (C) A real estate project in New York City | (D) A famous theme park

Answer: A research and development undertaking that produced the first nuclear weapons — Answer follows the verified answer field of the supplied workbook.

Q15 · U2 Light I: Nature of Light, Reflection & Mirrors · **source-key** · core

The image formed by concave mirror is real, inverted and of the same size as that of the object. The position of the object should be

(A) at the focus | (B) at the centre of curvature | (C) between the focus and centre of curvature | (D) beyond the centre of curvature

Answer: at the centre of curvature — Answer follows the verified answer field of the supplied workbook.

Q16 · U16 Modern & Nuclear Physics · **source-key** · core

A photoelectric cell is a device which

(A) converts light energy into electric energy | (B) converts electric energy into light energy | (C) stores light energy | (D) None of the above

Answer: converts light energy into electric energy — Answer follows the verified answer field of the supplied workbook.

Q19 · U13 Electric Charge, Field & Current Electricity · **source-key** · core

Which of the following liquids is a bad conductor of electricity?

(A) Salted water | (B) Orange juice | (C) Lemon juice | (D) None of the above

Answer: None of the above — Key is (D): salted water, orange juice and lemon juice are all electrolytes, so none is a bad conductor.

Q21 · U11 Circular Motion & Gravitation · **source-key** · core

Two objects of different masses falling freely near the surface of the Moon would

(A) have different accelerations | (B) undergo a change in their inertia | (C) have same velocity at any instant | (D) experience forces of same magnitude

Answer: have same velocity at any instant — Answer follows the verified answer field of the supplied workbook.

Q26 · U15 Magnetism & Electromagnetism · **source-key** · core

An AC current can be produced by

(A) choke coil | (B) dynamo | (C) transformer | (D) None of the above

Answer: dynamo — Answer follows the verified answer field of the supplied workbook.

Q27 · U13 Electric Charge, Field & Current Electricity · **source-key** · core

Current density is

(A) a scalar quantity | (B) a vector quantity | (C) dimensionless | (D) None of the above

Answer: a vector quantity — Answer follows the verified answer field of the supplied workbook.

Q40 · U2 Light I: Nature of Light, Reflection & Mirrors · **source-key** · core

Match List–I with List–II : a. Magenta b. Teal c. Mauve d. Cyan 1. Green and blue 2. Red and blue 3. Blue, green and white 4. Blue, red and white

(A) a-2, b-3, c-4, d-1 | (B) a-2, b-4, c-3, d-1 | (C) a-4, b-2, c-1, d-3 | (D) a-3, b-4, c-2, d-1

Answer: a-2, b-3, c-4, d-1 — Answer follows the verified answer field of the supplied workbook.

70th BPSC — 10 physics questions

Q6 · U1 Units, Dimensions & Measuring Instruments · **source-key** · core

The meter that is used to measure the distance moved by the vehicle is known as

(A) Ammeter | (B) Speedometer | (C) Chronometer | (D) Odometer

Answer: Odometer — Answer follows the verified answer field of the supplied workbook.

Q15 · U7 Heat, Temperature & Thermometry · **source-key** · core

Heat transfer that does not require a medium is called

(A) Radiation | (B) Reflection | (C) Conduction | (D) Convection

Answer: Radiation — Answer follows the verified answer field of the supplied workbook.

Q20 · U7 Heat, Temperature & Thermometry · **source-key** · extended

Which of the following represent the suitable condition for the liquefaction of gases?

(A) Low temperature, high pressure | (B) Low temperature, low pressure | (C) High temperature, high pressure | (D) High temperature, low pressure

Answer: Low temperature, high pressure — Answer follows the verified answer field of the supplied workbook.

Q45 · U5 Waves & the Electromagnetic Spectrum · **source-key** · core

Which of the following radiation is used to get relief from body aches?

(A) Infra-red radiation | (B) UV radiation | (C) Visible radiation | (D) None of these

Answer: Infra-red radiation — Answer follows the verified answer field of the supplied workbook.

Q70 · U11 Circular Motion & Gravitation · **source-key** · extended

Which of the following has highly elongated elliptical orbit?

(A) Asteroid | (B) Comet | (C) Meteorite | (D) Meteor

Answer: Comet — Answer follows the verified answer field of the supplied workbook.

Q85 · U10 Laws of Motion, Friction, Work-Power-Energy · **source-key** · core

Which of the following is a non-contact force?

(A) Magnetic force | (B) Frictional force | (C) Impact force | (D) None of these

Answer: Magnetic force — Answer follows the verified answer field of the supplied workbook.

Q104 · U13 Electric Charge, Field & Current Electricity · **source-key** · core

Which of the following is the resistance of the wire?

(A) R = IV | (B) R = 1 / 2V | (C) R = V / I | (D) R = I / V

Answer: R = V / I — Answer follows the verified answer field of the supplied workbook.

Q113 · U8 Thermal Expansion, Radiation Laws & Thermodynamics · **source-key** · core

Which of the following is the value of solar constant?

(A) 1.6 kW/m^2 | (B) 1.4 kW/m^2 | (C) 1.2 kW/m^2 | (D) 1.8 kW/m^2

Answer: 1.4 kW/m^2 — Answer follows the verified answer field of the supplied workbook.

Q123 · U15 Magnetism & Electromagnetism · **source-key** · core

Which of the following is equivalent to tesla?

(A) Ampere per Newton | (B) Newton per Coulomb | (C) Newton per ampere-second | (D) Newton per ampere meter

Answer: Newton per ampere meter — Answer follows the verified answer field of the supplied workbook.

Q150 · U16 Modern & Nuclear Physics · **source-key** · extended

Who was the first one to propose a model for the structure of an atom?

(A) E. Goldstein | (B) Rutherford | (C) Neils Bohr | (D) J. J. Thomson

Answer: J. J. Thomson — Answer follows the verified answer field of the supplied workbook.

71st BPSC (2025) — 10 physics questions

Q121 · U10 Laws of Motion, Friction, Work-Power-Energy · **added-answer** · core

We apply a force of 200 Newtons on a wooden Box and push it on the floor at constant velocity. The marginal friction force will be:

(A) 100 Newtons | (B) 200 Newtons | (C) 300 Newtons | (D) 400 Newtons

Answer: 200 Newtons — No key in workbook. Constant velocity ⇒ net force zero ⇒ friction balances the 200 N push.

Q122 · U10 Laws of Motion, Friction, Work-Power-Energy · **added-answer** · core

A truck starts from rest down a hill with a constant acceleration. It achieves 400 meters in 20 seconds. If the weight of the truck is 7 tons then what will be the force acting on it?

(A) 11000 Newtons | (B) 12000 Newtons | (C) 13000 Newtons | (D) 14000 Newtons

Answer: 14000 Newtons — No key in workbook. $s = \frac{1}{2}at^2 \Rightarrow a = 2\times400/20^2 = 2\ m/s^2$; $F = ma = 7000\ kg \times 2 = 14\ 000\ N$.

Q123 · U14 Electrical Power, Devices & Domestic Electricity · **added-answer** · core

Energy consumed in a home is 250 units then the total energy in Joule will be:

(A) $9x10^5$ | (B) $8x10^6$ | (C) $9x10^8$ | (D) 10^5

Answer: $9x10^8$ — No key in workbook. 1 unit = 1 kWh = 3.6×10^6 J ⇒ 250 units = 9×10^8 J.

Q124 · U6 Sound · **added-answer** · core

In which medium, the speed of sound is maximum?

(A) Steel | (B) Water | (C) Air | (D) Hydrogen

Answer: Steel — No key in workbook. v(solid) > v(liquid) > v(gas); steel ≈ 5,000–6,000 m/s.

Q125 · U10 Laws of Motion, Friction, Work-Power-Energy · **added-answer** · core

The potential energy of a freely falling body continuously decreases.

(A) The principle of conservation of energy is violated | (B) The principle of conservation of energy is not violated | (C) Gravitational force is violated | (D) Gravitational force is not violated

Answer: The principle of conservation of energy is not violated — No key in workbook. Lost PE appears as KE; total mechanical energy is conserved.

Q126 · U10 Laws of Motion, Friction, Work-Power-Energy · **added-answer** · core

A pair of Oxen exerts a force of 140 newton while ploughing the field. The field ploughed is 15 meter long. The work done in ploughing the length of the field is:

(A) 1900 Joule | (B) 2000 Joule | (C) 2100 Joule | (D) 2200 Joule

Answer: 2100 Joule — No key in workbook. $W = F \cdot s = 140\ N \times 15\ m = 2100\ J$.

Q127 · U2 Light I: Nature of Light, Reflection & Mirrors · **added-answer** · core

The focal distance of a plane mirror is

(A) #ERROR! | (B) -1 cm | (C) 2 cm | (D) Infinity

Answer: Infinity — No key in workbook (option A is garbled in the source). Plane mirror: $R = \infty \Rightarrow f = R/2 = \infty$; power zero.

Q128 · U14 Electrical Power, Devices & Domestic Electricity · **added-answer** · core

Which term in the following does not denote the electric power in electric circuit?

(A) I^2R | (B) IR^2 | (C) VI | (D) V^2/R

Answer: IR^2 — No key in workbook. $P = VI = I^2R = V^2/R$; IR^2 is not a power expression.

Q129 · U14 Electrical Power, Devices & Domestic Electricity · **added-answer** · core

In short circuit, the value of electric current in a time circuit:

(A) Very low | (B) Do not change | (C) Increases very high | (D) Continuously changing

Answer: Increases very high — No key in workbook. Short circuit ⇒ resistance ≈ 0 ⇒ $I = V/R$ becomes very large (fuse melts).

Q130 · U14 Electrical Power, Devices & Domestic Electricity · **added-answer** · core

What is the device to protect equipments from the electric shock?

(A) Fuse | (B) Generator | (C) Motor | (D) Current controller

Answer: Fuse — No key in workbook. Among the options the fuse is the protective device (strictly, earthing protects people from shock; the fuse protects equipment from over-current).